Egyptian Baccalaureate Mathematics

Student Edition • Open Workbook / Guided Practice • Part1

Eng Eslam Ahmed | 01120009622

Assistant Lecturer • Faculty of Engineering • Cairo University

Concept 1 | Binomial and Multinomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Binomial

$$\binom{n}{r}a^r b^{n-r}$$

Multinomial

$$\binom{n}{r}\binom{n-r}{s}a^r b^s c^{n-r-s}$$

Exponent check

$$r + (n - r) = n$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Binomial and Multinomial Theorems**

Q001 **Challenge**

For $(2x-3)^5$, find the coefficient of x^3 and the sum of all coefficients without writing the full expansion:

Solution

1. Use the binomial term rule: $q = 3$ and $r = 2$, so the coefficient is 5C2 times 72.
2. For the sum of coefficients, evaluate the expression at $x = 1$: -1.

Correct option: A

Final answer

Coefficient = 720; sum = -1.

Worked Solutions **Binomial and Multinomial Theorems**

Q002 **Challenge**

Find the coefficient of x^2y^3z in $(x+y+z)^6$, then find the coefficient of x^2yz^3 in $(x-2y+z)^6$:

Solution

1. The first exponents are (2,3,1), so use 6!/(2!3!1!) = 60.
2. The second exponents are (2,1,3); the factor (-2) is raised to the first power of y, giving −120.

Correct option: A

Final answer

First coefficient = 60; second coefficient = -120.

Worked Solutions **Binomial and Multinomial Theorems**

Q003 **Challenge**

For $\left(x^2+x+1\right)^4$, find the greatest coefficient and verify it by identifying the coefficient of x^4:

Solution

1. Expand and collect like powers: the central coefficient is 19.
2. The coefficient of x^4 is also 19; it is therefore the greatest.

Correct option: A

Final answer

Greatest coefficient = 19.

Worked Solutions **Binomial and Multinomial Theorems**

Q004 Written

Expand $(x+2)^5$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+2)^5$=$x^5+10x^4+40x^3+80x^2+80x+32$.

Final answer

$x^5+10x^4+40x^3+80x^2+80x+32$

Q005 MCQ

Expand $(3x-2)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x-2)^4$=$81x^4-216x^3+216x^2-96x+16$.

Correct option: D

Final answer

$81x^4-216x^3+216x^2-96x+16$

Worked Solutions **Binomial and Multinomial Theorems**

Q006 Written

Expand $(x+2)^6$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+2)^6$=$x^6+12x^5+60x^4+160x^3+240x^2+192x+64$.

Final answer

$x^6+12x^5+60x^4+160x^3+240x^2+192x+64$

Q007 MCQ

Expand $(x+1)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+1)^4$=$x^4+4x^3+6x^2+4x+1$.

Correct option: B

Final answer

$x^4+4x^3+6x^2+4x+1$

Worked Solutions **Binomial and Multinomial Theorems**

Q008 MCQ

Expand $(a+b)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$.

Correct option: C

Final answer

$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

Q009 Written

Expand $(a-2b)^4$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a-2b)^4 = a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$.

Final answer

$a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$

Worked Solutions **Binomial and Multinomial Theorems**

Q010 MCQ

Expand $(x-2)^5$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x-2)^5 = x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$.

Correct option: A

Final answer

$x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$

Q011 Written

Expand $(3x+2y)^5$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x+2y)^5 = 243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$.

Final answer

$243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$

Worked Solutions **Binomial and Multinomial Theorems**

Q012 Written

Expand $(2a^2 - 1)^5$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a^2 - 1)^5 = 32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$.

Final answer

$32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$

Q013 Written

Expand $(x - 3)^6$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 3)^6 = x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$.

Final answer

$x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$

Worked Solutions **Binomial and Multinomial Theorems**

Q014 **Written**

Expand $(2x - 3y)^6$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x - 3y)^6 = 64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$.

Final answer

$64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$

Q015 **Written**

Expand $\left(x + \frac{1}{3}\right)^6$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $\left(x + \frac{1}{3}\right)^6 = x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$.

Final answer

$$x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$$

Worked Solutions **Binomial and Multinomial Theorems**

Q016 **MCQ**

Expand $(x+3)^3$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+3)^3 = x^3 + 9x^2 + 27x + 27$.

Correct option: B

Final answer

$x^3 + 9x^2 + 27x + 27$

Worked Solutions **Binomial and Multinomial Theorems**

Q017 **MCQ**

Expand $(a+2b)^5$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a+2b)^5$=$a^5+10a^4b+40a^3b^2+80a^2b^3+80ab^4+32b^5$.

Correct option: D

Final answer

$a^5+10a^4b+40a^3b^2+80a^2b^3+80ab^4+32b^5$

Worked Solutions **Binomial and Multinomial Theorems**

Q018 MCQ

Expand $(x+2)^2$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+2)^2 = x^2 + 4x + 4$.

Correct option: D

Final answer

$x^2 + 4x + 4$

Q019 MCQ

Expand $(a-b)^3$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$.

Correct option: A

Final answer

$a^3 - 3a^2b + 3ab^2 - b^3$

Worked Solutions **Binomial and Multinomial Theorems**

Q020 MCQ

Expand $(2x+1)^5$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x+1)^5 = 32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$.

Correct option: B

Final answer

$32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$

Q021 Written

Expand $(3a-2b)^5$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3a-2b)^5 = 243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$.

Final answer

$243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$

Worked Solutions **Binomial and Multinomial Theorems**

Q022 MCQ

Expand $(2x-1)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x-1)^4$=$16x^4 - 32x^3 + 24x^2 - 8x + 1$.

Correct option: C

Final answer

$16x^4 - 32x^3 + 24x^2 - 8x + 1$

Q023 MCQ

Expand $(x-3)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x-3)^4$=$x^4 - 12x^3 + 54x^2 - 108x + 81$.

Correct option: B

Final answer

$x^4 - 12x^3 + 54x^2 - 108x + 81$

Worked Solutions **Binomial and Multinomial Theorems**

Q024 MCQ

Expand $(2a+b)^3$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a+b)^3$=$8a^3+12a^2b+6ab^2+b^3$.

Correct option: C

Final answer

$8a^3+12a^2b+6ab^2+b^3$

Q025 MCQ

Expand $\left(2a^2+3\right)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $\left(2a^2+3\right)^4$=$16a^8+96a^6+216a^4+216a^2+81$.

Correct option: C

Final answer

$16a^8+96a^6+216a^4+216a^2+81$

Worked Solutions **Binomial and Multinomial Theorems**

Q026 MCQ

Expand $\left(3a^2-2\right)^3$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $\left(3a^2-2\right)^3$=$27a^6-54a^4+36a^2-8$.

Correct option: B

Final answer

$27a^6-54a^4+36a^2-8$

Q027 Written

Find the coefficient of a^3b^2 in the expansion of $(a+b)^5$.

Solution

1. The target power uses $q=3$ copies of the first term, so $r=n-q=2$.
2. Use 5C2 times the first factor to power q and the second factor to power r.
3. The required coefficient is 10.

Final answer

10

Worked Solutions **Binomial and Multinomial Theorems**

Q028 Written

Find the coefficient of x^4y^2 in the expansion of $(x-2y)^6$.

Solution

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 2$.
2. Use 6C2 times the first factor to power q and the second factor to power r.
3. The required coefficient is 60.

Final answer

60

Q029 Written

Find the coefficient of x^4 in the expansion of $\left(2x^2-3\right)^7$.

Solution

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 5$.
2. Use 7C5 times the first factor to power q and the second factor to power r.
3. The required coefficient is -20412.

Final answer

-20412

Worked Solutions **Binomial and Multinomial Theorems**

Q030 Written

Find the coefficient of x^6y in the expansion of $(2x-3y)^7$.

Solution

1. The target power uses $q=6$ copies of the first term, so $r=n-q=1$.
2. Use 7C1 times the first factor to power q and the second factor to power r.
3. The required coefficient is -1344.

Final answer

-1344

Q031 MCQ

Find the coefficient of x^2 in the expansion of $(3x-2)^5$:

Solution

1. The target power uses $q=2$ copies of the first term, so $r=n-q=3$.
2. Use 5C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is -720.

Correct option: B

Final answer

-720

Worked Solutions **Binomial and Multinomial Theorems**

Q032 Written

Find the coefficient of a^6 in the expansion of $\left(3a^2-2\right)^8$.

Solution

1. The target power uses $q=3$ copies of the first term, so $r=n-q=5$.
2. Use 8C5 times the first factor to power q and the second factor to power r.
3. The required coefficient is -48384.

Final answer

-48384

Q033 MCQ

Find the coefficient of x^4y^4 in the expansion of $(2x-y)^8$:

Solution

1. The target power uses $q=4$ copies of the first term, so $r=n-q=4$.
2. Use 8C4 times the first factor to power q and the second factor to power r.
3. The required coefficient is 1120.

Correct option: A

Final answer

1120

Worked Solutions **Binomial and Multinomial Theorems**

Q034 MCQ

Find the coefficient of ab^3 in the expansion of $(3a+b)^4$:

Solution

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use 4C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is 12.

Correct option: B

Final answer

12

Q035 MCQ

Find the coefficient of x^3y^2 in the expansion of $(x-4y)^5$:

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C2 times the first factor to power q and the second factor to power r.
3. The required coefficient is 160.

Correct option: C

Final answer

160

Worked Solutions **Binomial and Multinomial Theorems**

Q036 MCQ

Find the coefficient of x^4 in the expansion of $(3x+1)^5$:

Solution

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 1$.
2. Use 5C1 times the first factor to power q and the second factor to power r.
3. The required coefficient is 405.

Correct option: D

Final answer

405

Q037 Written

Find the coefficient of a in the expansion of $(2a-3)^4$.

Solution

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use 4C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is -216.

Final answer

-216

Worked Solutions **Binomial and Multinomial Theorems**

Q038 **Written**

Find the coefficient of x^3 in the expansion of $(3x+2)^5$.

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C2 times the first factor to power q and the second factor to power r.
3. The required coefficient is 1080.

Final answer

1080

Q039 **Written**

Find the coefficient of x^6 in the expansion of $\left(2x^2-1\right)^6$.

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use 6C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is -160.

Final answer

-160

Worked Solutions **Binomial and Multinomial Theorems**

Q040 MCQ

Find the coefficient of x^4 in the expansion of $(3x^2 - 1)^5$:

Solution

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 3$.
2. Use 5C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is -90.

Correct option: D

Final answer

-90

Q041 Written

Find the coefficient of x^6 in the expansion of $(3x^2 + 2)^6$.

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use 6C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is 4320.

Final answer

4320

Worked Solutions **Binomial and Multinomial Theorems**

Q042 Written

If $(x+1)^5 = a_0 + a_x + \ldots$, find the sum of its coefficients.

Solution

1. Put $x = 1$; the sum of the coefficients is the value of the expansion at $x = 1$.
2. Therefore the required value is 32.

Final answer

32

Q043 MCQ

Find the coefficient of xyz in the expansion of $(x+y+z)^3$.:

Solution

1. The target powers are $p = 1,\ q = 1,\ r = 1$, and $p + q + r = 3$.
2. Coefficient = $\dfrac{3!}{1!\ \times\ 1!\ \times\ 1!} \times (1)^1 \times (1)^1 \times (1)^1 = 6$.
3. Therefore, the required coefficient is 6.

Correct option: A

Final answer

6

Worked Solutions **Binomial and Multinomial Theorems**

Q044 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a+b+c)^8$:

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is 280.

Correct option: C

Final answer

280

Q045 MCQ

Find the coefficient of a^3bc^2 in the expansion of $(a+b+c)^6$:

Solution

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 60$ on the signed factors.
3. The required coefficient is 60.

Correct option: A

Final answer

60

Worked Solutions **Binomial and Multinomial Theorems**

Q046 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a+b+c)^6$:

Solution

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 90.

Correct option: B

Final answer

90

Q047 MCQ

Find the coefficient of xy^5z in the expansion of $(x+y+z)^7$:

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is 42.

Correct option: A

Final answer

42

Worked Solutions **Binomial and Multinomial Theorems**

Q048 **Written**

Find the coefficient of x^4y^2z in the expansion of $(x+y+z)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 105$ on the signed factors.
3. The required coefficient is 105.

Final answer

105

Q049 **Written**

Find the coefficient of $x^2y^3z^2$ in the expansion of $(x+y+z)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 210$ on the signed factors.
3. The required coefficient is 210.

Final answer

210

Worked Solutions **Binomial and Multinomial Theorems**

Q050 **Written**

Find the coefficient of x^6yz^2 in the expansion of $(x+y+z)^9$.

Solution

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 252$ on the signed factors.
3. The required coefficient is 252.

Final answer

252

Q051 **Written**

Find the coefficient of $x^3y^3z^3$ in the expansion of $(x+y+z)^9$.

Solution

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 1680$ on the signed factors.
3. The required coefficient is 1680.

Final answer

1680

Worked Solutions **Binomial and Multinomial Theorems**

Q052 Written

Find the coefficient of $x^2y^4z^3$ in the expansion of $(x+y+z)^9$.

Solution

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 1260$ on the signed factors.
3. The required coefficient is 1260.

Final answer

1260

Q053 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a-2b+3c)^8$:

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is -6720.

Correct option: D

Final answer

-6720

Worked Solutions **Binomial and Multinomial Theorems**

Q054 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a+3b-2c)^6$:

Solution

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 3240.

Correct option: C

Final answer

3240

Q055 MCQ

Find the coefficient of ab^2c^3 in the expansion of $(a+3b-2c)^6$:

Solution

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 60$ on the signed factors.
3. The required coefficient is -4320.

Correct option: D

Final answer

-4320

Worked Solutions **Binomial and Multinomial Theorems**

Q056 **Written**

Find the coefficient of $a^2b^3c^2$ in the expansion of $(a + 2b - 3c)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 210$ on the signed factors.
3. The required coefficient is 15120.

Final answer

15120

Q057 **Written**

Find the coefficient of abc^5 in the expansion of $(a + 2b - 3c)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is -20412.

Final answer

-20412

Worked Solutions **Binomial and Multinomial Theorems**

Q058 Written

Find the coefficient of a^2b^4c in the expansion of $(a + 2b - 3c)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 105$ on the signed factors.
3. The required coefficient is -5040.

Final answer

-5040

Q059 Written

Find the coefficient of $a^2b^2c^4$ in the expansion of $(2a - b + 3c)^8$.

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 420$ on the signed factors.
3. The required coefficient is 136080.

Final answer

136080

Worked Solutions **Binomial and Multinomial Theorems**

Q060 Written

Find the coefficient of a^4bc^3 in the expansion of $(2a - b + 3c)^8$.

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is -120960.

Final answer

-120960

Q061 Written

Find the coefficient of $a^3b^3c^2$ in the expansion of $(2a - b + 3c)^8$.

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 560$ on the signed factors.
3. The required coefficient is -40320.

Final answer

-40320

Worked Solutions **Binomial and Multinomial Theorems**

Q062 Written

Find the greatest coefficient in the expansion of $\left(x^2 + x + 1\right)^4$.

Solution

1. Expand and collect like powers: $x^8 + 4x^7 + 10x^6 + 16x^5 + 19x^4 + 16x^3 + 10x^2 + 4x + 1$.
2. The greatest coefficient is 19.

Final answer

19

Quiz MCQ 1 Concept Quiz · MCQ

The coefficient of x^4 in $(2x - 3)^5$ is:

Solution

1. $q = 4$ and $r = 1$, so the coefficient is -240.

Correct option: A

Final answer

-240

Worked Solutions **Binomial and Multinomial Theorems**

Quiz MCQ 2 **Concept Quiz · MCQ**

The coefficient of $x^2y^2z^2$ in $(x+y+z)^6$ is:

Solution

1. Use 6!/(2!2!2!).

Correct option: C

Final answer

90

Quiz MCQ 3 **Concept Quiz · MCQ**

The coefficient of x^6 in $\left(3x^2-2\right)^6$ is:

Solution

1. Choose the first factor three times.

Correct option: A

Final answer

-4320

Worked Solutions **Binomial and Multinomial Theorems**

Quiz MCQ 4 **Concept Quiz · MCQ**

The coefficient of x^5 in $\left(2x^2+1\right)^4$ is:

Solution

1. All powers of x are even, so the coefficient is zero.

Correct option: A

Final answer

0

Quiz WRITTEN 5 **Concept Quiz · Written**

Expand $(2x-1)^4$ fully.

Solution

1. The expansion is $16x^4-32x^3+24x^2-8x+1$.

Final answer

$16x^4-32x^3+24x^2-8x+1$

Worked Solutions **Binomial and Multinomial Theorems**

Quiz WRITTEN 6 Concept Quiz · Written

Find the coefficient of x^4 in $\left(3x^2+2\right)^5$. Show the exponent check.

Solution

1. The first-term exponent is $q=2$ and $r=3$, giving 5C3 $\times$ 9 $\times$ 8=720.

Final answer

720

Quiz WRITTEN 7 Concept Quiz · Written

Find the coefficient of x^2y^3z in $(x-2y+z)^6$.

Solution

1. Use the multinomial coefficient and the signed factor; the result is -480.

Final answer

-480

Worked Solutions **Binomial and Multinomial Theorems**

Quiz WRITTEN 8 **Concept Quiz · Written**

Find the greatest coefficient in $(x^2 + x + 1)^4$. Show enough work to justify the maximum.

Solution

1. Expand: $x^8 + 4x^7 + 10x^6 + 16x^5 + 19x^4 + 16x^3 + 10x^2 + 4x + 1$; the greatest coefficient is 19.

Final answer

19

Answer key

Use the question number shown on each page.

Q001 A

Q002 A

Q003 A

Q004 $x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$

Q005 D

Q006 $x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$

Q007 B

Q008 C

Q009 $a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$

Q010 A

Q011 $243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$

Q012 $32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$

Q013 $x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$

Q014 $64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$

Q015 $x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$

Q016 B

Q017 D

Q018 D

Q019 A

Q020 B

Q021 $243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$

Q022 C

Q023 B

Q024 C

Q025 C

Q026 B

Q027 10

Q028 60

Q029 -20412

Q030 -1344

Q031 B

Q032 -48384

Q033 A

Q034 B

Q035 C

Q036 D

Q037 -216

Q038 1080

Q039 -160

Q040 D

Q041 4320

Q042 32

Q043 A

Q044 C

Q045 A

Q046 B

Q047 A

Q048 105

Q049 210

Q050 252

Q051 1680

Q052 1260

Q053 D

Q054 C

Q055 D

Q056 15120

Q057 -20412

Q058 -5040

Q059 136080

Q060 -120960

Q061 -40320

Q062 19

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $16x^4 - 32x^3 + 24x^2 - 8x + 1$

Quiz WRITTEN 6 720

Quiz WRITTEN 7 -480

Quiz WRITTEN 8 19

Concept 2 | Polynomial Division

English Student Edition • Focused formulas and guided practice

Formula opener

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder

$$R = P(a) \quad \text{for divisor } x - a$$

Degree check

$$\deg R < \deg B$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Polynomial Division**

Q001 Challenge

Find the quotient and the remainder when $2x^3+x^2-7x+3$ is divided by x^2+x-2. Then verify the result using $A = B \times Q+R$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3+x^2-7x+3 = \left(x^2+x-2\right) \times (2x-1)+(-2x+1)$.
4. Therefore, quotient $Q = 2x-1$ and remainder $R = -2x+1$.

Correct option: B

Final answer

$Q = 2x-1, \quad R = -2x+1$

Worked Solutions **Polynomial Division**

Q002 **Challenge**

Find the polynomial A when it is divided by $x^2 - 2x + 3$, the quotient is $3x - 1$, and the remainder is $x + 2$. Then divide your answer back to check it:

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = \left(x^2 - 2x + 3\right) \times (3x - 1) + (x + 2)$.
3. Expand and collect like powers: $A = 3x^3 - 7x^2 + 12x - 1$.
4. Therefore, $A = 3x^3 - 7x^2 + 12x - 1$.

Correct option: A

Final answer

$3x^3 - 7x^2 + 12x - 1$

Worked Solutions **Polynomial Division**

Q003 Challenge

When $x^3 + 5x^2 + 7x - 7$ is divided by B, the quotient is $x + 3$ and the remainder is $2x - 4$. Find B. Then verify the degree condition:

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3+5x^2+7x-7-(2x-4)}{x+3}$.
3. Divide and simplify: $B = x^2 + 2x - 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 + 5x^2 + 7x - 7$.

Correct option: D

Final answer

$x^2 + 2x - 1$

Worked Solutions **Polynomial Division**

Q004 Challenge

A cuboid has volume $2x^3 + 9x^2 + 4x - 15$ and base area $x^2 + 2x - 3$. Write its height in terms of x. State the mathematical check Which of the following is correct?:

Solution

1. Use height = volume ÷ base area: $h = \frac{2x^3+9x^2+4x-15}{x^2+2x-3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 5$.

Correct option: C

Final answer

$2x + 5$

Worked Solutions **Polynomial Division**

Q005 **Challenge**

Find the quotient and the remainder when $3x^4 - x^3 + 2x^2 - 4x + 1$ is divided by $x^2 - 2x + 2$. Decide whether the result is exact and justify from R:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - x^3 + 2x^2 - 4x + 1 = \left(x^2 - 2x + 2\right) \times \left(3x^2 + 5x + 6\right) + (-2x - 11)$.
4. Therefore, quotient $Q = 3x^2 + 5x + 6$ and remainder $R = -2x - 11$.

Correct option: B

Final answer

$Q = 3x^2 + 5x + 6, \quad R = -2x - 11$

Worked Solutions **Polynomial Division**

Q006 **Challenge**

Find the quotient and the remainder when $x^4 - 3x^2 + 2x + 6$ is divided by $x + 2$. Explain why a zero placeholder is needed:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^2 + 2x + 6 = (x + 2) \times \left(x^3 - 2x^2 + x\right) + (6)$.
4. Therefore, quotient $Q = x^3 - 2x^2 + x$ and remainder $R = 6$.

Correct option: A

Final answer

$Q = x^3 - 2x^2 + x, \quad R = 6$

Worked Solutions **Polynomial Division**

Q007 Written

Find the quotient and the remainder when $x^2 - 5x + 1$ is divided by $x - 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 - 5x + 1 = (x - 2) \times (x - 3) + (-5)$.
4. Therefore, quotient $Q = x - 3$ and remainder $R = -5$.

Final answer

$Q = x - 3, \quad R = -5$

Q008 Written

Find the quotient and the remainder when $x^3 - 4x + 1$ is divided by $x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 4x + 1 = (x - 3) \times \left(x^2 + 3x + 5\right) + (16)$.
4. Therefore, quotient $Q = x^2 + 3x + 5$ and remainder $R = 16$.

Final answer

$Q = x^2 + 3x + 5, \quad R = 16$

Worked Solutions **Polynomial Division**

Q009 MCQ

Find the quotient and the remainder when $3x^3 + 2x^2 - 7x + 5$ is divided by $x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 2x^2 - 7x + 5 = (x - 2) \times \left(3x^2 + 8x + 9\right) + (23)$.
4. Therefore, quotient $Q = 3x^2 + 8x + 9$ and remainder $R = 23$.

Correct option: D

Final answer

$Q = 3x^2 + 8x + 9, \quad R = 23$

Worked Solutions **Polynomial Division**

Q010 MCQ

Find the quotient and the remainder when $x^3 + 6x^2 - 4$ is divided by $2x - 3$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 6x^2 - 4 = (2x - 3) \times \left(\frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}\right) + \left(\frac{103}{8}\right)$.
4. Therefore, quotient $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}$ and remainder $R = \frac{103}{8}$.

Correct option: B

Final answer

$$Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, \quad R = \frac{103}{8}$$

Worked Solutions **Polynomial Division**

Q011 MCQ

Find the quotient and the remainder when $3x^3 - 4x^2 + 2x + 7$ is divided by $x - 1$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 4x^2 + 2x + 7 = (x - 1) \times \left(3x^2 - x + 1\right) + (8)$.
4. Therefore, quotient $Q = 3x^2 - x + 1$ and remainder $R = 8$.

Correct option: B

Final answer

$Q = 3x^2 - x + 1, \quad R = 8$

Q012 MCQ

Find the quotient and the remainder when $2x^4 + x^3 - 5x + 4$ is divided by $x + 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 + x^3 - 5x + 4 = (x + 2) \times \left(2x^3 - 3x^2 + 6x - 17\right) + (38)$.
4. Therefore, quotient $Q = 2x^3 - 3x^2 + 6x - 17$ and remainder $R = 38$.

Correct option: A

Final answer

$Q = 2x^3 - 3x^2 + 6x - 17, \quad R = 38$

Worked Solutions **Polynomial Division**

Q013 MCQ

Find the quotient and the remainder when $4x^4 - 3x^3 + x - 6$ is divided by $x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^4 - 3x^3 + x - 6 = (x - 2) \times \left(4x^3 + 5x^2 + 10x + 21\right) + (36)$.
4. Therefore, quotient $Q = 4x^3 + 5x^2 + 10x + 21$ and remainder $R = 36$.

Correct option: D

Final answer

$Q = 4x^3 + 5x^2 + 10x + 21, \quad R = 36$

Worked Solutions **Polynomial Division**

Q014 MCQ

Find the quotient and the remainder when $2x^3 - 9x + 1$ is divided by $3x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 9x + 1 = (3x - 2) \times \left(\frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}\right) + \left(-\frac{119}{27}\right)$.
4. Therefore, quotient $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}$ and remainder $R = -\frac{119}{27}$.

Correct option: B

Final answer

$$Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = -\frac{119}{27}$$

Worked Solutions **Polynomial Division**

Q015 Written

Find the quotient and the remainder when $2x^3 - 5x^2 - 2$ is divided by $2x^2 - x - 1$.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 5x^2 - 2 = \left(2x^2 - x - 1\right) \times (x - 2) + (-x - 4)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = -x - 4$.

Final answer

$Q = x - 2, \quad R = -x - 4$

Q016 Written

Find the quotient and the remainder when $2x^3 - 13x + 1$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 13x + 1 = \left(x^2 + 2x - 3\right) \times (2x - 4) + (x - 11)$.
4. Therefore, quotient $Q = 2x - 4$ and remainder $R = x - 11$.

Final answer

$Q = 2x - 4, \quad R = x - 11$

Worked Solutions **Polynomial Division**

Q017 Written

Find the quotient and the remainder when $2x^3 - 3x^2 + 3x + 4$ is divided by $x^2 - 2x + 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 3x^2 + 3x + 4 = \left(x^2 - 2x + 3\right) \times (2x + 1) + (-x + 1)$.
4. Therefore, quotient $Q = 2x + 1$ and remainder $R = -x + 1$.

Final answer

$Q = 2x + 1, \quad R = -x + 1$

Q018 Written

Find the quotient and the remainder when $x^3 - 7x + 6$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 7x + 6 = \left(x^2 + 2x - 3\right) \times (x - 2) + (0)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = 0$.

Final answer

$Q = x - 2, \quad R = 0$

Worked Solutions **Polynomial Division**

Q019 *MCQ*

Find the quotient and the remainder when $x^3 + 2x^2 - 5x + 4$ is divided by $x^2 + x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 2x^2 - 5x + 4 = \left(x^2 + x - 2\right) \times (x + 1) + (-4x + 6)$.
4. Therefore, quotient $Q = x + 1$ and remainder $R = -4x + 6$.

Correct option: D

Final answer

$Q = x + 1, \quad R = -4x + 6$

Q020 *MCQ*

Find the quotient and the remainder when $x^4 - 3x^3 + 2x - 5$ is divided by $x^2 + 2x - 1$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^3 + 2x - 5 = \left(x^2 + 2x - 1\right) \times \left(x^2 - 5x + 11\right) + (-25x + 6)$.
4. Therefore, quotient $Q = x^2 - 5x + 11$ and remainder $R = -25x + 6$.

Correct option: B

Final answer

$Q = x^2 - 5x + 11, \quad R = -25x + 6$

Worked Solutions **Polynomial Division**

Q021 MCQ

Find the quotient and the remainder when $2x^3 + x^2 - 4x + 5$ is divided by $x^2 - 2x + 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 + x^2 - 4x + 5 = \left(x^2 - 2x + 2\right) \times (2x + 5) + (2x - 5)$.
4. Therefore, quotient $Q = 2x + 5$ and remainder $R = 2x - 5$.

Correct option: D

Final answer

$Q = 2x + 5, \quad R = 2x - 5$

Q022 MCQ

Find the quotient and the remainder when $x^4 - 2x^3 + 5x^2 - x + 3$ is divided by $x^2 + 3x - 1$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 2x^3 + 5x^2 - x + 3 = \left(x^2 + 3x - 1\right) \times \left(x^2 - 5x + 21\right) + (-69x + 24)$.
4. Therefore, quotient $Q = x^2 - 5x + 21$ and remainder $R = -69x + 24$.

Correct option: C

Final answer

$Q = x^2 - 5x + 21, \quad R = -69x + 24$

Worked Solutions **Polynomial Division**

Q023 **MCQ**

Find the quotient and the remainder when $3x^3 - 5x + 2$ is divided by $x^2 - x - 1$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 5x + 2 = \left(x^2 - x - 1\right) \times (3x + 3) + (x + 5)$.
4. Therefore, quotient $Q = 3x + 3$ and remainder $R = x + 5$.

Correct option: B

Final answer

$Q = 3x + 3, \quad R = x + 5$

Q024 **MCQ**

Find the quotient and the remainder when $2x^4 - x^3 + 4x^2 - 7$ is divided by $x^2 + x + 3$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 - x^3 + 4x^2 - 7 = \left(x^2 + x + 3\right) \times \left(2x^2 - 3x + 1\right) + (8x - 10)$.
4. Therefore, quotient $Q = 2x^2 - 3x + 1$ and remainder $R = 8x - 10$.

Correct option: A

Final answer

$Q = 2x^2 - 3x + 1, \quad R = 8x - 10$

Worked Solutions **Polynomial Division**

Q025 Written

Find the quotient and the remainder when $6x^3 - 8x^2 + 5$ is divided by $2x^2 - 2x - 5$.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 8x^2 + 5 = \left(2x^2 - 2x - 5\right) \times (3x - 1) + (13x)$.
4. Therefore, quotient $Q = 3x - 1$ and remainder $R = 13x$.

Final answer

$Q = 3x - 1, \quad R = 13x$

Q026 Written

Find the quotient and the remainder when $3x^4 - 5x^3 + x - 1$ is divided by $-x^2 + 3x + 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - 5x^3 + x - 1 = \left(-x^2 + 3x + 2\right) \times \left(-3x^2 - 4x - 18\right) + (63x + 35)$.
4. Therefore, quotient $Q = -3x^2 - 4x - 18$ and remainder $R = 63x + 35$.

Final answer

$Q = -3x^2 - 4x - 18, \quad R = 63x + 35$

Worked Solutions **Polynomial Division**

Q027 MCQ

Find the quotient and the remainder when $4x^3 - x^2 + 8x - 3$ is divided by $2x^2 + x - 4$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^3 - x^2 + 8x - 3 = \left(2x^2 + x - 4\right) \times \left(2x - \frac{3}{2}\right) + \left(\frac{35x}{2} - 9\right)$.
4. Therefore, quotient $Q = 2x - \frac{3}{2}$ and remainder $R = \frac{35x}{2} - 9$.

Correct option: D

Final answer

$Q = 2x - \frac{3}{2}, \quad R = \frac{35x}{2} - 9$

Q028 MCQ

Find the quotient and the remainder when $x^3 - 8x + 2$ is divided by $x^2 - 4$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 8x + 2 = \left(x^2 - 4\right) \times (x) + (-4x + 2)$.
4. Therefore, quotient $Q = x$ and remainder $R = -4x + 2$.

Correct option: B

Final answer

$Q = x, \quad R = -4x + 2$

Worked Solutions **Polynomial Division**

Q029 MCQ

Find the quotient and the remainder when $5x^3 - 4x^2 + x - 6$ is divided by $2x^2 - x + 1$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $5x^3 - 4x^2 + x - 6 = (2x^2 - x + 1) \times \left(\frac{5x}{2} - \frac{3}{4}\right) + \left(-\frac{9x}{4} - \frac{21}{4}\right)$.
4. Therefore, quotient $Q = \frac{5x}{2} - \frac{3}{4}$ and remainder $R = -\frac{9x}{4} - \frac{21}{4}$.

Correct option: D

Final answer

$Q = \frac{5x}{2} - \frac{3}{4}, \quad R = -\frac{9x}{4} - \frac{21}{4}$

Worked Solutions **Polynomial Division**

Q030 MCQ

Find the quotient and the remainder when $3x^4 + 2x^3 - x^2 + 5$ is divided by $4x^2 + x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 + 2x^3 - x^2 + 5 = (4x^2 + x - 2) \times \left(\frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}\right) + \left(\frac{37x}{64} + \frac{163}{32}\right)$.
4. Therefore, quotient $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}$ and remainder $R = \frac{37x}{64} + \frac{163}{32}$.

Correct option: C

Final answer

$Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, \quad R = \frac{37x}{64} + \frac{163}{32}$

Worked Solutions **Polynomial Division**

Q031 MCQ

Find the quotient and the remainder when $3x^3 + 4x^2 - x + 9$ is divided by $2x^2 - x + 3$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 4x^2 - x + 9 = \left(2x^2 - x + 3\right) \times \left(\frac{3x}{2} + \frac{11}{4}\right) + \left(-\frac{11x}{4} + \frac{3}{4}\right)$.
4. Therefore, quotient $Q = \frac{3x}{2} + \frac{11}{4}$ and remainder $R = -\frac{11x}{4} + \frac{3}{4}$.

Correct option: D

Final answer

$Q = \frac{3x}{2} + \frac{11}{4}, \quad R = -\frac{11x}{4} + \frac{3}{4}$

Worked Solutions **Polynomial Division**

Q032 MCQ

Find the quotient and the remainder when $6x^3 - 5x + 1$ is divided by $2x^2 + 3x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 5x + 1 = (2x^2 + 3x - 2) \times \left(3x - \frac{9}{2}\right) + \left(\frac{29x}{2} - 8\right)$.
4. Therefore, quotient $Q = 3x - \frac{9}{2}$ and remainder $R = \frac{29x}{2} - 8$.

Correct option: B

Final answer

$$Q = 3x - \frac{9}{2}, \quad R = \frac{29x}{2} - 8$$

Q033 Written

Find the quotient and the remainder when $-9a^2 - 9ax + 4x^2$ is divided by $-3a + x$. Treat it as a polynomial in x; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power. Treat it as a polynomial in x; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-9a^2 - 9ax + 4x^2 = (-3a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$$Q = 3a + 4x, \quad R = 0$$

Worked Solutions **Polynomial Division**

Q034 Written

Find the quotient and the remainder when $-5a^3 + 5a^2x - 3ax^2 + x^3$ is divided by $-2a + x$. Treat it as a polynomial in x; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power. Treat it as a polynomial in x; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-5a^3 + 5a^2x - 3ax^2 + x^3 = (-2a + x) \times \left(3a^2 - ax + x^2\right) + \left(a^3\right)$.
4. Therefore, quotient $Q = 3a^2 - ax + x^2$ and remainder $R = a^3$.

Final answer

$Q = 3a^2 - ax + x^2, \quad R = a^3$

Q035 Written

Find the quotient and the remainder when $-6a^2 - 5ax + 4x^2$ is divided by $-2a + x$. Treat it as a polynomial in x; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power. Treat it as a polynomial in x; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-6a^2 - 5ax + 4x^2 = (-2a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$Q = 3a + 4x, \quad R = 0$

Worked Solutions **Polynomial Division**

Q036 Written

Find the quotient and the remainder when $-a^3+a^2x+ax^2+x^3$ is divided by $2a+x$. Treat it as a polynomial in x; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power. Treat it as a polynomial in x; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-a^3+a^2x+ax^2+x^3=(2a+x)\times\left(3a^2-ax+x^2\right)+\left(-7a^3\right)$.
4. Therefore, quotient $Q=3a^2-ax+x^2$ and remainder $R=-7a^3$.

Final answer

$Q=3a^2-ax+x^2, \quad R=-7a^3$

Q037 Written

Find the quotient and the remainder when $3a^3-2ax^2+x^3$ is divided by $a+x$. Treat it as a polynomial in x; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power. Treat it as a polynomial in x; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3a^3-2ax^2+x^3=(a+x)\times\left(3a^2-3ax+x^2\right)+(0)$.
4. Therefore, quotient $Q=3a^2-3ax+x^2$ and remainder $R=0$.

Final answer

$Q=3a^2-3ax+x^2, \quad R=0$

Worked Solutions **Polynomial Division**

Q038 MCQ

Find the quotient and the remainder when $-2a^3 + 7a^2x - 4ax^2 + x^3$ is divided by $-a + x$. Treat it as a polynomial in x; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power. Treat it as a polynomial in x; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-2a^3 + 7a^2x - 4ax^2 + x^3 = (-a + x) \times (4a^2 - 3ax + x^2) + (2a^3)$.
4. Therefore, quotient $Q = 4a^2 - 3ax + x^2$ and remainder $R = 2a^3$.

Correct option: D

Final answer

$Q = 4a^2 - 3ax + x^2, \quad R = 2a^3$

Q039 **MCQ**

Find the quotient and the remainder when $x^2 + 3xy - 4y^2$ is divided by $x + y$. Treat it as a polynomial in x; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power. Treat it as a polynomial in x; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 + 3xy - 4y^2 = (x + y) \times (x + 2y) + \left(-6y^2\right)$.
4. Therefore, quotient $Q = x + 2y$ and remainder $R = -6y^2$.

Correct option: B

Final answer

$Q = x + 2y, \quad R = -6y^2$

Worked Solutions **Polynomial Division**

Q040 Written

Find the quotient and the remainder when $x^3 + 3x^2y - 2y^3$ is divided by $x + y$. Treat it as a polynomial in x; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x; insert a zero coefficient for any missing power. Treat it as a polynomial in x; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 3x^2y - 2y^3 = (x + y) \times \left(x^2 + 2xy - 2y^2\right) + (0)$.
4. Therefore, quotient $Q = x^2 + 2xy - 2y^2$ and remainder $R = 0$.

Final answer

$Q = x^2 + 2xy - 2y^2, \quad R = 0$

Q041 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $x + 3$, and the remainder is $-2x + 5$.

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = \left(x^2 - 2x - 1\right) \times (x + 3) + (-2x + 5)$.
3. Expand and collect like powers: $A = x^3 + x^2 - 9x + 2$.
4. Therefore, $A = x^3 + x^2 - 9x + 2$.

Final answer

$x^3 + x^2 - 9x + 2$

Worked Solutions **Polynomial Division**

Q042 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $2x - 3$, and the remainder is $-2x$.

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = \left(x^2 - 2x - 1\right) \times (2x - 3) + (-2x)$.
3. Expand and collect like powers: $A = 2x^3 - 7x^2 + 2x + 3$.
4. Therefore, $A = 2x^3 - 7x^2 + 2x + 3$.

Final answer

$2x^3 - 7x^2 + 2x + 3$

Q043 Written

Find the polynomial A when it is divided by $x^2 - 3x + 2$, the quotient is $2x + 6$, and the remainder is $2x - 3$.

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = \left(x^2 - 3x + 2\right) \times (2x + 6) + (2x - 3)$.
3. Expand and collect like powers: $A = 2x^3 - 12x + 9$.
4. Therefore, $A = 2x^3 - 12x + 9$.

Final answer

$2x^3 - 12x + 9$

Worked Solutions **Polynomial Division**

Q044 Written

Find the polynomial A when it is divided by $2x^2 - x + 2$, the quotient is $2x - 1$, and the remainder is 2.

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = \left(2x^2 - x + 2\right) \times (2x - 1) + (2)$.
3. Expand and collect like powers: $A = 4x^3 - 4x^2 + 5x$.
4. Therefore, $A = 4x^3 - 4x^2 + 5x$.

Final answer

$4x^3 - 4x^2 + 5x$

Q045 Written

Find the polynomial A when it is divided by $2x^2 + x + 1$, the quotient is $3x + 2$, and the remainder is $-2x + 6$.

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = \left(2x^2 + x + 1\right) \times (3x + 2) + (-2x + 6)$.
3. Expand and collect like powers: $A = 6x^3 + 7x^2 + 3x + 8$.
4. Therefore, $A = 6x^3 + 7x^2 + 3x + 8$.

Final answer

$6x^3 + 7x^2 + 3x + 8$

Worked Solutions **Polynomial Division**

Q046 **Written**

When $4x^3 + 8x - 9$ is divided by B, the quotient is $2x - 3$ and the remainder is $x + 15$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{4x^3+8x-9-(x+15)}{2x-3}$.
3. Divide and simplify: $B = 2x^2 + 3x + 8$.
4. Check by multiplying B by Q and adding R to recover $A = 4x^3 + 8x - 9$.

Final answer

$2x^2 + 3x + 8$

Q047 **Written**

When $x^3 - x^2 + 3x + 1$ is divided by B, the quotient is $x^2 - 2x + 5$ and the remainder is -4. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3-x^2+3x+1-(-4)}{x^2-2x+5}$.
3. Divide and simplify: $B = x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 - x^2 + 3x + 1$.

Final answer

$x + 1$

Worked Solutions **Polynomial Division**

Q048 Written

When $2x^3 + x^2 - 3x + 5$ is divided by B, the quotient is $x + 1$ and the remainder is $-5x + 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+x^2-3x+5-(-5x+2)}{x+1}$.
3. Divide and simplify: $B = 2x^2 - x + 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + x^2 - 3x + 5$.

Final answer

$2x^2 - x + 3$

Q049 Written

When $6x^3 + 11x^2 - 2$ is divided by B, the quotient is $2x^2 + 3x - 1$ and the remainder is -1. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{6x^3+11x^2-2-(-1)}{2x^2+3x-1}$.
3. Divide and simplify: $B = 3x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = 6x^3 + 11x^2 - 2$.

Final answer

$3x + 1$

Worked Solutions **Polynomial Division**

Q050 Written

When $2x^3 + 4x^2 + 7$ is divided by B, the quotient is $x + 2$ and the remainder is $3x + 13$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+4x^2+7-(3x+13)}{x+2}$.
3. Divide and simplify: $B = 2x^2 - 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + 4x^2 + 7$.

Final answer

$2x^2 - 3$

Q051 Written

When $x^4 + x^2 + 3x - 1$ is divided by B, the quotient is $x^2 + x + 1$ and the remainder is $3x - 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^4+x^2+3x-1-(3x-2)}{x^2+x+1}$.
3. Divide and simplify: $B = x^2 - x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^4 + x^2 + 3x - 1$.

Final answer

$x^2 - x + 1$

Worked Solutions **Polynomial Division**

Q052 Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Write its height in terms of x.

Solution

1. Use height = volume ÷ base area: $h = \frac{4x^3+4x^2-9x-9}{2x^2-x-3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 3$.

Final answer

$2x + 3$

Quiz MCQ 1 Concept Quiz · MCQ

For $x^2 - 5x + 1$ divided by $x - 2$, what is R?:

Solution

1. Long division gives $R = -5$.

Correct option: A

Final answer

-5

Worked Solutions **Polynomial Division**

Quiz MCQ 2 Concept Quiz · MCQ

In a polynomial in x, a is treated as... Which of the following is correct?:

Solution

1. Only powers of x determine the division order.

Correct option: B

Final answer

constant

Quiz MCQ 3 Concept Quiz · MCQ

To recover B from $A = B \times Q + R$, use... Which of the following is correct?:

Solution

1. Isolate B after moving R to the other side.

Correct option: C

Final answer

$$\frac{A-R}{Q}$$

Worked Solutions **Polynomial Division**

Quiz MCQ 4 Concept Quiz · MCQ

If B has degree 2, a valid remainder can have degree...:

Solution

1. The remainder degree must be below 2.

Correct option: C

Final answer

1

Quiz WRITTEN 5 Concept Quiz · Written

Divide $3x^3 - 5x^2 + 2x - 7$ by $x - 2$. State Q, R, and verify $A = B \times Q + R$.

Solution

1. Long division gives $Q = 3x^2 + x + 4$, $R = 1$.
2. Substitution verifies $3x^3 - 5x^2 + 2x - 7 = (x-2) \times \left(3x^2 + x + 4\right) + (1)$.

Final answer

$Q = 3x^2 + x + 4, \quad R = 1$

Worked Solutions **Polynomial Division**

Quiz WRITTEN 6 Concept Quiz · Written

Recover A when B=$x^2 - x - 2$, Q=$2x + 3$, R=$-x + 1$.

Solution

1. Use $A = B \times Q + R$.
2. $A = 2x^3 + x^2 - 8x - 5$.

Final answer

$2x^3 + x^2 - 8x - 5$

Quiz WRITTEN 7 Concept Quiz · Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Find its height.

Solution

1. Divide volume by base area: $h = 2x + 3$.
2. The remainder is zero, so the height is polynomial.

Final answer

$2x + 3$

Worked Solutions **Polynomial Division**

Quiz WRITTEN 8 Concept Quiz · Written

Divide $x^4 - 2x^3 + 3x - 1$ by $x^2 - x + 1$ and explain why the remainder is valid.

Solution

1. Long division gives $Q = x^2 - x - 2$ and $R = 2x + 1$.
2. The remainder degree is less than 2, so it is valid even though it is nonzero.

Final answer

$Q = x^2 - x - 2, \quad R = 2x + 1$

Answer key

Use the question number shown on each page.

Q001 B
Q002 A
Q003 D
Q004 C
Q005 B
Q006 A
Q007 $Q = x - 3, \quad R = -5$
Q008 $Q = x^2 + 3x + 5, \quad R = 16$
Q009 D
Q010 B
Q011 B
Q012 A
Q013 D
Q014 B
Q015 $Q = x - 2, \quad R = -x - 4$
Q016 $Q = 2x - 4, \quad R = x - 11$
Q017 $Q = 2x + 1, \quad R = -x + 1$
Q018 $Q = x - 2, \quad R = 0$
Q019 D
Q020 B
Q021 D
Q022 C
Q023 B
Q024 A
Q025 $Q = 3x - 1, \quad R = 13x$
Q026 $Q = -3x^2 - 4x - 18, \quad R = 63x + 35$
Q027 D
Q028 B
Q029 D
Q030 C
Q031 D
Q032 B
Q033 $Q = 3a + 4x, \quad R = 0$
Q034 $Q = 3a^2 - ax + x^2, \quad R = a^3$
Q035 $Q = 3a + 4x, \quad R = 0$
Q036 $Q = 3a^2 - ax + x^2, \quad R = -7a^3$
Q037 $Q = 3a^2 - 3ax + x^2, \quad R = 0$
Q038 D
Q039 B
Q040 $Q = x^2 + 2xy - 2y^2, \quad R = 0$
Q041 $x^3 + x^2 - 9x + 2$
Q042 $2x^3 - 7x^2 + 2x + 3$
Q043 $2x^3 - 12x + 9$
Q044 $4x^3 - 4x^2 + 5x$
Q045 $6x^3 + 7x^2 + 3x + 8$
Q046 $2x^2 + 3x + 8$
Q047 $x + 1$
Q048 $2x^2 - x + 3$
Q049 $3x + 1$
Q050 $2x^2 - 3$
Q051 $x^2 - x + 1$
Q052 $2x + 3$
Quiz MCQ 1 A
Quiz MCQ 2 B
Quiz MCQ 3 C
Quiz MCQ 4 C
Quiz WRITTEN 5 $Q = 3x^2 + x + 4, \quad R = 1$
Quiz WRITTEN 6 $2x^3 + x^2 - 8x - 5$
Quiz WRITTEN 7 $2x + 3$
Quiz WRITTEN 8 $Q = x^2 - x - 2, \quad R = 2x + 1$

Concept 3 | Rational Expressions

English Student Edition • Focused formulas and guided practice

Formula opener

Restriction

$$\text{denominator} \neq 0$$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Rational Expressions**

Q001 Challenge

After simplifying this transfer chain before substitution, which result is correct for $2\,()$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2x(x+3)}{(x-3)(x-1)}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{2x\,(x+3)}{(x-3)\,(x-1)}$$

Worked Solutions **Rational Expressions**

Q002 **Challenge**

After simplifying this transfer chain before substitution, which result is correct for $3\,()$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3(x-3)(x+2)}{x(x+3)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{3\,(x-3)\,(x+2)}{x\,(x+3)}$$

Worked Solutions **Rational Expressions**

Q003 **Challenge**

After simplifying this transfer chain before substitution, which result is correct for $4\,()$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8(x-4)(x+1)}{(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{8\,(x-4)\,(x+1)}{(x-3)\,(x-2)}$$

Worked Solutions **Rational Expressions**

Q004 Challenge

After simplifying this transfer chain before substitution, which result is correct for $8\,()$**?:**

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8}{x}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8}{x}$$

Worked Solutions **Rational Expressions**

Q005 **Challenge**

After simplifying this transfer chain before substitution, which result is correct for $9\,()$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9\,(x-1)\left(x^4+x^3+x^2+x+1\right)}{2}$$

Worked Solutions **Rational Expressions**

Q006 **Challenge**

After simplifying this transfer chain before substitution, which result is correct for 10 ()?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{10\,(x-4)\,(x-1)\,(x+4)}{x\,(x-3)\,(x-2)}$$

Worked Solutions **Rational Expressions**

Q007 Written

Simplify $\frac{12ab^5}{9a^3b^2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{4b^3}{3a^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{4b^3}{3a^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$\frac{4b^3}{3a^2}$

Q008 Written

Simplify $\frac{6x^2y^3}{20x^3y}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3y^2}{10x}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3y^2}{10x}$.
4. Keep every value excluded by an original denominator.

Final answer

$\frac{3y^2}{10x}$

Worked Solutions **Rational Expressions**

Q009 Written

Simplify $\frac{6a^2bc^2}{8ab^4c^2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3a}{4b^3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3a}{4b^3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{3a}{4b^3}$$

Q010 Written

Simplify $\frac{2a^2b}{8a^3b^3}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{1}{4ab^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{1}{4ab^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{1}{4ab^2}$$

Worked Solutions **Rational Expressions**

Q011 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{12ab^5}{9a^3b^2}\right)$**?:**

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4b^3}{a^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{4b^3}{a^2}$$

Q012 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6x^2y^3}{20x^3y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9y^2}{10x}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9y^2}{10x}$$

Worked Solutions **Rational Expressions**

Q013 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{6a^2bc^2}{8ab^4c^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a}{b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{3a}{b^3}$$

Q014 **MCQ**

After reducing coefficients and powers, which result is correct for $2\left(\frac{2a^2b}{8a^3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{1}{2ab^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{1}{2ab^2}$$

Worked Solutions **Rational Expressions**

Q015 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{18a^3b^2}{24ab^5}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a^2}{2b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$\frac{3a^2}{2b^3}$

Q016 **MCQ**

After reducing coefficients and powers, which result is correct for $3\left(\frac{-15x^4y^3}{20x^2y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{9x^2y^2}{4}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{9x^2y^2}{4}$$

Worked Solutions **Rational Expressions**

Q017 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{45a^2b^4c}{60a^5bc^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $=\frac{3b^3}{a^3c}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{3b^3}{a^3c}$$

Q018 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{14x^5y^2}{21x^2y^4}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4x^3}{3y^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{4x^3}{3y^2}$$

Worked Solutions **Rational Expressions**

Q019 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{32a^4}{48ab^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2a^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$\frac{2a^3}{b^2}$

Q020 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{27x^3y^5}{36xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= 3x^2y^3$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$3x^2y^3$

Worked Solutions **Rational Expressions**

Q021 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-42a^2c^3}{56a^5c}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{3c^2}{2a^3}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$-\frac{3c^2}{2a^3}$

Q022 Written

Simplify $\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}$.

Solution

1. Factor each polynomial: $\frac{a^3b^2}{3c}$ and $\frac{9c^3}{a^4b^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{3c^2}{a}$.
4. Retain the original denominator restrictions.

Final answer

$\frac{3c^2}{a}$

Worked Solutions **Rational Expressions**

Q023 Written

Simplify $\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}$.

Solution

1. Factor each polynomial: $\frac{6bc^3}{5a^2}$ and $\frac{10a^4}{3b^3}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{4a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{4a^2c^3}{b^2}$$

Q024 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{12c^2}{a}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{12c^2}{a}$$

Worked Solutions **Rational Expressions**

Q025 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8a^2c^3}{b^2}$$

Q026 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6a^2b}{5c} \times \frac{15c^2}{4ab^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{27ac}{2b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{27ac}{2b^2}$$

Worked Solutions **Rational Expressions**

Q027 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{9x^3y}{14a^2} \times \frac{21a^3}{6xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9ax^2}{y}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9ax^2}{y}$$

Q028 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-8ab^2}{3c^2} \times \frac{12c^3}{-5a^2b}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{64bc}{5a}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{64bc}{5a}$$

Worked Solutions **Rational Expressions**

Q029 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{10x^2y^3}{9ab} \times \frac{27a^2b^2}{25xy}\right)$**?:**

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{18abxy^2}{5}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer
$\frac{18abxy^2}{5}$

Q030 Written

Simplify $\frac{x^2-4}{3x^2+7x+2}$**.**

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x-2}{3x+1}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x-2}{3x+1}$.
4. Keep every value excluded by an original denominator.

Final answer
$$\frac{x-2}{3x+1}$$

Worked Solutions **Rational Expressions**

Q031 Written

Simplify $\frac{2x^2-3x-5}{x^2+3x+2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2x-5}{x+2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2x-5}{x+2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2x-5}{x+2}$$

Q032 Written

Simplify $\frac{x^2+3x}{x^2-9}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x}{x-3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions **Rational Expressions**

Q033 **Written**

Simplify $\frac{x^2+x-2}{x^2-5x+4}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x+2}{x-4}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x+2}{x-4}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x+2}{x-4}$$

Q034 **Written**

Simplify $\frac{2x^2-4xy+2y^2}{x^3+3x^2y-4xy^2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2(x-y)}{x(x+4y)}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2(x-y)}{x(x+4y)}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2\left(x-y\right)}{x\left(x+4y\right)}$$

Worked Solutions **Rational Expressions**

Q035 Written

Simplify $\frac{x^3-1}{x^2+2x-3}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x^2+x+1}{x+3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x^2+x+1}{x+3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x^2+x+1}{x+3}$$

Q036 Written

Simplify $\frac{x^2-11x+24}{x^2-6x-16} \times \frac{x^2+2x}{x^2-6x+9}$.

Solution

1. Factor each polynomial: $\frac{(x-8)(x-3)}{(x-8)(x+2)}$ and $\frac{x(x+2)}{(x-3)^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions **Rational Expressions**

Q037 Written

Simplify $\frac{x+1}{x^2-4} \times \frac{x^2-2x-8}{x^2+x}$.

Solution

1. Factor each polynomial: $\frac{x+1}{(x-2)(x+2)}$ and $\frac{(x-4)(x+2)}{x(x+1)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x-4}{x(x-2)}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{x-4}{x(x-2)}$$

Q038 Written

Simplify $\frac{x^2-9}{x^2-2x} \times \frac{x^2-7x+10}{x^2+x-12}$.

Solution

1. Factor each polynomial: $\frac{(x-3)(x+3)}{x(x-2)}$ and $\frac{(x-5)(x-2)}{(x-3)(x+4)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{(x-5)(x+3)}{x(x+4)}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{(x-5)(x+3)}{x(x+4)}$$

Worked Solutions **Rational Expressions**

Q039 Written

Simplify $\frac{x^2+x-6}{2x^2+x-6} \times \frac{x^2+6x+8}{3x^2-5x-2}$.

Solution

1. Factor each polynomial: $\frac{(x-2)(x+3)}{(x+2)(2x-3)}$ and $\frac{(x+2)(x+4)}{(x-2)(3x+1)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{(x+3)(x+4)}{(2x-3)(3x+1)}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$$

Q040 Written

Simplify $\frac{x^2-4x-5}{2x^2+5x+2} \div \frac{x^2-x-20}{2x^2-5x-3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+1)}{(x+2)(2x+1)}$ and $\frac{(x-3)(2x+1)}{(x-5)(x+4)}$.
3. Cancel only common factors; the result is $= \frac{(x-3)(x+1)}{(x+2)(x+4)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(x+1)}{(x+2)(x+4)}$$

Worked Solutions **Rational Expressions**

Q041 Written

Simplify $\frac{x^2-9}{x^2-x-2} \div \frac{2x^2+5x-3}{x+1}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-3)(x+3)}{(x-2)(x+1)}$ and $\frac{x+1}{(x+3)(2x-1)}$.
3. Cancel only common factors; the result is $= \frac{x-3}{(x-2)(2x-1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{x-3}{(x-2)(2x-1)}$$

Q042 Written

Simplify $\frac{2xy}{3ab} \div \frac{8x^3y}{9a^2b^3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{2xy}{3ab}$ and $\frac{9a^2b^3}{8x^3y}$.
3. Cancel only common factors; the result is $= \frac{3ab^2}{4x^2}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3ab^2}{4x^2}$$

Worked Solutions **Rational Expressions**

Q043 Written

Simplify $\frac{2x^2-x-10}{x^2-2x-3} \div \frac{x^2+x-2}{x^2-6x+9}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x+2)(2x-5)}{(x-3)(x+1)}$ and $\frac{(x-3)^2}{(x-1)(x+2)}$.
3. Cancel only common factors; the result is $= \frac{(x-3)(2x-5)}{(x-1)(x+1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(2x-5)}{(x-1)(x+1)}$$

Q044 Written

Simplify $\frac{x^2-3x-10}{2x^2+3x-2} \div \frac{x^2-2x-15}{6x-3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+2)}{(x+2)(2x-1)}$ and $\frac{3(2x-1)}{(x-5)(x+3)}$.
3. Cancel only common factors; the result is $= \frac{3}{x+3}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3}{x+3}$$

Worked Solutions **Rational Expressions**

Q045 Written

Simplify $\frac{x^2-y^2}{x^2-2xy+y^2} \div \frac{x^2+xy}{x-y}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-y)(x+y)}{(x-y)^2}$ and $\frac{x-y}{x(x+y)}$.
3. Cancel only common factors; the result is $= \frac{1}{x}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$\frac{1}{x}$

Q046 Written

Simplify $\frac{x^3+8}{x^2-x-6} \times \frac{x^2-6x+9}{x^2-2x+4} \div \frac{2x-6}{x^5-1}$ **for** $x = \sqrt{123}$.

Solution

1. Factor every polynomial and postpone substitution until the symbolic cancellation is complete.
2. The chain simplifies to $= \frac{(x-1)(x^4+x^3+x^2+x+1)}{2}$.
3. Substitute $x = \sqrt{123}$ only after simplifying: $= -\frac{1}{2} + \frac{15129\sqrt{123}}{2}$.
4. Check all original denominators and divisor-nonzero conditions.

Final answer

$-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$

Worked Solutions **Rational Expressions**

Q047 **Challenge**

Simplify
$(\frac{x+1}{x-2} - \frac{2}{x+3} + \frac{3}{x+3} - \frac{1}{x-2})$; **state every excluded value.**

Solution

1. Group like denominators, then combine over the two-factor LCM.
2. The reduced expression is $= \frac{x^2+4x-2}{(x-2)(x+3)}$.
3. The original denominators still control the domain.

Final answer
$\frac{x^2+4x-2}{(x-2)(x+3)}$

Worked Solutions **Rational Expressions**

Q048 **Written**

Simplify

$$\frac{x^2-4}{x+1}+\frac{3}{x+1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= x - 1$.
4. Retain all values excluded by the original denominators.

Final answer

$x - 1$

Worked Solutions **Rational Expressions**

Q049 MCQ

Simplify

$$\frac{x^2+2x+3}{x+1}+\frac{2x-1}{x+1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{x^2+4x+2}{x+1}$.
5. Keep every value excluded by an original denominator.

Correct option: C

Final answer

$$\frac{x^2+4x+2}{x+1}$$

Worked Solutions **Rational Expressions**

Q050 MCQ

Simplify

$\dfrac{2x^2-5}{x^2-4}+\dfrac{x+6}{x^2-4}$:

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{2x^2+x+1}{(x-2)(x+2)}$.
5. Keep every value excluded by an original denominator.

Correct option: D

Final answer

$$\frac{2x^2+x+1}{(x-2)(x+2)}$$

Worked Solutions **Rational Expressions**

Q051 MCQ

Simplify
$$\frac{3x^2+x-2}{(x-2)(x+1)}+\frac{4x-3}{(x-2)(x+1)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{3x^2+5x-5}{(x-2)(x+1)}$.
5. Keep every value excluded by an original denominator.

Correct option: A

Final answer
$$\frac{3x^2+5x-5}{(x-2)(x+1)}$$

Worked Solutions **Rational Expressions**

Q052 **MCQ**

Simplify
$$\frac{x^3-1}{x+2}+\frac{2x^2+x+4}{x+2}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{x^3+2x^2+x+3}{x+2}$.
5. Keep every value excluded by an original denominator.

Correct option: B

Final answer
$$\frac{x^3+2x^2+x+3}{x+2}$$

Worked Solutions **Rational Expressions**

Q053 Written

Simplify

$$\frac{3}{x^2+x} - \frac{-x+3}{x^2+x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \dfrac{1}{x+1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{1}{x+1}$$

Worked Solutions **Rational Expressions**

Q054 MCQ

Simplify
$$\frac{x^2+4x+4}{x+2}-\frac{x+4}{x+2}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\frac{x(x+3)}{x+2}$.
5. Keep every value excluded by an original denominator.

Correct option: A

Final answer
$$\frac{x(x+3)}{x+2}$$

Worked Solutions **Rational Expressions**

Q055 MCQ

Simplify

$\dfrac{3x^2-2x+1}{x^2-1}-\dfrac{2x-5}{x^2-1}$:

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{3x^2-4x+6}{(x-1)(x+1)}$.
5. Keep every value excluded by an original denominator.

Correct option: B

Final answer

$\dfrac{3x^2-4x+6}{(x-1)(x+1)}$

Worked Solutions **Rational Expressions**

Q056 MCQ

Simplify

$$\frac{x^2-6x+8}{(x-1)(x+2)}-\frac{x^2-3x+2}{(x-1)(x+2)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=-\frac{3(x-2)}{(x-1)(x+2)}$.
5. Keep every value excluded by an original denominator.

Correct option: C

Final answer

$$-\frac{3(x-2)}{(x-1)(x+2)}$$

Worked Solutions **Rational Expressions**

Q057 **MCQ**

Simplify

$$\frac{2x^3+x}{x-3}-\frac{x^2+7}{x-3}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{2x^3-x^2+x-7}{x-3}$.
5. Keep every value excluded by an original denominator.

Correct option: C

Final answer

$$\frac{2x^3-x^2+x-7}{x-3}$$

Worked Solutions **Rational Expressions**

Q058 **Written**

Simplify
$$\frac{x+4}{x^2-2x}-\frac{3}{x^2-3x+2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x+2}{x\,(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer
$$\frac{x+2}{x\,(x-1)}$$

Q059 **Written**

Simplify
$$\frac{x-1}{x^2+4x+3}+\frac{2}{x^2+5x+6}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \dfrac{x}{(x+1)(x+2)}$.
4. Retain all values excluded by the original denominators.

Final answer
$$\frac{x}{(x+1)(x+2)}$$

Worked Solutions **Rational Expressions**

Q060 MCQ

Simplify

$$\frac{x-1}{(x+1)(x+3)}+\frac{2x+1}{(x+3)(x+5)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\frac{3x^2+7x-4}{(x+1)(x+3)(x+5)}$.
5. Keep every value excluded by an original denominator.

Correct option: B

Final answer

$$\frac{3x^2+7x-4}{(x+1)(x+3)(x+5)}$$

Worked Solutions **Rational Expressions**

Q061 MCQ

Simplify

$$\frac{x+3}{(x+2)(x+4)}+\frac{x-2}{(x+4)(x+6)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{2x^2+9x+14}{(x+2)(x+4)(x+6)}$.
5. Keep every value excluded by an original denominator.

Correct option: C

Final answer

$$\frac{2x^2+9x+14}{(x+2)(x+4)(x+6)}$$

Worked Solutions **Rational Expressions**

Q062 MCQ

Simplify
$$\frac{3x+2}{(x+1)(x+4)}+\frac{x+5}{(x+4)(x+7)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{4x^2+29x+19}{(x+1)(x+4)(x+7)}$.
5. Keep every value excluded by an original denominator.

Correct option: D

Final answer
$$\frac{4x^2+29x+19}{(x+1)(x+4)(x+7)}$$

Worked Solutions **Rational Expressions**

Q063 MCQ

Simplify

$$\frac{x+1}{(x+2)(x+5)}+\frac{4x-3}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{5x^2+14x+2}{(x+2)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: A

Final answer

$$\frac{5x^2+14x+2}{(x+2)(x+5)(x+8)}$$

Worked Solutions **Rational Expressions**

Q064 MCQ

Simplify
$$\frac{2x+1}{(x+3)(x+6)} + \frac{x+4}{(x+6)(x+9)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \dfrac{3x^2+26x+21}{(x+3)(x+6)(x+9)}$.
5. Keep every value excluded by an original denominator.

Correct option: B

Final answer
$$\frac{3x^2+26x+21}{(x+3)(x+6)(x+9)}$$

Worked Solutions **Rational Expressions**

Q065 **MCQ**

Simplify

$$\frac{x-2}{(x+1)(x+5)}+\frac{3x+1}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $=\dfrac{4x^2+10x-15}{(x+1)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: C

Final answer

$$\frac{4x^2+10x-15}{(x+1)(x+5)(x+8)}$$

Worked Solutions **Rational Expressions**

Q066 Written

Simplify

$$\frac{x}{x-1} - \frac{2}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \dfrac{x+2}{x+1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x+2}{x+1}$$

Q067 **Written**

Simplify
$$\frac{x+2}{x} - \frac{x-2}{x-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \dfrac{3x-2}{x\,(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer
$$\frac{3x-2}{x\,(x-1)}$$

Worked Solutions **Rational Expressions**

Q068 **Written**

Simplify
$$\frac{2}{x-1}+\frac{3}{x^2+5x-6}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $=\dfrac{2x+15}{(x-1)(x+6)}$.
4. Retain all values excluded by the original denominators.

Final answer
$$\frac{2x+15}{(x-1)(x+6)}$$

Worked Solutions **Rational Expressions**

Q069 MCQ

Simplify
$$\frac{x+2}{(x+1)(x+3)} - \frac{2}{x+1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= -\frac{x+4}{(x+1)(x+3)}$.
5. Keep every value excluded by an original denominator.

Correct option: C

Final answer
$$-\frac{x+4}{(x+1)(x+3)}$$

Worked Solutions **Rational Expressions**

Q070 Written

Simplify

$$\frac{1}{x+2}+\frac{1}{x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $=\dfrac{2x}{(x-2)(x+2)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x}{(x-2)(x+2)}$$

Q071 **Written**

Simplify

$$\frac{x+4}{x^2+2x}-\frac{x+5}{x^2+x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= -\dfrac{2}{x\,(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$-\frac{2}{x\,(x-1)}$$

Worked Solutions **Rational Expressions**

Q072 Written

Simplify

$$\frac{x+8}{x^2+x-2} - \frac{x+5}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \dfrac{2}{(x+1)(x+2)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x+1)(x+2)}$$

Q073 **Written**

Simplify

$$\frac{x-3}{x^2+x}+\frac{x}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \dfrac{2x^2-4x+3}{x\,(x-1)\,(x+1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x^2-4x+3}{x\,(x-1)\,(x+1)}$$

Worked Solutions **Rational Expressions**

Q074 Written

Simplify

$$\frac{4x}{x^2-1}-\frac{x-1}{x^2+x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \dfrac{3x-1}{x\,(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3x-1}{x\,(x-1)}$$

Q075 **Written**

Simplify

$$\frac{1}{x^2+5x+6}+\frac{1}{x^2+7x+12}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \dfrac{2}{(x+2)(x+4)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x+2)(x+4)}$$

Worked Solutions **Rational Expressions**

Q076 **Written**

Simplify

$$\frac{x-1}{x^2+3x+2}-\frac{x-3}{x^2+4x+3}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \dfrac{3}{(x+2)(x+3)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3}{(x+2)(x+3)}$$

Worked Solutions **Rational Expressions**

Q077 MCQ

A learner combines $\frac{2x-1}{(x-1)(x+2)} - \frac{x+4}{x+2}$. Correct the numerator line, then simplify the expression:

Solution

1. Keep one common denominator; never add the denominators.
2. Put the entire second numerator in parentheses before applying the sign.
3. After factoring and reducing, the correct result is $= -\frac{x^2+x-3}{(x-1)(x+2)}$.
4. Check the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{x^2+x-3}{(x-1)(x+2)}$$

Worked Solutions **Rational Expressions**

Q078 **Written**

Simplify
$$\frac{x^3+8x}{x^2-x-2}-\frac{8}{x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $=\dfrac{x^2+2x+4}{x+1}$.
4. Retain all values excluded by the original denominators.

Final answer
$$\frac{x^2+2x+4}{x+1}$$

Worked Solutions **Rational Expressions**

Q079 Challenge

Simplify the single-inner fraction together with the reciprocal-pair quotient; state the full domain.

Solution

1. Identify the two family triggers before manipulating the expression.
2. Simplify each component inside-out, keeping the original restrictions.
3. Combine the reduced components: $= -\dfrac{x^2 - 4x + 2}{2\,(x - 2)}$.
4. Audit every original inner and outer denominator.

Final answer

$$-\frac{x^2 - 4x + 2}{2\,(x - 2)}$$

Worked Solutions **Rational Expressions**

Q080 Written

Simplify

$$\frac{1}{1-\frac{1}{x-1}}.$$

Solution

1. The inner denominator requires x-1 $\neq$ 0.
2. Multiply the outer numerator and denominator by x-1.
3. The result is $\frac{x-1}{x-2}$.
4. The outer denominator also excludes $x = 2$.

Final answer

$$\frac{x-1}{x-2}$$

Q081 **Written**

Simplify

$$\frac{1}{1-\frac{x}{x+1}}.$$

Solution

1. The inner denominator requires x+1 $\neq$ 0.
2. Clear the inner denominator x+1 in the outer fraction.
3. The denominator becomes 1, so the result is $x + 1$.
4. Retain $x \neq -1$ from the original expression.

Final answer

$x + 1$

Worked Solutions **Rational Expressions**

Q082 **MCQ**

Simplify

$$\frac{1}{1-\frac{x+2}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 1$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$-x-1$

Q083 **MCQ**

Simplify

$$\frac{1}{1 - \dfrac{x+3}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 2$.
4. State every restriction from the original denominators.

Correct option: A

Final answer

$-x - 2$

Worked Solutions **Rational Expressions**

Q084 MCQ

Simplify

$$\frac{1}{1-\frac{2x+1}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $=-\frac{x+1}{x}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x}$$

Q085 **MCQ**

Simplify

$$\frac{1}{1-\dfrac{2x+3}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $=-\dfrac{x+1}{x+2}$.
4. State every restriction from the original denominators.

Correct option: C

Final answer

$$-\frac{x+1}{x+2}$$

Worked Solutions **Rational Expressions**

Q086 MCQ

Simplify
$$\frac{1}{1-\dfrac{3x+1}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\dfrac{x+2}{2x-1}$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$$-\frac{x+2}{2x-1}$$

Q087 **MCQ**

Simplify

$$\frac{1}{1-\dfrac{x+4}{x+3}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 3$.
4. State every restriction from the original denominators.

Correct option: A

Final answer

$-x - 3$

Worked Solutions **Rational Expressions**

Q088 MCQ

Simplify
$$\frac{1}{1-\frac{2x+5}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+1}{x+4}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x+4}$$

Q089 **Written**

Simplify

$$\frac{1}{1-x}+\frac{1}{1+x}\Big/\left(\frac{1}{1-x}-\frac{1}{1+x}\right).$$

Solution

1. Multiply both outer parts by (1-x)(1+x).
2. The numerator pair becomes 2; the denominator pair becomes 2x.
3. Reduce to $1/x$.
4. The original inner denominators and the outer denominator give $x \neq -1, 0, 1$.

Final answer

$1/x$

Worked Solutions **Rational Expressions**

Q090 MCQ

Simplify
$$\frac{1}{x+1}+\frac{1}{x+4}\Big/\left(\frac{1}{x+1}-\frac{1}{x+4}\right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $=\dfrac{2x+5}{3}$.
4. State every restriction from the original denominators.

Correct option: A

Final answer
$$\frac{2x+5}{3}$$

Q091 MCQ

Simplify
$\frac{1}{x+1}+\frac{1}{x+5}\Big/\left(\frac{1}{x+1}-\frac{1}{x+5}\right)$:

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $=\frac{x+3}{2}$.
4. State every restriction from the original denominators.

Correct option: A

Final answer
$\frac{x+3}{2}$

Worked Solutions **Rational Expressions**

Q092 Written

Simplify
$$\frac{1}{x+2}+\frac{1}{x-2}\Big/\left(\frac{1}{x+2}-\frac{1}{x-2}\right).$$

Solution

1. Multiply numerator and denominator by (x+2)(x-2).
2. The numerator pair becomes 2x; the denominator pair becomes -4.
3. The result is $-x/2$.
4. Keep $x \neq -2$ and $x \neq 2$.

Final answer

$$-\frac{x}{2}$$

Q093 **MCQ**

Simplify
$$\frac{1}{x+1}+\frac{1}{x+3}\Big/\left(\frac{1}{x+1}+\frac{1}{x+3}\right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= 1$.
4. State every restriction from the original denominators.

Correct option: C

Final answer

1

Worked Solutions **Rational Expressions**

Q094 MCQ

Simplify

$$\frac{1}{x+2} + \frac{1}{x+5} \Big/ \left(\frac{1}{x+2} + \frac{1}{x+5}\right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= 1$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

1

Q095 **MCQ**

Simplify
$$\frac{1}{x+3}+\frac{1}{x+6}\bigg/\left(\frac{1}{x+3}+\frac{1}{x+6}\right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= 1$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

1

Worked Solutions **Rational Expressions**

Q096 Written

Simplify

$$\frac{1}{x+y}+\frac{1}{x-y}\Big/\left(\frac{1}{x+y}-\frac{1}{x-y}\right).$$

Solution

1. Clear both inner denominators using (x+y)(x-y).
2. The numerator pair becomes 2x; the denominator pair becomes -2y.
3. Reduce to $-x/y$.
4. Keep $y \neq 0$ and $x \neq \pm y$.

Final answer

$$-\frac{x}{y}$$

Q097 **Written**

Simplify

$$\frac{1}{1-\frac{1}{1-\frac{1}{x}}}.$$

Solution

1. Start with the innermost 1-1/x.
2. Substitute that result into the next reciprocal layer.
3. The outer result reduces to $1 - x$.
4. The original denominators exclude $x = 0$ and $x = 1$.

Final answer

$1 - x$

Worked Solutions **Rational Expressions**

Q098 Written

Simplify

$$\frac{1}{1-\frac{1}{1+\frac{1}{x}}}.$$

Solution

1. Simplify 1+1/x before entering the next layer.
2. Substitute the result into 1-1/(...).
3. The final result is $x+1$.
4. Retain $x \neq 0$ and $x \neq -1$.

Final answer

$x+1$

Q099 **Written**

Simplify

$\left(1+\dfrac{x-y}{x+y}\right)\Big/\left(1-\dfrac{x-y}{x+y}\right).$

Solution

1. Let $u = (x - y)/(x + y)$, then combine 1+u and 1-u.
2. Clear x+y in the resulting numerator and denominator.
3. The reduced result is $\dfrac{x}{y}$.
4. Require $y \neq 0$ and $x + y \neq 0$.

Final answer

$\dfrac{x}{y}$

Worked Solutions **Rational Expressions**

Q100 Written

Simplify

$\left(x - \dfrac{2}{x-1}\right) \Big/ \left(\dfrac{x}{x-1} - 2\right)$.

Solution

1. Combine the polynomial pieces over x-1 in the numerator and denominator.
2. Form the outer quotient only after both pieces are simplified.
3. Cancel complete factors to obtain $-(x+1)$.
4. The original denominators retain $x \neq 1$ and $x \neq 2$.

Final answer

$-(x+1)$

Q101 Written

Simplify

$\left(\dfrac{x+1}{x-1} - \dfrac{x-1}{x+1}\right) \Big/ \left(\dfrac{x+1}{x-1} + \dfrac{x-1}{x+1}\right)$.

Solution

1. Use (x-1)(x+1) inside both the numerator and denominator.
2. The difference gives 4x; the sum gives 2(x²+1).
3. The simplified result is $\dfrac{2x}{x^2+1}$.
4. The original factors require $x \neq \pm 1$.

Final answer

$\dfrac{2x}{x^2+1}$

Worked Solutions **Rational Expressions**

Q102 Written

Simplify

$$\frac{1}{1+\frac{1}{1+\frac{1}{1+x}}}.$$

Solution

1. Translate the three booster stages as an inside-out reciprocal chain.
2. Simplify the innermost stage 1+1/(1+x).
3. Continuing outward gives $\frac{x+2}{2x+3}$.
4. The stated model condition is x>0.

Final answer

$$\frac{x+2}{2x+3}$$

Quiz MCQ 1 Concept Quiz · MCQ

Simplify $\frac{x^2-4}{3x^2+7x+2}$:

Solution

1. Factor both polynomials and cancel x+2.

Correct option: A

Final answer

$$\frac{x-2}{3x+1}$$

Worked Solutions **Rational Expressions**

Quiz MCQ 2 **Concept Quiz · MCQ**

Simplify
$\dfrac{x+1}{x-2}+\dfrac{2}{x-2}$:

Solution

1. The denominators match, so add the numerators only.

Correct option: A

Final answer
$\dfrac{x+3}{x-2}$

Quiz MCQ 3 **Concept Quiz · MCQ**

In
$\dfrac{3}{x^2+x}-\dfrac{3-x}{x^2+x}$**, the numerator after the minus sign is... Which of the following is correct?:**

Solution

1. Parentheses preserve the sign of every term in the second numerator.

Correct option: A

Final answer
$3-(3-x)$

Worked Solutions **Rational Expressions**

Quiz MCQ 4 Concept Quiz · MCQ

Which statement is safe when simplifying a compound fraction?:

Solution

1. A common multiplier preserves the quotient and exposes the structure.

Correct option: A

Final answer

multiply top and bottom by the same inner denominator

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $\frac{x^2-9}{x^2-2x-3} \times \frac{x^2-4}{x^2+x-6}$ and state the original restrictions.

Solution

1. Factor all four polynomials and cancel common factors.
2. Final form: $= \frac{x+2}{x+1}$; retain original denominator exclusions.

Final answer

$$\frac{x+2}{x+1}$$

Worked Solutions **Rational Expressions**

Quiz WRITTEN 6 Concept Quiz · Written

Simplify
$\frac{x^2-4}{x+1}+\frac{3}{x+1}$ **and state the original restriction.**

Solution

1. Use the common denominator x+1 and combine the numerators.
2. Final form: $= x-1$; $x \neq -1$ remains excluded.

Final answer

$x-1$

Quiz WRITTEN 7 Concept Quiz · Written

Simplify
$\frac{x^2-y^2}{x^2-2xy+y^2}-\frac{x+y}{x-y}$ **in x and y.**

Solution

1. Factor x^2-y^2, use the common (x-y) structure, and apply the minus sign to x+y.
2. Final form: $= 0$.

Final answer

0

Worked Solutions **Rational Expressions**

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $(x^2-16)/(x-4)=x+4$ is not an identity for every real x.

Solution

1. The cancellation is valid only when $x - 4 \neq 0$.
2. At $x = 4$ the original expression is undefined, so the graph has a hole.

Final answer

$x \neq 4;$ removable hole at $(4, 8)$

Answer key

Use the question number shown on each page.

Q001 A

Q002 B

Q003 C

Q004 D

Q005 A

Q006 B

Q007 $\frac{4b^3}{3a^2}$

Q008 $\frac{3y^2}{10x}$

Q009 $\frac{3a}{4b^3}$

Q010 $\frac{1}{4ab^2}$

Q011 B

Q012 A

Q013 A

Q014 B

Q015 A

Q016 B

Q017 C

Q018 D

Q019 A

Q020 B

Q021 C

Q022 $\frac{3c^2}{a}$

Q023 $\frac{4a^2c^3}{b^2}$

Q024 C

Q025 D

Q026 D

Q027 A

Q028 B

Q029 C

Q030 $\frac{x-2}{3x+1}$

Q031 $\frac{2x-5}{x+2}$

Q032 $\frac{x}{x-3}$

Q033 $\frac{x+2}{x-4}$

Q034 $\frac{2(x-y)}{x(x+4y)}$

Q035 $\frac{x^2+x+1}{x+3}$

Q036 $\frac{x}{x-3}$

Q037 $\frac{x-4}{x(x-2)}$

Q038 $\frac{(x-5)(x+3)}{x(x+4)}$

Q039 $\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$

Q040 $\frac{(x-3)(x+1)}{(x+2)(x+4)}$

Q041 $\frac{x-3}{(x-2)(2x-1)}$

Q042 $\frac{3ab^2}{4x^2}$

Q043 $\frac{(x-3)(2x-5)}{(x-1)(x+1)}$

Q044 $\frac{3}{x+3}$

Q045 $\frac{1}{x}$

Q046 $-\frac{1}{2}+\frac{15129\sqrt{123}}{2}$

Q047 $\frac{x^2+4x-2}{(x-2)(x+3)}$

Q048 $x-1$

Q049 C

Q050 D

Q051 A

Q052 B

Q053 $\frac{1}{x+1}$

Q054 A

Q055 B

Q056 C

Q057 C

Q058 $\frac{x+2}{x(x-1)}$

Q059 $\frac{x}{(x+1)(x+2)}$

Q060 B

Q061 C

Q062 D

Q063 A

Q064 B

Q065 C

Q066 $\frac{x+2}{x+1}$

Q067 $\frac{3x-2}{x(x-1)}$

Q068 $\frac{2x+15}{(x-1)(x+6)}$

Q069 C

Q070 $\frac{2x}{(x-2)(x+2)}$

Q071 $-\frac{2}{x(x-1)}$

Q072 $\frac{2}{(x+1)(x+2)}$

Q073 $\frac{2x^2-4x+3}{x(x-1)(x+1)}$

Q074 $\frac{3x-1}{x(x-1)}$

Q075 $\frac{2}{(x+2)(x+4)}$

Q076 $\frac{3}{(x+2)(x+3)}$

Q077 B

Q078 $\frac{x^2+2x+4}{x+1}$

Q079 $-\frac{x^2-4x+2}{2(x-2)}$

Q080 $\frac{x-1}{x-2}$

Q081 $x+1$

Q082 D

Q083 A

Q084 B

Q085 C

Q086 D

Q087 A

Q088 B

Q089 $1/x$

Q090 A

Q091 A

Q092 $-\frac{x}{2}$

Q093 C

Q094 D

Q095 B

Q096 $-\frac{x}{y}$

Q097 $1-x$

Q098 $x+1$

Q099 $\frac{x}{y}$

Q100 $-(x+1)$

Q101 $\frac{2x}{x^2+1}$

Q102 $\frac{x+2}{2x+3}$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{x+2}{x+1}$

Quiz WRITTEN 6 $x-1$

Quiz WRITTEN 7 0

Quiz WRITTEN 8 $x \neq 4;$ removable hole at $(4, 8)$

Concept 4 | Identities and Proofs

English Student Edition • Focused formulas and guided practice

Formula opener

Proof route

$$\text{LHS} \longrightarrow \text{RHS}$$

Square

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference

$$a^2 - b^2 = (a - b)(a + b)$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Identities and Proofs**

Q001 Challenge

$2x^2 + 3x + 1 = (x + 1)(ax + b) + c$. **Find a,b,c.**

Solution

1. Decide that this is a polynomial identity and expand the product.
2. Match the x^2, x, and constant coefficients.
3. The constants are $a = 2$, $b = 1$, $c = 0$; substitute back.

Final answer

$a = 2,\ b = 1,\ c = 0$

Worked Solutions **Identities and Proofs**

Q002 **Written**

$x^2 + 6 = a(x+1)^2 + b(x+1) + c$

Solution

1. Expand the shifted square: the right side is ax^2+(2a+b)x+(a+b+c).
2. Match coefficients with $x^2 + 0x + 6$: $a = 1$, $2a + b = 0$, $a + b + c = 6$.
3. Solve in order to obtain $a = 1$, $b = -2$, $c = 7$.
4. Substitution reproduces the original identity.

Final answer

$a = 1,\ b = -2,\ c = 7$

Q003 **Written**

$x^2 - x - 3 = a(x-1)^2 + b(x-1) + c$

Solution

1. Expand and collect the right side as ax^2+(-2a+b)x+(a-b+c).
2. Match with $x^2 - x - 3$ and solve $a = 1$, $b = 1$, $c = -3$.
3. Check all three coefficients in the original identity.

Final answer

$a = 1,\ b = 1,\ c = -3$

Worked Solutions **Identities and Proofs**

Q004 *MCQ*

For the identity $x^2 + 6x + 5 = a(x+2)^2 + b(x+2) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2, x, and constant coefficients.
3. Read off $a = 1$, $b = 2$, $c = -3$.
4. Substitute back to check.

Correct option: C

Final answer

$(1,\ 2,\ -3)$

Q005 *MCQ*

For the identity $2x^2 + 9x + 8 = a(x+3)^2 + b(x+3) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2, x, and constant coefficients.
3. Read off $a = 2$, $b = -3$, $c = -1$.
4. Substitute back to check.

Correct option: C

Final answer

$(2,\ -3,\ -1)$

Worked Solutions **Identities and Proofs**

Q006 MCQ

For the identity $3x^2 + 28x + 65 = a(x+4)^2 + b(x+4) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2, x, and constant coefficients.
3. Read off $a = 3$, $b = 4$, $c = 1$.
4. Substitute back to check.

Correct option: D

Final answer

$(3,\ 4,\ 1)$

Q007 MCQ

For the identity $x^2 + 5x + 3 = a(x+5)^2 + b(x+5) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2, x, and constant coefficients.
3. Read off $a = 1$, $b = -5$, $c = 3$.
4. Substitute back to check.

Correct option: A

Final answer

$(1,\ -5,\ 3)$

Worked Solutions **Identities and Proofs**

Q008 Written

$a(x+1) - b(x+5) = 8x$

Solution

1. Collect coefficients: $(a-b)x + (a-5b) = 8x$.
2. Solve $a - b = 8$ and $a - 5b = 0$.
3. Hence $a = 10$ and $b = 2$; the constant term checks as zero.

Final answer

$a = 10,\ b = 2$

Q009 MCQ

For the identity $a(x+1) - b(x+3) = x - 1$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 2$, $b = 1$.
4. Check both terms.

Correct option: A

Final answer

$(2,\ 1)$

Worked Solutions **Identities and Proofs**

Q010 MCQ

For the identity $a(x+2) - b(x+4) = 5x + 14$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 3$, $b = -2$.
4. Check both terms.

Correct option: B

Final answer

$(3,\ -2)$

Q011 MCQ

For the identity $a(x+3) - b(x+5) = x - 3$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 4$, $b = 3$.
4. Check both terms.

Correct option: D

Final answer

$(4,\ 3)$

Worked Solutions **Identities and Proofs**

Q012 MCQ

For the identity $a(x+4) - b(x+6) = 9x + 44$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 5$, $b = -4$.
4. Check both terms.

Correct option: D

Final answer

$(5, \ -4)$

Q013 Written

$(x+a)(b+3) = 4x + 12$

Solution

1. Expand as (b+3)x+a(b+3).
2. Match $b + 3 = 4$ and $a(b+3) = 12$.
3. Therefore $b = 1$ and $a = 3$; c is not present in the printed identity and is not applicable.

Final answer

$a = 3, \ b = 1$

Q014 MCQ

For the identity $(x+a)(b+4) = 3x+6$, the ordered tuple (a, b) is:

Solution

1. Expand as (b+d)x+a(b+d).
2. Match the coefficient of x and the constant term.
3. Obtain $a = 2$, $b = -1$; no extra parameter is present.
4. Substitute back.

Correct option: B

Final answer

$(2,\ -1)$

Q015 MCQ

For the identity $(x+a)(b+5) = 5x+15$, the ordered tuple (a, b) is:

Solution

1. Expand as (b+d)x+a(b+d).
2. Match the coefficient of x and the constant term.
3. Obtain $a = 3$, $b = 0$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer

$(3,\ 0)$

Worked Solutions **Identities and Proofs**

Q016 MCQ

For the identity $(x+a)(b+6) = 7x + 28$, the ordered tuple (a, b) is:

Solution

1. Expand as (b+d)x+a(b+d).
2. Match the coefficient of x and the constant term.
3. Obtain $a = 4$, $b = 1$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer

$(4,\ 1)$

Q017 Written

$2x^2 + 5x + 4 = (x+1)(ax+b) + c$

Solution

1. Expand the right side to ax^2+(a+b)x+(b+c).
2. Match $(a, a+b, b+c) = (2, 5, 4)$.
3. Solve $a = 2$, $b = 3$, $c = 1$ and check the constant term.

Final answer

$a = 2,\ b = 3,\ c = 1$

Worked Solutions **Identities and Proofs**

Q018 **Written**

$(ax + b)(x + 1) = 4x^2 + cx + 1$

Solution

1. Expand: ax^2+(a+b)x+b.
2. Match the x^2 and constant coefficients to get $a = 4$ and $b = 1$.
3. Then $c = a + b = 5$.

Final answer

$a = 4,\ b = 1,\ c = 5$

Q019 **Written**

$2x^2 - 7x + 8 = (x - 3)(ax + b) + c$

Solution

1. Expand to ax^2+(b-3a)x+(-3b+c).
2. Match with (2,-7,8).
3. Solve $a = 2$, $b = -1$, $c = 5$ and substitute back.

Final answer

$a = 2,\ b = -1,\ c = 5$

Worked Solutions **Identities and Proofs**

Q020 MCQ

For the identity $(ax-1)(x+1)+c = x^2-3$, the ordered tuple (a, c) is:

Solution

1. Expand the product to ax²+(ah+b)x+bh+c.
2. Match the x^2, x, and constant coefficients.
3. Use $a = 1$ and $c = -2$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B

Final answer

$(1,\ -2)$

Q021 MCQ

For the identity $(ax+2)(x+2)+c = 2x^2+6x+3$, the ordered tuple (a, c) is:

Solution

1. Expand the product to ax²+(ah+b)x+bh+c.
2. Match the x^2, x, and constant coefficients.
3. Use $a = 2$ and $c = -1$, then verify the middle coefficient.
4. Check the full identity.

Correct option: C

Final answer

$(2,\ -1)$

Worked Solutions **Identities and Proofs**

Q022 MCQ

For the identity $(ax+3)(x+3)+c = 3x^2+12x+9$, the ordered tuple (a, c) is:

Solution

1. Expand the product to ax^2+(ah+b)x+bh+c.
2. Match the x^2, x, and constant coefficients.
3. Use $a = 3$ and $c = 0$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B

Final answer

$(3,\ 0)$

Q023 Written

$$\frac{3x+1}{(x+2)(x-3)} = \frac{a}{x+2} + \frac{b}{x-3}$$

Solution

1. Exclude $x = -2$ and $x = 3$ from the original denominator.
2. Multiply by (x+2)(x-3): $3x + 1 = a(x-3) + b(x+2)$.
3. Match coefficients: $a + b = 3$ and $-3a + 2b = 1$, giving $a = 1$, $b = 2$.
4. Substitute a and b back into the rational identity.

Final answer

$a = 1,\ b = 2$

Worked Solutions **Identities and Proofs**

Q024 **Written**

$$\frac{x+8}{(x-2)(x+3)}=\frac{a}{x-2}+\frac{b}{x+3}$$

Solution

1. Exclude $x=2$ and $x=-3$.
2. Clear denominators: $x+8=a(x+3)+b(x-2)$.
3. Match $a+b=1$ and $3a-2b=8$, giving $a=2$, $b=-1$.
4. Substitute and verify.

Final answer

$a=2,\ b=-1$

Q025 **Written**

$$\frac{1}{(2x-1)(x+1)}=\frac{a}{2x-1}+\frac{b}{x+1}$$

Solution

1. Exclude $x=1/2$ and $x=-1$.
2. Clear denominators: $1=a(x+1)+b(2x-1)$.
3. Match $a+2b=0$ and $a-b=1$, giving $a=2/3$, $b=-1/3$.
4. Verify in the original decomposition.

Final answer

$a=\frac{2}{3},\ b=-\frac{1}{3}$

Worked Solutions **Identities and Proofs**

Q026 **Written**

$$\frac{a}{x-1}+\frac{b}{x-5}=\frac{5x-1}{(x-1)(x-5)}$$

Solution

1. Exclude $x=1$ and $x=5$.
2. Clear denominators: $a(x-5)+b(x-1)=5x-1$.
3. Match $a+b=5$ and $-5a-b=-1$, giving $a=-1$, $b=6$.
4. Check the original rational identity.

Final answer

$a=-1,\ b=6$

Q027 **Written**

$$\frac{3x-8}{3x^2+5x-2}=\frac{a}{x+2}+\frac{b}{3x-1}$$

Solution

1. Factor $3x^2+5x-2=(x+2)(3x-1)$.
2. Exclude $x=-2$ and $x=1/3$, then clear denominators.
3. Match $3a+b=3$ and $-a+2b=-8$, giving $a=2$, $b=-3$.
4. Check the factored rational identity.

Final answer

$a=2,\ b=-3$

Worked Solutions **Identities and Proofs**

Q028 **Written**

$$f(t) = \frac{22t + 15}{8t^2 + 10t + 3} = \frac{a}{2t + 1} + \frac{b}{4t + 3},\ t > 0$$

Solution

1. Factor $(8t^2 + 10t + 3 = (2t + 1)(4t + 3))$.
2. The stated domain is (t>0); the algebraic exclusions lie outside it.
3. Clear denominators: $(22t + 15 = a(4t + 3) + b(2t + 1))$.
4. Match $(4a + 2b = 22)$ and $(3a + b = 15)$ to get $(a = 4, b = 3)$.

Final answer

$a = 4,\ b = 3$

Q029 **Written**

Prove that $(x + y)^2 - (x - y)^2 = 4xy$.

Solution

1. Expand both conjugate squares on the left.
2. Remove the parentheses: $x^2 + 2xy + y^2 - x^2 + 2xy - y^2$.
3. Collect like terms to obtain 4xy, which is the right-hand side.

Final answer

$4xy$

Worked Solutions **Identities and Proofs**

Q030 Written

Prove that $(2a+b)^2-(a+2b)^2=3\left(a^2-b^2\right)$.

Solution

1. Recognise a difference of squares: $U^2-V^2=(U-V)(U+V)$.
2. Set $U=2a+b$ and $V=a+2b$; then $U-V=a-b$ and $U+V=3a+3b$.
3. Multiply to get $(a-b)(3a+3b)=3(a^2-b^2)$.

Final answer

$3a^2-3b^2$

Q031 MCQ

For the identity $-(x-2y+1)^2+(x+2y+1)^2=2y\,(4x+4)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy+8y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A

Final answer

$8xy+8y$

Worked Solutions **Identities and Proofs**

Q032 MCQ

For the identity $-(x-3y+2)^2+(x+3y+2)^2=3y(4x+8)$**, the value of the simplified left-hand side is:**

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $12xy+24y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$12xy+24y$

Q033 MCQ

For the identity $-(x-y+3)^2+(x+y+3)^2=y(4x+12)$**, the value of the simplified left-hand side is:**

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $4xy+12y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$4xy+12y$

Worked Solutions **Identities and Proofs**

Q034 MCQ

For the identity $-(x-2y+4)^2+(x+2y+4)^2=2y\,(4x+16)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy+32y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A

Final answer

$8xy+32y$

Q035 Written

Prove that $(x+y)^2+(x-y)^2=2x^2+2y^2$.

Solution

1. Expand both squares on the left.
2. Add $x^2+2xy+y^2$ and $x^2-2xy+y^2$.
3. The mixed terms cancel, leaving $2x^2+2y^2$.

Final answer

$2x^2+2y^2$

Worked Solutions **Identities and Proofs**

Q036 MCQ

For the identity $(x-2y+1)^2+(x+2y+1)^2=8y^2+2(x+1)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2+4x+8y^2+2$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$2x^2+4x+8y^2+2$

Q037 MCQ

For the identity $(x-3y+2)^2+(x+3y+2)^2=18y^2+2(x+2)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2+8x+18y^2+8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$2x^2+8x+18y^2+8$

Worked Solutions **Identities and Proofs**

Q038 MCQ

For the identity $(x - y + 3)^2 + (x + y + 3)^2 = 2y^2 + 2(x + 3)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 12x + 2y^2 + 18$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$2x^2 + 12x + 2y^2 + 18$

Q039 MCQ

For the identity $(x - 2y + 4)^2 + (x + 2y + 4)^2 = 8y^2 + 2(x + 4)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 16x + 8y^2 + 32$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$2x^2 + 16x + 8y^2 + 32$

Worked Solutions **Identities and Proofs**

Q040 Written

Prove that $(a+2b)^2 - 4ab = (a-2b)^2 + 4ab$.

Solution

1. Expand the left-hand side: $a^2 + 4ab + 4b^2 - 4ab = a^2 + 4b^2$.
2. Expand the right-hand side: $a^2 - 4ab + 4b^2 + 4ab = a^2 + 4b^2$.
3. Both sides simplify to the same expression.

Final answer

$a^2 + 4b^2$

Q041 MCQ

For the identity $-(x-2y+3)^2 + (x+2y-1)^2 = (8x+8)(y-1)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy - 8x + 8y - 8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$8xy - 8x + 8y - 8$

Worked Solutions **Identities and Proofs**

Q042 MCQ

For the identity $-(2x-3y+5)^2+(2x+3y-1)^2=(24x+24)(y-1)$**, the value of the simplified left-hand side is:**

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $24xy-24x+24y-24$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$24xy-24x+24y-24$

Q043 Written

Prove that $\left(x^2+x+1\right)\left(x^2-x+1\right)=x^4+x^2+1$**.**

Solution

1. Group the factors as $[(x^2+1)-x][(x^2+1)+x]$.
2. Use $(U-V)(U+V)=U^2-V^2$ with $U=x^2+1$ and $V=x$.
3. Simplify $(x^2+1)^2-x^2$ to x^4+x^2+1.

Final answer

x^4+x^2+1

Worked Solutions **Identities and Proofs**

Q044 Written

Prove that $\left(a^2 - ab + b^2\right)\left(a^2 + ab + b^2\right) = \left(a^2 + b^2\right)^2 - (ab)^2$.

Solution

1. Group the factors as $[(a^2 + b^2) - ab][(a^2 + b^2) + ab]$.
2. Apply $(U - V)(U + V) = U^2 - V^2$ with $U = a^2 + b^2$ and $V = ab$.
3. This gives $(a^2 + b^2)^2 - (ab)^2$, exactly the right-hand side.

Final answer

$a^4 + a^2b^2 + b^4$

Q045 Written

Prove that $\left(a^2 + b^2\right)\left(c^2 + d^2\right) = (ac - bd)^2 + (ad + bc)^2$.

Solution

1. Expand the left-hand product to $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$.
2. Expand the first right-hand square: $a^2c^2 - 2abcd + b^2d^2$.
3. Expand the second right-hand square: $a^2d^2 + 2abcd + b^2c^2$.
4. Add the two expansions; the -2abcd and +2abcd terms cancel, leaving the left-hand expression.

Final answer

$a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Worked Solutions **Identities and Proofs**

Q046 **Written**

Prove that $(x^2+1)(y^2+1)=(xy+1)^2+(x-y)^2$.

Solution

1. Expand the left-hand side to $x^2y^2+x^2+y^2+1$.
2. Expand the right-hand side: $x^2y^2+2xy+1+x^2-2xy+y^2$.
3. The +2xy and -2xy terms cancel, leaving the same expression as the left.

Final answer

$x^2y^2+x^2+y^2+1$

Q047 **Written**

Prove that $(a^2-b^2)(c^2-d^2)=(ac-bd)^2-(ad-bc)^2$.

Solution

1. Expand the first right-hand square: $a^2c^2-2abcd+b^2d^2$.
2. Expand the second right-hand square: $a^2d^2-2abcd+b^2c^2$.
3. Subtract; the mixed terms cancel, giving $a^2c^2-a^2d^2-b^2c^2+b^2d^2$.
4. Factor the result as $(a^2-b^2)(c^2-d^2)$.

Final answer

$a^2c^2-a^2d^2-b^2c^2+b^2d^2$

Worked Solutions **Identities and Proofs**

Q048 **Written**

Prove that $(a+b)^3 - 3ab\,(a+b) = a^3 + b^3$.

Solution

1. Expand $(a+b)^3$ to $a^3 + 3a^2b + 3ab^2 + b^3$.
2. Expand 3ab(a+b) to $3a^2b + 3ab^2$.
3. Subtract the matching cross terms; the result is $a^3 + b^3$.

Final answer

$a^3 + b^3$

Q049 **Written**

Prove that $(a+b+c)\left(a^2 + b^2 + c^2 - ab - bc - ca\right) = a^3 + b^3 + c^3 - 3abc$.

Solution

1. Distribute a+b+c across the second bracket.
2. Write all products in a fixed order: a^3, b^3, c^3, then the mixed terms.
3. Each mixed term cancels with its opposite, while -abc appears three times.
4. The remaining expression is $a^3 + b^3 + c^3 - 3abc$.

Final answer

$a^3 - 3abc + b^3 + c^3$

Q050 Challenge

For the integrated identity $\left(x^2+1\right)\left(-(x-2y+1)^2+(x+2y+1)^2\right)=2y\,(4x+4)\left(x^2+1\right)$, the value of the simplified left-hand side is:

Solution

1. Recognise the cubic or symmetric pattern before expanding.
2. Expand in descending degree and collect like terms.
3. The cross terms cancel, leaving $8x^3y+8x^2y+8xy+8y$.
4. Therefore the proposed equality is proved.

Correct option: B

Final answer

$8x^3y+8x^2y+8xy+8y$

Worked Solutions **Identities and Proofs**

Q051 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 + 1 = 2(x + y - xy)$.

Solution

1. Use $x + y = 1$, so $(x + y)^2 = 1$.
2. Expand: $x^2 + 2xy + y^2 = 1$, hence $x^2 + y^2 + 1 = 2 - 2xy$.
3. Since $x + y = 1$, $2 - 2xy = 2(x + y - xy)$. Therefore the equality holds.

Final answer

$x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 = 1 - 2xy$.

Solution

1. Square the condition: $(x + y)^2 = 1$.
2. Expand to $x^2 + 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 1 - 2xy$.

Final answer

$x^2 + y^2 = 1 - 2xy$

Worked Solutions **Identities and Proofs**

Q053 MCQ

Under the condition $x + y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 2$.
2. $y = 2 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$-2xy + 4$

Q054 MCQ

Under the condition $x + y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 3$.
2. $y = 3 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$-2xy + 9$

Worked Solutions **Identities and Proofs**

Q055 MCQ

Under the condition $x + y = -1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = -1$.
2. $y = -1 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$-2xy + 1$

Q056 MCQ

Under the condition $x + y = 4$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 4$.
2. $y = 4 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 16$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: A

Final answer

$-2xy + 16$

Worked Solutions **Identities and Proofs**

Q057 Written

Under the condition $x - y = 1$, prove that $x^2 + y^2 = 2xy + 1$.

Solution

1. Square the condition: $(x - y)^2 = 1$.
2. Expand to $x^2 - 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 2xy + 1$.

Final answer

$x^2 + y^2 = 2xy + 1$

Q058 MCQ

Under the condition $x - y = 1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 1$.
2. $x = y + 1$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B

Final answer

$2xy + 1$

Worked Solutions **Identities and Proofs**

Q059 MCQ

Under the condition $x - y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 2$.
2. $x = y + 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B

Final answer

$2xy + 4$

Q060 MCQ

Under the condition $x - y = -2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = -2$.
2. $x = y - 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$2xy + 4$

Worked Solutions **Identities and Proofs**

Q061 MCQ

Under the condition $x - y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 3$.
2. $x = y + 3$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$2xy + 9$

Q062 Written

Under the condition $x + y = 1$, prove that $x^2 - x = y^2 - y$.

Solution

1. Move the right-hand side to the left: $x^2 - y^2 - x + y$.
2. Factor: $x^2 - y^2 - x + y = (x - y)(x + y - 1)$.
3. The condition makes $x + y - 1 = 0$, so the difference is zero.

Final answer

$x^2 - x = y^2 - y$

Worked Solutions **Identities and Proofs**

Q063 Written

Under the condition $a + b + c = 0$, prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

Solution

1. From $a + b + c = 0$, write $c = -a - b$.
2. Substitute into the left-hand side: $a^2 + b^2 + (-a - b)^2 = 2a^2 + 2ab + 2b^2$.
3. Substitute into the right-hand side: $-2[ab + b(-a - b) + (-a - b)a] = 2a^2 + 2ab + 2b^2$.
4. Both sides reduce to the same expression, so the equality is proved.

Final answer

$a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 Written

Under the condition $a + b + c = 0$, prove that $a^3 + b^3 + c^3 = 3abc$.

Solution

1. Use the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.
2. The condition gives $a + b + c = 0$, so the product on the right is zero.
3. Therefore $a^3 + b^3 + c^3 - 3abc = 0$, which proves $a^3 + b^3 + c^3 = 3abc$.

Final answer

$a^3 + b^3 + c^3 = 3abc$

Worked Solutions **Identities and Proofs**

Q065 Written

Under the condition $a + b + c = 0$, prove that $a^2 + ca = b^2 + bc$.

Solution

1. Subtract the right-hand side: $a^2 + ca - b^2 - bc$.
2. Factor by grouping: $(a - b)(a + b) + (a - b)c = (a - b)(a + b + c)$.
3. Since $a + b + c = 0$, the difference is zero; hence the two sides are equal.

Final answer

$a^2 + ca = b^2 + bc$

Q066 Written

Under the condition $a + b + c = 0$, prove that $(a + b)(b + c)(c + a) + abc = 0$.

Solution

1. From $a + b + c = 0$, write $a + b = -c$, $b + c = -a$, and $c + a = -b$.
2. Then $(a + b)(b + c)(c + a) = (-c)(-a)(-b) = -abc$.
3. Adding abc gives zero, so the required equality is proved.

Final answer

$(a + b)(b + c)(c + a) + abc = 0$

Worked Solutions **Identities and Proofs**

Q067 **Written**

Under the condition $a + b + c = 0, \quad abc \neq 0$, prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$.

Solution

1. Because $a + b + c = 0$, $b + c = -a$, $c + a = -b$, and $a + b = -c$.
2. The non-zero condition ensures that all three denominators are non-zero.
3. Therefore each fraction equals -1: $a/(b + c) = b/(c + a) = c/(a + b) = -1$.
4. Their sum is $-1 - 1 - 1 = -3$.

Final answer

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$$

Worked Solutions **Identities and Proofs**

Q068 Challenge

Under the condition $x + y = 3$, the expression $x^2 + y^2 + 1$ is equal to:

Solution

1. Use $x + y = 3$ to replace each cyclic denominator.
2. The non-zero condition keeps every original denominator defined.
3. Each term simplifies and the total becomes $-2xy + 10$.
4. Thus the required constant sum is proved.

Correct option: C

Final answer

$-2xy + 10$

Worked Solutions **Identities and Proofs**

Q069 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$.

Solution

1. Let $a/b = c/d = k$, so $a = bk$ and $c = dk$.
2. Substitute into the left-hand side: ab/(a^2+b^2)=k/(k^2+1).
3. Substitute into the right-hand side: cd/(c^2+d^2)=k/(k^2+1).
4. The two sides are equal.

Final answer

$$\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$$

Q070 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2+d^2}$$

Worked Solutions **Identities and Proofs**

Q071 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer
$$\frac{cd}{c^2 + d^2}$$

Q072 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer
$$\frac{cd}{c^2 + d^2}$$

Worked Solutions **Identities and Proofs**

Q073 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions **Identities and Proofs**

Q074 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions **Identities and Proofs**

Q075 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions **Identities and Proofs**

Q076 **Written**

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$.

Solution

1. Let $a = bk$ and $c = dk$.
2. The left-hand side becomes $k(b^2 + d^2)/(b^2 - d^2)$.
3. The right-hand side becomes $k^2(b^2 + d^2)/[k^2(b^2 - d^2)]$, which is the same expression.
4. Hence the equality is proved.

Final answer

$$\frac{ab + cd}{ab - cd} = \frac{a^2 + c^2}{a^2 - c^2}$$

Q077 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions **Identities and Proofs**

Q078 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions **Identities and Proofs**

Q079 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions **Identities and Proofs**

Q080 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions **Identities and Proofs**

Q081 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Solution

1. Write $a = bk$ and $c = dk$.
2. Then $(a+b)/(a-b) = (bk+b)/(bk-b) = (k+1)/(k-1)$.
3. Similarly $(c+d)/(c-d) = (dk+d)/(dk-d) = (k+1)/(k-1)$.
4. The equality follows.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q082 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer

$$\frac{c+d}{c-d}$$

Worked Solutions **Identities and Proofs**

Q083 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer
$$\frac{c+d}{c-d}$$

Q084 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer
$$\frac{c+d}{c-d}$$

Worked Solutions **Identities and Proofs**

Q085 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer
$$\frac{c+d}{c-d}$$

Q086 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$.

Solution

1. Substitute $a = bk$ and $c = dk$.
2. The left-hand side becomes (3k+ $k)/(3 + 1) = k$.
3. The right-hand side becomes $(2k - k)/(2 - 1) = k$.
4. Thus both sides are equal.

Final answer
$$\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$$

Worked Solutions **Identities and Proofs**

Q087 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{2a-c}{2b-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{2a-c}{2b-d}$$

Q088 Written

Under the condition $x/a = y/b$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b}, \quad \frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$.

Solution

1. Let $x = ak$ and $y = bk$.
2. The left-hand side is $(a^2 + a^2k^2)/(a^2 - a^2k^2) = (1 + k^2)/(1 - k^2)$.
3. The right-hand side reduces to the same value.
4. Hence the equality is proved.

Final answer

$$\frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$$

Worked Solutions **Identities and Proofs**

Q089 **Written**

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Solution

1. Use the equal concentration condition to write $a = bk$ and $c = dk$.
2. The first ratio becomes (k+1)/(k-1).
3. The second ratio becomes (k+1)/(k-1).
4. The required equality follows, provided the denominators are non-zero.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q090 **Written**

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$.

Solution

1. Let the common ratio be k, so $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also xy+yz+zx=k^2(ab+bc+ca).
4. Both ratios therefore equal k^2.

Final answer

$$\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$$

Worked Solutions **Identities and Proofs**

Q091 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x+y+z}{a+b+c} = \frac{x}{a}$.

Solution

1. Let the common ratio be k.
2. Then $x + y + z = k(a + b + c)$.
3. Therefore $(x + y + z)/(a + b + c) = k = x/a$.

Final answer

$$\frac{x+y+z}{a+b+c} = \frac{x}{a}$$

Q092 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$.

Solution

1. Set $x = ak$, $y = bk$ and $z = ck$.
2. The left numerator is k(a+2b+3c), so the left ratio is k.
3. The right ratio is also k because $x + y + z = k(a + b + c)$.
4. The two ratios are equal.

Final answer

$$\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$$

Worked Solutions **Identities and Proofs**

Q093 Written

Under the condition $x : y : z = 2 : 3 : 5$, prove or find the required expression: $x : y : z = 2 : 3 : 5$, $\frac{2xy+yz+3zx}{x^2+y^2+z^2}$.

Solution

1. Let $x = 2k$, $y = 3k$ and $z = 5k$.
2. The numerator is $(12 + 15 + 30)k^2 = 57k^2$.
3. The denominator is $(4 + 9 + 25)k^2 = 38k^2$.
4. Therefore the value is 57/38.

Final answer
$$\frac{57}{38}$$

Q094 Written

Under the condition $a : b : c = 2 : 3 : 4$, prove or find the required expression: $a : b : c = 2 : 3 : 4$, $\frac{ab+bc+ca}{a^2+b^2+c^2}$.

Solution

1. Let $a = 2k$, $b = 3k$ and $c = 4k$.
2. The numerator is $(6 + 12 + 8)k^2 = 26k^2$.
3. The denominator is $(4 + 9 + 16)k^2 = 29k^2$.
4. Therefore the value is 26/29.

Final answer
$$\frac{26}{29}$$

Worked Solutions **Identities and Proofs**

Q095 Written

Under the condition $a : b : c = 3 : 5 : 2$, prove or find the required expression: $a : b : c = 3 : 5 : 2$, $\frac{a^2+b^2-c^2}{a^2-b^2+c^2}$.

Solution

1. Let $a = 3k$, $b = 5k$ and $c = 2k$.
2. The numerator is $(9 + 25 - 4)k^2 = 30k^2$.
3. The denominator is $(9 - 25 + 4)k^2 = -12k^2$.
4. Therefore the value is $-5/2$.

Final answer

$-\frac{5}{2}$

Q096 Written

Under the condition $x : y : z = 2 : -3 : 4$, prove or find the required expression: $x : y : z = 2 : -3 : 4$, $\frac{x^2+y^2+z^2}{xy+yz+zx}$.

Solution

1. Let $x = 2k$, $y = -3k$ and $z = 4k$.
2. The numerator is $(4 + 9 + 16)k^2 = 29k^2$.
3. The denominator is $(-6 - 12 + 8)k^2 = -10k^2$.
4. Therefore the value is $-29/10$.

Final answer

$-\frac{29}{10}$

Worked Solutions **Identities and Proofs**

Q097 Written

Under the condition $x/7 = y/3 \neq 0$, prove or find the required expression: $\frac{x}{7} = \frac{y}{3} \neq 0, \quad \frac{x^2+5y^2}{x^2+xy+y^2}$.

Solution

1. Let $x = 7k$ and $y = 3k$, where k is non-zero.
2. The numerator is $(49 + 45)k^2 = 94k^2$.
3. The denominator is $(49 + 21 + 9)k^2 = 79k^2$.
4. Therefore the value is 94/79.

Final answer

$\frac{94}{79}$

Q098 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad (a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Solution

1. Let $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also ax+by+cz=k(a^2+b^2+c^2).
4. Both sides equal $k^2(a^2 + b^2 + c^2)^2$.

Final answer

$(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Worked Solutions **Identities and Proofs**

Quiz MCQ 1 **Concept Quiz · MCQ**

If $x^2 + 4x + 1 = (x+1)(ax+b) + c$, then (a, b, c) is... Which of the following is correct?:

Solution

1. Expand to $ax^2 + (a+b)x + b + c$.
2. Match $a = 1,\ a + b = 4,\ b + c = 1$, so $b = 3,\ c = -2$.

Correct option: A

Final answer

$(1, 3, -2)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For the identity $\left(x^2 - 2x + 2\right)\left(x^2 + 2x + 2\right) = -4x^2 + \left(x^2 + 2\right)^2$, the value of the simplified left-hand side is:

Solution

1. Rewrite the factors as (U-V)(U+V).
2. Apply U^2-V^2 before carrying out any long expansion.
3. Simplify to $x^4 + 4$.
4. This is exactly the stated right-hand side.

Correct option: A

Final answer

$x^4 + 4$

Worked Solutions **Identities and Proofs**

Quiz MCQ 3 Concept Quiz · MCQ

Under the condition $a + b + c = 0$, the expression $3a^2 + 3ac$ is equal to:

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $3b^2 + 3bc$.
4. Hence the two sides are equal under the given condition.

Correct option: A

Final answer

$3b^2 + 3bc$

Quiz MCQ 4 Concept Quiz · MCQ

Under the condition $x : y : z = 2 : 3 : 5$, the required expression is equal to:

Solution

1. Write the variables as the stated multiples of k.
2. Substitute into the numerator and denominator.
3. Cancel the common factor and simplify to $\frac{3}{2}$.
4. Hence the required value is obtained.

Correct option: B

Final answer

$$\frac{3}{2}$$

Worked Solutions **Identities and Proofs**

Quiz WRITTEN 5 Concept Quiz · Written

Find a, b, c if $x^2 - 3x + 5 = (x - 2)(ax + b) + c$.

Solution

1. Expand to $ax^2 + (b - 2a)x - 2b + c$.
2. Match $a = 1,\ b - 2 = -3,\ -2b + c = 5$.
3. Therefore $b = -1,\ c = 3$.

Final answer

$a = 1,\ b = -1,\ c = 3$

Quiz WRITTEN 6 Concept Quiz · Written

Prove that $\left(x^2 - 3x + 4\right)\left(x^2 + 3x + 4\right) = -9x^2 + \left(x^2 + 4\right)^2$.

Solution

1. Rewrite the factors as (U-V)(U+V).
2. Apply U^2-V^2 before carrying out any long expansion.
3. Simplify to $x^4 - x^2 + 16$.
4. This is exactly the stated right-hand side.

Final answer

$x^4 - x^2 + 16$

Worked Solutions **Identities and Proofs**

Quiz WRITTEN 7 Concept Quiz · Written

Under the condition $a + b + c = 0$, prove that $4a^2 + 4ac = 4b^2 + 4bc$.

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $4b^2 + 4bc$.
4. Hence the two sides are equal under the given condition.

Final answer

$4b^2 + 4bc$

Quiz WRITTEN 8 Concept Quiz · Written

Under the condition $x/a = y/b = z/c$, prove that $\left(a^2 + b^2 + c^2\right)\left(x^2 + y^2 + z^2\right) = (ax + by + cz)^2$.

Solution

1. Use the common multiplier from $x/a = y/b = z/c$.
2. Substitute and collect the common factor.
3. Both sides simplify to $(ax + by + cz)^2$.
4. Thus the identity is proved.

Final answer

$(ax + by + cz)^2$

Answer key

Use the question number shown on each page.

Q001 $a = 2,\ b = 1,\ c = 0$

Q002 $a = 1,\ b = -2,\ c = 7$

Q003 $a = 1,\ b = 1,\ c = -3$

Q004 C

Q005 C

Q006 D

Q007 A

Q008 $a = 10,\ b = 2$

Q009 A

Q010 B

Q011 D

Q012 D

Q013 $a = 3,\ b = 1$

Q014 B

Q015 C

Q016 C

Q017 $a = 2,\ b = 3,\ c = 1$

Q018 $a = 4,\ b = 1,\ c = 5$

Q019 $a = 2,\ b = -1,\ c = 5$

Q020 B

Q021 C

Q022 B

Q023 $a = 1,\ b = 2$

Q024 $a = 2,\ b = -1$

Q025 $a = \frac{2}{3},\ b = -\frac{1}{3}$

Q026 $a = -1,\ b = 6$

Q027 $a = 2,\ b = -3$

Q028 $a = 4,\ b = 3$

Q029 $4xy$

Q030 $3a^2 - 3b^2$

Q031 A

Q032 D

Q033 D

Q034 A

Q035 $2x^2 + 2y^2$

Q036 C

Q037 C

Q038 C

Q039 D

Q040 $a^2 + 4b^2$

Q041 D

Q042 C

Q043 $x^4 + x^2 + 1$

Q044 $a^4 + a^2b^2 + b^4$

Q045 $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Q046 $x^2y^2 + x^2 + y^2 + 1$

Q047 $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$

Q048 $a^3 + b^3$

Q049 $a^3 - 3abc + b^3 + c^3$

Q050 B

Q051 $x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 $x^2 + y^2 = 1 - 2xy$

Q053 C

Q054 D

Q055 C

Q056 A

Q057 $x^2 + y^2 = 2xy + 1$

Q058 B

Q059 B

Q060 C

Q061 D

Q062 $x^2 - x = y^2 - y$

Q063 $a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 $a^3 + b^3 + c^3 = 3abc$

Q065 $a^2 + ca = b^2 + bc$

Q066 $(a + b)(b + c)(c + a) + abc = 0$

Q067 $\frac{a}{b + c} + \frac{b}{c + a} + \frac{c}{a + b} = -3$

Q068 C

Q069 $\frac{ab}{a^2 + b^2} = \frac{cd}{c^2 + d^2}$

Q070 C

Q071 D

Q072 C

Q073 D

Q074 D

Q075 C

Q076 $\frac{ab + cd}{ab - cd} = \frac{a^2 + c^2}{a^2 - c^2}$

Q077 A

Q078 D

Q079 D

Q080 A

Q081 $\frac{a + b}{a - b} = \frac{c + d}{c - d}$

Q082 B

Q083 B

Q084 A

Q085 A

Q086 $\frac{3a + c}{3b + d} = \frac{2a - c}{2b - d}$

Q087 A

Q088 $\frac{a^2 + x^2}{a^2 - x^2} = \frac{b^2 + y^2}{b^2 - y^2}$

Q089 $\frac{a + b}{a - b} = \frac{c + d}{c - d}$

Q090 $\frac{x^2 + y^2 + z^2}{a^2 + b^2 + c^2} = \frac{xy + yz + zx}{ab + bc + ca}$

Q091 $\frac{x + y + z}{a + b + c} = \frac{x}{a}$

Q092 $\frac{x + 2y + 3z}{a + 2b + 3c} = \frac{x + y + z}{a + b + c}$

Q093 $\frac{57}{38}$

Q094 $\frac{26}{29}$

Q095 $-\frac{5}{2}$

Q096 $-\frac{29}{10}$

Q097 $\frac{94}{79}$

Q098 $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a = 1,\ b = -1,\ c = 3$

Quiz WRITTEN 6 $x^4 - x^2 + 16$

Quiz WRITTEN 7 $4b^2 + 4bc$

Quiz WRITTEN 8 $(ax + by + cz)^2$

Concept 5 | Inequalities

English Student Edition • Focused formulas and guided practice

Formula opener

Square rule

$$(x-y)^2 \ge 0$$

Order

$$0 < a < b \Rightarrow a^2 < b^2$$

AM–GM

$$\frac{a+b}{2} \ge \sqrt{ab}$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Inequalities**

Q001 Written

Under the condition $a > b, \quad x > y$, prove that $2(ax + by) > (a + b)(x + y)$.

Solution

1. Start with LHS-RHS: 2(ax+by)-(a+b)(x+y).
2. Expand and factor: ax-ay-bx+by=(a-b)(x-y).
3. Both factors are positive, so (a-b)(x-y)>0; hence the required inequality holds.

Final answer

$2(ax + by) > (a + b)(x + y)$

Q002 MCQ

If x>y, prove that $3x + y > 2\,(x + y)$.
If x>y, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition x>y makes each factor positive.
4. Hence $3x + y$>$2\,(x + y)$.

Correct option: A

Final answer

$x - y$

Worked Solutions **Inequalities**

Q003 MCQ

If x>y, prove that $3x + 2y > 2x + 3y$**.**
If x>y, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition x>y makes each factor positive.
4. Hence $3x + 2y$>$2x + 3y$.

Correct option: C

Final answer

$x - y$

Q004 MCQ

If x>y, prove that $3(x+y) > 2(x+2y)$.
If x>y, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition x>y makes each factor positive.
4. Hence $3(x+y)$>$2(x+2y)$.

Correct option: D

Final answer

$x - y$

Worked Solutions **Inequalities**

Q005 MCQ

If x>y, prove that $3x + 4y > 2x + 5y$.
If x>y, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition x>y makes each factor positive.
4. Hence $3x + 4y$>$2x + 5y$.

Correct option: D

Final answer

$x - y$

Q006 **MCQ**

If x>y, prove that $3x + 5y > 2(x + 3y)$.
If x>y, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition x>y makes each factor positive.
4. Hence $3x + 5y>2(x + 3y)$.

Correct option: B

Final answer

$x - y$

Worked Solutions **Inequalities**

Q007 Written

Under the condition $a > b, \quad c > d$, prove that $ac + bd > ad + bc$.

Solution

1. Subtract the right-hand side: ac+bd-ad-bc.
2. Factor: $a(c-d) - b(c-d) = (a-b)(c-d)$.
3. Both factors are positive, so the difference is positive; therefore ac+bd>ad+bc.

Final answer

$ac + bd > ad + bc$

Q008 Written

Under the condition $a > c, \quad b > d$, prove that $ab + cd > ad + bc$.

Solution

1. Subtract the right-hand side: ab+cd-ad-bc.
2. Factor: $a(b-d) - c(b-d) = (a-c)(b-d)$.
3. Both factors are positive, so the difference is positive; hence ab+cd>ad+bc.

Final answer

$ab + cd > ad + bc$

Worked Solutions **Inequalities**

Q009 MCQ

If a>b and c>d, prove that $ac + bd > ad + bc$.
If a>b and c>d, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - b)(c - d)$.
2. Factor the difference as $(a - b)(c - d)$.
3. The condition a>b and c>d makes each factor positive.
4. Hence $ac + bd > ad + bc$.

Correct option: D

Final answer

$(a - b)(c - d)$

Q010 MCQ

If a>2b and c>d, prove that $ac + 2bd > ad + 2bc$.
If a>2b and c>d, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 2b)(c - d)$.
2. Factor the difference as $(a - 2b)(c - d)$.
3. The condition a>2b and c>d makes each factor positive.
4. Hence $ac + 2bd > ad + 2bc$.

Correct option: C

Final answer

$(a - 2b)(c - d)$

Worked Solutions **Inequalities**

Q011 MCQ

If a>3b and c>d, prove that $ac + 3bd > ad + 3bc$.
If a>3b and c>d, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 3b)(c - d)$.
2. Factor the difference as $(a - 3b)(c - d)$.
3. The condition a>3b and c>d makes each factor positive.
4. Hence $ac + 3bd$>$ad + 3bc$.

Correct option: A

Final answer

$(a - 3b)(c - d)$

Q012 Written

Under the condition $x > y$, prove that $3x - 4y > 2x - 3y$.

Solution

1. Subtract the right-hand side: (3x-4y)-(2x-3y).
2. Simplify to x-y.
3. Since x>y, x-y>0; therefore 3x-4y>2x-3y.

Final answer

$3x - 4y > 2x - 3y$

Worked Solutions **Inequalities**

Q013 Written

Under the condition $x > 3, \quad y > -5$, prove that $xy + 5x > 3y + 15$.

Solution

1. Start with LHS-RHS: xy+5x-3y-15.
2. Factor by grouping: $x(y + 5) - 3(y + 5) = (x - 3)(y + 5)$.
3. Since x-3>0 and y+5>0, the difference is positive; hence the inequality holds.

Final answer

$xy + 5x > 3y + 15$

Q014 Written

Under the condition $x > 2, \quad y > 1$, prove that $xy + 2 > x + 2y$.

Solution

1. Start with LHS-RHS: xy+2-x-2y.
2. Factor: $x(y - 1) - 2(y - 1) = (x - 2)(y - 1)$.
3. Both factors are positive, so the difference is positive; hence xy+2>x+2y.

Final answer

$xy + 2 > x + 2y$

Worked Solutions **Inequalities**

Q015 Written

Under the condition $a > 1, \quad b > 1$, prove that $ab + 1 > a + b$.

Solution

1. Start with LHS-RHS: ab+1-a-b.
2. Factor: ab-a-b+1=(a-1)(b-1).
3. Since a-1>0 and b-1>0, the difference is positive; hence ab+1>a+b.

Final answer

$ab + 1 > a + b$

Q016 Written

Under the condition $a > b > x > y$, prove that $ax + by > ay + bx$.

Solution

1. Subtract the right-hand side: ax+by-ay-bx.
2. Factor: $a(x - y) - b(x - y) = (a - b)(x - y)$.
3. Both factors are positive because a>b and x>y; therefore the difference is positive.

Final answer

$ax + by > ay + bx$

Worked Solutions **Inequalities**

Q017 **Written**

Under the condition $a > b > c > 0, \quad x > y > z > 0$, prove that $ax + by + cz > ay + bz + cx$.

Solution

1. Subtract the right-hand side and regroup: ax+by+cz-ay-bz-cx.
2. Rewrite it as (a-b)(x-y)+(b-c)(x-z).
3. Every factor in both products is positive, so the sum is positive; hence ax+by+cz>ay+bz+cx.

Final answer

$ax + by + cz > ay + bz + cx$

Worked Solutions **Inequalities**

Q018 Challenge

If a>b and x>y, prove that $2(ax+by) > 2(ay+bx)$.
If a>b and x>y, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $2(a-b)(x-y)$.
2. Factor the difference as $2(a-b)(x-y)$.
3. The condition a>b and x>y makes each factor positive.
4. Hence $2(ax+by)$>$2(ay+bx)$.

Correct option: D

Final answer

$2(a-b)(x-y)$

Worked Solutions **Inequalities**

Q019 Written

Prove that $a^2 + 4b^2 \geq 4ab$.

Solution

1. Start with LHS - RHS: $a^2 + 4b^2 - 4ab$.
2. Complete the square: $a^2 + 4b^2 - 4ab = (a - 2b)^2 \geq 0$.
3. Therefore $a^2 + 4b^2 \geq 4ab$.
4. Equality holds when $a = 2b$.

Final answer

$a^2 + 4b^2 \geq 4ab$

Q020 MCQ

Under a,b real, prove that $a^2 + 4b^2 \geq 4ab$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $a^2 + 4b^2$ - $4ab$.
2. Factor/complete the square to obtain $(a - 2b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 4b^2 \geq 4ab$.

Correct option: D

Final answer

$(a - 2b)^2$

Worked Solutions **Inequalities**

Q021 MCQ

Under a,b real, prove that $a^2 + 9b^2 \geq 6ab$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $a^2 + 9b^2$ - $6ab$.
2. Factor/complete the square to obtain $(a - 3b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 9b^2 \geq 6ab$.

Correct option: C

Final answer

$(a - 3b)^2$

Q022 MCQ

Under a,b real, prove that $a^2 + 16b^2 \geq 8ab$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $a^2 + 16b^2$ - $8ab$.
2. Factor/complete the square to obtain $(a - 4b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 16b^2 \geq 8ab$.

Correct option: B

Final answer

$(a - 4b)^2$

Worked Solutions **Inequalities**

Q023 MCQ

Under a,b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2$ - $10ab$.
2. Factor/complete the square to obtain $(a-5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A

Final answer

$(a-5b)^2$

Q024 MCQ

Under a,b real, prove that $a^2 + 36b^2 \geq 12ab$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $a^2 + 36b^2$ - $12ab$.
2. Factor/complete the square to obtain $(a-6b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 36b^2 \geq 12ab$.

Correct option: C

Final answer

$(a-6b)^2$

Worked Solutions **Inequalities**

Q025 MCQ

Under a,b real, prove that $a^2 + 49b^2 \geq 14ab$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $a^2 + 49b^2$ - $14ab$.
2. Factor/complete the square to obtain $(a - 7b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 49b^2 \geq 14ab$.

Correct option: B

Final answer

$(a - 7b)^2$

Q026 Written

Prove that $a^2 + 5b^2 \geq 4ab$.

Solution

1. Start with LHS - RHS: $a^2 + 5b^2 - 4ab$.
2. Complete the square: $a^2 + 5b^2 - 4ab = (a - 2b)^2 + b^2 \geq 0$.
3. Therefore $a^2 + 5b^2 \geq 4ab$.
4. Equality holds when $a = b = 0$.

Final answer

$a^2 + 5b^2 \geq 4ab$

Worked Solutions **Inequalities**

Q027 Written

Prove that $9x^2 \geq y(6x - y)$.

Solution

1. Subtract the right-hand side: $9x^2 - y(6x - y)$.
2. Complete the square: $9x^2 - y(6x - y) = (3x - y)^2 \geq 0$.
3. Therefore $9x^2 \geq y(6x - y)$.
4. Equality holds when $y = 3x$.

Final answer

$9x^2 \geq y(6x - y)$

Q028 Written

Prove that $2(x^2 + y^2) \geq (x + y)^2$.

Solution

1. Subtract the right-hand side: $2(x^2 + y^2) - (x + y)^2$.
2. This is $(x - y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = y$.

Final answer

$2(x^2 + y^2) \geq (x + y)^2$

Worked Solutions **Inequalities**

Q029 Written

Prove that $(2x-y)^2 \geq 3(x^2-y^2)$.

Solution

1. Subtract the right-hand side: $(2x-y)^2 - 3(x^2-y^2)$.
2. Simplify to $(x-2y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = 2y$.

Final answer

$(2x-y)^2 \geq 3(x^2-y^2)$

Q030 Written

Prove that $(2a+3b)^2 \geq 12ab$.

Solution

1. Subtract the right-hand side: $(2a+3b)^2 - 12ab$.
2. The cross terms cancel, giving $4a^2 + 9b^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $a = b = 0$.

Final answer

$(2a+3b)^2 \geq 12ab$

Q031 MCQ

Under x,y real, prove that $4x^2 \geq y(4x - y)$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $4x^2$ - $y(4x - y)$.
2. Factor/complete the square to obtain $(2x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $4x^2 \geq y(4x - y)$.

Correct option: A

Final answer

$(2x - y)^2$

Q032 MCQ

Under x,y real, prove that $9x^2 \geq y(6x - y)$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $9x^2$ - $y(6x - y)$.
2. Factor/complete the square to obtain $(3x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $9x^2 \geq y(6x - y)$.

Correct option: C

Final answer

$(3x - y)^2$

Worked Solutions **Inequalities**

Q033 *MCQ*

Under x,y real, prove that $16x^2 \geq y\,(8x - y)$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $16x^2$ - $y\,(8x - y)$.
2. Factor/complete the square to obtain $(4x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $16x^2 \geq y\,(8x - y)$.

Correct option: D

Final answer

$(4x - y)^2$

Q034 *MCQ*

Under x,y real, prove that $25x^2 \geq y\,(10x - y)$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $25x^2$ - $y\,(10x - y)$.
2. Factor/complete the square to obtain $(5x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $25x^2 \geq y\,(10x - y)$.

Correct option: D

Final answer

$(5x - y)^2$

Worked Solutions **Inequalities**

Q035 *MCQ*

Under x,y real, prove that $36x^2 \geq y\,(12x - y)$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $36x^2$ - $y\,(12x - y)$.
2. Factor/complete the square to obtain $(6x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $36x^2 \geq y\,(12x - y)$.

Correct option: A

Final answer

$(6x - y)^2$

Q036 *MCQ*

Under x,y real, prove that $x^2 + y^2 \geq 2xy$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $x^2 + y^2$ - $2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Correct option: C

Final answer

$(x - y)^2$

Worked Solutions **Inequalities**

Q037 **Written**

Prove that $x^2 + y^2 \geq 2(x - y - 1)$.

Solution

1. Start with LHS - RHS: $x^2 + y^2 - 2(x - y - 1)$.
2. Complete the squares: $x^2 + y^2 - 2(x - y - 1) = (x - 1)^2 + (y + 1)^2 \geq 0$.
3. Therefore $x^2 + y^2 \geq 2(x - y - 1)$.
4. Equality holds when $x = 1,\ y = -1$.

Final answer

$x^2 + y^2 \geq 2(x - y - 1)$

Q038 **Written**

Prove that $x^2 + 2 > 2x$.

Solution

1. Subtract the right-hand side: $x^2 + 2 - 2x$.
2. Complete the square: $x^2 + 2 - 2x = (x - 1)^2 + 1 > 0$.
3. Therefore $x^2 + 2 > 2x$.
4. There is no equality case.

Final answer

$x^2 + 2 > 2x$

Worked Solutions **Inequalities**

Q039 Written

Prove that $x^2 + y^2 - 6x + 8y + 25 \geq 0$.

Solution

1. Complete the squares: $x^2 + y^2 - 6x + 8y + 25 = (x-3)^2 + (y+4)^2 \geq 0$.
2. Therefore the inequality holds.
3. Equality holds when $x = 3,\ y = -4$.

Final answer

$x^2 + y^2 - 6x + 8y + 25 \geq 0$

Q040 Written

Prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Solution

1. Complete the squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x-5)^2 + 2(y+3)^2 \geq 0$.
2. Therefore the expression is never negative.
3. Equality holds when $x = 5,\ y = -3$.

Final answer

$x^2 + 2y^2 - 10x + 12y + 43 \geq 0$

Worked Solutions **Inequalities**

Q041 **Written**

Prove that $(a^2+b^2)(x^2+y^2) \geq (ax+by)^2$.

Solution

1. Subtract the right-hand side: $(a^2+b^2)(x^2+y^2)-(ax+by)^2$.
2. Use $(a^2+b^2)(x^2+y^2)-(ax+by)^2=(ay-bx)^2 \geq 0$.
3. Therefore the required inequality holds.
4. Equality holds when $ay=bx$.

Final answer

$(a^2+b^2)(x^2+y^2) \geq (ax+by)^2$

Worked Solutions **Inequalities**

Q042 **Challenge**

For real x, y, prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Solution

1. Complete both squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x-5)^2 + 2(y+3)^2 \geq 0$.
2. Therefore $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.
3. Equality holds when $x = 5,\ y = -3$.

Final answer

0

Worked Solutions **Inequalities**

Q043 Written

Prove that $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

Solution

1. Both sides are positive. Compare their squares.
2. $\left(\sqrt{\frac{a^2+b^2}{2}}\right)^2 - \left(\frac{a+b}{2}\right)^2 = \frac{(a-b)^2}{4} \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$$

Worked Solutions **Inequalities**

Q044 MCQ

Under a>0 and b>0, prove that $\sqrt{\frac{a^2}{2}+\frac{b^2}{2}} \geq \frac{a}{2}+\frac{b}{2}$.
After squaring the positive sides, the value of $\mathrm{LHS}^2 - \mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2 - \mathrm{RHS}^2 = \frac{(a-b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2}+\frac{b^2}{2}} \geq \frac{a}{2}+\frac{b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-b)^2}{4}$$

Worked Solutions **Inequalities**

Q045 MCQ

Under a>0 and b>0, prove that $\sqrt{\frac{a^2}{2}+2b^2} \geq \frac{a}{2}+b$.
After squaring the positive sides, the value of $\mathrm{LHS}^2-\mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2-\mathrm{RHS}^2=\frac{(a-2b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2}+2b^2} \geq \frac{a}{2}+b$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-2b)^2}{4}$$

Worked Solutions **Inequalities**

Q046 MCQ

Under a>0 and b>0, prove that $\sqrt{\frac{a^2}{2}+\frac{9b^2}{2}} \geq \frac{a}{2}+\frac{3b}{2}$.
After squaring the positive sides, the value of $\mathrm{LHS}^2 - \mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2 - \mathrm{RHS}^2 = \frac{(a-3b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2}+\frac{9b^2}{2}} \geq \frac{a}{2}+\frac{3b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: C

Final answer

$$\frac{(a-3b)^2}{4}$$

Worked Solutions **Inequalities**

Q047 MCQ

Under a>0 and b>0, prove that $\sqrt{\frac{a^2}{2}+8b^2} \geq \frac{a}{2}+2b$.
After squaring the positive sides, the value of $\mathrm{LHS}^2-\mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2-\mathrm{RHS}^2=\frac{(a-4b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2}+8b^2}\geq\frac{a}{2}+2b$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-4b)^2}{4}$$

Worked Solutions **Inequalities**

Q048 MCQ

Under a>0 and b>0, prove that $\sqrt{\frac{a^2}{2}+\frac{25b^2}{2}} \ge \frac{a}{2}+\frac{5b}{2}$.
After squaring the positive sides, the value of $\mathrm{LHS}^2 - \mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2 - \mathrm{RHS}^2 = \frac{(a-5b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2}+\frac{25b^2}{2}} \ge \frac{a}{2}+\frac{5b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-5b)^2}{4}$$

Worked Solutions **Inequalities**

Q049 MCQ

Under a>0 and b>0, prove that $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.
After squaring the positive sides, the value of $\mathrm{LHS}^2 - \mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2 - \mathrm{RHS}^2 = \frac{(a-6b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.
5. Equality occurs when the displayed square is zero.

Correct option: B

Final answer

$$\frac{(a-6b)^2}{4}$$

Worked Solutions **Inequalities**

Q050 MCQ

Under a>0 and b>0, prove that $\sqrt{\frac{a^2}{2}+\frac{49b^2}{2}} \geq \frac{a}{2}+\frac{7b}{2}$.
After squaring the positive sides, the value of $\mathrm{LHS}^2 - \mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2 - \mathrm{RHS}^2 = \frac{(a-7b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2}+\frac{49b^2}{2}} \geq \frac{a}{2}+\frac{7b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions **Inequalities**

Q051 MCQ

Under a>0 and b>0, prove that $\sqrt{\frac{a^2}{2}+32b^2} \geq \frac{a}{2}+4b$.
After squaring the positive sides, the value of $\mathrm{LHS}^2-\mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2-\mathrm{RHS}^2=\frac{(a-8b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2}+32b^2} \geq \frac{a}{2}+4b$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-8b)^2}{4}$$

Worked Solutions **Inequalities**

Q052 **Written**

Prove that $\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$.

Solution

1. Both sides are positive. Compare their squares.
2. $(\sqrt{2(a+b)})^2 - (\sqrt{a} + \sqrt{b})^2 = (\sqrt{a} - \sqrt{b})^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$

Q053 **Written**

Prove that $a + 2 \geq \sqrt{3 + 6a}$.

Solution

1. Both sides are positive. Compare their squares.
2. $(a+2)^2 - (\sqrt{3+6a})^2 = (a-1)^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = 1$.

Final answer

$a + 2 \geq \sqrt{3 + 6a}$

Worked Solutions **Inequalities**

Q054 **Written**

Prove that $\sqrt{2(a^2+b^2)} \geq a+b$.

Solution

1. Both sides are positive. Compare their squares.
2. $(\sqrt{2(a^2+b^2)})^2 - (a+b)^2 = (a-b)^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$\sqrt{2(a^2+b^2)} \geq a+b$

Q055 **Written**

Prove that $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a}+\sqrt{b}}{2}$.

Solution

1. Both sides are positive. Compare their squares.
2. $\left(\sqrt{\frac{a+b}{2}}\right)^2 - \left(\frac{\sqrt{a}+\sqrt{b}}{2}\right)^2 = \frac{(\sqrt{a}-\sqrt{b})^2}{4} \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a}+\sqrt{b}}{2}$$

Worked Solutions **Inequalities**

Q056 MCQ

Under a>0 and b>0, prove that $\sqrt{a}+\sqrt{b}>\sqrt{a+b}$.
After squaring the positive sides, the value of $\mathrm{LHS}^2-\mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2-\mathrm{RHS}^2=2\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a}+\sqrt{b}>\sqrt{a+b}$.

Correct option: B

Final answer

$2\sqrt{a}\sqrt{b}$

Q057 MCQ

Under a>0 and b>0, prove that $\sqrt{a} + 2\sqrt{b} > \sqrt{a+4b}$.
After squaring the positive sides, the value of $\mathrm{LHS}^2 - \mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2 - \mathrm{RHS}^2 = 4\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 2\sqrt{b} > \sqrt{a+4b}$.

Correct option: A

Final answer

$4\sqrt{a}\sqrt{b}$

Worked Solutions **Inequalities**

Q058 MCQ

Under a>0 and b>0, prove that $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.
After squaring the positive sides, the value of $\mathrm{LHS}^2 - \mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2 - \mathrm{RHS}^2 = 6\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 3\sqrt{b}>\sqrt{a + 9b}$.

Correct option: C

Final answer
$6\sqrt{a}\sqrt{b}$

Q059 MCQ

Under a>0 and b>0, prove that $\sqrt{a}+4\sqrt{b}>\sqrt{a+16b}$.
After squaring the positive sides, the value of $\mathrm{LHS}^2-\mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2-\mathrm{RHS}^2=8\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a}+4\sqrt{b}>\sqrt{a+16b}$.

Correct option: B

Final answer

$8\sqrt{a}\sqrt{b}$

Worked Solutions **Inequalities**

Q060 **Written**

Prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Solution

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$

Q061 **Written**

Prove that $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$.

Solution

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + \sqrt{b})^2 - (\sqrt{4a + b})^2 = 4\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$

Worked Solutions **Inequalities**

Q062 **Written**

Prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

Solution

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + \sqrt{b})^2 - (\sqrt{a+b})^2 = 2\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$\sqrt{a} + \sqrt{b} > \sqrt{a+b}$

Q063 **Written**

Prove that $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$.

Solution

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + 3\sqrt{b})^2 - (\sqrt{4a+9b})^2 = 12\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$

Worked Solutions **Inequalities**

Q064 **Written**

Prove that $\sqrt{a-b} > \sqrt{a} - \sqrt{b}$.

Solution

1. Both sides are positive because $a > b > 0$. Compare their squares.
2. $(\sqrt{a-b})^2 - (\sqrt{a} - \sqrt{b})^2 = 2\sqrt{b}(\sqrt{a} - \sqrt{b}) > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$\sqrt{a-b} > \sqrt{a} - \sqrt{b}$

Q065 **Written**

If $a > b > 0$, prove that $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.

Solution

1. Both sides are positive. Compare their squares.
2. $\left(\sqrt{a} - \frac{b}{2\sqrt{a}}\right)^2 - (\sqrt{a-b})^2 = \frac{b^2}{4a} > 0$.
3. Therefore $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.
4. There is no equality case.

Final answer

$$\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$$

Worked Solutions **Inequalities**

Q066 **Challenge**

For $a, b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Solution

1. Both sides are positive under $a, b > 0$; compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Therefore $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Final answer

$\sqrt{a + 16b}$

Worked Solutions **Inequalities**

Q067 Written

Prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Solution

1. By AM–GM, $\frac{b}{a} + \frac{a}{b} \geq 2\sqrt{\frac{b}{a} \cdot \frac{a}{b}} = 2$.
2. Equality holds when $a = b$.

Final answer

$$\frac{b}{a} + \frac{a}{b} \geq 2$$

Q068 Written

Prove that $\frac{b}{3a} + \frac{12a}{b} \geq 4$.

Solution

1. Apply AM–GM to the two positive terms: $\frac{b}{3a} + \frac{12a}{b} \geq 2\sqrt{4} = 4$.
2. Equality holds when $b = 6a$.

Final answer

$$\frac{b}{3a} + \frac{12a}{b} \geq 4$$

Worked Solutions **Inequalities**

Q069 MCQ

Under a>0 and b>0, prove that $\frac{a^2+b^2}{ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{a^2+b^2}{ab}$ - 2.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{a^2+b^2}{ab} \geq 2$.

Correct option: A

Final answer

$$\frac{(a-b)^2}{ab}$$

Q070 MCQ

Under a>0 and b>0, prove that $\frac{4a^2+b^2}{2ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{4a^2+b^2}{2ab}$ - 2.
2. Factor/complete the square to obtain $\frac{(2a-b)^2}{2ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{4a^2+b^2}{2ab} \geq 2$.

Correct option: C

Final answer

$$\frac{(2a-b)^2}{2ab}$$

Worked Solutions **Inequalities**

Q071 MCQ

Under a>0 and b>0, prove that $\frac{9a^2+b^2}{3ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{9a^2+b^2}{3ab}$ - 2.
2. Factor/complete the square to obtain $\frac{(3a-b)^2}{3ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{9a^2+b^2}{3ab} \geq 2$.

Correct option: B

Final answer

$$\frac{(3a-b)^2}{3ab}$$

Q072 **MCQ**

Under a>0 and b>0, prove that $\frac{16a^2+b^2}{4ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{16a^2+b^2}{4ab}$ - 2.
2. Factor/complete the square to obtain $\frac{(4a-b)^2}{4ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{16a^2+b^2}{4ab} \geq 2$.

Correct option: A

Final answer

$$\frac{(4a-b)^2}{4ab}$$

Worked Solutions **Inequalities**

Q073 MCQ

Under a>0 and b>0, prove that $\frac{25a^2+b^2}{5ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{25a^2+b^2}{5ab}$ - 2.
2. Factor/complete the square to obtain $\frac{(5a-b)^2}{5ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{25a^2+b^2}{5ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(5a-b)^2}{5ab}$$

Q074 MCQ

Under a>0 and b>0, prove that $\frac{36a^2+b^2}{6ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{36a^2+b^2}{6ab}$ - 2.
2. Factor/complete the square to obtain $\frac{(6a-b)^2}{6ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{36a^2+b^2}{6ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(6a-b)^2}{6ab}$$

Worked Solutions **Inequalities**

Q075 MCQ

Under a>0 and b>0, prove that $\frac{49a^2+b^2}{7ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{49a^2+b^2}{7ab}$ - 2.
2. Factor/complete the square to obtain $\frac{(7a-b)^2}{7ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{49a^2+b^2}{7ab} \geq 2$.

Correct option: C

Final answer

$$\frac{(7a-b)^2}{7ab}$$

Q076 **MCQ**

Under a>0 and b>0, prove that $\frac{64a^2+b^2}{8ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{64a^2+b^2}{8ab}$ - 2.
2. Factor/complete the square to obtain $\frac{(8a-b)^2}{8ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{64a^2+b^2}{8ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(8a-b)^2}{8ab}$$

Worked Solutions **Inequalities**

Q077 **Written**

Prove that $(a+b)\left(\frac{1}{a}+\frac{1}{b}\right) \geq 4$.

Solution

1. Expand: $(a+b)(\frac{1}{a}+\frac{1}{b}) = \frac{a}{b}+\frac{b}{a}+2$.
2. Use $\frac{a}{b}+\frac{b}{a} \geq 2$ to obtain the lower bound 4.
3. Equality holds when $a=b$.

Final answer

$(a+b)\left(\frac{1}{a}+\frac{1}{b}\right) \geq 4$

Q078 **Written**

Prove that $\left(x+\frac{1}{x}\right)\left(x+\frac{4}{x}\right) \geq 9$.

Solution

1. Expand to $x^2+5+\frac{4}{x^2}$.
2. Since $x^2+\frac{4}{x^2} \geq 4$, the product is at least 9.
3. Equality holds when $x=\sqrt{2}$.

Final answer

$\left(x+\frac{1}{x}\right)\left(x+\frac{4}{x}\right) \geq 9$

Worked Solutions **Inequalities**

Q079 Written

Prove that $(a+8b)\left(\frac{2}{a}+\frac{1}{b}\right) \geq 18$.

Solution

1. Expand to $10+\frac{a}{b}+\frac{16b}{a}$.
2. AM–GM gives $\frac{a}{b}+\frac{16b}{a} \geq 8$, so the product is at least 18.
3. Equality holds when $a=4b$.

Final answer

$(a+8b)\left(\frac{2}{a}+\frac{1}{b}\right) \geq 18$

Q080 Written

Prove that $\left(a+\frac{3}{b}\right)\left(b+\frac{3}{a}\right) \geq 12$.

Solution

1. Expand to $ab+6+\frac{9}{ab}$.
2. AM–GM gives $ab+\frac{9}{ab} \geq 6$, so the product is at least 12.
3. Equality holds when $ab=3$.

Final answer

$\left(a+\frac{3}{b}\right)\left(b+\frac{3}{a}\right) \geq 12$

Worked Solutions **Inequalities**

Q081 MCQ

Under a>0 and b>0, prove that $\frac{(a+b)^2}{ab} \geq 4$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+b)^2}{ab}$ - 4.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+b)^2}{ab} \geq 4$.

Correct option: D

Final answer

$$\frac{(a-b)^2}{ab}$$

Q082 **MCQ**

Under a>0 and b>0, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab}$ - 25.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4\,(a-b)^2}{ab}$$

Worked Solutions **Inequalities**

Q083 MCQ

Under a>0 and b>0, prove that $\frac{(a+9b)(9a+b)}{ab} \geq 100$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+9b)(9a+b)}{ab}$ - 100.
2. Factor/complete the square to obtain $\frac{9(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+9b)(9a+b)}{ab} \geq 100$.

Correct option: C

Final answer

$$\frac{9\left(a-b\right)^2}{ab}$$

Q084 **MCQ**

Under a>0 and b>0, prove that $\frac{(a+16b)(16a+b)}{ab} \geq 289$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+16b)(16a+b)}{ab}$ - 289.
2. Factor/complete the square to obtain $\frac{16(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+16b)(16a+b)}{ab} \geq 289$.

Correct option: D

Final answer

$$\frac{16\,(a-b)^2}{ab}$$

Worked Solutions **Inequalities**

Q085 Written

Prove that $\left(a+\frac{1}{b}\right)\left(b+\frac{9}{a}\right) \geq 16$.

Solution

1. Expand to $ab + 10 + \frac{9}{ab}$.
2. AM–GM gives $ab + \frac{9}{ab} \geq 6$, so the product is at least 16.
3. Equality holds when $ab = 3$.

Final answer

$$\left(a+\frac{1}{b}\right)\left(b+\frac{9}{a}\right) \geq 16$$

Q086 Written

Prove that $a + \frac{4}{a} \geq 4$.

Solution

1. Apply AM–GM: $a + \frac{4}{a} \geq 2\sqrt{4} = 4$.
2. Equality holds when $a = 2$.

Final answer

$$a + \frac{4}{a} \geq 4$$

Worked Solutions **Inequalities**

Q087 Written

Prove that $3ab + \frac{4}{ab} \geq 4\sqrt{3}$.

Solution

1. Apply AM–GM to $3ab$ and $\frac{4}{ab}$: the sum is at least $2\sqrt{12} = 4\sqrt{3}$.
2. Equality holds when $ab = \frac{2}{\sqrt{3}}$.

Final answer

$$3ab + \frac{4}{ab} \geq 4\sqrt{3}$$

Q088 Written

Prove that $a + b + \frac{9}{a+b} \geq 6$.

Solution

1. Let $t = a + b > 0$. AM–GM gives $t + \frac{9}{t} \geq 6$.
2. Equality holds when $a + b = 3$.

Final answer

$$a + b + \frac{9}{a + b} \geq 6$$

Worked Solutions **Inequalities**

Q089 **Written**

A rectangle has perimeter 200 cm. Prove that its area $LW \le 2500\,\text{cm}^2$.

Solution

1. From the perimeter, $L + W = 100$.
2. AM–GM gives $\frac{L+W}{2} \ge \sqrt{LW}$, so $50 \ge \sqrt{LW}$.
3. Therefore $LW \le 2500$.
4. Equality holds when $L = W = 50$.

Final answer

$LW \le 2500$

Worked Solutions **Inequalities**

Q090 Challenge

For $a, b > 0$, prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Solution

1. Put the terms over a positive denominator: $\frac{b}{a} + \frac{a}{b} - 2 = \frac{(a-b)^2}{ab} \geq 0$.
2. Hence $\frac{b}{a} + \frac{a}{b} \geq 2$.
3. Equality holds when $a = b$.

Final answer

2

Worked Solutions **Inequalities**

Quiz MCQ 1 Concept Quiz · MCQ

If x>y, prove that $3x + y > 2(x + y)$.
If x>y, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition x>y makes each factor positive.
4. Hence $3x + y$>$2(x + y)$.

Correct option: A

Final answer

$x - y$

Quiz MCQ 2 Concept Quiz · MCQ

Under a,b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2$ - $10ab$.
2. Factor/complete the square to obtain $(a - 5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A

Final answer

$(a - 5b)^2$

Worked Solutions **Inequalities**

Quiz MCQ 3 **Concept Quiz · MCQ**

Under a>0 and b>0, prove that $\sqrt{\frac{a^2}{2}+\frac{49b^2}{2}} \geq \frac{a}{2}+\frac{7b}{2}$.
After squaring the positive sides, the value of $\mathrm{LHS}^2 - \mathrm{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2 - \mathrm{RHS}^2 = \frac{(a-7b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2}+\frac{49b^2}{2}} \geq \frac{a}{2}+\frac{7b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions **Inequalities**

Quiz MCQ 4 Concept Quiz · MCQ

Under a>0 and b>0, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab}$ - 25.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4\,(a-b)^2}{ab}$$

Quiz WRITTEN 5 Concept Quiz · Written

If x>y, prove that $3x + 4y > 2x + 5y$.

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition x>y makes each factor positive.
4. Hence $3x + 4y$>$2x + 5y$.

Final answer

$2x + 5y$

Worked Solutions **Inequalities**

Quiz WRITTEN 6 Concept Quiz · Written

Under x,y real, prove that $x^2 + y^2 \geq 2xy$.

Solution

1. Start with LHS - RHS: $x^2 + y^2$ - $2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Final answer

$2xy$

Quiz WRITTEN 7 Concept Quiz · Written

Under a>0 and b>0, prove that $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\mathrm{LHS}^2 - \mathrm{RHS}^2 = 12\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 6\sqrt{b}$>$\sqrt{a + 36b}$.

Final answer

$\sqrt{a + 36b}$

Worked Solutions **Inequalities**

Quiz WRITTEN 8 **Concept Quiz · Written**

Under t>0, prove that $\frac{t^2+1}{t} \geq 2$.

Solution

1. Start with LHS - RHS: $\frac{t^2+1}{t}$ - 2.
2. Factor/complete the square to obtain $\frac{(t-1)^2}{t}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{t^2+1}{t} \geq 2$.

Final answer

2

Answer key

Use the question number shown on each page.

Q001 $2(ax+by) > (a+b)(x+y)$

Q002 A

Q003 C

Q004 D

Q005 D

Q006 B

Q007 $ac+bd > ad+bc$

Q008 $ab+cd > ad+bc$

Q009 D

Q010 C

Q011 A

Q012 $3x-4y > 2x-3y$

Q013 $xy+5x > 3y+15$

Q014 $xy+2 > x+2y$

Q015 $ab+1 > a+b$

Q016 $ax+by > ay+bx$

Q017 $ax+by+cz > ay+bz+cx$

Q018 D

Q019 $a^2+4b^2 \geq 4ab$

Q020 D

Q021 C

Q022 B

Q023 A

Q024 C

Q025 B

Q026 $a^2+5b^2 \geq 4ab$

Q027 $9x^2 \geq y(6x-y)$

Q028 $2(x^2+y^2) \geq (x+y)^2$

Q029 $(2x-y)^2 \geq 3(x^2-y^2)$

Q030 $(2a+3b)^2 \geq 12ab$

Q031 A

Q032 C

Q033 D

Q034 D

Q035 A

Q036 C

Q037 $x^2+y^2 \geq 2(x-y-1)$

Q038 $x^2+2 > 2x$

Q039 $x^2+y^2-6x+8y+25 \geq 0$

Q040 $x^2+2y^2-10x+12y+43 \geq 0$

Q041 $(a^2+b^2)(x^2+y^2) \geq (ax+by)^2$

Q042 0

Q043 $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$

Q044 A

Q045 A

Q046 C

Q047 A

Q048 A

Q049 B

Q050 A

Q051 A

Q052 $\sqrt{2(a+b)} \geq \sqrt{a}+\sqrt{b}$

Q053 $a+2 \geq \sqrt{3+6a}$

Q054 $\sqrt{2(a^2+b^2)} \geq a+b$

Q055 $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a}+\sqrt{b}}{2}$

Q056 B

Q057 A

Q058 C

Q059 B

Q060 $\sqrt{a}+4\sqrt{b} > \sqrt{a+16b}$

Q061 $2\sqrt{a}+\sqrt{b} > \sqrt{4a+b}$

Q062 $\sqrt{a}+\sqrt{b} > \sqrt{a+b}$

Q063 $2\sqrt{a}+3\sqrt{b} > \sqrt{4a+9b}$

Q064 $\sqrt{a-b} > \sqrt{a}-\sqrt{b}$

Q065 $\sqrt{a-b} < \sqrt{a}-\frac{b}{2\sqrt{a}}$

Q066 $\sqrt{a+16b}$

Q067 $\frac{b}{a}+\frac{a}{b} \geq 2$

Q068 $\frac{b}{3a}+\frac{12a}{b} \geq 4$

Q069 A

Q070 C

Q071 B

Q072 A

Q073 D

Q074 D

Q075 C

Q076 D

Q077 $(a+b)\left(\frac{1}{a}+\frac{1}{b}\right) \geq 4$

Q078 $\left(x+\frac{1}{x}\right)\left(x+\frac{4}{x}\right) \geq 9$

Q079 $(a+8b)\left(\frac{2}{a}+\frac{1}{b}\right) \geq 18$

Q080 $\left(a+\frac{3}{b}\right)\left(b+\frac{3}{a}\right) \geq 12$

Q081 D

Q082 B

Q083 C

Q084 D

Q085 $\left(a+\frac{1}{b}\right)\left(b+\frac{9}{a}\right) \geq 16$

Q086 $a+\frac{4}{a} \geq 4$

Q087 $3ab+\frac{4}{ab} \geq 4\sqrt{3}$

Q088 $a+b+\frac{9}{a+b} \geq 6$

Q089 $LW \leq 2500$

Q090 2

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $2x+5y$

Quiz WRITTEN 6 $2xy$

Quiz WRITTEN 7 $\sqrt{a+36b}$

Quiz WRITTEN 8 2

Concept 1 | Arithmetic Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = a + (n-1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Test

$$a_{n+1} - a_n = d$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Arithmetic Sequences**

Q001 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = 2n - 3$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -1,\ a_2 = 1,\ a_3 = 3$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = -1,\ a_2 = 1,\ a_3 = 3$

Q002 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 7n$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -6,\ a_2 = -10,\ a_3 = -12$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = -6,\ a_2 = -10,\ a_3 = -12$

Worked Solutions **Arithmetic Sequences**

Q003 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 5n - 8$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3,\ a_2 = 2,\ a_3 = 7$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = -3,\ a_2 = 2,\ a_3 = 7$

Q004 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -3n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -2,\ a_2 = -5,\ a_3 = -8$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = -2,\ a_2 = -5,\ a_3 = -8$

Worked Solutions **Arithmetic Sequences**

Q005 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = -2n - 4$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -6,\ a_2 = -8,\ a_3 = -10$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = -6,\ a_2 = -8,\ a_3 = -10$

Q006 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 1$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 0,\ a_2 = 3,\ a_3 = 8$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = 0,\ a_2 = 3,\ a_3 = 8$

Worked Solutions **Arithmetic Sequences**

Q007 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3,\ a_2 = 7,\ a_3 = 13$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = 3,\ a_2 = 7,\ a_3 = 13$

Q008 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + 2n + 4$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 7,\ a_2 = 12,\ a_3 = 19$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = 7,\ a_2 = 12,\ a_3 = 19$

Worked Solutions **Arithmetic Sequences**

Q009 MCQ

For $a_n = 3n - 5$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2,\ a_2 = 1,\ a_3 = 4$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(-2, 1, 4)$

Q010 MCQ

For $a_n = 4n + 7$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 11,\ a_2 = 15,\ a_3 = 19$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(11, 15, 19)$

Worked Solutions **Arithmetic Sequences**

Q011 MCQ

For $a_n = n^2 - 4n + 6$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 3,\ a_2 = 2,\ a_3 = 3$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(3, 2, 3)$

Q012 MCQ

For $a_n = 2n^2 + 3$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 5,\ a_2 = 11,\ a_3 = 21$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(5, 11, 21)$

Worked Solutions **Arithmetic Sequences**

Q013 MCQ

For $a_n = n^2 + 5n - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 4,\ a_2 = 10,\ a_3 = 16$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(4, 10, 16)$

Q014 MCQ

For $a_n = 5 - 2n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 3,\ a_2 = 1,\ a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(3, 1, -1)$

Worked Solutions **Arithmetic Sequences**

Q015 MCQ

For $a_n = 2n^2 - 7n + 9$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 4,\ a_2 = 3,\ a_3 = 6$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(4, 3, 6)$

Q016 MCQ

For $a_n = n^2 - 3n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -3,\ a_2 = -3,\ a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(-3, -3, -1)$

Worked Solutions **Arithmetic Sequences**

Q017 MCQ

For $a_n = 3n^2 - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 1,\ a_2 = 10,\ a_3 = 25$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(1, 10, 25)$

Q018 MCQ

For $a_n = n^2 + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 3,\ a_2 = 6,\ a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(3, 6, 11)$

Worked Solutions **Arithmetic Sequences**

Q019 MCQ

For $a_n = 7n - 10$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -3,\ a_2 = 4,\ a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-3, 4, 11)$

Q020 MCQ

For $a_n = 2n^2 + n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2,\ a_2 = 9,\ a_3 = 20$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 9, 20)$

Worked Solutions **Arithmetic Sequences**

Q021 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = 2 \times 3^n$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 6,\ a_2 = 18,\ a_3 = 54$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = 6,\ a_2 = 18,\ a_3 = 54$

Q022 **Written**

When the general term of the sequence $\{a_n\}$ is $a_n = 3^n - 1$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2,\ a_2 = 8,\ a_3 = 26$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = 2,\ a_2 = 8,\ a_3 = 26$

Worked Solutions **Arithmetic Sequences**

Q023 Written

When the general term of the sequence $\{a_n\}$ is $a_n = (-3)^n$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -3,\ a_2 = 9,\ a_3 = -27$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = -3,\ a_2 = 9,\ a_3 = -27$

Q024 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3 \times (-2)^n$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -6,\ a_2 = 12,\ a_3 = -24$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = -6,\ a_2 = 12,\ a_3 = -24$

Worked Solutions **Arithmetic Sequences**

Q025 *MCQ*

For $a_n = 2^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2,\ a_2 = 4,\ a_3 = 8$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 4, 8)$

Q026 *MCQ*

For $a_n = (-2)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2,\ a_2 = 4,\ a_3 = -8$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(-2, 4, -8)$

Worked Solutions **Arithmetic Sequences**

Q027 MCQ

For $a_n = 2 \times 3^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 6,\ a_2 = 18,\ a_3 = 54$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(6, 18, 54)$

Q028 MCQ

For $a_n = 5 \times (-3)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -15,\ a_2 = 45,\ a_3 = -135$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-15, 45, -135)$

Worked Solutions **Arithmetic Sequences**

Q029 MCQ

For $a_n = 4^n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 3,\ a_2 = 15,\ a_3 = 63$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(3, 15, 63)$

Q030 MCQ

For $a_n = 2 \times 3^{n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2,\ a_2 = 6,\ a_3 = 18$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(2, 6, 18)$

Worked Solutions **Arithmetic Sequences**

Q031 MCQ

For $a_n = (-1)^{n-1}5^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 5,\ a_2 = -25,\ a_3 = 125$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(5, -25, 125)$

Q032 MCQ

For $a_n = (-3)^n + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -1,\ a_2 = 11,\ a_3 = -25$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-1, 11, -25)$

Worked Solutions **Arithmetic Sequences**

Q033 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{n+1}$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2},\ a_2 = \frac{2}{3},\ a_3 = \frac{3}{4}$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = \dfrac{1}{2},\ a_2 = \dfrac{2}{3},\ a_3 = \dfrac{3}{4}$

Q034 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{2n+1}$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{3},\ a_2 = \frac{2}{5},\ a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = \dfrac{1}{3},\ a_2 = \dfrac{2}{5},\ a_3 = \dfrac{3}{7}$

Worked Solutions **Arithmetic Sequences**

Q035 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{1}{3n-2}$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 1,\ a_2 = \frac{1}{4},\ a_3 = \frac{1}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 1,\ a_2 = \frac{1}{4},\ a_3 = \frac{1}{7}$$

Q036 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{2n}{n+3}$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2},\ a_2 = \frac{4}{5},\ a_3 = 1$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{2},\ a_2 = \frac{4}{5},\ a_3 = 1$$

Worked Solutions **Arithmetic Sequences**

Q037 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n+1}{2n-1}$, find the first term through the 3rd term.

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2,\ a_2 = 1,\ a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Final answer

$a_1 = 2,\ a_2 = 1,\ a_3 = \frac{4}{5}$

Q038 MCQ

For $a_n = \frac{n+1}{n+2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = \frac{2}{3},\ a_2 = \frac{3}{4},\ a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(\frac{2}{3}, \frac{3}{4}, \frac{4}{5})$

Worked Solutions **Arithmetic Sequences**

Q039 MCQ

For $a_n = \frac{2n-1}{3n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{4},\ a_2 = \frac{3}{7},\ a_3 = \frac{1}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer
$(\frac{1}{4}, \frac{3}{7}, \frac{1}{2})$

Q040 MCQ

For $a_n = \frac{n^2+1}{n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 1,\ a_2 = \frac{5}{3},\ a_3 = \frac{5}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer
$(1, \frac{5}{3}, \frac{5}{2})$

Worked Solutions **Arithmetic Sequences**

Q041 MCQ

For $a_n = \frac{3n}{2n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 3,\ a_2 = 2,\ a_3 = \frac{9}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer
$(3, 2, \frac{9}{5})$

Q042 MCQ

For $a_n = \frac{2n+3}{n+4}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 1,\ a_2 = \frac{7}{6},\ a_3 = \frac{9}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer
$(1, \frac{7}{6}, \frac{9}{7})$

Worked Solutions **Arithmetic Sequences**

Q043 MCQ

For $a_n = \frac{n}{3n-2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 1,\ a_2 = \frac{1}{4},\ a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(1, \frac{1}{4}, \frac{3}{7})$

Q044 Written

For the sequence $\{a_n\} : 0,\ 1,\ 4,\ 9,\ 16,\ \ldots$, deduce the general term.

Solution

1. The terms are $(\text{term number} - 1)^2$, so $a_n = (n-1)^2$.
2. Checking $n = 1, 2, 3$ gives 0,1,4.

Final answer

$a_n = (n-1)^2$

Worked Solutions **Arithmetic Sequences**

Q045 Written

For the sequence $\{a_n\}$: 0, 1, 8, 27, 64, ..., deduce the general term.

Solution

1. The terms are successive cubes starting with 0: $(n-1)^3$.
2. The formula gives 0,1,8,27,64 for $n = 1, 2, 3, 4, 5$.

Final answer

$a_n = (n-1)^3$

Q046 Written

For the sequence $\{a_n\}$: 1, 4, 9, 16, 25, ..., deduce the general term.

Solution

1. These are the squares $1^2, 2^2, 3^2, \ldots$, so $a_n = n^2$.

Final answer

$a_n = n^2$

Worked Solutions **Arithmetic Sequences**

Q047 Written

For the sequence $\{a_n\} : 7^2,\ 8^2,\ 9^2,\ 10^2,\ 11^2,\ \ldots$, deduce the general term.

Solution

1. The bases are $n + 6$, so $a_n = (n + 6)^2$.

Final answer

$a_n = (n + 6)^2$

Q048 MCQ

Which formula gives the general term of $\{a_n\} : 1,\ 8,\ 27,\ 64,\ \ldots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = n^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B

Final answer

n^3

Worked Solutions **Arithmetic Sequences**

Q049 *MCQ*

Which formula gives the general term of $\{a_n\}$: $9,\ 16,\ 25,\ 36,\ \ldots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = (n+2)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C

Final answer

$(n+2)^2$

Q050 *MCQ*

Which formula gives the general term of $\{a_n\}$: $-1,\ 0,\ 1,\ 8,\ \ldots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = (n-2)^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D

Final answer

$(n-2)^3$

Worked Solutions **Arithmetic Sequences**

Q051 MCQ

Which formula gives the general term of $\{a_n\}$: $36,\ 49,\ 64,\ 81,\ \ldots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = (n+5)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A

Final answer

$(n+5)^2$

Q052 MCQ

Which formula gives the general term of $\{a_n\}$: $1,\ 9,\ 25,\ 49,\ \ldots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = (2n-1)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B

Final answer

$(2n-1)^2$

Worked Solutions **Arithmetic Sequences**

Q053 MCQ

Which formula gives the general term of $\{a_n\}: 0,\ 1,\ 16,\ 81,\ \ldots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = (n-1)^4$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C

Final answer

$(n-1)^4$

Q054 Written

For the sequence $\{a_n\}: 2,\ 5,\ 8,\ 11,\ 14,\ \ldots$, deduce the general term.

Solution

1. The common increase is 3 and the first term is 2, so a_n=2+3(n-1)=3n-1.

Final answer

$a_n = 3n - 1$

Worked Solutions **Arithmetic Sequences**

Q055 Written

For the sequence $\{a_n\}: 0,\ 2,\ 4,\ 6,\ 8,\ \ldots$, deduce the general term.

Solution

1. The common difference is 2 and a_1=0, so a_n=2(n-1)=2n-2.

Final answer

$a_n = 2n - 2$

Q056 Written

For the sequence $\{a_n\}: 0,\ 3,\ 6,\ 9,\ 12,\ \ldots$, deduce the general term.

Solution

1. The common difference is 3 and a_1=0, so a_n=3(n-1)=3n-3.

Final answer

$a_n = 3n - 3$

Worked Solutions **Arithmetic Sequences**

Q057 Written

For the sequence $\{a_n\}$: 4, 9, 14, 19, 24, ..., deduce the general term.

Solution

1. The common difference is 5 and a_1=4, so a_n=4+5(n-1)=5n-1.

Final answer

$a_n = 5n - 1$

Q058 Written

For the sequence $\{a_n\}$: 6, 13, 20, 27, 34, ..., deduce the general term.

Solution

1. The common difference is 7 and a_1=6, so a_n=6+7(n-1)=7n-1.

Final answer

$a_n = 7n - 1$

Worked Solutions **Arithmetic Sequences**

Q059 MCQ

Which formula gives the general term of $\{a_n\}$: 7, 12, 17, 22, ...?:

Solution

1. Separate the visible pattern: the rule is $a_n = 5n + 2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A

Final answer

$5n + 2$

Q060 MCQ

Which formula gives the general term of $\{a_n\}$: −4, −1, 2, 5, ...?:

Solution

1. Separate the visible pattern: the rule is $a_n = 3n - 7$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B

Final answer

$3n - 7$

Worked Solutions **Arithmetic Sequences**

Q061 MCQ

Which formula gives the general term of $\{a_n\}$: $10,\ 6,\ 2,\ -2,\ \ldots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = 14 - 4n$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C

Final answer

$14 - 4n$

Q062 MCQ

Which formula gives the general term of $\{a_n\}$: $\frac{1}{2},\ 1,\ \frac{3}{2},\ 2,\ \ldots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = \frac{n}{2}$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D

Final answer

$\frac{n}{2}$

Worked Solutions **Arithmetic Sequences**

Q063 Written

For the sequence $\{a_n\}$: $2,\ -4,\ 6,\ -8,\ 10,\ \ldots$, deduce the general term.

Solution

1. The magnitude is $2n$ and the first sign is positive with alternation, so the sign factor is $(-1)^{n-1}$.
2. Therefore $a_n = (-1)^{n-1}2n$.

Final answer

$a_n = (-1)^{n-1}\, 2n$

Q064 Written

For the sequence $\{a_n\}$: $-3,\ 6,\ -9,\ 12,\ -15,\ \ldots$, deduce the general term.

Solution

1. The magnitude is $3n$; the first term is negative and signs alternate, giving $(-1)^n$.
2. Therefore $a_n = (-1)^n 3n$.

Final answer

$a_n = (-1)^n\, 3n$

Worked Solutions **Arithmetic Sequences**

Q065 Written

For the sequence $\{a_n\}: -5,\ 10,\ -15,\ 20,\ -25,\ \ldots$, deduce the general term.

Solution

1. The magnitude is $5n$ and the first sign is negative, so $a_n = (-1)^n 5n$.

Final answer

$a_n = (-1)^n\, 5n$

Q066 Written

For the sequence $\{a_n\}: -1,\ 4,\ -7,\ 10,\ -13,\ \ldots$, deduce the general term.

Solution

1. The magnitudes 1,4,7,10,13 are $3n-2$; the first sign is negative, so $a_n = (-1)^n(3n-2)$.

Final answer

$a_n = (-1)^n(3n-2)$

Worked Solutions **Arithmetic Sequences**

Q067 **Written**

For the sequence $\{a_n\}$: $1,\ \frac{2}{3},\ \frac{3}{5},\ \frac{4}{7},\ \frac{5}{9},\ \ldots$, deduce the general term.

Solution

1. The numerator is n and the denominator is the odd number 2n-1.
2. Therefore a_n=n/(2n-1).

Final answer

$$a_n = \frac{n}{2n-1}$$

Q068 **Written**

For the sequence $\{a_n\}$: $1,\ \frac{3}{4},\ \frac{5}{9},\ \frac{7}{16},\ \frac{9}{25},\ \ldots$, deduce the general term.

Solution

1. The numerator is the odd number $2n-1$ and the denominator is n^2.
2. Therefore $a_n = (2n-1)/n^2$.

Final answer

$$a_n = \frac{2n-1}{n^2}$$

Worked Solutions **Arithmetic Sequences**

Q069 Written

For the sequence $\{a_n\}$: 1, $-\frac{1}{3}$, $\frac{1}{5}$, $-\frac{1}{7}$, $\frac{1}{9}$, ..., deduce the general term.

Solution

1. The denominator is $2n-1$ and the signs alternate from a positive first term, so $a_n = (-1)^{n-1}/(2n-1)$.

Final answer

$$a_n = \frac{(-1)^{n-1}}{2n-1}$$

Q070 Written

For the sequence $\{a_n\}$: $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, $\frac{4}{5}$, $\frac{5}{6}$, ..., deduce the general term.

Solution

1. The numerator is n and the denominator is n+1, so a_n=n/(n+1).

Final answer

$$a_n = \frac{n}{n+1}$$

Worked Solutions **Arithmetic Sequences**

Q071 Written

For the sequence $\{a_n\} : 1 \times 3,\ 2 \times 9,\ 3 \times 27,\ 4 \times 81,\ 5 \times 243,\ \ldots$, deduce the general term.

Solution

1. The first factor is n and the second factor is 3^n, so $a_n = n3^n$.

Final answer

$a_n = n\,3^n$

Q072 Written

For the sequence $\{a_n\} : 1 \times \frac{1}{2},\ 2 \times \frac{1}{3},\ 3 \times \frac{1}{4},\ 4 \times \frac{1}{5},\ 5 \times \frac{1}{6},\ \ldots$, deduce the general term then find the 20th term.

Solution

1. The first factor is the term number n and the second factor is 1/(n+1).
2. Hence the general term is a_n=n/(n+1).
3. Substitute $n = 20$: a_{20}=20/(20+1)=20/21.

Final answer

$a_n = \dfrac{n}{n+1}; \quad a_{20} = \dfrac{20}{21}$

Worked Solutions **Arithmetic Sequences**

Q073 Written

Find the first term and the common difference of the arithmetic sequences: (a) $2, 5, 8, 11, \ldots$; **(b)** $3, -2, -7, -12, -17, \ldots$.

Solution

1. For (a), $5 - 2 = 8 - 5 = 11 - 8 = 3$, so $a = 2$, $d = 3$.
2. For (b), each successive difference is -5, so $a = 3$, $d = -5$.

Final answer

$(a) \quad a = 2,\ d = 3(b) \quad a = 3,\ d = -5$

Q074 Written

Find the first term and common difference of: (a) $1, 5, 9, 13, 17, \ldots$; **(b)** $23, 19, 15, 11, 7, \ldots$.

Solution

1. (a) The constant increase is (4), so $a = 1, d = 4$.
2. (b) The constant change is -4, so $a = 23, d = -4$.

Final answer

$(a) \quad a = 1,\ d = 4(b) \quad a = 23,\ d = -4$

Worked Solutions **Arithmetic Sequences**

Q075 MCQ

The listed terms $\{a_n\} = (4, 9, 14, 19, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d:

Solution

1. The first term is $a = 4$.
2. Subtract consecutive terms: $d = 5$.

Correct option: B

Final answer

$a = 4, d = 5$

Q076 MCQ

The listed terms $\{a_n\} = (-7, -3, 1, 5, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d:

Solution

1. The first term is $a = -7$.
2. Subtract consecutive terms: $d = 4$.

Correct option: A

Final answer

$a = -7, d = 4$

Worked Solutions **Arithmetic Sequences**

Q077 MCQ

The listed terms $\{a_n\} = (18, 13, 8, 3, \ldots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d:

Solution

1. The first term is $a = 18$.
2. Subtract consecutive terms: $d = -5$.

Correct option: C

Final answer

$a = 18, d = -5$

Q078 MCQ

The listed terms $\{a_n\} = (12, 8, 4, 0, \ldots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d:

Solution

1. The first term is $a = 12$.
2. Subtract consecutive terms: $d = -4$.

Correct option: D

Final answer

$a = 12, d = -4$

Worked Solutions **Arithmetic Sequences**

Q079 Written

Find the first term and common difference of: (a) $3, 5, 7, 9, 11, \ldots$; **(b)** $-13, -9, -5, -1, 3, \ldots$; **(c)** $-2, 1, 4, 7, 10, \ldots$; **(d)** $10, 7, 4, 1, -2, \ldots$; **(e)** $6, \frac{16}{3}, \frac{14}{3}, 4, \frac{10}{3}, \ldots$; **(f)** $-4, -\frac{9}{2}, -5, -\frac{11}{2}, -6, \ldots$.

Solution

1. Subtract consecutive terms in each list. The six constant differences are $2, 4, 3, -3, -\frac{2}{3}, -\frac{1}{2}$, respectively; the first terms are the first entries.

Final answer

$(a)(3,2); (b)(-13,4); (c)(-2,3); (d)(10,-3); (e)(6,-\frac{2}{3}); (f)(-4,-\frac{1}{2})$.

Worked Solutions **Arithmetic Sequences**

Q080 MCQ

The listed terms $\{a_n\} = (\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \ldots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d:

Solution

1. The first term is $a = \frac{1}{2}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: B

Final answer

$a = \dfrac{1}{2}, d = \dfrac{1}{3}$

Q081 MCQ

The listed terms $\{a_n\} = (-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \ldots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d:

Solution

1. The first term is $a = -\frac{3}{4}$.
2. Subtract consecutive terms: $d = \frac{1}{2}$.

Correct option: A

Final answer

$a = -\dfrac{3}{4}, d = \dfrac{1}{2}$

Worked Solutions **Arithmetic Sequences**

Q082 MCQ

The listed terms $\{a_n\} = (\frac{5}{3}, 2, \frac{7}{3}, \frac{8}{3}, \ldots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d:

Solution

1. The first term is $a = \frac{5}{3}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: C

Final answer

$a = \frac{5}{3}, d = \frac{1}{3}$

Q083 MCQ

The listed terms $\{a_n\} = (-\frac{5}{2}, -1, \frac{1}{2}, 2, \ldots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d:

Solution

1. The first term is $a = -\frac{5}{2}$.
2. Subtract consecutive terms: $d = \frac{3}{2}$.

Correct option: B

Final answer

$a = -\frac{5}{2}, d = \frac{3}{2}$

Worked Solutions **Arithmetic Sequences**

Q084 Written

The sequences are arithmetic. Find the common difference and the numbers that fit in (A) and (B): (a) $A, 2, -1, B, \ldots$; **(b)** $13, A, B, -2, \ldots$.

Solution

1. (a) $d = -1 - 2 = -3$; hence $A = 2 - d = 5$ and $B = -1 + d = -4$.
2. (b) The change from 13 to -2 covers three equal steps: $3d = -15$, so $d = -5$; therefore $A = 8, B = 3$.

Final answer

$(a) \quad d = -3,\ A = 5,\ B = -4 (b) \quad d = -5,\ A = 8,\ B = 3$

Q085 Written

The sequences are arithmetic. Find the common difference and the missing numbers: (a) $2, 9, A, B, \ldots$; **(b)** $A, 17, B, 5, \ldots$.

Solution

1. (a) $d = 9 - 2 = 7$, so $A = 16, B = 23$.
2. (b) $5 - 17 = 2d$, hence $d = -6$; then $A = 23, B = 11$.

Final answer

$(a) \quad d = 7,\ A = 16,\ B = 23 (b) \quad d = -6,\ A = 23,\ B = 11$

Worked Solutions **Arithmetic Sequences**

Q086 MCQ

The sequence $\{a_n\} = (6, A, 14, B, \ldots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 10, B = 18$.

Correct option: A

Final answer

$d = 4, A = 10, B = 18$

Q087 MCQ

The sequence $\{a_n\} = (A, 5, 9, B, \ldots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 1, B = 13$.

Correct option: C

Final answer

$d = 4, A = 1, B = 13$

Worked Solutions **Arithmetic Sequences**

Q088 **MCQ**

The sequence $\{a_n\} = (11, A, B, 23, \ldots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 15, B = 19$.

Correct option: B

Final answer

$d = 4, A = 15, B = 19$

Worked Solutions **Arithmetic Sequences**

Q089 Written

Find the common difference and missing terms: (a) $8, 15, A, B, \ldots$; **(b)** $7, 13, A, B, \ldots$; **(c)** $A, 9, 4, B, \ldots$; **(d)** $A, 15, B, 9, \ldots$; **(e)** $11, A, B, 20, \ldots$; **(f)** $17, A, B, 5, C, \ldots$.

Solution

1. Use the equal-step condition. For example, in (d), $9 - 15 = 2d$, so $d = -2$, giving $A = 17, B = 13$. Applying the same calculation to (a)–(f) gives the listed values.

Final answer

$(a) d = 7, A = 22, B = 29; (b) d = 6, A = 19, B = 25; (c) d = -5, A = 14, B = -1; (d) d = -2, A = 17, B = 13; (e) d = 3, A = 14, B = 17; (f) d = -4, A = 13, B = 9, C = 1.$

Worked Solutions **Arithmetic Sequences**

Q090 Written

For each arithmetic sequence, find the general term and the 7th term: (a) first term (4), common difference (3); (b) first term (2), common difference -3.

Solution

1. (a) $a_n = 4 + 3(n-1) = 3n + 1$; $a_7 = 22$.
2. (b) $a_n = 2 - 3(n-1) = 5 - 3n$; $a_7 = -16$.

Final answer

$(a) \quad a_n = 3n + 1,\ a_7 = 22 (b) \quad a_n = 5 - 3n,\ a_7 = -16$

Q091 Written

For each arithmetic sequence, find the general term and the 7th term: (a) $a = 5, d = 4$; (b) $a = -2, d = 3$; (c) $a = 10, d = -3$; (d) $a = -\frac{1}{2}, d = -\frac{1}{2}$.

Solution

1. Substitute each pair into $a_n = a + (n-1)d$, then put $n = 7$. The results are $4n + 1, 3n - 5, 13 - 3n, -\frac{n}{2}$, with 7th terms $29, 16, -8, -\frac{7}{2}$.

Final answer

$(a) a_n = 4n + 1, a_7 = 29; (b) a_n = 3n - 5, a_7 = 16; (c) a_n = 13 - 3n, a_7 = -8; (d) a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}.$

Worked Solutions **Arithmetic Sequences**

Q092 Written

For the arithmetic sequence a_n with first term (8) and common difference -6, find the general term and the 20th term.

Solution

1. Use $a_n = a + (n-1)d$: $a_n = 8 - 6(n-1) = 14 - 6n$.
2. Substitute $n = 20$: $a_{20} = 14 - 120 = -106$.

Final answer

$a_n = 8 + (n-1)(-6) = 14 - 6n; \quad a_{20} = -106$

Q093 Written

For each arithmetic sequence, find the general term and the 10th term: (a) $a = -7, d = 3$; **(b)** $a = -2, d = -3$; **(c)** $a = 7, d = \frac{1}{2}$; **(d)** $a = \frac{2}{3}, d = -\frac{1}{3}$.

Solution

1. Use $a_n = a + (n-1)d$ and substitute $n = 10$. Simplification gives the four displayed formulas and terms.

Final answer

$(a)a_n = 3n - 10, a_{10} = 20; (b)a_n = 1 - 3n, a_{10} = -29; (c)a_n = \frac{n+13}{2}, a_{10} = \frac{23}{2}; (d)a_n = \frac{3-n}{3}, a_{10} = -\frac{7}{3}.$

Worked Solutions **Arithmetic Sequences**

Q094 Written

In an arithmetic sequence, $a_3 + a_7 = 38$ and $\frac{a_4}{a_{12}} = \frac{2}{5}$. Find the first term and common difference.

Solution

1. $a_3 + a_7 = (a + 2d) + (a + 6d) = 2a + 8d = 38$, so $a + 4d = 19$.
2. The ratio gives $5(a + 3d) = 2(a + 11d)$, hence $3a = 7d$.
3. Solving yields $d = 3$ and $a = 7$.

Final answer

$a = 7, d = 3$.

Q095 Written

Find the general term of the arithmetic sequence whose first term is -5 and whose 17th term is 75.

Solution

1. $75 = -5 + 16d$, so $d = 5$.
2. $a_n = -5 + 5(n - 1) = 5n - 10$.

Final answer

$a_n = 5n - 10$

Worked Solutions **Arithmetic Sequences**

Q096 Written

The first term is 6, and the 4th term is 18. Find a_n.

Solution

1. $18 = 6 + 3d$, so $d = 4$.
2. $a_n = 6 + 4(n - 1) = 4n + 2$.

Final answer

$a_n = 4n + 2$

Q097 Written

The first term is 20, and the 8th term is -1. Find a_n.

Solution

1. $-1 = 20 + 7d$, so $d = -3$.
2. $a_n = 20 - 3(n - 1) = 23 - 3n$.

Final answer

$a_n = 23 - 3n$

Worked Solutions **Arithmetic Sequences**

Q098 **Written**

The first term is -8, and the 8th term is 55. Find a_n.

Solution

1. $55 = -8 + 7d$, so $d = 9$.
2. $a_n = -8 + 9(n-1) = 9n - 17$.

Final answer

$a_n = 9n - 17$

Q099 **Written**

The first term is -18, and the 10th term is -63. Find a_n.

Solution

1. $-63 = -18 + 9d$, so $d = -5$.
2. $a_n = -18 - 5(n-1) = -5n - 13$.

Final answer

$a_n = -5n - 13$

Worked Solutions **Arithmetic Sequences**

Q100 MCQ

An arithmetic sequence has $a = 4$ and $a_7 = 22$. Select the correct pair (a_n, d):

Solution

1. $22 = 4 + 6d$, so $d = 3$.
2. $a_n = 4 + 3(n-1) = 3n + 1$.

Correct option: D

Final answer

$(a_n, d) = (3n + 1, 3)$

Q101 MCQ

An arithmetic sequence has $a = -3$ and $a_9 = 29$. Select the correct pair (a_n, d):

Solution

1. $29 = -3 + 8d$, so $d = 4$.
2. $a_n = -3 + 4(n-1) = 4n - 7$.

Correct option: B

Final answer

$(a_n, d) = (4n - 7, 4)$

Worked Solutions **Arithmetic Sequences**

Q102 *MCQ*

An arithmetic sequence has $a = 12$ and $a_5 = -4$. Select the correct pair (a_n, d):

Solution

1. $-4 = 12 + 4d$, so $d = -4$.

2. $a_n = 12 - 4(n-1) = 16 - 4n$.

Correct option: A

Final answer

$(a_n, d) = (16 - 4n, -4)$

Q103 *MCQ*

An arithmetic sequence has $a = 5$ and $a_{11} = 45$. Select the correct pair (a_n, d):

Solution

1. $45 = 5 + 10d$, so $d = 4$.

2. $a_n = 5 + 4(n-1) = 4n + 1$.

Correct option: B

Final answer

$(a_n, d) = (4n + 1, 4)$

Worked Solutions **Arithmetic Sequences**

Q104 Written

Find the general term of the arithmetic sequence whose common difference is 3 and whose 16th term is 81.

Solution

1. $81 = a + 15(3)$, so $a = 36$.
2. $a_n = 36 + 3(n - 1) = 3n + 33$.

Final answer

$a_n = 3n + 33$

Q105 Written

The common difference is 2, and the 8th term is 4. Find a_n.

Solution

1. $4 = a + 7(2)$, so $a = -10$.
2. $a_n = -10 + 2(n - 1) = 2n - 12$.

Final answer

$a_n = 2n - 12$

Worked Solutions **Arithmetic Sequences**

Q106 Written

The common difference is -4, and the 5th term is -7. Find a_n.

Solution

1. $-7 = a + 4(-4)$, so $a = 9$.
2. $a_n = 9 - 4(n-1) = 13 - 4n$.

Final answer

$a_n = 13 - 4n$

Q107 Written

The common difference is -7, and the 6th term is -24. Find a_n.

Solution

1. $-24 = a + 5(-7)$, so $a = 11$.
2. $a_n = 11 - 7(n-1) = 18 - 7n$.

Final answer

$a_n = 18 - 7n$

Worked Solutions **Arithmetic Sequences**

Q108 MCQ

An arithmetic sequence has common difference -2 and $a_4 = 1$. Select the correct pair (a_n, a):

Solution

1. $1 = a + 3(-2)$, so $a = 7$.
2. $a_n = 7 - 2(n-1) = 9 - 2n$.

Correct option: A

Final answer

$(a_n, a) = (9 - 2n, 7)$

Q109 MCQ

An arithmetic sequence has common difference 5 and $a_6 = 17$. Select the correct pair (a_n, a):

Solution

1. $17 = a + 5(5)$, so $a = -8$.
2. $a_n = -8 + 5(n-1) = 5n - 13$.

Correct option: D

Final answer

$(a_n, a) = (5n - 13, -8)$

Worked Solutions **Arithmetic Sequences**

Q110 *MCQ*

An arithmetic sequence has common difference 4 and $a_{10} = 38$. Select the correct pair (a_n, a):

Solution

1. $38 = a + 9(4)$, so $a = 2$.
2. $a_n = 2 + 4(n-1) = 4n - 2$.

Correct option: C

Final answer

$(a_n, a) = (4n - 2, 2)$

Q111 *MCQ*

An arithmetic sequence has common difference -3 and $a_8 = -18$. Select the correct pair (a_n, a):

Solution

1. $-18 = a + 7(-3)$, so $a = 3$.
2. $a_n = 3 - 3(n-1) = 6 - 3n$.

Correct option: D

Final answer

$(a_n, a) = (6 - 3n, 3)$

Worked Solutions **Arithmetic Sequences**

Q112 Written

Find the general term of the arithmetic sequence $\{a_n\}$ whose 3rd term is 10 and the 8th term is 50.

Solution

1. Use $a_n = a + (n-1)d$. The data give $10 = a + 2d$ and $50 = a + 7d$.
2. Subtracting gives $5d = 40$, so $d = 8$ and $a = -6$.
3. Therefore $a_n = -6 + 8(n-1) = 8n - 14$.

Final answer

$a_n = 8n - 14$

Q113 Written

Find the general term of the arithmetic sequence whose 6th term is 2 and whose 15th term is -25.

Solution

1. $2 = a + 5d$ and $-25 = a + 14d$, so $d = -3,\ a = 17$.
2. $a_n = 17 - 3(n-1) = 20 - 3n$.

Final answer

$a_n = 20 - 3n$

Worked Solutions **Arithmetic Sequences**

Q114 Written

The 4th term is 14, and the 10th term is 62. Find a_n.

Solution

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8,\ a = -10$.
2. $a_n = 8n - 18$.

Final answer

$a_n = 8n - 18$

Q115 Written

The 3rd term is 30, and the 10th term is 2. Find a_n.

Solution

1. $30 = a + 2d$ and $2 = a + 9d$, so $d = -4,\ a = 38$.
2. $a_n = 42 - 4n$.

Final answer

$a_n = 42 - 4n$

Q116 Written

The 10th term is 49, and the 23rd term is 10. Find a_n.

Solution

1. $49 = a + 9d$ and $10 = a + 22d$, so $d = -3,\ a = 76$.
2. $a_n = 79 - 3n$.

Final answer

$a_n = 79 - 3n$

Q117 Written

The 3rd term is 6, and the 10th term is 62. Find a_n.

Solution

1. $6 = a + 2d$ and $62 = a + 9d$, so $d = 8,\ a = -10$.
2. $a_n = 8n - 18$.

Final answer

$a_n = 8n - 18$

Worked Solutions **Arithmetic Sequences**

Q118 MCQ

An arithmetic sequence has $a_3 = 5$ and $a_9 = 17$. Select the correct pair (a_n, a):

Solution

1. $5 = a + 2d$ and $17 = a + 8d$, so $d = 2,\ a = 1$.
2. $a_n = 1 + 2(n-1) = 2n - 1$.

Correct option: C

Final answer

$(a_n, a) = (2n - 1, 1)$

Q119 MCQ

An arithmetic sequence has $a_4 = 7$ and $a_{10} = -5$. Select the correct pair (a_n, a):

Solution

1. $7 = a + 3d$ and $-5 = a + 9d$, so $d = -2,\ a = 13$.
2. $a_n = 13 - 2(n-1) = 15 - 2n$.

Correct option: C

Final answer

$(a_n, a) = (15 - 2n, 13)$

Worked Solutions **Arithmetic Sequences**

Q120 MCQ

An arithmetic sequence has $a_7 = 31$ and $a_{12} = 56$. Select the correct pair (a_n, a):

Solution

1. $31 = a + 6d$ and $56 = a + 11d$, so $d = 5,\ a = 1$.
2. $a_n = 1 + 5(n-1) = 5n - 4$.

Correct option: B

Final answer

$(a_n, a) = (5n - 4, 1)$

Q121 MCQ

An arithmetic sequence has $a_2 = -4$ and $a_8 = 14$. Select the correct pair (a_n, a):

Solution

1. $-4 = a + d$ and $14 = a + 7d$, so $d = 3,\ a = -7$.
2. $a_n = -7 + 3(n-1) = 3n - 10$.

Correct option: A

Final answer

$(a_n, a) = (3n - 10, -7)$

Worked Solutions **Arithmetic Sequences**

Q122 MCQ

An arithmetic sequence has $a_5 = 3$ and $a_{11} = 9$. Select the correct pair (a_n, a):

Solution

1. $3 = a + 4d$ and $9 = a + 10d$, so $d = \frac{1}{2},\ a = -\frac{1}{2}$.
2. $a_n = -\frac{1}{2} + \frac{1}{2}(n-1) = \frac{n-2}{2}$.

Correct option: A

Final answer

$\left(a_n = \dfrac{n-2}{2}, -\dfrac{1}{2}\right)$

Q123 MCQ

An arithmetic sequence has $a_2 = 6$ and $a_9 = 20$. Select the correct pair (a_n, a):

Solution

1. $6 = a + d$ and $20 = a + 8d$, so $d = 2,\ a = 4$.
2. $a_n = 4 + 2(n-1) = 2n + 2$.

Correct option: B

Final answer

$(a_n = 2n + 2, 4)$

Worked Solutions **Arithmetic Sequences**

Q124 Written

The common difference is $\frac{2}{5}$, and the 30th term is $\frac{23}{5}$. Find a_n.

Solution

1. $\frac{23}{5} = a + 29(\frac{2}{5})$, so $a = -7$.
2. $a_n = -7 + \frac{2}{5}(n-1) = \frac{2n-37}{5}$.

Final answer

$$a_n = \frac{2n-37}{5}$$

Q125 MCQ

An arithmetic sequence has $d = -\frac{3}{2}$ and $a_5 = -9$. Select the correct pair (a_n, a):

Solution

1. $-9 = a + 4(-\frac{3}{2})$, so $a = -3$.
2. $a_n = -3 - \frac{3}{2}(n-1) = -\frac{3(n+1)}{2}$.

Correct option: A

Final answer

$$\left(a_n = -\frac{3(n+1)}{2}, -3\right)$$

Worked Solutions **Arithmetic Sequences**

Q126 MCQ

An arithmetic sequence has $d = \frac{2}{3}$ and $a_{12} = 10$. Select the correct pair (a_n, a):

Solution

1. $10 = a + 11(\frac{2}{3})$, so $a = \frac{8}{3}$.
2. $a_n = \frac{8}{3} + \frac{2}{3}(n-1) = \frac{2n+6}{3}$.

Correct option: D

Final answer

$\left(a_n = \dfrac{2n+6}{3}, \dfrac{8}{3}\right)$

Q127 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, which term is 113?

Solution

1. First find a: $33 = a + 7(4)$, so $a = 5$.
2. Then $a_n = 5 + 4(n-1) = 4n + 1$.
3. Solve $4n + 1 = 113$: $n = 28$.

Final answer

$n = 28$, the 28th term

Q128 Written

For the arithmetic sequence with first term 1 and common difference 3, determine which term is 40.

Solution

1. $a_n = 1 + 3(n-1) = 3n - 2$.
2. $3n - 2 = 40$ gives $n = 14$.

Final answer

$n = 14$, the 14th term

Q129 Written

For the arithmetic sequence with common difference -5 and 13th term -7, determine which term is -52.

Solution

1. $-7 = a + 12(-5)$, so $a = 53$ and $a_n = 58 - 5n$.
2. $58 - 5n = -52$ gives $n = 22$.

Final answer

$n = 22$, the 22nd term

Worked Solutions **Arithmetic Sequences**

Q130 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50, determine which term is -110.

Solution

1. $-20 = a + 9d$ and $-50 = a + 19d$, so $d = -3,\ a = 7$.
2. $a_n = 10 - 3n$, and $10 - 3n = -110$ gives $n = 40$.

Final answer

$n = 40$, the 40th term

Q131 MCQ

For $a = 3, d = 4$, which term of the arithmetic sequence is 59?:

Solution

1. $a_n = 3 + 4(n - 1) = 4n - 1$.
2. $4n - 1 = 59$ gives $n = 15$.

Correct option: C

Final answer

$n = 15$

Worked Solutions **Arithmetic Sequences**

Q132 MCQ

For $a = -5, d = 6$, which term of the arithmetic sequence is 55?:

Solution

1. $a_n = -5 + 6(n-1) = 6n - 11$.
2. $6n - 11 = 55$ gives $n = 11$.

Correct option: D

Final answer

$n = 11$

Q133 MCQ

For $a = 20, d = -3$, which term of the arithmetic sequence is -16?:

Solution

1. $a_n = 20 - 3(n-1) = 23 - 3n$.
2. $23 - 3n = -16$ gives $n = 13$.

Correct option: A

Final answer

$n = 13$

Worked Solutions **Arithmetic Sequences**

Q134 MCQ

For $a = 1, d = 5$, which term of the arithmetic sequence is 86?:

Solution

1. $a_n = 1 + 5(n-1) = 5n - 4$.
2. $5n - 4 = 86$ gives $n = 18$.

Correct option: B

Final answer

$n = 18$

Q135 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, find the first term greater than 500.

Solution

1. From the given data, $a_n = 4n + 1$.
2. Require $4n + 1 > 500$, hence $n > 124.75$.
3. The least natural index is $n = 125$.

Final answer

$n = 125$, the 125th term

Worked Solutions **Arithmetic Sequences**

Q136 Written

For the arithmetic sequence with first term 1 and common difference 3, find the first term greater than 100.

Solution

1. $a_n = 3n - 2$.
2. $3n - 2 > 100$ gives $n > 34$.
3. The first valid index is $n = 35$.

Final answer

$n = 35$, the 35th term

Q137 MCQ

For $a = 2, d = 7$, find the first term greater than 100:

Solution

1. $a_n = 7n - 5$.
2. $7n - 5 > 100$ gives $n > 15$, so $n = 16$.

Correct option: A

Final answer

$n = 16$

Worked Solutions **Arithmetic Sequences**

Q138 MCQ

For $a = -9, d = 4$, find the first term greater than 50:

Solution

1. $a_n = 4n - 13$.
2. $4n - 13 > 50$ gives $n > 15.75$, so $n = 16$.

Correct option: D

Final answer

$n = 16$

Q139 MCQ

For $a = 6, d = 3$, find the first term greater than 200:

Solution

1. $a_n = 3n + 3$.
2. $3n + 3 > 200$ gives $n > 65\frac{2}{3}$, so $n = 66$.

Correct option: D

Final answer

$n = 66$

Worked Solutions **Arithmetic Sequences**

Q140 MCQ

For $a = -1, d = 9$, find the first term greater than 300:

Solution

1. $a_n = 9n - 10$.
2. $9n - 10 > 300$ gives $n > 34\frac{4}{9}$, so $n = 35$.

Correct option: B

Final answer

$n = 35$

Q141 Written

For the arithmetic sequence with common difference -5 and 13th term -7, find the first term smaller than -300.

Solution

1. $a_n = 58 - 5n$.
2. $58 - 5n < -300$ gives $n > 71.6$.
3. The first valid index is $n = 72$.

Final answer

$n = 72$, the 72nd term

Worked Solutions **Arithmetic Sequences**

Q142 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50, find the first term smaller than -500.

Solution

1. $a_n = 10 - 3n$.
2. $10 - 3n < -500$ gives $n > 170$.
3. The first valid index is $n = 171$.

Final answer

$n = 171$, the 171st term

Q143 MCQ

For $a = 10, d = -4$, find the first term smaller than -100:

Solution

1. $a_n = 14 - 4n$.
2. $14 - 4n < -100$ gives $n > 28.5$, so $n = 29$.

Correct option: B

Final answer

$n = 29$

Worked Solutions **Arithmetic Sequences**

Q144 MCQ

For $a = 25, d = -6$, find the first term smaller than -250:

Solution

1. $a_n = 31 - 6n$.
2. $31 - 6n < -250$ gives $n > 46\frac{5}{6}$, so $n = 47$.

Correct option: B

Final answer

$n = 47$

Q145 MCQ

For $a = -3, d = -5$, find the first term smaller than -200:

Solution

1. $a_n = 2 - 5n$.
2. $2 - 5n < -200$ gives $n > 40.4$, so $n = 41$.

Correct option: C

Final answer

$n = 41$

Worked Solutions **Arithmetic Sequences**

Q146 MCQ

For $a = 4, d = 6$, find the first term that is at least 100:

Solution

1. $a_n = 6n - 2$.
2. $6n - 2 \geq 100$ gives $n \geq 17$, so $n = 17$.

Correct option: A

Final answer

$n = 17$

Q147 MCQ

For the expression For $a = 5, d = 4$, does the sequence contain the term 42?, the required value is:

Solution

1. $a_n = 4n + 1$.
2. $4n + 1 = 42$ gives $n = \frac{41}{4}$, not a natural index.
3. Therefore 42 is not a term.

Correct option: B

Final answer

No term

Q148 Written

When $a, 6, 2a$ form an arithmetic sequence in this order, find a.

Solution

1. $2(6) = a + 2a$.
2. $3a = 12$, so $a = 4$.

Final answer

$a = 4$

Q149 Written

When the three numbers $x, 6, x^2$ form an arithmetic sequence in this order, find x.

Solution

1. Use $2b = a + c$: $2(6) = x + x^2$.
2. Thus $x^2 + x - 12 = 0 = (x + 4)(x - 3)$.
3. Therefore $x = -4$ or $x = 3$.

Final answer

$x = -4$ or $x = 3$

Worked Solutions **Arithmetic Sequences**

Q150 Written

When $x^2, x, -8$ form an arithmetic sequence in this order, find x.

Solution

1. $2x = x^2 - 8$.
2. $x^2 - 2x - 8 = (x-4)(x+2) = 0$.
3. $x = -2$ or $x = 4$.

Final answer

$x = -2$ or $x = 4$

Q151 Written

When $-3, x, x^2$ form an arithmetic sequence in this order, find x.

Solution

1. $2x = -3 + x^2$.
2. $x^2 - 2x - 3 = (x-3)(x+1) = 0$.
3. $x = -1$ or $x = 3$.

Final answer

$x = -1$ or $x = 3$

Worked Solutions **Arithmetic Sequences**

Q152 Challenge

When $2, x, y, z$ form an arithmetic sequence in this order, and $2x, y+1, z-y$ also form an arithmetic sequence in this order, find x, y, z.

Solution

1. Let the common difference of $2, x, y, z$ be d: $x = 2 + d,\ y = 2 + 2d,\ z = 2 + 3d$.
2. $2(y+1) = 2x + (z - y)$ gives $6 + 4d = 4 + 3d$, hence $d = -2$.
3. $(x, y, z) = (0, -2, -4)$.

Final answer

$x = 0, \quad y = -2, \quad z = -4$

Worked Solutions **Arithmetic Sequences**

Q153 Written

Find the sum (S_7) of a sequence with first term (8), last term (22), and (7) terms.

Solution

1. Use $(S_n = \frac{n}{2}(a + l))$.
2. $(S_7 = \frac{7}{2}(8 + 22) = 105)$.

Final answer

105

Q154 Written

Find (S_{10}) for a sequence with first term (7), last term (13), and (10) terms.

Solution

1. $(S_{10} = \frac{10}{2}(7 + 13) = 100)$.

Final answer

100

Worked Solutions **Arithmetic Sequences**

Q155 Written

Find (S_{10}) for a sequence with first term (3), last term (21), and (10) terms.

Solution

1. $(S_{10} = \frac{10}{2}(3 + 21) = 120)$.

Final answer

120

Q156 Written

Find (S_8) for a sequence with first term (5), last term (33), and (8) terms.

Solution

1. $(S_8 = \frac{8}{2}(5 + 33) = 152)$.

Final answer

152

Worked Solutions **Arithmetic Sequences**

Q157 Written

Find (S_{10}) for a sequence with first term (2), last term (29), and (10) terms.

Solution

1. ($S_{10} = \frac{10}{2}(2 + 29) = 155$).

Final answer

155

Q158 MCQ

A sequence has first term 6, last term 30, and 7 terms. Find its sum:

Solution

1. Use $S_7 = \frac{7}{2}(6 + 30)$.
2. Thus $S_7 = 126$.

Correct option: B

Final answer

126

Worked Solutions **Arithmetic Sequences**

Q159 MCQ

A sequence has first term -4, last term 20, and 9 terms. Find its sum:

Solution

1. $S_9 = \frac{9}{2}(-4 + 20) = 72.$

Correct option: B

Final answer

72

Q160 MCQ

Find the sum of 10 terms whose first term is 15 and last term is 3:

Solution

1. $S_{10} = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions **Arithmetic Sequences**

Q161 *MCQ*

The first term is 7, the last term is 42, and there are 8 terms. Find S_8:

Solution

1. $S_8 = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q162 *MCQ*

Find the sum of 6 terms with first term -12 and last term 8:

Solution

1. $S_6 = \frac{6}{2}(-12 + 8) = -12.$

Correct option: B

Final answer

-12

Worked Solutions **Arithmetic Sequences**

Q163 MCQ

A sequence has $S_9 = 72$ and first term -4. Find its last term:

Solution

1. From $72 = \frac{9}{2}(-4 + l)$, $l = 20$.

Correct option: C

Final answer

20

Q164 Written

Find (S_{13}) when the first term is (20), the common difference is (-5), and the number of terms is (13).

Solution

1. Use $(S_n = \frac{n}{2}\{2a + (n-1)d\})$.
2. $(S_{13} = \frac{13}{2}\{40 + 12(-5)\} = -130)$.

Final answer

-130

Worked Solutions **Arithmetic Sequences**

Q165 Written

Find (S_{12}) when the first term is (-6), the common difference is (3), and there are (12) terms.

Solution

1. $(S_{12} = \frac{12}{2}\{-12 + 11(3)\} = 144)$.

Final answer

144

Q166 Written

Find (S_{15}) when the first term is (10), the common difference is (-3), and there are (15) terms.

Solution

1. $(S_{15} = \frac{15}{2}\{20 + 14(-3)\} = -165)$.

Final answer

-165

Worked Solutions **Arithmetic Sequences**

Q167 Written

Find (S_{11}) when the first term is (-4), the common difference is (4), and there are (11) terms.

Solution

1. ($S_{11} = \frac{11}{2}\{-8 + 10(4)\} = 176$).

Final answer

176

Q168 Written

Find (S_{18}) when the first term is (3), the common difference is (2), and there are (18) terms.

Solution

1. ($S_{18} = \frac{18}{2}\{6 + 17(2)\} = 360$).

Final answer

360

Worked Solutions **Arithmetic Sequences**

Q169 MCQ

For $a = 4, d = 3, n = 12$, find S_{12}:

Solution

1. $S_{12} = \frac{12}{2}\{8 + 11(3)\} = 246.$

Correct option: C

Final answer

246

Q170 MCQ

For $a = 9, d = -2, n = 10$, find S_{10}:

Solution

1. $S_{10} = 5\{18 + 9(-2)\} = 0.$

Correct option: B

Final answer

0

Worked Solutions **Arithmetic Sequences**

Q171 MCQ

For $a = -5, d = 4, n = 9$, find S_9:

Solution

1. $S_9 = \frac{9}{2}\{-10 + 8(4)\} = 88.$

Correct option: C

Final answer

88

Q172 MCQ

For $a = 2, d = -3, n = 15$, find S_{15}:

Solution

1. $S_{15} = \frac{15}{2}\{4 + 14(-3)\} = -285.$

Correct option: B

Final answer

-285

Worked Solutions **Arithmetic Sequences**

Q173 MCQ

For $a = 11, d = 0, n = 8$, find S_8:

Solution

1. All terms equal 11, so $S_8 = 88$.

Correct option: C

Final answer

88

Q174 MCQ

For $a = -3, d = -2, n = 7$, find S_7:

Solution

1. $S_7 = \frac{7}{2}\{-6 + 6(-2)\} = -54$.

Correct option: B

Final answer

-54

Worked Solutions **Arithmetic Sequences**

Q175 Written

Find the sum (S) of a sequence with first term (2), common difference (3), and last term (47).

Solution

1. Find (n) from $(47 = 2 + (n-1)3)$, giving $(n = 16)$.
2. Then $(S = \frac{16}{2}(2+47) = 392)$.

Final answer

392

Q176 Written

Find (S) for a sequence with first term (5), common difference (3), and last term (44).

Solution

1. Solve $(44 = 5 + (n-1)3)$, so $(n = 14)$.
2. $(S = \frac{14}{2}(5+44) = 343)$.

Final answer

343

Worked Solutions **Arithmetic Sequences**

Q177 Written

Find (S) for a sequence with first term (3), common difference (4), and last term (75).

Solution

1. ($75 = 3 + (n-1)4$) gives ($n = 19$).
2. ($S = \frac{19}{2}(3 + 75) = 741$).

Final answer

741

Q178 Written

Find (S) for a sequence with first term (2), common difference (5), and last term (57).

Solution

1. ($57 = 2 + (n-1)5$) gives ($n = 12$).
2. ($S = \frac{12}{2}(2 + 57) = 354$).

Final answer

354

Worked Solutions **Arithmetic Sequences**

Q179 Written

Find (S) for a sequence with first term (4), common difference (3), and last term (100).

Solution

1. $(100 = 4 + (n-1)3)$ gives $(n = 33)$.
2. $(S = \frac{33}{2}(4 + 100) = 1716)$.

Final answer

1716

Q180 MCQ

A sequence has $a = 5, d = 4, l = 41$. Find its sum:

Solution

1. $41 = 5 + (n-1)4$ gives $n = 10$.
2. $S = \frac{10}{2}(5 + 41) = 230$.

Correct option: C

Final answer

230

Worked Solutions **Arithmetic Sequences**

Q181 MCQ

A sequence has $a = -7, d = 5, l = 48$. Find its sum:

Solution

1. $48 = -7 + (n-1)5$ gives $n = 12$.
2. $S = 6(-7 + 48) = 246$.

Correct option: C

Final answer

246

Q182 MCQ

A sequence has $a = 12, d = 2, l = 50$. Find its sum:

Solution

1. $50 = 12 + (n-1)2$ gives $n = 20$.
2. $S = 10(12 + 50) = 620$.

Correct option: C

Final answer

620

Worked Solutions **Arithmetic Sequences**

Q183 MCQ

A sequence has $a = -10, d = 6, l = 50$. Find its sum:

Solution

1. $50 = -10 + (n-1)6$ gives $n = 11$.
2. $S = \frac{11}{2}(-10 + 50) = 220$.

Correct option: C

Final answer

220

Q184 MCQ

A sequence has $a = 18, d = -3, l = -9$. Find its sum:

Solution

1. Solve $-9 = 18 + (n-1)(-3)$, giving $n = 10$.
2. $S = \frac{10}{2}(18 - 9) = 45$.

Correct option: C

Final answer

45

Worked Solutions **Arithmetic Sequences**

Q185 MCQ

A sequence has $a = 40, d = -4, l = -12$. Find its sum:

Solution

1. $n = 14$ from $-12 = 40 + (n-1)(-4)$.
2. $S = 7(40 - 12) = 196$.

Correct option: B

Final answer

196

Q186 Written

For the arithmetic sequence with first term (48) and common difference (-2), find the sum from the 20th term to the 30th term.

Solution

1. Use $(S_{30} - S_{19})$, since the range is (a_{20}) through (a_{30}).
2. $(a_{20} = 10, a_{30} = -10)$, so $(11 \times \frac{10+(-10)}{2} = 0)$.

Final answer

0

Worked Solutions **Arithmetic Sequences**

Q187 Written

For the arithmetic sequence with first term (-53) and common difference (4), find the sum from the 10th term to the 25th term.

Solution

1. $(a_{10} = -53 + 9(4) = -17)$ and $(a_{25} = -53 + 24(4) = 43)$.
2. There are (16) terms, so the sum is $(16 \times \frac{-17+43}{2} = 208)$.

Final answer

208

Q188 MCQ

For $a = 3, d = 4$, find the sum from a_5 to a_{12}:

Solution

1. $a_5 = 19, a_{12} = 47$, with 8 terms.
2. Sum $= 8(19 + 47)/2 = 264$.

Correct option: B

Final answer

264

Worked Solutions **Arithmetic Sequences**

Q189 MCQ

For $a = 20, d = -3$, find the sum from a_4 to a_{10}:

Solution

1. $a_4 = 11, a_{10} = -7$, with 7 terms.
2. Sum $= 7(11 - 7)/2 = 14$.

Correct option: C

Final answer

14

Q190 MCQ

For $a = 60, d = -3$, find the sum from a_8 to a_{16}:

Solution

1. $a_8 = 39, a_{16} = 15$, with 9 terms.
2. Sum $= 9(39 + 15)/2 = 243$.

Correct option: C

Final answer

243

Worked Solutions **Arithmetic Sequences**

Q191 MCQ

For $a = 1, d = 7$, find the sum from a_3 to a_9:

Solution

1. $a_3 = 15, a_9 = 57$, with 7 terms.
2. Sum $= 7(15 + 57)/2 = 252$.

Correct option: C

Final answer

252

Q192 MCQ

For $a = 60, d = -3$, find the sum from a_9 to a_{17}:

Solution

1. $a_9 = 36, a_{17} = 12$, with 9 terms.
2. Sum $= 9(36 + 12)/2 = 216$.

Correct option: B

Final answer

216

Worked Solutions **Arithmetic Sequences**

Q193 MCQ

For $a = 12, d = 5$, find the sum from a_2 to a_8:

Solution

1. $a_2 = 17, a_8 = 47$, with 7 terms.
2. Sum $= 7(17 + 47)/2 = 224$.

Correct option: C

Final answer

224

Q194 Written

For the arithmetic sequence $(90, 84, 78, \ldots)$, find the sum from the 10th term to the 20th term.

Solution

1. Here $(a = 90, d = -6)$, so the required sum is $(S_{20} - S_9)$.
2. $(S_{20} = 660)$ and $(S_9 = 594)$, hence the sum is $(660 - 594 = 66)$.

Final answer

66

Worked Solutions **Arithmetic Sequences**

Q195 Written

For the arithmetic sequence $(-9, -3, 3, \ldots)$, find the sum from the 11th term to the 20th term.

Solution

1. Here $a = -9, d = 6$, so use $S_{20} - S_{10}$.
2. $a_{10} = 45$ and $a_{20} = 105$; there are 10 terms.
3. Required sum = $\frac{10(45+105)}{2} = 750$.

Final answer

750

Q196 Written

For the arithmetic sequence $(74, 69, 64, \ldots)$, find the sum from the 8th term to the 18th term.

Solution

1. Here $(a = 74, d = -5)$, so $(a_8 = 39)$ and $(a_{18} = -11)$.
2. There are (11) terms; the sum is $(11 \times \frac{39-11}{2} = 154)$.

Final answer

154

Worked Solutions **Arithmetic Sequences**

Q197 MCQ

For $a = 4, d = 5$, find the sum from a_6 to a_{13}:

Solution

1. $a_6 = 29, a_{13} = 64$, with 8 terms.
2. Sum $= 8(29 + 64)/2 = 372$.

Correct option: C

Final answer

372

Q198 MCQ

For $a = -25, d = 6$, find the sum from a_5 to a_{11}:

Solution

1. $a_5 = -1, a_{11} = 35$, with 7 terms.
2. Sum $= 7(-1 + 35)/2 = 119$.

Correct option: C

Final answer

119

Worked Solutions **Arithmetic Sequences**

Q199 Written

Find (S_n) for a sequence with first term (1) and common difference (4).

Solution

1. Substitute ($a = 1, d = 4$) into ($S_n = \frac{n}{2}\{2a + (n-1)d\}$).
2. ($S_n = \frac{n}{2}(2 + 4n - 4) = 2n^2 - n$).

Final answer

$2n^2 - n$

Q200 Written

Find (S_n) for a sequence with first term (2) and common difference (-3).

Solution

1. ($S_n = \frac{n}{2}\{4 + (n-1)(-3)\}$).
2. Therefore ($S_n = \frac{n(7-3n)}{2}$).

Final answer

$$\frac{n(7-3n)}{2}$$

Worked Solutions **Arithmetic Sequences**

Q201 Written

Find (S_n) for a sequence with first term (0) and common difference (5).

Solution

1. ($S_n = \frac{n}{2}\{0 + 5(n-1)\}$).
2. Hence ($S_n = \frac{5n(n-1)}{2}$).

Final answer

$$\frac{5n(n-1)}{2}$$

Q202 Written

Find (S_n) for a sequence with first term (3) and common difference (−4).

Solution

1. ($S_n = \frac{n}{2}\{6 - 4(n-1)\}$).
2. Therefore ($S_n = 5n - 2n^2$).

Final answer

$5n - 2n^2$

Worked Solutions **Arithmetic Sequences**

Q203 Written

Find (S_n) for a sequence with first term (-4) and common difference (-2).

Solution

1. ($S_n = \frac{n}{2}\{-8 - 2(n-1)\}$).
2. Therefore ($S_n = -n^2 - 3n$).

Final answer

$-n^2 - 3n$

Worked Solutions **Arithmetic Sequences**

Q204 Challenge

For the arithmetic sequence $(94, 89, 84, \ldots)$, find the sum of the odd-numbered terms from the first term to the 11th term.

Solution

1. The odd-numbered terms are $(a_1, a_3, a_5, a_7, a_9, a_{11})$, a new arithmetic sequence with first term (94), common difference $(2(-5) = -10)$, and (6) terms.
2. Its last term is (44), so the sum is $(\frac{6}{2}(94 + 44) = 414)$.

Final answer

414

Worked Solutions **Arithmetic Sequences**

Q205 Written

Find the sum of the natural numbers from 1 to 100.

Solution

1. There are $n = 100$ terms, with $a = 1$ and $l = 100$.
2. $S_{100} = \frac{100}{2}(1 + 100) = 5050$.

Final answer

5050

Q206 Written

Find the sum of the natural numbers from 1 to 200.

Solution

1. Use $n = 200, a = 1, l = 200$.
2. $S = \frac{200}{2}(1 + 200) = 20100$.

Final answer

20100

Worked Solutions **Arithmetic Sequences**

Q207 Written

Find the sum of the natural numbers from 1 to 150.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 150$.
2. $Therefore S = 11325$.

Final answer

11325

Q208 Written

Find the sum of the natural numbers from 1 to 70.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 70$.
2. $Therefore S = 2485$.

Final answer

2485

Worked Solutions **Arithmetic Sequences**

Q209 Written

Find the sum of the natural numbers from 1 to 100.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 100$.
2. $Therefore S = 5050$.

Final answer

5050

Q210 Written

Find the sum of the natural numbers from 1 to 400.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 400$.
2. $Therefore S = 80200$.

Final answer

80200

Worked Solutions **Arithmetic Sequences**

Q211 **Written**

Find the sum of the natural numbers from 1 to 250.

Solution

1. Use $S_n = \frac{n}{2}(1+n)$ with $n = 250$.
2. $ThereforeS = 31375$.

Final answer

31375

Q212 **Written**

Find the sum of the natural numbers from 1 to 81.

Solution

1. Use $S_n = \frac{n}{2}(1+n)$ with $n = 81$.
2. $ThereforeS = 8085$.

Final answer

8085

Worked Solutions **Arithmetic Sequences**

Q213 Written

Find the sum of the natural numbers from 1 to 301.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 301$.
2. $Therefore S = 24535$.

Final answer

24535

Q214 Written

Find the sum of the natural numbers from 1 to 99.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 99$.
2. $Therefore S = 4950$.

Final answer

4950

Worked Solutions **Arithmetic Sequences**

Q215 MCQ

Find the sum of the natural numbers from 1 to 30:

Solution

1. $S = \frac{30}{2}(1 + 30) = 465.$

Correct option: B

Final answer

465

Q216 MCQ

Find the sum of the natural numbers from 1 to 45:

Solution

1. $S = \frac{45}{2}(1 + 45) = 1035.$

Correct option: B

Final answer

1035

Worked Solutions **Arithmetic Sequences**

Q217 MCQ

Find the sum of the natural numbers from 1 to 60:

Solution

1. $S = \frac{60}{2}(1 + 60) = 1830.$

Correct option: B

Final answer

1830

Q218 MCQ

Find the sum of the natural numbers from 1 to 75:

Solution

1. $S = \frac{75}{2}(1 + 75) = 2850.$

Correct option: B

Final answer

2850

Worked Solutions **Arithmetic Sequences**

Q219 MCQ

Find the sum of the natural numbers from 1 to 90:

Solution

1. $S = \frac{90}{2}(1 + 90) = 4095.$

Correct option: B

Final answer

4095

Q220 MCQ

Find the sum of the natural numbers from 1 to 125:

Solution

1. $S = \frac{125}{2}(1 + 125) = 7875.$

Correct option: B

Final answer

7875

Worked Solutions **Arithmetic Sequences**

Q221 MCQ

Find the sum of the natural numbers from 1 to 180:

Solution

1. $S = \frac{180}{2}(1 + 180) = 16290.$

Correct option: B

Final answer

16290

Q222 MCQ

Find the sum of the natural numbers from 1 to 220:

Solution

1. $S = \frac{220}{2}(1 + 220) = 24310.$

Correct option: B

Final answer

24310

Worked Solutions **Arithmetic Sequences**

Q223 MCQ

Find the sum of the natural numbers from 1 to 275:

Solution

1. $S = \frac{275}{2}(1 + 275) = 37950.$

Correct option: B

Final answer

37950

Q224 MCQ

Find the sum of the natural numbers from 1 to 350:

Solution

1. $S = \frac{350}{2}(1 + 350) = 61425.$

Correct option: B

Final answer

61425

Worked Solutions **Arithmetic Sequences**

Q225 **Written**

Find the sum of the natural numbers from 101 to 200.

Solution

1. There are $n = 100$ terms, with $a = 101$ and $l = 200$.
2. $S = \frac{100}{2}(101 + 200) = 15050$.

Final answer

15050

Q226 **Written**

Find the sum of the natural numbers from 201 to 300.

Solution

1. There are 100 terms from 201 to 300.
2. $S = \frac{100}{2}(201 + 300) = 25050$.

Final answer

25050

Worked Solutions **Arithmetic Sequences**

Q227 Written

Find the sum of all two-digit natural numbers.

Solution

1. $Use 10 + 11 + \cdots + 99 : n = 90.$
2. $S = \frac{90}{2}(10 + 99) = 4905.$

Final answer

4905

Q228 MCQ

A sequence has first term 5, last term 35, and 7 terms. Select its sum:

Solution

1. $S = \frac{7}{2}(5 + 35) = 140.$

Correct option: B

Final answer

140

Worked Solutions **Arithmetic Sequences**

Q229 MCQ

A sequence has first term 12, last term 68, and 9 terms. Select its sum:

Solution

1. $S = \frac{9}{2}(12 + 68) = 360.$

Correct option: B

Final answer

360

Q230 MCQ

A sequence has first term -4, last term 20, and 9 terms. Select its sum:

Solution

1. $S = \frac{9}{2}(-4 + 20) = 72.$

Correct option: B

Final answer

72

Worked Solutions **Arithmetic Sequences**

Q231 MCQ

A sequence has first term 7, last term 42, and 8 terms. Select its sum:

Solution

1. $S = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q232 MCQ

A sequence has first term 15, last term 3, and 10 terms. Select its sum:

Solution

1. $S = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions **Arithmetic Sequences**

Q233 MCQ

A sequence has first term 10, last term 70, and 13 terms. Select its sum:

Solution

1. $S = \frac{13}{2}(10 + 70) = 520.$

Correct option: B

Final answer

520

Q234 MCQ

A sequence has first term 2, last term 57, and 12 terms. Select its sum:

Solution

1. $S = \frac{12}{2}(2 + 57) = 354.$

Correct option: B

Final answer

354

Worked Solutions **Arithmetic Sequences**

Q235 MCQ

A sequence has first term 25, last term 100, and 16 terms. Select its sum:

Solution

1. $S = \frac{16}{2}(25 + 100) = 1000.$

Correct option: B

Final answer

1000

Q236 MCQ

A sequence has first term 101, last term 200, and 100 terms. Select its sum:

Solution

1. $S = \frac{100}{2}(101 + 200) = 15050.$

Correct option: B

Final answer

15050

Worked Solutions **Arithmetic Sequences**

Q237 MCQ

A sequence has first term 31, last term 91, and 16 terms. Select its sum:

Solution

1. $S = \frac{16}{2}(31 + 91) = 976.$

Correct option: B

Final answer

976

Q238 Written

Find the sum of the multiples of 3 among the two-digit natural numbers.

Solution

1. The sequence is $12, 15, 18, \ldots, 99$, so $a = 12, d = 3, l = 99$.
2. $99 = 12 + (n - 1)3$ gives $n = 30$.
3. $S = \frac{30}{2}(12 + 99) = 1665.$

Final answer

1665

Worked Solutions **Arithmetic Sequences**

Q239 Written

Find the sum of all two-digit natural numbers that are multiples of 5.

Solution

1. The sequence is $10, 15, \ldots, 95$; $n = (95 - 10)/5 + 1 = 18$.
2. $S = \frac{18}{2}(10 + 95) = 945$.

Final answer

945

Q240 Written

Find the sum of the multiples of 4 from 1 to 100.

Solution

1. The sequence is $4, 8, \ldots, 100$; $n = 25$.
2. $S = \frac{25}{2}(4 + 100) = 1300$.

Final answer

1300

Worked Solutions **Arithmetic Sequences**

Q241 Written

Find the sum of all multiples of 6 from 10 to 100.

Solution

1. The sequence is $12, 18, \dots, 96$; $n = 15$.
2. $S = \frac{15}{2}(12 + 96) = 810$.

Final answer

810

Q242 MCQ

Select the sum of the multiples of 3 from 3 to 99:

Solution

1. $n = (99 - 3)/3 + 1 = 33; S = \frac{33}{2}(3 + 99) = 1683$.

Correct option: B

Final answer

1683

Worked Solutions **Arithmetic Sequences**

Q243 MCQ

Select the sum of the multiples of 4 from 4 to 100:

Solution

1. $n = (100 - 4)/4 + 1 = 25; S = \frac{25}{2}(4 + 100) = 1300.$

Correct option: B

Final answer

1300

Q244 Written

Find the sum of the natural numbers from 1 to 100 that leave a remainder of 1 when divided by 4.

Solution

1. The sequence is $1, 5, \dots, 97$; $n = 25$.
2. $S = \frac{25}{2}(1 + 97) = 1225.$

Final answer

1225

Worked Solutions **Arithmetic Sequences**

Q245 **Written**

Find the sum of all two-digit natural numbers that leave a remainder of 3 when divided by 5.

Solution

1. The sequence is $13, 18, \ldots, 98$; $n = (98 - 13)/5 + 1 = 18$.
2. $S = \frac{18}{2}(13 + 98) = 999$.

Final answer

999

Q246 **Written**

Find the sum of all two-digit odd natural numbers.

Solution

1. The sequence is $11, 13, \ldots, 99$; $n = 45$.
2. $S = \frac{45}{2}(11 + 99) = 2475$.

Final answer

2475

Worked Solutions **Arithmetic Sequences**

Q247 Written

Find the sum of all natural numbers from 10 to 100 that are not divisible by 6.

Solution

1. Total 10 to 100 is 5005; multiples of 6 are 12 to 96 with sum 810.
2. Required sum=5005-810=4195.

Final answer

4195

Worked Solutions **Arithmetic Sequences**

Q248 Challenge

Find the sum of the numbers from 1 to 100 which are not divisible by 4.

Solution

1. Total $S_{100} = 5050$. Multiples of 4 have sum 1300.
2. Required sum=5050-1300=3750.

Final answer

3750

Worked Solutions **Arithmetic Sequences**

Q249 Written

In an arithmetic sequence with first term 4 and common difference 3, find how many terms from the first term have sum 50.

Solution

1. $50 = \frac{n}{2}\{2(4) + (n-1)3\} \Rightarrow 3n^2 + 5n - 100 = 0.$
2. The natural root is $n = 5$; $S_5 = 50$.

Final answer

5

Q250 Written

Find n when $a = 3, d = 2$ and $S_n = 63$.

Solution

1. $S_n = n(n+2) = 63 \Rightarrow n = 7.$

Final answer

7

Worked Solutions **Arithmetic Sequences**

Q251 MCQ

Select the correct option: Find n if $S_n = 48$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n+2) = 48 \Rightarrow n = 6.$

Correct option: B

Final answer

6

Q252 MCQ

Select the correct option: Find n if $S_n = 80$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n+2) = 80 \Rightarrow n = 8.$

Correct option: B

Final answer

8

Worked Solutions **Arithmetic Sequences**

Q253 MCQ

Select the correct option: Find n if $S_n = 120$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 120 \Rightarrow n = 10.$

Correct option: B

Final answer

10

Q254 MCQ

Select the correct option: Find n if $S_n = 168$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 168 \Rightarrow n = 12.$

Correct option: B

Final answer

12

Worked Solutions **Arithmetic Sequences**

Q255 MCQ

Select the correct option: Find n if $S_n = 255$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 255 \Rightarrow n = 15.$

Correct option: B

Final answer

15

Q256 MCQ

Select the correct option: Find n if $S_n = 360$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 360 \Rightarrow n = 18.$

Correct option: B

Final answer

18

Worked Solutions **Arithmetic Sequences**

Q257 MCQ

Select the correct option: Find n if $S_n = 440$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n+2) = 440 \Rightarrow n = 20.$

Correct option: B

Final answer

20

Q258 MCQ

Select the correct option: Find n if $S_n = 624$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n+2) = 624 \Rightarrow n = 24.$

Correct option: B

Final answer

24

Worked Solutions **Arithmetic Sequences**

Q259 MCQ

Select the correct option: Find n if $S_n = 840$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n+2) = 840 \Rightarrow n = 28.$

Correct option: B

Final answer

28

Q260 MCQ

Select the correct option: Find n if $S_n = 1088$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n+2) = 1088 \Rightarrow n = 32.$

Correct option: B

Final answer

32

Worked Solutions **Arithmetic Sequences**

Q261 MCQ

Select the correct option: Find n if $S_n = 1368$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n+2) = 1368 \Rightarrow n = 36.$

Correct option: B

Final answer

36

Q262 MCQ

Select the correct option: Find n if $S_n = 1680$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n+2) = 1680 \Rightarrow n = 40.$

Correct option: B

Final answer

40

Worked Solutions **Arithmetic Sequences**

Q263 Written

For first term 17 and common difference -2, find how many terms have sum -63.

Solution

1. $S_n = n(18 - n)$.
2. $n(18 - n) = -63 \Rightarrow n^2 - 18n - 63 = 0$; hence $n = 21$.

Final answer

21

Q264 Written

Find n when $a = 24, d = -4$ and $S_n = -28$.

Solution

1. $S_n = n(26 - 2n) = -28 \Rightarrow n^2 - 13n - 14 = 0$; $n = 14$.

Final answer

14

Worked Solutions **Arithmetic Sequences**

Q265 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 80$:

Solution

1. $S_n = \frac{n}{2}(17 + 3n) = 80$; positive root $n = 5$.

Correct option: C

Final answer

5

Q266 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 133$:

Solution

1. $S_n = \frac{n}{2}(17 + 3n) = 133$; positive root $n = 7$.

Correct option: C

Final answer

7

Worked Solutions **Arithmetic Sequences**

Q267 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 198$:

Solution

1. $S_n = \frac{n}{2}(17 + 3n) = 198$; positive root $n = 9$.

Correct option: C

Final answer

9

Q268 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 275$:

Solution

1. $S_n = \frac{n}{2}(17 + 3n) = 275$; positive root $n = 11$.

Correct option: C

Final answer

11

Worked Solutions **Arithmetic Sequences**

Q269 **Written**

Find the general term of $\{a_n\}$ with first term -60 and sum of the first 15 terms -60.

Solution

1. $-60 = \frac{15}{2}\{2(-60) + 14d\} \Rightarrow d = 8.$
2. $a_n = -60 + 8(n-1) = 8n - 68.$

Final answer

$a_n = 8n - 68$

Q270 **Written**

Find the general term with first term 8 and sum of the first 13 terms -130.

Solution

1. $-130 = \frac{13}{2}(16 + 12d) \Rightarrow d = -3.$
2. $a_n = 8 - 3(n-1) = 11 - 3n.$

Final answer

$a_n = 11 - 3n$

Q271 Written

Find the general term with first term 4 and $S_6 = 114$.

Solution

1. $\frac{6}{2}(8 + 5d) = 114 \Rightarrow d = 6$.
2. $a_n = 4 + 6(n - 1) = 6n - 2$.

Final answer

$a_n = 6n - 2$

Q272 Written

Find the general term when $a_3 = 10$ and $S_{10} = 175$.

Solution

1. $a + 2d = 10, 2a + 9d = 35 \Rightarrow d = 3,\ a = 4$.
2. $a_n = 4 + 3(n - 1) = 3n + 1$.

Final answer

$a_n = 3n + 1$

Worked Solutions **Arithmetic Sequences**

Q273 **Written**

For first term 65 and common difference -4, find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 69 - 4n$; $a_n > 0 \Rightarrow n < 17.25$.
2. The last positive term is $a_{17} = 1$; $S_{17} = \frac{17}{2}(65 + 1) = 561$.

Final answer

$n = 17,\ S = 561$

Q274 **Written**

For first term 70 and common difference -4, find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 74 - 4n$; $a_{18} = 2$ is the last positive term.
2. $S_{18} = \frac{18}{2}(70 + 2) = 648$.

Final answer

$n = 18,\ S = 648$

Worked Solutions **Arithmetic Sequences**

Q275 Written

For $a = 20, d = -6$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 26 - 6n$; the last positive term is $a_4 = 2$.
2. $S_4 = \frac{4}{2}(20 + 2) = 44$.

Final answer

$n = 4,\ S = 44$

Q276 Written

For $a = 126, d = -13$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 139 - 13n$; the last positive term is $a_{10} = 9$.
2. $S_{10} = \frac{10}{2}(126 + 9) = 675$.

Final answer

$n = 10,\ S = 675$

Worked Solutions **Arithmetic Sequences**

Q277 **Written**

Find the general term where $S_{10} = 10$ and $S_{20} = 40$.

Solution

1. $2a + 9d = 2, 2a + 19d = 4 \Rightarrow d = \frac{1}{5},\ a = \frac{1}{10}$.
2. $a_n = \frac{1}{10} + (n-1)\frac{1}{5} = \frac{2n-1}{10}$.

Final answer

$$a_n = \frac{2n-1}{10}$$

Q278 **Written**

Find the general term where $S_6 = 3$ and $S_{10} = 65$.

Solution

1. $2a + 5d = 1, 2a + 9d = 13 \Rightarrow d = 3,\ a = -7$.
2. $a_n = -7 + 3(n-1) = 3n - 10$.

Final answer

$$a_n = 3n - 10$$

Worked Solutions **Arithmetic Sequences**

Q279 **Written**

Find the general term where $S_{20} = 400$ and $S_{30} = 900$.

Solution

1. $2a + 19d = 40,\ 2a + 29d = 60 \Rightarrow d = 2,\ a = 1.$
2. $a_n = 1 + 2(n-1) = 2n - 1.$

Final answer

$a_n = 2n - 1$

Worked Solutions **Arithmetic Sequences**

Q280 Challenge

The 15th term is 33 and the 50th term is 103. Find the sum of the first 79 terms.

Solution

1. $35d = 70 \Rightarrow d = 2,\ a = 5.$
2. $S_{79} = \frac{79}{2}\{2(5) + 78(2)\} = 6557.$

Final answer

6557

Worked Solutions **Arithmetic Sequences**

Quiz MCQ 1 Concept Quiz · MCQ

For $a_n = \frac{2n+1}{n+2}$, what is a_5?:

Solution

1. Substitute $n = 5$: a_5=(2(5)+1)/(5+2)=11/7.

Correct option: A

Final answer

$\frac{11}{7}$

Quiz MCQ 2 Concept Quiz · MCQ

An arithmetic sequence has $a_3 = 10$ and $a_8 = 30$. Select its general term:

Solution

1. $10 = a + 2d$ and $30 = a + 7d$, so $d = 4,\ a = -2$.
2. $a_n = 4n - 2$.

Correct option: B

Final answer

$a_n = 4n - 2$

Worked Solutions **Arithmetic Sequences**

Quiz MCQ 3 Concept Quiz · MCQ

For $a = 12, d = -2$, select the sum of odd-numbered terms through a_9:

Solution

1. Odd terms are $12, 8, 4, 0, -4$; their sum is 20.

Correct option: C

Final answer

20

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: If $S_{10} = 10$ and $S_{20} = 40$, find d:

Solution

1. $d = \frac{1}{5}$.

Correct option: B

Final answer

$\frac{1}{5}$

Worked Solutions **Arithmetic Sequences**

Quiz WRITTEN 5 Concept Quiz · Written

If $a_n = 3n^2 - 2n + 1$, find a_1, a_2, a_3.

Solution

1. Substitution gives a_1=2, a_2=9, and a_3=22.

Final answer

$a_1 = 2,\ a_2 = 9,\ a_3 = 22$

Quiz WRITTEN 6 Concept Quiz · Written

Find a_n for an arithmetic sequence with $a_4 = 14$ and $a_{10} = 62$.

Solution

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8,\ a = -10$.
2. $a_n = 8n - 18$.

Final answer

$a_n = 8n - 18$

Worked Solutions **Arithmetic Sequences**

Quiz WRITTEN 7 Concept Quiz · Written

For $a = 9, d = 4$, find the sum of odd-numbered terms from a_1 to a_{15}.

Solution

1. Selected terms have 8 terms, first 9, last 65.
2. Sum $= 8(9 + 65)/2 = 296$.

Final answer

296

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term when $S_{20} = 400$ and $S_{30} = 900$.

Solution

1. $d = 2, a = 1$; therefore $a_n = 2n - 1$.

Final answer

$a_n = 2n - 1$

Answer key

Use the question number shown on each page.

Q001 $a_1 = -1,\ a_2 = 1,\ a_3 = 3$

Q002 $a_1 = -6,\ a_2 = -10,\ a_3 = -12$

Q003 $a_1 = -3,\ a_2 = 2,\ a_3 = 7$

Q004 $a_1 = -2,\ a_2 = -5,\ a_3 = -8$

Q005 $a_1 = -6,\ a_2 = -8,\ a_3 = -10$

Q006 $a_1 = 0,\ a_2 = 3,\ a_3 = 8$

Q007 $a_1 = 3,\ a_2 = 7,\ a_3 = 13$

Q008 $a_1 = 7,\ a_2 = 12,\ a_3 = 19$

Q009 B

Q010 C

Q011 D

Q012 A

Q013 B

Q014 C

Q015 D

Q016 A

Q017 B

Q018 C

Q019 D

Q020 A

Q021 $a_1 = 6,\ a_2 = 18,\ a_3 = 54$

Q022 $a_1 = 2,\ a_2 = 8,\ a_3 = 26$

Q023 $a_1 = -3,\ a_2 = 9,\ a_3 = -27$

Q024 $a_1 = -6,\ a_2 = 12,\ a_3 = -24$

Q025 A

Q026 B

Q027 C

Q028 D

Q029 A

Q030 B

Q031 C

Q032 D

Q033 $a_1 = \frac{1}{2},\ a_2 = \frac{2}{3},\ a_3 = \frac{3}{4}$

Q034 $a_1 = \frac{1}{3},\ a_2 = \frac{2}{5},\ a_3 = \frac{3}{7}$

Q035 $a_1 = 1,\ a_2 = \frac{1}{4},\ a_3 = \frac{1}{7}$

Q036 $a_1 = \frac{1}{2},\ a_2 = \frac{4}{5},\ a_3 = 1$

Q037 $a_1 = 2,\ a_2 = 1,\ a_3 = \frac{4}{5}$

Q038 C

Q039 D

Q040 A

Q041 B

Q042 C

Q043 D

Q044 $a_n = (n-1)^2$

Q045 $a_n = (n-1)^3$

Q046 $a_n = n^2$

Q047 $a_n = (n+6)^2$

Q048 B

Q049 C

Q050 D

Q051 A

Q052 B

Q053 C

Q054 $a_n = 3n - 1$

Q055 $a_n = 2n - 2$

Q056 $a_n = 3n - 3$

Q057 $a_n = 5n - 1$

Q058 $a_n = 7n - 1$

Q059 A

Q060 B

Q061 C

Q062 D

Q063 $a_n = (-1)^{n-1}\, 2n$

Q064 $a_n = (-1)^n\, 3n$

Q065 $a_n = (-1)^n\, 5n$

Q066 $a_n = (-1)^n (3n-2)$

Q067 $a_n = \dfrac{n}{2n-1}$

Q068 $a_n = \dfrac{2n-1}{n^2}$

Q069 $a_n = \dfrac{(-1)^{n-1}}{2n-1}$

Q070 $a_n = \dfrac{n}{n+1}$

Q071 $a_n = n\, 3^n$

Q072 $a_n = \dfrac{n}{n+1}; \quad a_{20} = \dfrac{20}{21}$

Q073 $(a) \quad a = 2,\ d = 3 (b) \quad a = 3,\ d = -5$

Q074 $(a) \quad a = 1,\ d = 4 (b) \quad a = 23,\ d = -4$

Q075 B

Q076 A

Q077 C

Q078 D

Q079 $(a)(3,2); (b)(-13,4); (c)(-2,3); (d)(10,-3); (e)(6,-\frac{2}{3}); (f)(-4,-\frac{1}{2})$.

Q080 B

Q081 A

Q082 C

Q083 B

Q084 $(a) \quad d = -3,\ A = 5,\ B = -4 (b) \quad d = -5,\ A = 8,\ B = 3$

Q085 $(a) \quad d = 7,\ A = 16,\ B = 23 (b) \quad d = -6,\ A = 23,\ B = 11$

Q086 A

Q087 C

Q088 B

Q089 $(a) d = 7, A = 22, B = 29; (b) d = 6, A = 19, B = 25; (c) d = -5, A = 14, B = -1; (d) d = -2, A = 17, B = 13; (e) d = 3, A = 14, B = 17; (f) d = -4, A = 13, B = 9, C = 1$.

Q090 $(a) \quad a_n = 3n + 1,\ a_7 = 22 (b) \quad a_n = 5 - 3n,\ a_7 = -16$

Q091 $(a) a_n = 4n + 1, a_7 = 29; (b) a_n = 3n - 5, a_7 = 16; (c) a_n = 13 - 3n, a_7 = -8; (d) a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}$.

Q092 $a_n = 8 + (n-1)(-6) = 14 - 6n; \quad a_{20} = -106$

Q093 $(a) a_n = 3n - 10, a_{10} = 20; (b) a_n = 1 - 3n, a_{10} = -29; (c) a_n = \frac{n+13}{2}, a_{10} = \frac{23}{2}; (d) a_n = \frac{3-n}{3}, a_{10} = -\frac{7}{3}$.

Q094 $a = 7, d = 3.$

Q095 $a_n = 5n - 10$

Q096 $a_n = 4n + 2$

Q097 $a_n = 23 - 3n$

Q098 $a_n = 9n - 17$

Q099 $a_n = -5n - 13$

Q100 D

Q101 B

Q102 A

Q103 B

Q104 $a_n = 3n + 33$

Q105 $a_n = 2n - 12$

Q106 $a_n = 13 - 4n$

Q107 $a_n = 18 - 7n$

Q108 A

Q109 D

Q110 C

Q111 D

Q112 $a_n = 8n - 14$

Q113 $a_n = 20 - 3n$

Q114 $a_n = 8n - 18$

Q115 $a_n = 42 - 4n$

Q116 $a_n = 79 - 3n$

Q117 $a_n = 8n - 18$

Q118 C

Q119 C

Q120 B

Q121 A

Q122 A

Q123 B

Q124 $a_n = \dfrac{2n - 37}{5}$

Q125 A

Q126 D

Q127 $n = 28$, the 28th term

Q128 $n = 14$, the 14th term

Q129 $n = 22$, the 22nd term

Q130 $n = 40$, the 40th term

Q131 C

Q132 D

Q133 A

Q134 B

Q135 $n = 125$, the 125th term

Q136 $n = 35$, the 35th term

Q137 A

Q138 D

Q139 D

Q140 B

Q141 $n = 72$, the 72nd term

Q142 $n = 171$, the 171st term

Q143 B

Q144 B

Q145 C

Q146 A

Q147 B

Q148 $a = 4$

Q149 $x = -4$ or $x = 3$

Q150 $x = -2$ or $x = 4$

Q151 $x = -1$ or $x = 3$

Q152 $x = 0, \quad y = -2, \quad z = -4$

Q153 105

Q154 100

Q155 120

Q156 152

Q157 155

Q158 B

Q159 B

Q160 B

Q161 B

Q162 B

Q163 C

Q164 -130

Q165 144

Q166 -165

Q167 176

Q168 360

Q169 C

Q170 B

Q171 C

Q172 B

Q173 C

Q174 B

Q175 392

Q176 343

Q177 741

Q178 354

Q179 1716

Q180 C

Q181 C

Q182 C

Q183 C

Q184 C

Q185 B

Q186 0

Q187 208

Q188 B

Q189 C

Q190 C

Q191 C

Q192 B

Q193 C

Q194 66

Q195 750

Q196 154

Q197 C

Q198 C

Q199 $2n^2 - n$

Q200 $\dfrac{n(7 - 3n)}{2}$

Q201 $\dfrac{5n(n - 1)}{2}$

Q202 $5n - 2n^2$

Q203 $-n^2 - 3n$

Q204 414

Q205 5050

Q206 20100

Q207 11325

Q208 2485

Q209 5050

Q210 80200

Q211 31375

Q212 8085

Q213 24535

Q214 4950

Q215 B

Q216 B

Q217 B

Q218 B

Q219 B

Q220 B

Q221 B

Q222 B

Q223 B

Q224 B

Q225 15050

Q226 25050

Q227 4905

Q228 B

Q229 B

Q230 B

Q231 B

Q232 B

Q233 B

Q234 B

Q235 B

Q236 B

Q237 B

Q238 1665

Q239 945

Q240 1300

Q241 810

Q242 B

Q243 B

Q244 1225

Q245 999

Q246 2475

Q247 4195

Q248 3750

Q249 5

Q250 7

Q251 B

Q252 B

Q253 B

Q254 B

Q255 B

Q256 B

Q257 B

Q258 B

Q259 B

Q260 B

Q261 B

Q262 B

Q263 21

Q264 14

Q265 C

Q266 C

Q267 C

Q268 C

Q269 $a_n = 8n - 68$

Q270 $a_n = 11 - 3n$

Q271 $a_n = 6n - 2$

Q272 $a_n = 3n + 1$

Q273 $n = 17,\ S = 561$

Q274 $n = 18,\ S = 648$

Q275 $n = 4,\ S = 44$

Q276 $n = 10,\ S = 675$

Q277 $a_n = \dfrac{2n-1}{10}$

Q278 $a_n = 3n - 10$

Q279 $a_n = 2n - 1$

Q280 6557

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 $a_1 = 2,\ a_2 = 9,\ a_3 = 22$

Quiz WRITTEN 6 $a_n = 8n - 18$

Quiz WRITTEN 7 296

Quiz WRITTEN 8 $a_n = 2n - 1$

Concept 2 | Geometric Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a\frac{r^n-1}{r-1}$$

Infinite sum

$$S_\infty = \frac{a}{1-r}\ (|r| < 1)$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Geometric Sequences**

Q001 Written

Find the general term of a geometric sequence whose first term is 2 and common ratio is 3.

Solution

1. Here $a = 2$ and $r = 3$.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$2 \times 3^{n-1}$

Q002 Written

Find the general term when the first term is 3 and the common ratio is 4.

Solution

1. $a_n = 3 \times 4^{n-1}$

Final answer

$3 \times 4^{n-1}$

Worked Solutions **Geometric Sequences**

Q003 Written

Find the general term of the geometric sequence in part a. The first term is 5, common ratio 2.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5 \times 2^{n-1}$.

Final answer

$5 \times 2^{n-1}$

Q004 Written

Find the general term of the geometric sequence in part b. The first term is 2, common ratio 5.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2 \times 5^{n-1}$.

Final answer

$2 \times 5^{n-1}$

Worked Solutions **Geometric Sequences**

Q005 Written

Find the general term of the geometric sequence in part c. The first term is 7, common ratio 4.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $7 \times 4^{n-1}$.

Final answer

$7 \times 4^{n-1}$

Q006 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = 3$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = 3$.

Correct option: A

Final answer

$a_n = 2\left(3\right)^4$

Worked Solutions **Geometric Sequences**

Q007 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = 2$, and $n = 6$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = 2$.

Correct option: A

Final answer

$a_n = 3\,(2)^5$

Q008 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 4$, and $n = 4$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 4$.

Correct option: A

Final answer

$a_n = 5\,(4)^3$

Worked Solutions **Geometric Sequences**

Q009 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = 5$, and $n = 3$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = 5$.

Correct option: A

Final answer

$a_n = 4\,(5)^2$

Q010 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 2$, and $n = 7$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 2$.

Correct option: A

Final answer

$a_n = 5\,(2)^6$

Worked Solutions **Geometric Sequences**

Q011 MCQ

Select the correct option: For first term $a = 7$, common ratio $r = 4$, and $n = 3$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 7$ and $r = 4$.

Correct option: A

Final answer

$a_n = 7\,(4)^2$

Q012 Written

Find the general term when the first term is 5 and the common ratio is -2.

Solution

1. $a_n = 5(-2)^{n-1}$

Final answer

$5(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q013 **Written**

Find the general term of the geometric sequence in part d. The first term is 2, common ratio -3.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2(-3)^{n-1}$.

Final answer

$2(-3)^{n-1}$

Q014 **Written**

Find the general term of the geometric sequence in part e. The first term is 3, common ratio -5.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3(-5)^{n-1}$.

Final answer

$3(-5)^{n-1}$

Worked Solutions **Geometric Sequences**

Q015 Written

Find the general term of the geometric sequence in part f. The first term is 4, common ratio −2.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4(-2)^{n-1}$.

Final answer

$4(-2)^{n-1}$

Q016 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = -3$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = -3$.

Correct option: A

Final answer

$a_n = 2\,(-3)^4$

Worked Solutions **Geometric Sequences**

Q017 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -2$, and $n = 6$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -2$.

Correct option: A

Final answer

$a_n = 3\left(-2\right)^5$

Q018 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = -1$, and $n = 4$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = -1$.

Correct option: A

Final answer

$a_n = 6\left(-1\right)^3$

Worked Solutions **Geometric Sequences**

Q019 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -2$, and $n = 3$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -2$.

Correct option: A

Final answer

$a_n = 8\,(-2)^2$

Q020 MCQ

Select the correct option: For first term $a = 9$, common ratio $r = -2$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 9$ and $r = -2$.

Correct option: A

Final answer

$a_n = 9\,(-2)^4$

Worked Solutions **Geometric Sequences**

Q021 **Written**

Find the general term when the first term is 8 and the common ratio is $-\frac{1}{3}$.

Solution

1. $a_n = 8\left(-\frac{1}{3}\right)^{n-1}$

Final answer

$8\left(-\frac{1}{3}\right)^{n-1}$

Q022 **Written**

Find the general term of the geometric sequence in part g. The first term is 4, common ratio $\frac{1}{3}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4\left(\frac{1}{3}\right)^{n-1}$.

Final answer

$4\left(\frac{1}{3}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q023 Written

Find the general term of the geometric sequence in part h. The first term is 3**, common ratio** $-\frac{1}{2}$**.**

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{1}{2}\right)^{n-1}$.

Final answer

$3\left(-\frac{1}{2}\right)^{n-1}$

Q024 Written

Find the general term of the geometric sequence in part i. The first term is 6**, common ratio** $\frac{1}{2}$**.**

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $6\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$6\left(\frac{1}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q025 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = \frac{1}{3}$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = \frac{1}{3}$.

Correct option: A

Final answer

$a_n = 4\left(\frac{1}{3}\right)^4$

Q026 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -\frac{1}{2}$, and $n = 6$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -\frac{1}{2}$.

Correct option: A

Final answer

$a_n = 3\left(-\frac{1}{2}\right)^5$

Worked Solutions **Geometric Sequences**

Q027 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = \frac{1}{2}$, and $n = 4$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = \frac{1}{2}$.

Correct option: A

Final answer

$a_n = 6\left(\frac{1}{2}\right)^3$

Q028 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = \frac{3}{2}$, and $n = 5$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = \frac{3}{2}$.

Correct option: A

Final answer

$a_n = 2\left(\frac{3}{2}\right)^4$

Worked Solutions **Geometric Sequences**

Q029 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -\frac{3}{2}$, and $n = 4$, which expression gives a_n?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -\frac{3}{2}$.

Correct option: A

Final answer

$a_n = 8\left(-\frac{3}{2}\right)^3$

Q030 Written

Find the general term of $5, 10, 20, 40, \ldots$.

Solution

1. The common ratio is $r = 10/5 = 2$.

2. $a_n = 5 \times 2^{n-1}$

Final answer

$5 \times 2^{n-1}$

Worked Solutions **Geometric Sequences**

Q031 Written

Find the general term of $2, 6, 18, 54, \ldots$**.**

Solution

1. The common ratio is 3.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$2 \times 3^{n-1}$

Q032 Written

Find the general term of the geometric sequence in part j. The sequence is $3, 6, 12, 24, \ldots$**.**

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3 \times 2^{n-1}$.

Final answer

$3 \times 2^{n-1}$

Worked Solutions **Geometric Sequences**

Q033 *MCQ*

Select the correct option: Find the general term of the geometric sequence $3, 6, 12, 24, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$3 \times 2^{n-1}$

Q034 *MCQ*

Select the correct option: Find the general term of the geometric sequence $2, 10, 50, 250, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$2 \times 5^{n-1}$

Worked Solutions **Geometric Sequences**

Q035 MCQ

Select the correct option: Find the general term of the geometric sequence $4, 8, 16, 32, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$4 \times 2^{n-1}$

Q036 MCQ

Select the correct option: Find the general term of the geometric sequence $7, 21, 63, 189, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$7 \times 3^{n-1}$

Worked Solutions **Geometric Sequences**

Q037 MCQ

Select the correct option: Find the general term of the geometric sequence $1, 4, 16, 64, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

4^{n-1}

Q038 Written

Find the general term of the geometric sequence in part k. The sequence is $2, -6, 18, -54, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2(-3)^{n-1}$.

Final answer

$2(-3)^{n-1}$

Worked Solutions **Geometric Sequences**

Q039 Written

Find the general term of the geometric sequence in part I. The sequence is $5, -10, 20, -40, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5(-2)^{n-1}$.

Final answer

$5(-2)^{n-1}$

Q040 MCQ

Select the correct option: Find the general term of the geometric sequence $5, -10, 20, -40, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$5(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q041 *MCQ*

Select the correct option: Find the general term of the geometric sequence $2, -6, 18, -54, \ldots$**:**

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$2(-3)^{n-1}$

Q042 *MCQ*

Select the correct option: Find the general term of the geometric sequence $-4, 8, -16, 32, \ldots$**:**

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$-4(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q043 MCQ

Select the correct option: Find the general term of the geometric sequence $6, -18, 54, -162, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$6(-3)^{n-1}$

Q044 Written

Find the general term of $-4, 6, -9, \frac{27}{2}, \ldots$.

Solution

1. The common ratio is $r = 6/(-4) = -3/2$.
2. $a_n = -4\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$-4\left(-\frac{3}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q045 Written

Find the general term of $-8, 12, -18, 27, \ldots$.

Solution

1. The common ratio is $-3/2$.
2. $a_n = -8\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 Written

Find the general term of $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \ldots$.

Solution

1. The common ratio is $1/3$.
2. $a_n = 2\left(\frac{1}{3}\right)^{n-1}$

Final answer

$2\left(\frac{1}{3}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q047 **Written**

Find the general term of the geometric sequence in part m. The sequence is $48, 24, 12, 6, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $48\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$$48\left(\frac{1}{2}\right)^{n-1}$$

Q048 **Written**

Find the general term of the geometric sequence in part n. The sequence is $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$.

Final answer

$$\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$$

Worked Solutions **Geometric Sequences**

Q049 Written

Find the general term of the geometric sequence in part o. The sequence is $8, 16\sqrt{2}, 64, 128\sqrt{2}, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $8\left(2\sqrt{2}\right)^{n-1}$.

Final answer

$8\left(2\sqrt{2}\right)^{n-1}$

Q050 Written

Find the general term of the geometric sequence in part p. The sequence is $3, -\frac{15}{2}, \frac{75}{4}, -\frac{375}{8}, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{5}{2}\right)^{n-1}$.

Final answer

$3\left(-\frac{5}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q051 Written

Find the general term of the geometric sequence in part q. The sequence is $9, -3\sqrt{3}, 3, -\sqrt{3}, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$.

Final answer

$$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$$

Q052 Written

Find the general term of the geometric sequence in part r. The sequence is $5, -\frac{10}{3}, \frac{20}{9}, -\frac{40}{27}, \ldots$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5\left(-\frac{2}{3}\right)^{n-1}$.

Final answer

$$5\left(-\frac{2}{3}\right)^{n-1}$$

Worked Solutions **Geometric Sequences**

Q053 *MCQ*

Select the correct option: Find the general term of the geometric sequence $-8, 12, -18, 27, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$-8\left(-\frac{3}{2}\right)^{n-1}$

Q054 *MCQ*

Select the correct option: Find the general term of the geometric sequence $48, 24, 12, 6, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$48\left(\frac{1}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q055 MCQ

Select the correct option: Find the general term of the geometric sequence $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q056 MCQ

Select the correct option: Find the general term of the geometric sequence $8, 16\sqrt{2}, 64, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$8\left(2\sqrt{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q057 MCQ

Select the correct option: Find the general term of the geometric sequence $9, -3\sqrt{3}, 3, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q058 MCQ

Select the correct option: Find the general term of the geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$3\left(-\frac{1}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q059 MCQ

Select the correct option: Find the general term of the geometric sequence $10, 5, \frac{5}{2}, \ldots$:

Solution

1. Divide consecutive terms to obtain r, then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$10\left(\frac{1}{2}\right)^{n-1}$

Q060 Written

Find the common ratio and missing values in $A, -18, 6, B, \ldots$.

Solution

1. $r = 6/(-18) = -\frac{1}{3}$

2. $A = 54, \quad B = 6\left(-\frac{1}{3}\right) = -2$

Final answer

$r = -\frac{1}{3},\ A = 54,\ B = -2$

Worked Solutions **Geometric Sequences**

Q061 MCQ

Select the correct option: The sequence $A, 12, 6, B, \ldots$ is geometric. Find r, A, B:

Solution

1. Divide adjacent known terms to find r, then extend one step in each direction.

Correct option: A

Final answer

$r = \frac{1}{2},\ A = 24,\ B = 3$

Q062 Written

Find the common ratio and the missing values in $A, 24, 12, B, \ldots$.

Solution

1. $24r = 12 \Rightarrow r = \frac{1}{2}$
2. $A\left(\frac{1}{2}\right) = 24 \Rightarrow A = 48, \quad B = 12\left(\frac{1}{2}\right) = 6$

Final answer

$r = \frac{1}{2},\ A = 48,\ B = 6$

Worked Solutions **Geometric Sequences**

Q063 **Written**

Find the common ratio and missing values in $A, -3, 9, B$.

Solution

1. $r = -3, \quad A = 1, \quad B = -27$

Final answer

$r = -3,\ A = 1,\ B = -27$

Q064 **Written**

Find the common ratio and missing values in $A, \frac{1}{3}, \frac{1}{9}, B$.

Solution

1. $r = \frac{1}{3}, \quad A = 1, \quad B = \frac{1}{27}$

Final answer

$r = \dfrac{1}{3},\ A = 1,\ B = \dfrac{1}{27}$

Worked Solutions **Geometric Sequences**

Q065 Written

Find the common ratio and missing values in $A, -\frac{1}{4}, \frac{1}{8}, B$.

Solution

1. $r = -\frac{1}{2}, \quad A = \frac{1}{2}, \quad B = -\frac{1}{16}$

Final answer

$r = -\dfrac{1}{2}$, $A = \dfrac{1}{2}$, $B = -\dfrac{1}{16}$

Worked Solutions **Geometric Sequences**

Q066 **Challenge**

In an increasing geometric sequence, if $a_3 + a_5 = 80$ and $a_4 = 24$, find a_7.

Solution

1. $a_3 = \frac{24}{r}, \quad a_5 = 24r, \quad 24\left(r + \frac{1}{r}\right) = 80$
2. $3r^2 - 10r + 3 = 0 \Rightarrow r = 3$ or $\frac{1}{3}$
3. The sequence is increasing, so $r = 3$; hence $a_7 = 24(3)^3 = 648$.

Final answer

$a_7 = 648$

Worked Solutions **Geometric Sequences**

Q067 **Written**

Find the general term of a geometric sequence whose 4th term is 24 and 6th term is 96.

Solution

1. $r^2 = 96/24 = 4 \Rightarrow r = \pm 2$
2. $(a, r) = (3, 2)$ or $(-3, -2)$
3. Therefore both real branches are retained.

Final answer

$a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 **Written**

Find the general term when the 3rd term is 18 and the 5th term is 162.

Solution

1. $r^2 = 162/18 = 9 \Rightarrow r = \pm 3$
2. $a = 18/r^2 = 2$

Final answer

$a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Worked Solutions **Geometric Sequences**

Q069 Written

Find the requested values. The 2nd term is 6 and the 4th term is 54.

Solution

1. $r^2 = 54/6 = 9 \Rightarrow r = \pm 3$
2. $a = 6/r$

Final answer

$a_n = 2(3)^{n-1}$ or $-2(-3)^{n-1}$

Q070 Written

Find the requested values. The 3rd term is 27 and the 5th term is 243.

Solution

1. $r^2 = 243/27 = 9 \Rightarrow r = \pm 3$
2. $a = 3$

Final answer

$a_n = 3(3)^{n-1}$ or $3(-3)^{n-1}$

Worked Solutions **Geometric Sequences**

Q071 **Written**

Find the requested values. The 2nd term is 14 and the 5th term is 112.

Solution

1. $r^3 = 112/14 = 8 \Rightarrow r = 2$
2. $a = 7$

Final answer

$a_n = 7(2)^{n-1}$

Q072 **Written**

Find the general term when the 3rd term is 12 and the 6th term is −96.

Solution

1. $r^3 = -96/12 = -8 \Rightarrow r = -2$
2. $a = 12/r^2 = 3$
3. $a_n = 3(-2)^{n-1}$

Final answer

$a_n = 3(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q073 Written

Find the general term when the 2nd term is -6 and the 5th term is 48.

Solution

1. $r^3 = 48/(-6) = -8 \Rightarrow r = -2$
2. $a = -6/r = 3$

Final answer

$a_n = 3(-2)^{n-1}$

Q074 Written

Find the general term when the 3rd term is 12 and the 7th term is 192.

Solution

1. $r^4 = 192/12 = 16 \Rightarrow r = \pm 2$
2. $a = 12/r^2 = 3$

Final answer

$a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q075 Written

Find the requested values. The 3rd term is -28 and the 5th term is -112.

Solution

1. $r^2 = 4 \Rightarrow r = \pm 2$
2. $a = -28/4 = -7$

Final answer

$a_n = -7(2)^{n-1} \text{ or } -7(-2)^{n-1}$

Q076 Written

Find the requested values. The 4th term is 16 and the 7th term is -128.

Solution

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -2$

Final answer

$a_n = -2(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q077 Written

Find the requested values. The 3rd term is -20 and the 6th term is 160.

Solution

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -5$

Final answer

$a_n = -5(-2)^{n-1}$

Q078 Written

Find the requested values. The 2nd term is 3 and the 6th term is 768.

Solution

1. $r^4 = 768/3 = 256 \Rightarrow r = \pm 4$
2. $a = 3/r$

Final answer

$a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Worked Solutions **Geometric Sequences**

Q079 **Written**

Find the requested values. The 3rd term is 9 and the 7th term is 729/16.

Solution

1. $r^4 = 81/16 \Rightarrow r = \pm\frac{3}{2}$
2. $a = 4$

Final answer

$a_n = 4(\frac{3}{2})^{n-1}$ or $4(-\frac{3}{2})^{n-1}$

Q080 **Written**

Find the requested values. The 4th term is 2/9 and the 8th term is 18.

Solution

1. $r^4 = 81 \Rightarrow r = \pm 3$
2. $a = \frac{2/9}{r^3} = \pm\frac{2}{243}$

Final answer

$a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Worked Solutions **Geometric Sequences**

Q081 MCQ

Select the correct option: For $a = 2, r = 3$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 2\,(3)^{n-1}$

Q082 MCQ

Select the correct option: For $a = 3, r = 2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 3\,(2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q083 MCQ

Select the correct option: For $a = 5$, $r = 4$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 5\,(4)^{n-1}$

Q084 MCQ

Select the correct option: For $a = 4$, $r = 5$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 4\,(5)^{n-1}$

Worked Solutions **Geometric Sequences**

Q085 MCQ

Select the correct option: For $a = 2$, $r = -3$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 2\left(-3\right)^{n-1}$

Q086 MCQ

Select the correct option: For $a = 3$, $r = -2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 3\left(-2\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q087 MCQ

Select the correct option: For $a = 6$, $r = -1$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 6\,(-1)^{n-1}$

Q088 MCQ

Select the correct option: For $a = 8$, $r = -2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 8\,(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q089 MCQ

Select the correct option: For $a = 4$, $r = \frac{1}{3}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 4\left(\frac{1}{3}\right)^{n-1}$

Q090 MCQ

Select the correct option: For $a = 3$, $r = -\frac{1}{2}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 3\left(-\frac{1}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q091 MCQ

Select the correct option: For $a = 6$, $r = \frac{1}{2}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 6\left(\frac{1}{2}\right)^{n-1}$

Q092 MCQ

Select the correct option: For $a = 2$, $r = \frac{3}{2}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 2\left(\frac{3}{2}\right)^{n-1}$

Worked Solutions **Geometric Sequences**

Q093 MCQ

Select the correct option: For $a = 8, r = -\frac{3}{2}$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 8\left(-\frac{3}{2}\right)^{n-1}$

Q094 MCQ

Select the correct option: For $a = 5, r = 2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 5\,(2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q095 MCQ

Select the correct option: For $a = 7$, $r = 4$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 7\,(4)^{n-1}$

Q096 MCQ

Select the correct option: For $a = 9$, $r = -2$, which expression gives the general term a_n:

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$a_n = 9\,(-2)^{n-1}$

Worked Solutions **Geometric Sequences**

Q097 Written

When $2, x, 8$ form a geometric sequence in this order, find x.

Solution

1. $x^2 = 2 \times 8 = 16$
2. $x = \pm 4$

Final answer

$x = \pm 4$

Q098 Written

Find the requested values. When $1, x, 25$ form a geometric sequence, find x.

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$x = \pm 5$

Worked Solutions **Geometric Sequences**

Q099 Written

Find the requested values. When $1/2, x, 8$ form a geometric sequence, find x.

Solution

1. $x^2 = \frac{1}{2} \times 8 = 4 \Rightarrow x = \pm 2$

Final answer

$x = \pm 2$

Q100 Written

When $1, x, 12$ form a geometric sequence in this order, find x.

Solution

1. $x^2 = 1 \times 12 = 12$
2. $x = \pm 2\sqrt{3}$

Final answer

$x = \pm 2\sqrt{3}$

Worked Solutions **Geometric Sequences**

Q101 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 8$, find x:

Solution

1. Use $x^2 = (2)(8)$.

Correct option: A

Final answer

± 4

Q102 MCQ

Select the correct option: If x is the middle term of the geometric triple $1, x, 25$, find x:

Solution

1. Use $x^2 = (1)(25)$.

Correct option: A

Final answer

± 5

Worked Solutions **Geometric Sequences**

Q103 MCQ

Select the correct option: If x is the middle term of the geometric triple $\frac{1}{2}, x, 8$, find x:

Solution

1. Use $x^2 = (\frac{1}{2})(8)$.

Correct option: A

Final answer

± 2

Q104 MCQ

Select the correct option: If x is the middle term of the geometric triple $3, x, 12$, find x:

Solution

1. Use $x^2 = (3)(12)$.

Correct option: A

Final answer

± 6

Worked Solutions **Geometric Sequences**

Q105 MCQ

Select the correct option: If x is the middle term of the geometric triple $4, x, 9$, find x:

Solution

1. Use $x^2 = (4)(9)$.

Correct option: A

Final answer

± 6

Q106 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 18$, find x:

Solution

1. Use $x^2 = (2)(18)$.

Correct option: A

Final answer

± 6

Worked Solutions **Geometric Sequences**

Q107 Written

The numbers $6, a, b$ form an arithmetic sequence and $a, b, 16$ form a geometric sequence. Find a, b.

Solution

1. $2a = 6 + b \Rightarrow b = 2a - 6$
2. $b^2 = 16a \Rightarrow 4a^2 - 40a + 36 = 0$
3. $(a, b) = (9, 12)$ or $(1, -4)$

Final answer

$(a, b) = (9, 12)$ or $(1, -4)$

Q108 Written

Find the requested values. The numbers $a, b, 2$ form an arithmetic sequence and $a, 2, b$ form a geometric sequence. Find a, b.

Solution

1. $2b = a + 2$
2. $b^2 = 2a$
3. $(a, b) = (2, 2)$ or $(-4, -1)$

Final answer

$(a, b) = (2, 2)$ or $(-4, -1)$

Worked Solutions **Geometric Sequences**

Q109 Written

The numbers $-2, a, b$ form an arithmetic sequence and $a, b, 18$ form a geometric sequence. Find a, b.

Solution

1. $b = 2a + 2$
2. $b^2 = 18a \Rightarrow (a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$

Final answer

$(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$

Worked Solutions **Geometric Sequences**

Q110 Challenge

A planning model gives $-5, a, b$ as an arithmetic sequence, while $a, b, 45$ form a geometric sequence. Find a, b.

Solution

1. $b = 2a + 5$
2. $b^2 = 45a \Rightarrow 4a^2 - 25a + 25 = 0$
3. $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$

Final answer

$(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$

Worked Solutions **Geometric Sequences**

Q111 Written

Find the sum S of a geometric sequence with first term 4, common ratio 3, and 5 terms.

Solution

1. Use $S_5 = \dfrac{4(3^5 - 1)}{3 - 1} = 2(3^5 - 1)$.
2. Therefore $S_5 = 484$.

Final answer

484

Q112 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 5 terms.

Solution

1. $S_5 = \dfrac{3(2^5 - 1)}{2 - 1} = 3(31)$.
2. Therefore $S_5 = 93$.

Final answer

93

Worked Solutions **Geometric Sequences**

Q113 Written

Find the sum S of a geometric sequence with first term 5, common ratio 3, and 4 terms.

Solution

1. $S_4 = \dfrac{5(3^4 - 1)}{3 - 1} = 5(40)$.
2. Therefore $S_4 = 200$.

Final answer

200

Q114 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 10 terms.

Solution

1. $S_{10} = \dfrac{3(2^{10} - 1)}{2 - 1} = 3(1023)$.
2. Therefore $S_{10} = 3069$.

Final answer

3069

Worked Solutions **Geometric Sequences**

Q115 MCQ

For a geometric sequence with $a = 4$, $r = 3$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{4(1-(3)^4)}{1-(3)}$.
2. Therefore $S_4 = 160$.

Correct option: D

Final answer

160

Q116 MCQ

For a geometric sequence with $a = 5$, $r = 2$, and $n = 6$, calculate S_n:

Solution

1. $S_n = \dfrac{5(1-(2)^6)}{1-(2)}$.
2. Therefore $S_6 = 315$.

Correct option: C

Final answer

315

Worked Solutions **Geometric Sequences**

Q117 *MCQ*

For a geometric sequence with $a = 2$, $r = 4$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \dfrac{2(1-(4)^5)}{1-(4)}$.
2. Therefore $S_5 = 682$.

Correct option: B

Final answer

682

Q118 *MCQ*

For a geometric sequence with $a = 3$, $r = 5$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{3(1-(5)^4)}{1-(5)}$.
2. Therefore $S_4 = 468$.

Correct option: A

Final answer

468

Worked Solutions **Geometric Sequences**

Q119 MCQ

For a geometric sequence with $a = 7$, $r = 2$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \dfrac{7(1-(2)^5)}{1-(2)}$.
2. Therefore $S_5 = 217$.

Correct option: D

Final answer

217

Q120 MCQ

For a geometric sequence with $a = 6$, $r = 3$, and $n = 3$, calculate S_n:

Solution

1. $S_n = \dfrac{6(1-(3)^3)}{1-(3)}$.
2. Therefore $S_3 = 78$.

Correct option: C

Final answer

78

Worked Solutions **Geometric Sequences**

Q121 **Written**

Find the sum S of a geometric sequence with first term 8, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. Use $S_6 = \dfrac{8(1-(1/2)^6)}{1-1/2} = 16\{1-(1/2)^6\}$.
2. Therefore $S_6 = \dfrac{63}{4}$.

Final answer

$\dfrac{63}{4}$

Q122 **Written**

Find the sum S of a geometric sequence with first term 6, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. $S_6 = \dfrac{6(1-(1/2)^6)}{1-1/2} = 12\{1-(1/2)^6\}$.
2. Therefore $S_6 = \dfrac{189}{16}$.

Final answer

$\dfrac{189}{16}$

Worked Solutions **Geometric Sequences**

Q123 **Written**

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{3}$, and 3 terms.

Solution

1. $S_3 = \dfrac{1-(1/3)^3}{1-1/3} = \dfrac{26}{27} \cdot \dfrac{3}{2}$.
2. Therefore $S_3 = \dfrac{13}{9}$.

Final answer

$\dfrac{13}{9}$

Q124 **Written**

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{2}$, and 8 terms.

Solution

1. $S_8 = \dfrac{1-(1/2)^8}{1-1/2} = 2\{1-(1/2)^8\}$.
2. Therefore $S_8 = \dfrac{255}{128}$.

Final answer

$\dfrac{255}{128}$

Worked Solutions **Geometric Sequences**

Q125 MCQ

For a geometric sequence with $a = 8$, $r = \frac{1}{2}$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{8(1 - (\frac{1}{2})^4)}{1 - (\frac{1}{2})}$.
2. Therefore $S_4 = 15$.

Correct option: B

Final answer

15

Q126 MCQ

For a geometric sequence with $a = 9$, $r = \frac{1}{3}$, and $n = 3$, calculate S_n:

Solution

1. $S_n = \dfrac{9(1 - (\frac{1}{3})^3)}{1 - (\frac{1}{3})}$.
2. Therefore $S_3 = 13$.

Correct option: A

Final answer

13

Worked Solutions **Geometric Sequences**

Q127 MCQ

For a geometric sequence with $a = 12$, $r = \frac{1}{2}$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \dfrac{12(1-(\frac{1}{2})^5)}{1-(\frac{1}{2})}$.
2. Therefore $S_5 = \dfrac{93}{4}$.

Correct option: D

Final answer

$\dfrac{93}{4}$

Q128 MCQ

For a geometric sequence with $a = 5$, $r = \frac{1}{4}$, and $n = 3$, calculate S_n:

Solution

1. $S_n = \dfrac{5(1 - (\frac{1}{4})^3)}{1 - (\frac{1}{4})}$.
2. Therefore $S_3 = \dfrac{105}{16}$.

Correct option: C

Final answer

$\dfrac{105}{16}$

Worked Solutions **Geometric Sequences**

Q129 **MCQ**

For a geometric sequence with $a = 10$, $r = \frac{1}{2}$, and $n = 6$, calculate S_n:

Solution

1. $S_n = \dfrac{10(1-(\frac{1}{2})^6)}{1-(\frac{1}{2})}$.
2. Therefore $S_6 = \dfrac{315}{16}$.

Correct option: B

Final answer

$\dfrac{315}{16}$

Q130 **MCQ**

For a geometric sequence with $a = 4$, $r = \frac{1}{3}$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{4(1 - (\frac{1}{3})^4)}{1 - (\frac{1}{3})}$.
2. Therefore $S_4 = \dfrac{160}{27}$.

Correct option: A

Final answer

$\dfrac{160}{27}$

Worked Solutions **Geometric Sequences**

Q131 **Written**

Find the sum S of a geometric sequence with first term 6, common ratio -2, and 4 terms.

Solution

1. $S_4 = \dfrac{6\{1-(-2)^4\}}{1-(-2)} = \dfrac{6(-15)}{3}$.
2. Therefore $S_4 = -30$.

Final answer

-30

Q132 **Written**

Find the sum S of a geometric sequence with first term 1, common ratio -3, and 4 terms.

Solution

1. $S_4 = \dfrac{1-(-3)^4}{1-(-3)} = \dfrac{-80}{4}$.
2. Therefore $S_4 = -20$.

Final answer

-20

Worked Solutions **Geometric Sequences**

Q133 Written

Find the sum S of a geometric sequence with first term 2, common ratio -3, and 6 terms.

Solution

1. $S_6 = \frac{2\{1-(-3)^6\}}{1-(-3)} = \frac{2(-728)}{4}$.
2. Therefore $S_6 = -364$.

Final answer

-364

Q134 MCQ

For a geometric sequence with $a = 4$, $r = -2$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \frac{4(1-(-2)^5)}{1-(-2)}$.
2. Therefore $S_5 = 44$.

Correct option: D

Final answer

44

Worked Solutions **Geometric Sequences**

Q135 MCQ

For a geometric sequence with $a = 3$, $r = -3$, and $n = 4$, calculate S_n:

Solution

1. $S_n = \dfrac{3(1-(-3)^4)}{1-(-3)}$.
2. Therefore $S_4 = -60$.

Correct option: C

Final answer

-60

Q136 MCQ

For a geometric sequence with $a = 2$, $r = -1$, and $n = 6$, calculate S_n:

Solution

1. $S_n = \dfrac{2(1-(-1)^6)}{1-(-1)}$.
2. Therefore $S_6 = 0$.

Correct option: B

Final answer

0

Worked Solutions **Geometric Sequences**

Q137 **MCQ**

For a geometric sequence with $a = 5$, $r = -2$, and $n = 6$, calculate S_n:

Solution

1. $S_n = \dfrac{5(1-(-2)^6)}{1-(-2)}$.
2. Therefore $S_6 = -105$.

Correct option: A

Final answer

-105

Q138 **MCQ**

For a geometric sequence with $a = 7$, $r = -1$, and $n = 5$, calculate S_n:

Solution

1. $S_n = \dfrac{7(1-(-1)^5)}{1-(-1)}$.
2. Therefore $S_5 = 7$.

Correct option: D

Final answer

7

Worked Solutions **Geometric Sequences**

Q139 MCQ

For a geometric sequence with $a = 6$, $r = -3$, and $n = 3$, calculate S_n:

Solution

1. $S_n = \dfrac{6(1-(-3)^3)}{1-(-3)}$.
2. Therefore $S_3 = 42$.

Correct option: C

Final answer

42

Q140 Written

Find S_n from the first term to the n-th term when the first term is 3 and the common ratio is -2.

Solution

1. Here $a = 3$ and $r = -2$.
2. $S_n = \dfrac{3\{1-(-2)^n\}}{1-(-2)} = 1-(-2)^n$.

Final answer

$1-(-2)^n$

Worked Solutions **Geometric Sequences**

Q141 **Written**

Find S_n from the first term to the n-th term when the first term is 7 and the common ratio is 2.

Solution

1. $S_n = \dfrac{7(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 7(2^n - 1)$.

Final answer

$7(2^n - 1)$

Q142 **Written**

Find S_n from the first term to the n-th term when the first term is 2 and the common ratio is $\dfrac{1}{3}$.

Solution

1. $S_n = \dfrac{2\{1 - (1/3)^n\}}{1 - 1/3}$.
2. Hence $S_n = 3\{1 - (1/3)^n\}$.

Final answer

$3\{1 - (\frac{1}{3})^n\}$

Worked Solutions **Geometric Sequences**

Q143 Written

Find S_n from the first term to the n-th term when the first term is 4 and the common ratio is 2.

Solution

1. $S_n = \dfrac{4(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 4(2^n - 1)$.

Final answer

$4(2^n - 1)$

Q144 MCQ

For the geometric sequence with $a = 3, r = 2$, find the sum S_n from the first term to the n-th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \dfrac{a(1 - r^n)}{1 - r}$ to obtain $S_n = 3(2^n - 1)$.

Correct option: C

Final answer

$3(2^n - 1)$

Worked Solutions **Geometric Sequences**

Q145 MCQ

For the geometric sequence with $a = 5, r = \frac{1}{2}$, find the sum S_n from the first term to the n-th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \dfrac{a(1-r^n)}{1-r}$ to obtain $S_n = 10\{1-(\frac{1}{2})^n\}$.

Correct option: D

Final answer

$10\{1-(\frac{1}{2})^n\}$

Q146 MCQ

For the geometric sequence with $a = 4, r = -2$, find the sum S_n from the first term to the n-th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \dfrac{a(1-r^n)}{1-r}$ to obtain $S_n = \dfrac{4\{1-(-2)^n\}}{3}$.

Correct option: A

Final answer

$\dfrac{4\{1-(-2)^n\}}{3}$

Worked Solutions **Geometric Sequences**

Q147 **Written**

Find S_n from the first term to the n-th term of $2, 6, 18, 54, \ldots$.

Solution

1. The sequence has $a = 2$ and $r = 3$.
2. $S_n = \dfrac{2(3^n - 1)}{3 - 1} = 3^n - 1$.

Final answer

$3^n - 1$

Q148 **Written**

Find S_n from the first term to the n-th term of $2, -6, 18, -54, \ldots$.

Solution

1. The sequence has $a = 2$ and $r = -3$.
2. $S_n = \dfrac{2\{1 - (-3)^n\}}{1 - (-3)} = \dfrac{1 - (-3)^n}{2}$.

Final answer

$\dfrac{1 - (-3)^n}{2}$

Worked Solutions **Geometric Sequences**

Q149 Written

Find S_n from the first term to the n-th term of $8, -4, 2, -1, \ldots$.

Solution

1. The sequence has $a = 8$ and $r = -\dfrac{1}{2}$.
2. $S_n = \dfrac{8\{1 - (-1/2)^n\}}{1 + 1/2} = \dfrac{16}{3}\{1 - (-1/2)^n\}$.

Final answer
$\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$

Q150 Written

Find S_n from the first term to the n-th term of $1, -3, 9, -27, \ldots$.

Solution

1. The sequence has $a = 1$ and $r = -3$.
2. $S_n = \dfrac{1 - (-3)^n}{1 + 3} = \dfrac{1 - (-3)^n}{4}$.

Final answer
$\frac{1 - (-3)^n}{4}$

Worked Solutions **Geometric Sequences**

Q151 Written

Find S_n from the first term to the n-th term when the first term is 16 and the common ratio is $-\dfrac{1}{2}$.

Solution

1. $S_n = \dfrac{16\{1-(-1/2)^n\}}{1+1/2}$.
2. Hence $S_n = \dfrac{32}{3}\{1-(-1/2)^n\}$.

Final answer

$\frac{32}{3}\{1-(-\frac{1}{2})^n\}$

Q152 Written

Find S_n from the first term to the n-th term of $1, -2, 4, -8, \ldots$.

Solution

1. The sequence has $a = 1$ and $r = -2$.
2. $S_n = \dfrac{1-(-2)^n}{1+2} = \dfrac{1-(-2)^n}{3}$.

Final answer

$\dfrac{1-(-2)^n}{3}$

Worked Solutions **Geometric Sequences**

Q153 *MCQ*

A culture grows by $3, 9, 27, 81$ units. Find the four-stage total:

Solution

1. Recognise the listed stages as a finite geometric sequence and identify its first term and ratio.
2. Apply the finite-GP formula to obtain $S_n = 120$.

Correct option: A

Final answer

120

Worked Solutions **Geometric Sequences**

Q154 **Challenge**

A filtration scale is geometric: $3, 1, \frac{1}{3}, \frac{1}{9}, \dots$ mg/L removed per stage. Calculate the total removed in 4 stages.

Solution

1. Here $a = 3$, $r = \frac{1}{3}$, and $n = 4$.
2. $S_4 = \dfrac{3\{1-(1/3)^4\}}{1-1/3} = \dfrac{40}{9}$ mg/L.

Final answer

$\frac{40}{9}$

Worked Solutions **Geometric Sequences**

Q155 *MCQ*

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is 28, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 28$, so $r^2 = 4$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 1$ or $a = \frac{7}{3}$, respectively.

Correct option: D

Final answer

$(a, r) = (1, 2)$ or $(a, r) = \left(\frac{7}{3}, -2\right)$

Worked Solutions **Geometric Sequences**

Q156 MCQ

Given $S_3 = 26$ and the sum from the 3rd to the 5th terms is 234, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 234$, so $r^2 = 9$; then $a(1 + r + r^2) = 26$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 2$ or $a = \frac{26}{7}$, respectively.

Correct option: C

Final answer

$(a, r) = (2, 3)$ or $(a, r) = \left(\frac{26}{7}, -3\right)$

Worked Solutions **Geometric Sequences**

Q157 MCQ

Given $S_3 = 21$ and the sum from the 3rd to the 5th terms is 84, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$; then $a(1 + r + r^2) = 21$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 3$ or $a = 7$, respectively.

Correct option: B

Final answer

$(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

Worked Solutions **Geometric Sequences**

Q158 MCQ

Given $S_3 = 28$ and the sum from the 3rd to the 5th terms is 112, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 112$, so $r^2 = 4$; then $a(1 + r + r^2) = 28$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: A

Final answer

$$(a, r) = (4, 2) \text{ or } (a, r) = \left(\frac{28}{3}, -2\right)$$

Worked Solutions **Geometric Sequences**

Q159 MCQ

Given $S_3 = 13$ and the sum from the 3rd to the 5th terms is 117, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 117$, so $r^2 = 9$; then $a(1 + r + r^2) = 13$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 1$ or $a = \frac{13}{7}$, respectively.

Correct option: D

Final answer

$(a, r) = (1, 3)$ or $(a, r) = \left(\frac{13}{7}, -3\right)$

Worked Solutions **Geometric Sequences**

Q160 MCQ

Given $S_3 = 35$ and the sum from the 3rd to the 5th terms is 140, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$; then $a(1 + r + r^2) = 35$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 5$ or $a = \frac{35}{3}$, respectively.

Correct option: C

Final answer

$(a, r) = (5, 2) \text{ or } (a, r) = \left(\frac{35}{3}, -2\right)$

Worked Solutions **Geometric Sequences**

Q161 *MCQ*

Given $S_3 = \dfrac{19}{2}$ and the sum from the 3rd to the 5th terms is $\dfrac{171}{8}$, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = \dfrac{171}{8}$, so $r^2 = \dfrac{9}{4}$; then $a(1 + r + r^2) = \dfrac{19}{2}$.
2. Solving in the real numbers gives $r = \dfrac{3}{2}$ or $r = -\dfrac{3}{2}$.
3. Back-substitution gives $a = 2$ or $a = \dfrac{38}{7}$, respectively.

Correct option: D

Final answer

$(a, r) = \left(2, \dfrac{3}{2}\right)$ or $(a, r) = \left(\dfrac{38}{7}, -\dfrac{3}{2}\right)$

Worked Solutions **Geometric Sequences**

Q162 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 4th to the 6th term is -72.

Solution

1. The 4th+6th block is $r^3 S_3 = -72$, so $r^3 = -8$ and $r = -2$.
2. Then $a(1 - 2 + 4) = 9$, giving $a = 3$.

Final answer

$(a, r) = (3, -2)$

Worked Solutions **Geometric Sequences**

Q163 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is $\frac{7}{4}$, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = \frac{7}{4}$, so $r^2 = \frac{1}{4}$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = -\frac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: C

Final answer

$(a, r) = \left(4, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{28}{3}, -\frac{1}{2}\right)$

Worked Solutions **Geometric Sequences**

Q164 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 35$ and the sum from the 3rd to the 5th term is 140.

Solution

1. Write $S_3 = a(1 + r + r^2) = 35$. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$ and $r = \pm 2$.
2. For $r = 2$, $7a = 35$, giving $a = 5$. For $r = -2$, $3a = 35$, giving $a = \frac{35}{3}$.

Final answer

$(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$

Q165 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and the sum from the 3rd to the 5th term is 84.

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$ and $r = \pm 2$.
2. Using $a(1 + r + r^2) = 21$ gives $a = 3$ for $r = 2$ and $a = 7$ for $r = -2$.

Final answer

$(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

Worked Solutions **Geometric Sequences**

Q166 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 3rd to the 5th term is 36.

Solution

1. The shifted block gives $r^2 S_3 = 36$, hence $r = \pm 2$.
2. Substitution into $a(1 + r + r^2) = 9$ gives $a = \frac{9}{7}$ for $r = 2$, or $a = 3$ for $r = -2$.

Final answer

$(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$

Q167 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 78$ and the sum from the 3rd to the 5th term is 702.

Solution

1. The shifted block gives $r^2 S_3 = 702$, so $r^2 = 9$ and $r = \pm 3$.
2. Using $a(1 + r + r^2) = 78$ gives $a = 6$ for $r = 3$, or $a = \frac{78}{7}$ for $r = -3$.

Final answer

$(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$

Worked Solutions **Geometric Sequences**

Q168 MCQ

Given $S_3 = 6$ and the sum from the 3rd to the 5th terms is 24, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 24$, so $r^2 = 4$; then $a(1 + r + r^2) = 6$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 2$ or $a = \frac{6}{7}$, respectively.

Correct option: B

Final answer

$(a, r) = (2, -2)$ or $(a, r) = \left(\frac{6}{7}, 2\right)$

Worked Solutions **Geometric Sequences**

Q169 MCQ

Given $S_3 = 9$ and the sum from the 3rd to the 5th terms is 36, find all real pairs (a, r):

Solution

1. The shifted block is $r^2 S_3 = 36$, so $r^2 = 4$; then $a(1 + r + r^2) = 9$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 3$ or $a = \frac{9}{7}$, respectively.

Correct option: A

Final answer

$$(a, r) = (3, -2) \text{ or } (a, r) = \left(\frac{9}{7}, 2\right)$$

Worked Solutions **Geometric Sequences**

Q170 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 3 and $S_3 = 13$.

Solution

1. From $ar = 3$, write $a = 3/r$. Then $3/r + 3 + 3r = 13$, so $3r^2 - 10r + 3 = 0$.
2. Hence $r = 3$ or $r = \frac{1}{3}$, giving $a = 1$ or $a = 9$, respectively.

Final answer

$(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

Q171 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 6, and $S_3 = 21$.

Solution

1. From $ar = 6$, $a = 6/r$. Substitution into $S_3 = 21$ gives $6/r + 6 + 6r = 21$, hence $2r^2 - 5r + 2 = 0$.
2. Therefore $r = 2$ or $r = \frac{1}{2}$, giving $a = 3$ or $a = 12$, respectively.

Final answer

$(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

Worked Solutions **Geometric Sequences**

Q172 MCQ

The second term is 6 and $S_3 = 21$. Find all real pairs (a, r):

Solution

1. Use $ar = 6$ and $a(1 + r + r^2) = 21$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = 2$ or $r = \frac{1}{2}$.
3. Back-substitution gives $a = 3$ or $a = 12$, respectively.

Correct option: C

Final answer

$(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

Worked Solutions **Geometric Sequences**

Q173 MCQ

The second term is 6 and $S_3 = 26$. Find all real pairs (a, r):

Solution

1. Use $ar = 6$ and $a(1 + r + r^2) = 26$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 2$ or $a = 18$, respectively.

Correct option: D

Final answer

$(a, r) = (2, 3)$ or $(a, r) = \left(18, \frac{1}{3}\right)$

Worked Solutions **Geometric Sequences**

Q174 MCQ

The second term is 8 and $S_3 = 28$. Find all real pairs (a, r):

Solution

1. Use $ar = 8$ and $a(1 + r + r^2) = 28$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = 2$ or $r = \dfrac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = 16$, respectively.

Correct option: B

Final answer

$(a, r) = (4, 2)$ or $(a, r) = \left(16, \dfrac{1}{2}\right)$

Worked Solutions **Geometric Sequences**

Q175 MCQ

The second term is 3 and $S_3 = 13$. Find all real pairs (a, r):

Solution

1. Use $ar = 3$ and $a(1 + r + r^2) = 13$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Correct option: C

Final answer

$(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

Worked Solutions **Geometric Sequences**

Q176 *MCQ*

The second term is $\frac{5}{2}$ and $S_3 = \frac{35}{4}$. Find all real pairs (a, r):

Solution

1. Use $ar = \frac{5}{2}$ and $a(1 + r + r^2) = \frac{35}{4}$; after substitution, solve the quadratic in r.
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = 2$.
3. Back-substitution gives $a = 5$ or $a = \frac{5}{4}$, respectively.

Correct option: D

Final answer

$(a, r) = \left(5, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{5}{4}, 2\right)$

Worked Solutions **Geometric Sequences**

Q177 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 4 and $S_3 = 7$.

Solution

1. The data give $ar^2 = 4$ and $a + ar + ar^2 = 7$. Substitute $a = 4/r^2$: $4 + 4r + 4r^2 = 7r^2$, so $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Final answer

$(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$

Q178 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 12, and $S_3 = 9$.

Solution

1. Use $ar^2 = 12$ and $a + ar + ar^2 = 9$. After multiplying by r^2, $r^2 + 4r + 4 = 0$, so $r = -2$.
2. Then $a = 12/(-2)^2 = 3$.

Final answer

$(a, r) = (3, -2)$

Worked Solutions **Geometric Sequences**

Q179 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 20, and $S_3 = 35$.

Solution

1. Use $ar^2 = 20$ and $a + ar + ar^2 = 35$. Multiplication by r^2 gives $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$, and $a = 5$ or $a = 45$, respectively.

Final answer

$(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$

Q180 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and $S_6 = 189$.

Solution

1. Use $S_6 = S_3 + r^3 S_3$: $189 = 21(1 + r^3)$, hence $r^3 = 8$ and $r = 2$ in the real numbers.
2. Then $a(1 + 2 + 4) = 21$, so $a = 3$.

Final answer

$(a, r) = (3, 2)$

Worked Solutions **Geometric Sequences**

Q181 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 26$ and $S_6 = 728$.

Solution

1. Since $S_6 = S_3 + r^3 S_3$, $728 = 26(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $a(1 + 3 + 9) = 26$, giving $a = 2$.

Final answer

$(a, r) = (2, 3)$

Q182 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 65$ and $S_6 = 1820$.

Solution

1. Use $S_6 = S_3(1 + r^3)$: $1820 = 65(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $13a = 65$, giving $a = 5$.

Final answer

$(a, r) = (5, 3)$

Worked Solutions **Geometric Sequences**

Q183 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is 3, and $S_4 = 51$.

Solution

1. Let $a + ar = 3$. Factor $S_4 = (a + ar)(1 + r^2)$, so $3(1 + r^2) = 51$ and $r = \pm 4$.
2. Then $a = 3/(1 + r)$, giving $a = \dfrac{3}{5}$ for $r = 4$, or $a = -1$ for $r = -4$.

Final answer

$(a, r) = \left(\dfrac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$

Q184 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is -8, and $S_4 = -80$.

Solution

1. Factor $S_4 = (a + ar)(1 + r^2)$. Thus $-8(1 + r^2) = -80$, so $r = \pm 3$.
2. Using $a(1 + r) = -8$ gives $a = -2$ for $r = 3$, or $a = 4$ for $r = -3$.

Final answer

$(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$

Worked Solutions **Geometric Sequences**

Q185 MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10}:

Solution

1. $a_7 = S_7 - S_6 = 64$ and $a_8 = S_8 - S_7 = 128$.
2. Thus $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 512$.

Correct option: D

Final answer

512

Worked Solutions **Geometric Sequences**

Q186 Challenge

If S_n is the sum of the first n terms of a geometric sequence, $S_8 = 255$, $S_6 = 63$, and $a_8 - a_7 = 64$, find a_{10}.

Solution

1. $S_8 - S_6 = a_7 + a_8 = 255 - 63 = 192$. Together with $a_8 - a_7 = 64$, this gives $a_7 = 64$ and $a_8 = 128$.
2. Hence $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 128(4) = 512$.

Final answer

$a_7 = 64,\ a_8 = 128,\ r = 2,\ a_{10} = 512$

Worked Solutions **Geometric Sequences**

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: For $a = 3, r = 2$, find a_6:

Solution

1. $a_6 = 3(2)^5 = 96$

Correct option: B

Final answer

96

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: If $2, x, 8$ are consecutive geometric terms, find x:

Solution

1. $x^2 = 16 \Rightarrow x = \pm 4$

Correct option: C

Final answer

± 4

Worked Solutions **Geometric Sequences**

Quiz MCQ 3 Concept Quiz · MCQ

For $1, 3, 9, \ldots$, find S_n:

Solution

1. $a = 1, r = 3$, so $S_n = (3^n - 1)/2$.

Correct option: B

Final answer

$$\frac{3^n - 1}{2}$$

Quiz MCQ 4 Concept Quiz · MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10}:

Solution

1. $a_7 = S_7 - S_6 = 64$, $a_8 = S_8 - S_7 = 128$, so $r = 2$.
2. Therefore $a_{10} = 128(2^2) = 512$.

Correct option: D

Final answer

512

Worked Solutions **Geometric Sequences**

Quiz WRITTEN 5 Concept Quiz · Written

Find the general term of $-4, 8, -16, 32, \dots$.

Solution

1. The ratio is -2.
2. $a_n = -4(-2)^{n-1}$

Final answer

$-4(-2)^{n-1}$

Quiz WRITTEN 6 Concept Quiz · Written

If $1, x, 25$ are consecutive geometric terms, find x.

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$x = \pm 5$

Worked Solutions **Geometric Sequences**

Quiz WRITTEN 7 Concept Quiz · Written

A four-stage process has outputs $5, 15, 45, 135$. Find the total output.

Solution

1. $S_4 = 5(1 + 3 + 9 + 27) = 200.$

Final answer

200

Quiz WRITTEN 8 Concept Quiz · Written

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10}.

Solution

1. $a_7 = 64$, $a_8 = 128$, hence $r = 2$.
2. $a_{10} = a_8 r^2 = 512.$

Final answer

512

Answer key

Use the question number shown on each page.

Q001 $2 \times 3^{n-1}$

Q002 $3 \times 4^{n-1}$

Q003 $5 \times 2^{n-1}$

Q004 $2 \times 5^{n-1}$

Q005 $7 \times 4^{n-1}$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 $5(-2)^{n-1}$

Q013 $2(-3)^{n-1}$

Q014 $3(-5)^{n-1}$

Q015 $4(-2)^{n-1}$

Q016 A

Q017 A

Q018 A

Q019 A

Q020 A

Q021 $8\left(-\frac{1}{3}\right)^{n-1}$

Q022 $4\left(\frac{1}{3}\right)^{n-1}$

Q023 $3\left(-\frac{1}{2}\right)^{n-1}$

Q024 $6\left(\frac{1}{2}\right)^{n-1}$

Q025 A

Q026 A

Q027 A

Q028 A

Q029 A

Q030 $5 \times 2^{n-1}$

Q031 $2 \times 3^{n-1}$

Q032 $3 \times 2^{n-1}$

Q033 A

Q034 A

Q035 A

Q036 A

Q037 A

Q038 $2(-3)^{n-1}$

Q039 $5(-2)^{n-1}$

Q040 A

Q041 A

Q042 A

Q043 A

Q044 $-4\left(-\frac{3}{2}\right)^{n-1}$

Q045 $-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 $2\left(\frac{1}{3}\right)^{n-1}$

Q047 $48\left(\frac{1}{2}\right)^{n-1}$

Q048 $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q049 $8\left(2\sqrt{2}\right)^{n-1}$

Q050 $3\left(-\frac{5}{2}\right)^{n-1}$

Q051 $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q052 $5\left(-\frac{2}{3}\right)^{n-1}$

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $r = -\frac{1}{3},\ A = 54,\ B = -2$

Q061 A

Q062 $r = \frac{1}{2},\ A = 48,\ B = 6$

Q063 $r = -3,\ A = 1,\ B = -27$

Q064 $r = \frac{1}{3},\ A = 1,\ B = \frac{1}{27}$

Q065 $r = -\frac{1}{2},\ A = \frac{1}{2},\ B = -\frac{1}{16}$

Q066 $a_7 = 648$

Q067 $a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 $a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Q069 $a_n = 2(3)^{n-1}$ or $-2(-3)^{n-1}$

Q070 $a_n = 3(3)^{n-1}$ or $3(-3)^{n-1}$

Q071 $a_n = 7(2)^{n-1}$

Q072 $a_n = 3(-2)^{n-1}$

Q073 $a_n = 3(-2)^{n-1}$

Q074 $a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Q075 $a_n = -7(2)^{n-1}$ or $-7(-2)^{n-1}$

Q076 $a_n = -2(-2)^{n-1}$

Q077 $a_n = -5(-2)^{n-1}$

Q078 $a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Q079 $a_n = 4(\frac{3}{2})^{n-1}$ or $4(-\frac{3}{2})^{n-1}$

Q080 $a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Q081 A

Q082 A

Q083 A

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A

Q090 A

Q091 A

Q092 A

Q093 A

Q094 A

Q095 A

Q096 A

Q097 $x = \pm 4$

Q098 $x = \pm 5$

Q099 $x = \pm 2$

Q100 $x = \pm 2\sqrt{3}$

Q101 A

Q102 A

Q103 A

Q104 A

Q105 A

Q106 A

Q107 $(a, b) = (9, 12)$ or $(1, -4)$

Q108 $(a, b) = (2, 2)$ or $(-4, -1)$

Q109 $(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$

Q110 $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$

Q111 484

Q112 93

Q113 200

Q114 3069

Q115 D

Q116 C

Q117 B

Q118 A

Q119 D

Q120 C

Q121 $\frac{63}{4}$

Q122 $\frac{189}{16}$

Q123 $\frac{13}{9}$

Q124 $\frac{255}{128}$

Q125 B

Q126 A

Q127 D

Q128 C

Q129 B

Q130 A

Q131 -30

Q132 -20

Q133 -364

Q134 D

Q135 C

Q136 B

Q137 A

Q138 D

Q139 C

Q140 $1 - (-2)^n$

Q141 $7(2^n - 1)$

Q142 $3\{1 - (\frac{1}{3})^n\}$

Q143 $4(2^n - 1)$

Q144 C

Q145 D

Q146 A

Q147 $3^n - 1$

Q148 $\frac{1 - (-3)^n}{2}$

Q149 $\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$

Q150 $\frac{1 - (-3)^n}{4}$

Q151 $\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$

Q152 $\frac{1 - (-2)^n}{3}$

Q153 A

Q154 $\frac{40}{9}$

Q155 D

Q156 C

Q157 B

Q158 A

Q159 D

Q160 C

Q161 D

Q162 $(a, r) = (3, -2)$

Q163 C

Q164 $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$

Q165 $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

Q166 $(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$

Q167 $(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$

Q168 B

Q169 A

Q170 $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

Q171 $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

Q172 C

Q173 D

Q174 B

Q175 C

Q176 D

Q177 $(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$

Q178 $(a, r) = (3, -2)$

Q179 $(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$

Q180 $(a, r) = (3, 2)$

Q181 $(a, r) = (2, 3)$

Q182 $(a, r) = (5, 3)$

Q183 $(a, r) = \left(\frac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$

Q184 $(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$

Q185 D

Q186 $a_7 = 64,\ a_8 = 128,\ r = 2,\ a_{10} = 512$

Quiz MCQ 1 B

Quiz MCQ 2 C

Quiz MCQ 3 B

Quiz MCQ 4 D

Quiz WRITTEN 5 $-4(-2)^{n-1}$

Quiz WRITTEN 6 $x = \pm 5$

Quiz WRITTEN 7 200

Quiz WRITTEN 8 512

Concept 3 | Summation

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$\sum(au_k + bv_k) = a\sum u_k + b\sum v_k$$

Standard sums

$$\sum k = \frac{n(n+1)}{2}$$

Telescoping

$$\sum_{k=1}^{n}(u_k - u_{k+1}) = u_1 - u_{n+1}$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Summation**

Q001 Written

Write $\sum_{k=1}^{5} 2k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^{5} 2k = 2 + 4 + 6 + 8 + 10$

Final answer

$2 + 4 + 6 + 8 + 10$

Q002 Written

Write $\sum_{k=1}^{5} (2k+1)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^{5} (2k+1) = 3 + 5 + 7 + 9 + 11$

Final answer

$3 + 5 + 7 + 9 + 11$

Worked Solutions **Summation**

Q003 Written

Write $\sum_{k=1}^{5} 3k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^{5} 3k = 3 + 6 + 9 + 12 + 15$

Final answer

$3 + 6 + 9 + 12 + 15$

Q004 Written

Write $\sum_{k=1}^{4} (4k - 2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^{4} (4k - 2) = 2 + 6 + 10 + 14$

Final answer

$2 + 6 + 10 + 14$

Worked Solutions **Summation**

Q005 Written

Write $\sum_{k=1}^{5}(2k-3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^{5}(2k-3) = -1 + 1 + 3 + 5 + 7$

Final answer

$-1 + 1 + 3 + 5 + 7$

Q006 MCQ

Select the correct option: Expand $\sum_{k=1}^{3} 4k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$4 + 8 + 12$

Worked Solutions **Summation**

Q007 MCQ

Select the correct option: Expand $\sum_{k=1}^{4} 2^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$2 + 4 + 8 + 16$

Q008 MCQ

Select the correct option: Expand $\sum_{k=1}^{3}(5^k - 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$4 + 24 + 124$

Worked Solutions **Summation**

Q009 *MCQ*

Select the correct option: Expand $\sum_{k=1}^{5}(k+4)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$5+6+7+8+9$

Q010 *MCQ*

Select the correct option: Expand $\sum_{k=2}^{5}(2k+3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$7+9+11+13$

Worked Solutions **Summation**

Q011 MCQ

Select the correct option: Expand $\sum_{k=1}^{3}(2^k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$3 + 5 + 9$

Q012 MCQ

Select the correct option: Expand $\sum_{k=1}^{3}(k^2 + 3k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$4 + 10 + 18$

Worked Solutions **Summation**

Q013 **MCQ**

Select the correct option: Expand $\sum_{k=2}^{5}(k^2+1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$5+10+17+26$

Q014 **Written**

Write $\sum_{k=1}^{3}(k^2-3)$ as a sum of terms.

Solution

1. Substitute $k=1,2,3$.

2. $\sum_{k=1}^{3}(k^2-3)=-2+1+6$

Final answer

$-2+1+6$

Worked Solutions **Summation**

Q015 **Written**

Write $\sum_{k=1}^{3}(3k^2+2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^{3}(3k^2+2) = 5 + 14 + 29$

Final answer

$5 + 14 + 29$

Q016 **Written**

Write $\sum_{k=1}^{4} k^2$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^{4} k^2 = 1 + 4 + 9 + 16$

Final answer

$1 + 4 + 9 + 16$

Worked Solutions **Summation**

Q017 **Written**

Write $\sum_{k=1}^{3}(2k^2+3)$ **as a sum of terms.**

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^{3}(2k^2+3) = 5 + 11 + 21$

Final answer

$5 + 11 + 21$

Q018 **Written**

Write $\sum_{k=1}^{4}(k^2-2k)$ **as a sum of terms.**

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^{4}(k^2-2k) = -1 + 0 + 3 + 8$

Final answer

$-1 + 0 + 3 + 8$

Worked Solutions **Summation**

Q019 MCQ

Select the correct option: Expand $\sum_{k=1}^{4}(2k+1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$3+5+7+9$

Q020 MCQ

Select the correct option: Expand $\sum_{k=2}^{5}(3k-1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$5+8+11+14$

Worked Solutions **Summation**

Q021 *MCQ*

Select the correct option: Expand $\sum_{k=1}^{4}(2k^2 - 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$-1 + 5 + 15 + 29$

Q022 *MCQ*

Select the correct option: Expand $\sum_{k=1}^{4}(3k^2 + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$4 + 13 + 28 + 49$

Worked Solutions **Summation**

Q023 MCQ

Select the correct option: Expand $\sum_{k=1}^{4}(k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$0 + 2 + 6 + 12$

Q024 MCQ

Select the correct option: Expand $\sum_{k=2}^{4} 5k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$10 + 15 + 20$

Worked Solutions **Summation**

Q025 MCQ

Select the correct option: Expand $\sum_{k=1}^{4} 3$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$3 + 3 + 3 + 3$

Q026 MCQ

Select the correct option: Expand $\sum_{k=1}^{4}(5k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$6 + 11 + 16 + 21$

Worked Solutions **Summation**

Q027 Written

Write $\sum_{k=1}^{4} 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^{4} 3^k = 3 + 9 + 27 + 81$

Final answer

$3 + 9 + 27 + 81$

Q028 Written

Write $\sum_{k=1}^{3} 2$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^{3} 2 = 2 + 2 + 2$

Final answer

$2 + 2 + 2$

Worked Solutions **Summation**

Q029 Written

Write $\sum_{k=1}^{4} 2^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^{4} 2^k = 2 + 4 + 8 + 16$

Final answer

$2 + 4 + 8 + 16$

Q030 Written

Write $\sum_{k=1}^{6} 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^{6} 3 = 3 + 3 + 3 + 3 + 3 + 3$

Final answer

$3 + 3 + 3 + 3 + 3 + 3$

Worked Solutions **Summation**

Q031 Written

Write $\sum_{k=1}^{3} 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^{3} 3^k = 3 + 9 + 27$

Final answer

$3 + 9 + 27$

Q032 Written

Write $\sum_{k=1}^{3} (5^k + 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^{3} (5^k + 3) = 8 + 28 + 128$

Final answer

$8 + 28 + 128$

Worked Solutions **Summation**

Q033 Written

Write $\sum_{k=1}^{4}(2^k - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^{4}(2^k - 3) = -1 + 1 + 5 + 13$

Final answer

$-1 + 1 + 5 + 13$

Q034 Written

Write $\sum_{k=1}^{4} 5$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^{4} 5 = 5 + 5 + 5 + 5$

Final answer

$5 + 5 + 5 + 5$

Worked Solutions **Summation**

Q035 **Written**

Write $\sum_{k=1}^{3} 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^{3} 3 = 3 + 3 + 3$

Final answer

$3 + 3 + 3$

Q036 **Written**

Write $\sum_{k=1}^{5} 4$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^{5} 4 = 4 + 4 + 4 + 4 + 4$

Final answer

$4 + 4 + 4 + 4 + 4$

Worked Solutions **Summation**

Q037 MCQ

Select the correct option: Expand $\sum_{k=1}^{3}(k^2+2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$3+6+11$

Q038 MCQ

Select the correct option: Expand $\sum_{k=1}^{5} 7$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$7+7+7+7+7$

Worked Solutions **Summation**

Q039 MCQ

Select the correct option: Expand $\sum_{k=3}^{6} k^2$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$9 + 16 + 25 + 36$

Q040 MCQ

Select the correct option: Expand $\sum_{k=1}^{3} 6^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$6 + 36 + 216$

Worked Solutions **Summation**

Q041 MCQ

Select the correct option: Expand $\sum_{k=1}^{5}(4k-3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$1+5+9+13+17$

Q042 MCQ

Select the correct option: Expand $\sum_{k=1}^{4}(3k-5)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$-2+1+4+7$

Worked Solutions **Summation**

Q043 MCQ

Select the correct option: Expand $\sum_{k=1}^{3}(4^k - 2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$2 + 14 + 62$

Q044 MCQ

Select the correct option: Expand $\sum_{k=1}^{3}(2k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$1 + 6 + 15$

Worked Solutions **Summation**

Q045 Written

Express $3 + 7 + 11 + \cdots + (4n - 1)$ using the summation symbol Σ.

Solution

1. The general term is $4k - 1$.
2. $\sum_{k=1}^{n}(4k - 1)$

Final answer

$$\sum_{k=1}^{n}(4k - 1)$$

Q046 Written

Express $5 + 8 + 11 + \cdots + (3n + 2)$ using Σ.

Solution

1. The general term is $3k + 2$.
2. $\sum_{k=1}^{n}(3k + 2)$

Final answer

$$\sum_{k=1}^{n}(3k + 2)$$

Worked Solutions **Summation**

Q047 Written

Write the sum of the first n terms of the sequence $a_n = 2n + 1$ using Σ.

Solution

1. Change the term index from n to k.
2. $\sum_{k=1}^{n}(2k + 1)$

Final answer

$$\sum_{k=1}^{n}(2k+1)$$

Q048 Written

Express $1 + 3 + 5 + \cdots + (2n - 1)$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{n}(2k - 1)$

Final answer

$$\sum_{k=1}^{n}(2k-1)$$

Worked Solutions **Summation**

Q049 Written

Write the sum of the first n terms of $a_n = 7n - 1$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{n}(7k-1)$

Final answer

$$\sum_{k=1}^{n}(7k-1)$$

Q050 Written

Write the sum of the first n terms of $a_n = 4n + 5$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{n}(4k+5)$

Final answer

$$\sum_{k=1}^{n}(4k+5)$$

Worked Solutions **Summation**

Q051 Written

Write the sum of the first n terms of $a_n = 3n - 2$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{n}(3k-2)$

Final answer

$$\sum_{k=1}^{n}(3k-2)$$

Q052 MCQ

Select the correct option: Express $2+5+8+\cdots+(3n-1)$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{n}(3k-1)$$

Worked Solutions **Summation**

Q053 MCQ

Select the correct option: Express $7 + 11 + 15 + \cdots + 43$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{10}(4k+3)$$

Q054 MCQ

Select the correct option: Express $6 + 10 + 14 + \cdots + (4n + 2)$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{n}(4k+2)$$

Worked Solutions **Summation**

Q055 MCQ

Select the correct option: Express $8 + 13 + 18 + \cdots + 103$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{20}(5k+3)$$

Q056 MCQ

Select the correct option: Express $11^2 + 12^2 + \cdots + 19^2$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=11}^{19} k^2$$

Worked Solutions **Summation**

Q057 *MCQ*

Select the correct option: Express $2^2 + 3^2 + \cdots + 10^2$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=2}^{10} k^2$$

Q058 *MCQ*

Select the correct option: Express $1 + 3 + 9 + \cdots + 3^{n-1}$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=0}^{n-1} 3^k$$

Worked Solutions **Summation**

Q059 MCQ

Select the correct option: Express $2 + 7 + 12 + \cdots + (5n - 3)$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{n}(5k - 3)$$

Q060 Written

Express $6^2 + 7^2 + 8^2 + \cdots + 15^2$ using Σ.

Solution

1. The index starts at 6 and ends at 15.
2. $\sum_{k=6}^{15} k^2$

Final answer

$$\sum_{k=6}^{15} k^2$$

Worked Solutions **Summation**

Q061 Written

Express $4^2 + 5^2 + 6^2 + \cdots + 13^2$ using Σ.

Solution

1. The index starts at 4 and ends at 13.
2. $\sum_{k=4}^{13} k^2$

Final answer

$$\sum_{k=4}^{13} k^2$$

Q062 Written

Express $1 + 7 + 25 + 79 + \cdots + (3^n - 2)$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{n}(3^k - 2)$

Final answer

$$\sum_{k=1}^{n}(3^k - 2)$$

Worked Solutions **Summation**

Q063 **Written**

Express $1 + 2 + 4 + 8 + \cdots + 2^{n-1}$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{n} 2^{k-1}$

Final answer

$$\sum_{k=1}^{n} 2^{k-1}$$

Q064 **Written**

Express $7^2 + 8^2 + \cdots + 21^2$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=7}^{21} k^2$

Final answer

$$\sum_{k=7}^{21} k^2$$

Worked Solutions **Summation**

Q065 Written

Express $12^2 + 13^2 + \cdots + 18^2$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=12}^{18} k^2$

Final answer

$$\sum_{k=12}^{18} k^2$$

Q066 Written

Express $10^2 + 11^2 + \cdots + 20^2$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=10}^{20} k^2$

Final answer

$$\sum_{k=10}^{20} k^2$$

Worked Solutions **Summation**

Q067 MCQ

Select the correct option: Express $4^2 + 5^2 + \cdots + 12^2$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=4}^{12} k^2$$

Q068 MCQ

Select the correct option: Express $3^1 + 3^2 + \cdots + 3^n$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{n} 3^k$$

Worked Solutions **Summation**

Q069 MCQ

Select the correct option: Express $2 + 4 + 8 + \cdots + 2^n$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{n} 2^k$$

Q070 MCQ

Select the correct option: Express $1 + 2^1 + 2^2 + \cdots + 2^{n-1}$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=0}^{n-1} 2^k$$

Worked Solutions **Summation**

Q071 MCQ

Select the correct option: Express $7 + 14 + 21 + \cdots + 7n$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{n} 7k$$

Q072 MCQ

Select the correct option: Express $10 + 15 + 20 + \cdots + 5n$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=2}^{n} 5k$$

Worked Solutions **Summation**

Q073 MCQ

Select the correct option: Express $6^2 + 7^2 + \cdots + 16^2$ using the summation symbol Σ:

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=6}^{16} k^2$$

Q074 Written

Express $2 + 5 + 8 + \cdots + 29$ using Σ.

Solution

1. The general term is $3k - 1$, and $29 = 3(10) - 1$.

2. $\sum_{k=1}^{10}(3k - 1)$

Final answer

$$\sum_{k=1}^{10}(3k - 1)$$

Worked Solutions **Summation**

Q075 **Written**

Express $8 + 16 + 24 + \cdots + 56$ **using** Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{7} 8k$

Final answer

$$\sum_{k=1}^{7} 8k$$

Q076 **Written**

Express $7 + 13 + 19 + \cdots + 73$ **using** Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{12}(6k+1)$

Final answer

$$\sum_{k=1}^{12}(6k+1)$$

Worked Solutions **Summation**

Q077 **Written**

Express $4 + 9 + 14 + \cdots + 119$ using Σ.

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{24}(5k-1)$

Final answer

$$\sum_{k=1}^{24}(5k-1)$$

Worked Solutions **Summation**

Q078 **Challenge**

A green-tech startup installs 1 solar panel in Week 1, 4 in Week 2, 7 in Week 3, and $3n - 2$ in Week n. Write one expression using Σ for the total panels installed from Week 1 through Week n.

Solution

1. The weekly pattern is arithmetic with general term $3k - 2$.
2. Total $= \sum_{k=1}^{n}(3k - 2)$

Final answer

$$\sum_{k=1}^{n}(3k - 2)$$

Worked Solutions **Summation**

Q079 Written

Find $\sum_{k=1}^{n}(-3)$.

Solution

1. There are n copies of -3.
2. $\sum_{k=1}^{n}(-3) = -3n$

Final answer

$-3n$

Q080 Written

Find $\sum_{k=1}^{n} 3$.

Solution

1. There are n copies of 3.
2. $\sum_{k=1}^{n} 3 = 3n$

Final answer

$3n$

Worked Solutions **Summation**

Q081 **Written**

Find $\sum\limits_{k=1}^{n} 6$.

Solution

1. $\sum_{k=1}^{n} 6 = 6n$

Final answer

$6n$

Q082 **Written**

Find $\sum\limits_{k=1}^{n} (-4)$.

Solution

1. $\sum_{k=1}^{n} (-4) = -4n$

Final answer

$-4n$

Worked Solutions **Summation**

Q083 Written

Find $\sum_{k=1}^{n} 1.$

Solution

1. $\sum_{k=1}^{n} 1 = n$

Final answer

n

Q084 MCQ

Select the correct option: Find $\sum_{k=1}^{n} 9$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$9n$

Worked Solutions **Summation**

Q085 MCQ

Select the correct option: Find $\sum_{k=1}^{n}(-2)$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$-2n$

Q086 MCQ

Select the correct option: Find $\sum_{k=1}^{7} 4$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

28

Worked Solutions **Summation**

Q087 MCQ

Select the correct option: Find $\sum_{k=1}^{12}(-5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

-60

Q088 MCQ

Select the correct option: Find $\sum_{k=1}^{n} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)}{2}$$

Worked Solutions **Summation**

Q089 MCQ

Select the correct option: Find $\sum_{k=1}^{n} 4k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$2n(n+1)$

Q090 MCQ

Select the correct option: Find $\sum_{k=1}^{10} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

55

Worked Solutions **Summation**

Q091 MCQ

Select the correct option: Find $\sum_{k=1}^{8} 3k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

108

Q092 MCQ

Select the correct option: Find $\sum_{k=1}^{6} 5k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

105

Worked Solutions **Summation**

Q093 MCQ

Select the correct option: Find $\sum_{k=1}^{20} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

210

Q094 MCQ

Select the correct option: Find $\sum_{k=1}^{n} k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(2n+1)}{6}$$

Worked Solutions **Summation**

Q095 MCQ

Select the correct option: Find $\sum_{k=1}^{n} 6k^2$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$n(n+1)(2n+1)$

Q096 MCQ

Select the correct option: Find $\sum_{k=1}^{5} k^2$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

55

Worked Solutions **Summation**

Q097 MCQ

Select the correct option: Find $\sum_{k=1}^{4} 2k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

60

Q098 MCQ

Select the correct option: Find $\sum_{k=1}^{3} 4k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

56

Worked Solutions **Summation**

Q099 MCQ

Select the correct option: Find $\sum_{k=1}^{7} k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

140

Q100 Written

Find $\sum_{k=1}^{n}(5k+4)$.

Solution

1. $\sum(5k+4) = 5\sum k + \sum 4$
2. $= \frac{5n(n+1)}{2} + 4n = \frac{n(5n+13)}{2}$

Final answer

$$\frac{n(5n+13)}{2}$$

Worked Solutions **Summation**

Q101 Written

Find $\sum_{k=1}^{n}(2k-3)$.

Solution

1. $2\sum k - \sum 3 = n(n+1) - 3n$
2. $= n(n-2)$

Final answer

$n(n-2)$

Q102 Written

Find $\sum_{k=1}^{n}(4k-1)$.

Solution

1. $4\sum k - \sum 1 = 2n(n+1) - n$
2. $= n(2n+1)$

Final answer

$n(2n+1)$

Worked Solutions **Summation**

Q103 Written

Find $\sum_{k=1}^{n}(3k+2)$.

Solution

1. $3\sum k + \sum 2 = \frac{3n(n+1)}{2} + 2n$
2. $= \frac{n(3n+7)}{2}$

Final answer

$$\frac{n(3n+7)}{2}$$

Q104 Written

Find $\sum_{k=1}^{n}(6k-3)$.

Solution

1. $6\sum k - \sum 3 = 3n(n+1) - 3n$
2. $= 3n^2$

Final answer

$3n^2$

Worked Solutions **Summation**

Q105 MCQ

Select the correct option: Find $\sum_{k=1}^{n}(3k-1)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(3n+1)}{2}$$

Q106 MCQ

Select the correct option: Find $\sum_{k=1}^{n}(2k+5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$n(n+6)$

Worked Solutions **Summation**

Q107 MCQ

Select the correct option: Find $\sum_{k=1}^{n}(8k-4)$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$4n^2$

Q108 MCQ

Select the correct option: Find $\sum_{k=1}^{n}(5k+3)$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$\frac{n(5n+11)}{2}$

Worked Solutions **Summation**

Q109 Written

Find $\sum_{k=1}^{n}(k-1)^2$.

Solution

1. $\sum(k^2-2k+1)=\sum k^2-2\sum k+\sum 1$
2. $=\frac{n(n-1)(2n-1)}{6}$

Final answer

$$\frac{n(n-1)(2n-1)}{6}$$

Q110 MCQ

Select the correct option: Find $\sum_{k=1}^{n}(k^2+k)$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions **Summation**

Q111 **MCQ**

Select the correct option: Find $\sum_{k=1}^{n}(2k^2 - k)$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(4n-1)}{6}$$

Q112 **MCQ**

Select the correct option: Find $\sum_{k=1}^{n}(k^2 - 1)$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(2n^2+3n-5)}{6}$$

Worked Solutions **Summation**

Q113 MCQ

Select the correct option: Find $\sum_{k=1}^{n}(3k^2 + 2k)$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Q114 MCQ

Select the correct option: Find $\sum_{k=1}^{n}(4k^2 - 3k + 2)$**:**

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(8n^2 + 3n + 7)}{6}$$

Worked Solutions **Summation**

Q115 Written

Find $1 \cdot 3 + 2 \cdot 8 + 3 \cdot 13 + \cdots + n(5n-2)$.

Solution

1. $\sum_{k=1}^{n} k(5k-2) = 5\sum k^2 - 2\sum k$
2. $= \frac{5n(n+1)(2n+1)}{6} - n(n+1) = \frac{n(n+1)(10n-1)}{6}$

Final answer

$$\frac{n(n+1)(10n-1)}{6}$$

Q116 Written

Find $\sum_{k=1}^{n}(k-1)(3k-1)$.

Solution

1. $(k-1)(3k-1) = 3k^2 - 4k + 1$
2. $3\sum k^2 - 4\sum k + \sum 1 = \frac{n(n-1)(2n+1)}{2}$

Final answer

$$\frac{n(n-1)(2n+1)}{2}$$

Worked Solutions **Summation**

Q117 Written

Find $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + n(2n+1)$.

Solution

1. $\sum_{k=1}^{n} k(2k+1) = 2\sum k^2 + \sum k$
2. $= \frac{n(n+1)(4n+5)}{6}$

Final answer

$$\frac{n(n+1)(4n+5)}{6}$$

Q118 Written

Find $\displaystyle\sum_{k=1}^{n} k(k-2)$.

Solution

1. $\sum(k^2 - 2k) = \sum k^2 - 2\sum k$
2. $= \frac{n(n+1)(2n-5)}{6}$

Final answer

$$\frac{n(n+1)(2n-5)}{6}$$

Worked Solutions **Summation**

Q119 Written

Find $\sum_{k=1}^{n} k(2k-2)$.

Solution

1. $2\sum k^2 - 2\sum k$
2. $= \frac{2n(n+1)(n-1)}{3}$

Final answer

$\frac{2n(n+1)(n-1)}{3}$

Q120 Written

Find $1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 + \cdots + n(n+3)$.

Solution

1. $\sum_{k=1}^{n} k(k+3) = \sum k^2 + 3\sum k$
2. $= \frac{n(n+1)(n+5)}{3}$

Final answer

$\frac{n(n+1)(n+5)}{3}$

Worked Solutions **Summation**

Q121 Written

Find $1 \cdot 5 + 2 \cdot 8 + 3 \cdot 11 + \cdots + n(3n + 2)$.

Solution

1. $\sum_{k=1}^{n} k(3k + 2) = 3\sum k^2 + 2\sum k$
2. $= \frac{n(n+1)(2n+3)}{2}$

Final answer

$$\frac{n(n + 1)(2n + 3)}{2}$$

Q122 Written

Find $2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 + \cdots + (n + 1)(n + 4)$.

Solution

1. $\sum_{k=1}^{n}(k + 1)(k + 4) = \sum(k^2 + 5k + 4)$
2. $= \frac{n(n+4)(n+5)}{3}$

Final answer

$$\frac{n(n + 4)(n + 5)}{3}$$

Worked Solutions **Summation**

Q123 Written

Find $\sum_{k=1}^{4}(2k-1)$.

Solution

1. $2\sum_{k=1}^{4}k-\sum_{k=1}^{4}1=2(10)-4$
2. $=16$

Final answer

16

Q124 Written

Find $\sum_{k=1}^{5}(3k-2)$.

Solution

1. $3\sum_{k=1}^{5}k-2\sum_{k=1}^{5}1=3(15)-10$
2. $=35$

Final answer

35

Worked Solutions **Summation**

Q125 Written

Find $\sum_{k=1}^{20}(2k-7)$.

Solution

1. $2\sum_{k=1}^{20} k - 7\sum_{k=1}^{20} 1 = 2(210) - 140$
2. $= 280$

Final answer

280

Q126 Written

Find $\sum_{k=1}^{8}(4k+5)$.

Solution

1. $4\sum_{k=1}^{8} k + 5\sum_{k=1}^{8} 1 = 4(36) + 40$
2. $= 184$

Final answer

184

Worked Solutions **Summation**

Q127 **Written**

Find $\sum_{k=1}^{10}(2k-1)$.

Solution

1. $2\sum_{k=1}^{10} k - \sum_{k=1}^{10} 1 = 2(55) - 10$
2. $= 100$

Final answer

100

Worked Solutions **Summation**

Q128 **Challenge**

Find $\sum_{r=4}^{8}(r^2 + r + 5)$.

Solution

1. $\sum_{r=4}^{8}(r^2 + r + 5) = \sum_{r=1}^{8}(r^2 + r + 5) - \sum_{r=1}^{3}(r^2 + r + 5)$
2. $= (204 + 36 + 40) - (14 + 6 + 15) = 245$

Final answer

245

Worked Solutions **Summation**

Q129 **Written**

Find the sum $\sum_{k=1}^{n} k^2(k+1)$**.**

Solution

1. Expand: $k^2(k+1) = k^3 + k^2$.
2. Use $\sum k^3 = \frac{n^2(n+1)^2}{4}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $\sum_{k=1}^{n} k^2(k+1) = \frac{1}{12}n(n+1)(n+2)(3n+1)$

Final answer

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q130 Written

Find the sum $\sum_{k=1}^{n} k^2(k-2)$**.**

Solution

1. Expand: $k^2(k-2) = k^3 - 2k^2$.
2. Substitute the formulae for $\sum k^3$ and $\sum k^2$, then factor.
3. $\sum_{k=1}^{n} k^2(k-2) = \dfrac{n(n+1)(3n^2-5n-4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions **Summation**

Q131 **Written**

Find the sum $\sum_{k=1}^{n} k^2(k+2)$**.**

Solution

1. Expand: $k^2(k+2) = k^3 + 2k^2$.
2. Substitute the standard sums and factor the common $n(n+1)$.
3. $\sum_{k=1}^{n} k^2(k+2) = \dfrac{n(n+1)(3n^2+11n+4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2+11n+4)}{12}$$

Q132 **Written**

Find the sum $\sum_{k=1}^{n} k^2(k-3)$.

Solution

1. Expand: $k^2(k-3) = k^3 - 3k^2$.
2. Substitute the standard sums and simplify the common factor.
3. $\sum_{k=1}^{n} k^2(k-3) = \dfrac{n(n+1)(n^2-3n-2)}{4}$

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions **Summation**

Q133 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k+1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q134 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k-2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions **Summation**

Q135 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k+2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+11n+4)}{12}$$

Q136 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k-3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions **Summation**

Q137 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k+3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+5n+2)}{4}$$

Q138 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k-4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-13n-8)}{12}$$

Worked Solutions **Summation**

Q139 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k+4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+19n+8)}{12}$$

Q140 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k-1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions **Summation**

Q141 **Written**

Find the sum $\sum_{k=1}^{n} k(k^2-1)$**.**

Solution

1. Expand: $k(k^2-1) = k^3 - k$.
2. Use $\sum k^3$ and $\sum k$, then factor $n^2+n-2 = (n-1)(n+2)$.
3. $\sum_{k=1}^{n} k(k^2-1) = \dfrac{n(n+1)(n-1)(n+2)}{4}$

Final answer

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Q142 **Written**

Find the sum $\sum_{k=1}^{n} k(k^2+2)$.

Solution

1. Expand: $k(k^2+2) = k^3 + 2k$.
2. Use $\sum k^3$ and $\sum k$, then factor.
3. $\sum_{k=1}^{n} k(k^2+2) = \dfrac{n(n+1)(n^2+n+4)}{4}$

Final answer

$$\frac{n(n+1)(n^2+n+4)}{4}$$

Worked Solutions **Summation**

Q143 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2-1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Q144 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2+2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+4)}{4}$$

Worked Solutions **Summation**

Q145 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2+3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+6)}{4}$$

Q146 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2-2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-4)}{4}$$

Worked Solutions **Summation**

Q147 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2+4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+8)}{4}$$

Q148 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2-3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-6)}{4}$$

Worked Solutions **Summation**

Q149 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k+5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+23n+10)}{12}$$

Q150 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k-5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-17n-10)}{12}$$

Worked Solutions **Summation**

Q151 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2+5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+10)}{4}$$

Q152 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2-5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-10)}{4}$$

Worked Solutions **Summation**

Q153 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k+6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+27n+12)}{12}$$

Q154 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2+6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+12)}{4}$$

Worked Solutions **Summation**

Q155 *MCQ*

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k-6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-25n-12)}{12}$$

Q156 *MCQ*

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2-6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-12)}{4}$$

Worked Solutions **Summation**

Q157 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k+7)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+31n+14)}{12}$$

Q158 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k^2(k+8)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+35n+16)}{12}$$

Worked Solutions **Summation**

Q159 **Written**

Find the sum $\sum_{k=1}^{n} 6 \times 3^{k-1}$.

Solution

1. Use $a = 6, r = 3$.
2. $S_n = \frac{6(3^n-1)}{3-1} = 3(3^n - 1)$

Final answer

$3(3^n - 1)$

Q160 **Written**

Find the sum $\sum_{k=1}^{n} 4 \times 2^{k-1}$.

Solution

1. Use $a = 4, r = 2$.
2. $S_n = \frac{4(2^n-1)}{2-1} = 4(2^n - 1)$

Final answer

$4(2^n - 1)$

Worked Solutions **Summation**

Q161 **Written**

Find the sum $\sum_{k=1}^{n} 2^k$.

Solution

1. $Rewriteas(2 \times 2^{k-1})$.
2. $S_n = \frac{2(2^n-1)}{2-1} = 2(2^n - 1)$

Final answer

$2(2^n - 1)$

Q162 **Written**

Find the sum $\sum_{k=1}^{n} 2 \times (-3)^{k-1}$.

Solution

1. Here $a = 2$ and $r = -3$.
2. $S_n = \frac{2(1-(-3)^n)}{1-(-3)} = \frac{1-(-3)^n}{2}$

Final answer

$$\frac{1-(-3)^n}{2}$$

Worked Solutions **Summation**

Q163 Written

Find the sum $\sum_{k=1}^{n} 3 \times (-2)^{k-1}$.

Solution

1. Use $a = 3, r = -2$.
2. $S_n = \frac{3(1-(-2)^n)}{1-(-2)} = 1 - (-2)^n$

Final answer

$1 - (-2)^n$

Q164 Written

Find the sum $\sum_{k=1}^{n} (-5)^k$.

Solution

1. Rewrite as $(-5) \times (-5)^{k-1}$, so $a = -5, r = -5$.
2. $S_n = \frac{-5(1-(-5)^n)}{6} = \frac{5((-5)^n-1)}{6}$

Final answer

$$\frac{5\big((-5)^n - 1\big)}{6}$$

Worked Solutions **Summation**

Q165 **Written**

Find the sum $\sum_{k=1}^{n} 2 \times (-1)^{k-1}$.

Solution

1. Use $a = 2, r = -1$.
2. $S_n = \frac{2(1-(-1)^n)}{2} = 1 - (-1)^n$

Final answer

$1 - (-1)^n$

Q166 **Written**

Find the sum $\sum_{k=1}^{n} 8 \times (-3)^{k-1}$.

Solution

1. Use $a = 8, r = -3$.
2. $S_n = \frac{8(1-(-3)^n)}{4} = 2(1 - (-3)^n)$

Final answer

$2(1 - (-3)^n)$

Worked Solutions **Summation**

Q167 Written

Find the sum $\sum_{k=1}^{n}(-2)^{k-1}$.

Solution

1. Use $(a = 1, r = -2)$.
2. $S_n = \frac{1-(-2)^n}{1-(-2)} = \frac{1-(-2)^n}{3}$

Final answer

$$\frac{1-(-2)^n}{3}$$

Q168 Written

Find the sum $\sum_{k=1}^{n}(-3)^{k}$.

Solution

1. $Rewriteas((-3)(-3)^{k-1})$.
2. $S_n = \frac{-3(1-(-3)^n)}{4} = \frac{3((-3)^n-1)}{4}$

Final answer

$$\frac{3\left((-3)^n-1\right)}{4}$$

Worked Solutions **Summation**

Q169 Written

Find the sum $\sum_{k=1}^{n} 4^k$.

Solution

1. Rewrite $4^k = 4 \times 4^{k-1}$, so $a = 4, r = 4$.
2. $S_n = \frac{4(4^n-1)}{4-1} = \frac{4(4^n-1)}{3}$

Final answer

$$\frac{4(4^n - 1)}{3}$$

Q170 Written

Find the sum $\sum_{k=1}^{n} 4^{k-1}$.

Solution

1. The first term is 1 and the common ratio is 4.
2. $S_n = \frac{4^n-1}{4-1} = \frac{4^n-1}{3}$

Final answer

$$\frac{4^n - 1}{3}$$

Worked Solutions **Summation**

Q171 **Written**

Find the sum $\sum_{k=1}^{n} 3^{k-1}$.

Solution

1. The first term is 1 and the ratio is 3.
2. $S_n = \frac{3^n-1}{3-1} = \frac{3^n-1}{2}$

Final answer

$$\frac{3^n - 1}{2}$$

Q172 **Written**

Find the sum $\sum_{k=1}^{4} (-2)^{k-1}$.

Solution

1. This is a finite geometric sum with $(a = 1, r = -2, n = 4)$.
2. $S_4 = 1 - 2 + 4 - 8 = -5$

Final answer

-5

Worked Solutions **Summation**

Q173 Written

Find the sum $\sum_{k=1}^{3} 3^k$.

Solution

1. Use $(a = 3, r = 3, n = 3)$.
2. $S_3 = 3 + 9 + 27 = 39$

Final answer

39

Q174 Written

Find the sum $\sum_{k=1}^{5} 2^k$.

Solution

1. Use $(a = 2, r = 2, n = 5)$.
2. $S_5 = 2 + 4 + 8 + 16 + 32 = 62$

Final answer

62

Worked Solutions **Summation**

Q175 **Written**

Find the sum $\sum_{k=1}^{4}(-3)^k$.

Solution

1. Use $(a = -3, r = -3, n = 4)$.
2. $S_4 = -3 + 9 - 27 + 81 = 60$

Final answer

60

Worked Solutions **Summation**

Q176 **Challenge**

Find the finite sum $\sum_{r=5}^{9}(2r+7)$.

Solution

1. Substitute ($r = 5, 6, 7, 8, 9$): (17+19+21+23+25).
2. $17 + 19 + 21 + 23 + 25 = 105$

Final answer

105

Worked Solutions **Summation**

Q177 **Written**

Find $\dfrac{1}{2\times 3}+\dfrac{1}{3\times 4}+\dfrac{1}{4\times 5}+\cdots+\dfrac{1}{(n+1)(n+2)}$.

Solution

1. Split each term as $\frac{1}{j}-\frac{1}{j+1}$.
2. $\frac{1}{2}-\frac{1}{n+2}=\frac{n}{2(n+2)}$

Final answer

$$\frac{n}{2(n+2)}$$

Q178 **Written**

Find $\dfrac{1}{1\times 2}+\dfrac{1}{2\times 3}+\dfrac{1}{3\times 4}+\cdots+\dfrac{1}{n(n+1)}$.

Solution

1. Use $\frac{1}{j(j+1)}=\frac{1}{j}-\frac{1}{j+1}$.
2. $1-\frac{1}{n+1}=\frac{n}{n+1}$

Final answer

$$\frac{n}{n+1}$$

Worked Solutions **Summation**

Q179 Written

Find $\frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \cdots + \frac{1}{(n+2)(n+3)}$.

Solution

1. The gap is 1 and the first term starts at 3.
2. $\frac{1}{3} - \frac{1}{n+3} = \frac{n}{3(n+3)}$

Final answer

$$\frac{n}{3(n+3)}$$

Q180 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 2} + \cdots + \frac{1}{n(n+1)}$**:**

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{n+1}$$

Worked Solutions **Summation**

Q181 MCQ

Select the correct option: Find the sum $\frac{1}{2\times3} + \cdots + \frac{1}{(n+1)(n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(n+2)}$$

Q182 MCQ

Select the correct option: Find the sum $\frac{1}{3\times4} + \cdots + \frac{1}{(n+2)(n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(n+3)}$$

Worked Solutions **Summation**

Q183 MCQ

Select the correct option: Find the sum $\frac{1}{4\times5}+\cdots+\frac{1}{(n+3)(n+4)}$**:**

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(n+4)}$$

Q184 MCQ

Select the correct option: Find the sum $\frac{1}{6\times7}+\cdots+\frac{1}{(n+5)(n+6)}$**:**

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{6(n+6)}$$

Worked Solutions **Summation**

Q185 MCQ

Select the correct option: Find the sum $\frac{1}{7\times8}+\cdots+\frac{1}{(n+6)(n+7)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{7(n+7)}$$

Q186 MCQ

Select the correct option: Find the sum $\frac{1}{9\times10}+\cdots+\frac{1}{(n+8)(n+9)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{9(n+9)}$$

Worked Solutions **Summation**

Q187 MCQ

Select the correct option: Find the sum $\frac{1}{10\times 11} + \cdots + \frac{1}{(n+9)(n+10)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{10(n+10)}$$

Q188 MCQ

Select the correct option: Find the sum $\frac{1}{11\times 12} + \cdots + \frac{1}{(n+10)(n+11)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{11(n+11)}$$

Worked Solutions **Summation**

Q189 Written

Find the following sum: $\dfrac{1}{2\times 5}+\dfrac{1}{5\times 8}+\dfrac{1}{8\times 11}+\cdots+\dfrac{1}{(3n-1)(3n+2)}$.

Solution

1. Use $\frac{1}{ab}=\frac{1}{b-a}(\frac{1}{a}-\frac{1}{b})$ with gap 3.
2. $\frac{1}{3}(\frac{1}{2}-\frac{1}{3n+2})=\frac{n}{2(3n+2)}$

Final answer

$\dfrac{n}{2(3n+2)}$

Q190 Written

Find $\dfrac{1}{1\times 3}+\dfrac{1}{3\times 5}+\dfrac{1}{5\times 7}+\cdots+\dfrac{1}{(2n-1)(2n+1)}$.

Solution

1. The gap is 2, so factor out $\frac{1}{2}$.
2. $\frac{1}{2}(1-\frac{1}{2n+1})=\frac{n}{2n+1}$

Final answer

$\dfrac{n}{2n+1}$

Worked Solutions **Summation**

Q191 Written

Find $\frac{1}{1\times 3}+\frac{1}{2\times 4}+\frac{1}{3\times 5}+\cdots+\frac{1}{n(n+2)}$.

Solution

1. Use the constant gap 2 decomposition; two end terms remain.
2. $\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$

Final answer
$$\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$$

Q192 Written

Find $\frac{1}{1\times 4}+\frac{1}{4\times 7}+\frac{1}{7\times 10}+\cdots+\frac{1}{(3n-2)(3n+1)}$.

Solution

1. Use gap 3 and the first denominator 1.
2. $\frac{1}{3}(1-\frac{1}{3n+1})=\frac{n}{3n+1}$

Final answer
$$\frac{n}{3n+1}$$

Worked Solutions **Summation**

Q193 Written

Find $\dfrac{1}{1\times 7}+\dfrac{1}{4\times 10}+\dfrac{1}{7\times 13}+\cdots+\dfrac{1}{(3n-2)(3n+4)}$.

Solution

1. The two factors differ by 6, while the first factors advance by 3; two terms remain at each end.
2. $\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$

Final answer

$\dfrac{1}{6}(1+\dfrac{1}{4}-\dfrac{1}{3n+1}-\dfrac{1}{3n+4})$

Q194 MCQ

Select the correct option: Find the sum $\frac{1}{1\times 3}+\cdots+\frac{1}{(2n-1)(2n+1)}$**:**

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$\dfrac{n}{2n+1}$

Worked Solutions **Summation**

Q195 MCQ

Select the correct option: Find the sum $\frac{1}{1\times4}+\cdots+\frac{1}{(3n-2)(3n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3n+1}$$

Q196 MCQ

Select the correct option: Find the sum $\frac{1}{1\times7}+\cdots+\frac{1}{(3n-2)(3n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$$

Worked Solutions **Summation**

Q197 MCQ

Select the correct option: Find the sum $\frac{1}{2\times6}+\cdots+\frac{1}{(4n-2)(4n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(4n+2)}$$

Q198 MCQ

Select the correct option: Find the sum $\frac{1}{3\times8}+\cdots+\frac{1}{(5n-2)(5n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(5n+3)}$$

Worked Solutions **Summation**

Q199 MCQ

Select the correct option: Find the sum $\frac{1}{1\times5}+\cdots+\frac{1}{(4n-3)(4n+1)}$**:**

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4n+1}$$

Q200 MCQ

Select the correct option: Find the sum $\frac{1}{2\times7}+\cdots+\frac{1}{(5n-3)(5n+2)}$**:**

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(5n+2)}$$

Worked Solutions **Summation**

Q201 MCQ

Select the correct option: Find the sum $\frac{1}{4\times10}+\cdots+\frac{1}{(6n-2)(6n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(6n+4)}$$

Q202 Written

Find $\frac{1}{4\times6}+\frac{1}{5\times7}+\frac{1}{6\times8}+\cdots+\frac{1}{(n+3)(n+5)}$.

Solution

1. Write the general term as $\frac{1}{(j+3)(j+5)}$, gap 2.

2. $\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$

Final answer

$$\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$$

Worked Solutions **Summation**

Q203 Written

Find $\dfrac{1}{2\times 4}+\dfrac{1}{4\times 6}+\dfrac{1}{6\times 8}+\cdots+\dfrac{1}{2n(2n+2)}$.

Solution

1. Factor each denominator as $4j(j+1)$.
2. $\frac{1}{4}(1-\frac{1}{n+1})=\frac{n}{4(n+1)}$

Final answer

$$\frac{n}{4(n+1)}$$

Q204 Written

Find the following sum: $\displaystyle\sum_{k=1}^{n}\frac{1}{\sqrt{k+2}+\sqrt{k+1}}$.

Solution

1. Multiply by the conjugate.
2. $\frac{\sqrt{k+2}-\sqrt{k+1}}{(k+2)-(k+1)}=\sqrt{k+2}-\sqrt{k+1}$
3. $\sum_{k=1}^{n}(\sqrt{k+2}-\sqrt{k+1})=\sqrt{n+2}-\sqrt{2}$

Final answer

$$\sqrt{n+2}-\sqrt{2}$$

Worked Solutions **Summation**

Q205 Written

Find $\sum_{k=1}^{n} \frac{1}{\sqrt{k+1}+\sqrt{k}}$.

Solution

1. Rationalise with $\sqrt{k+1}-\sqrt{k}$.
2. $\sum_{k=1}^{n}(\sqrt{k+1}-\sqrt{k}) = \sqrt{n+1}-1$

Final answer

$\sqrt{n+1}-1$

Q206 Written

Find $\sum_{k=1}^{n} \frac{1}{\sqrt{k}+\sqrt{k-1}}$.

Solution

1. Rationalise: the denominator difference is 1.
2. $\sum_{k=1}^{n}(\sqrt{k}-\sqrt{k-1}) = \sqrt{n}$

Final answer

$\sqrt{n}$

Worked Solutions **Summation**

Q207 Written

Find $\dfrac{1}{\sqrt{1}+\sqrt{2}} + \dfrac{1}{\sqrt{2}+\sqrt{3}} + \cdots + \dfrac{1}{\sqrt{n}+\sqrt{n+1}}$.

Solution

1. Use the conjugate to obtain $\sqrt{j+1}-\sqrt{j}$.
2. $\sum_{j=1}^{n}(\sqrt{j+1}-\sqrt{j}) = \sqrt{n+1}-1$

Final answer

$\sqrt{n+1}-1$

Q208 Written

Find $\displaystyle\sum_{k=1}^{n} \frac{1}{\sqrt{2k-1}+\sqrt{2k+1}}$.

Solution

1. Rationalise; the square-difference gap is 2.
2. $\frac{1}{2}\sum_{k=1}^{n}(\sqrt{2k+1}-\sqrt{2k-1}) = \frac{\sqrt{2n+1}-1}{2}$

Final answer

$$\frac{\sqrt{2n+1}-1}{2}$$

Worked Solutions **Summation**

Q209 Written

Find $\dfrac{1}{\sqrt{2}+\sqrt{3}} + \dfrac{1}{\sqrt{3}+\sqrt{4}} + \cdots + \dfrac{1}{\sqrt{n+1}+\sqrt{n+2}}$.

Solution

1. Multiply by each conjugate.
2. $\sum_{j=1}^{n}(\sqrt{j+2}-\sqrt{j+1}) = \sqrt{n+2}-\sqrt{2}$

Final answer

$\sqrt{n+2}-\sqrt{2}$

Worked Solutions **Summation**

Q210 **Challenge**

Find the sum, showing your steps: $\sum_{r=1}^{30} \frac{1}{r(r+2)}$.

Solution

1. Use $\frac{1}{r(r+2)} = \frac{1}{2}(\frac{1}{r} - \frac{1}{r+2})$.
2. $S = \frac{1}{2}(1 + \frac{1}{2} - \frac{1}{31} - \frac{1}{32})$
3. $S = \frac{1425}{1984}$

Final answer

$\frac{1425}{1984}$

Worked Solutions **Summation**

Q211 **Written**

Find the following sum $S = 3 \times 1 + 5 \times 3 + 7 \times 3^2 + 9 \times 3^3 + \cdots + (2n+1) \times 3^{n-1}$**.**

Solution

1. The arithmetic coefficient is $2k+1$ and the geometric ratio is $r = 3$.
2. Write S and $3S$, subtract with aligned powers, and simplify: $-2S = -2n \times 3^n$.
3. $S = n \times 3^n$

Final answer

$n \times 3^n$

Q212 **Written**

Find the sum $\sum_{k=1}^{n}(2k+1) \times (3)^{k-1}$**.**

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $3^n n$

Final answer

$3^n n$

Worked Solutions **Summation**

Q213 Written

Find the sum $\sum_{k=1}^{4}(2k+1)\times(3)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 4$).
3. 324

Final answer

324

Q214 Written

Find the sum $\sum_{k=1}^{n}(2k-1)\times(4)^{k-1}$.

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $\frac{6\,2^{2n}n-5\,2^{2n}+5}{9}$

Final answer

$$\frac{6\,2^{2n}n-5\,2^{2n}+5}{9}$$

Worked Solutions **Summation**

Q215 Written

Find the sum $\sum_{k=1}^{6}(2k-1)\times(4)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 6$).
3. 14109

Final answer

14109

Q216 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n}(2k+1)\times(3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$3^n n$

Worked Solutions **Summation**

Q217 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n}(2k-1)\times(4)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{6\,2^{2n}n-5\,2^{2n}+5}{9}$$

Q218 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n}(k+4)\times(2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$2^n n+3\,2^n-3$

Worked Solutions **Summation**

Q219 Written

Find the following sum $S = 2 \times 1 + 3 \times 3 + 4 \times 3^2 + 5 \times 3^3 + \cdots + (n+1) \times 3^{n-1}$.

Solution

1. Here the coefficient is $k+1$ and $r = 3$.
2. Subtract $3S$ from S; the interior coefficients reduce to a geometric sum of 1s.
3. $-2S = \frac{1-(2n+1)\times 3^n}{2}$
4. $S = \frac{(2n+1)\times 3^n - 1}{4}$

Final answer

$$\frac{(2n+1) \times 3^n - 1}{4}$$

Q220 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} (k+1) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2\,3^n n + 3^n - 1}{4}$$

Worked Solutions **Summation**

Q221 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n}(k-2)\times(3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2\,3^{n}n-5\,3^{n}+5}{4}$$

Q222 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n}(2k-3)\times(3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$3^{n}n-2\,3^{n}+2$

Worked Solutions **Summation**

Q223 **MCQ**

Select the correct option: Find the sum $\sum_{k=1}^{n}(2k+1)\times(-2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{6(-2)^n n+5(-2)^n-5}{9}$$

Worked Solutions **Summation**

Q224 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n}(k+2)\times(-3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{4\,(-3)^{n}\,n+9\,(-3)^{n}-9}{16}$$

Worked Solutions **Summation**

Q225 Written

Find the following sum $S = 1 \times 1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \cdots + n \times 2^{n-1}$.

Solution

1. Use $a_k = k$ and $r = 2$.
2. Applying $S - rS$ gives $-S = 1 - (n-1) \times 2^n$.
3. $S = (n-1) \times 2^n + 1$

Final answer

$(n-1) \times 2^n + 1$

Q226 Written

Find the following sum $S = 1 \times 3 + 3 \times 3^2 + 5 \times 3^3 + 7 \times 3^4 + \cdots + (2n-1) \times 3^n$.

Solution

1. Factor out the leading 3, leaving $\sum_{k=1}^{n}(2k-1)3^{k-1}$.
2. The $S - rS$ subtraction gives $\sum_{k=1}^{n}(2k-1)3^{k-1} = (n-1)3^n + 1$.
3. $S = (n-1) \times 3^{n+1} + 3$

Final answer

$(n-1) \times 3^{n+1} + 3$

Worked Solutions **Summation**

Q227 MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n}(3k)\times(3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{3\left(2\,3^{n}n-3^{n}+1\right)}{4}$$

Worked Solutions **Summation**

Q228 **Challenge**

Find the sum of the partial sums $5 + (5 + 9) + (5 + 9 + 13) + \cdots + [5 + 9 + \cdots + (4k + 1)]$**.**

Solution

1. The j-th inner arithmetic sum is $5 + 9 + \cdots + (4j + 1) = j(2j + 3)$.
2. Sum the partial sums: $\sum_{j=1}^{k}(2j^2 + 3j)$.
3. $\frac{k(k+1)(4k+11)}{6}$

Final answer

$$\frac{k(k + 1)(4k + 11)}{6}$$

Worked Solutions **Summation**

Q229 Written

For the sequence $3,\ 3+5,\ 3+5+7,\ 3+5+7+9, \ldots$, **find the** k**-th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 2.
2. $u_i = 2i + 1, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}\left[2(3) + (k-1)(2)\right] = k\,(k+2)$

Final answer

$a_k = k(k+2)$

Q230 Written

For the sequence $1,\ 1+3,\ 1+3+5,\ 1+3+5+7, \ldots$, **find the** k**-th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 2.
2. $u_i = 2i - 1, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}\left[2(1) + (k-1)(2)\right] = k^2$

Final answer

$a_k = k^2$

Worked Solutions **Summation**

Q231 Written

For the sequence $2,\ 2+4,\ 2+4+6,\ 2+4+6+8, \ldots$, find the k-th term.

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.
2. $u_i = 2i, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}[2(2) + (k-1)(2)] = k\,(k+1)$

Final answer

$a_k = k(k+1)$

Q232 Written

For the sequence $1,\ 1+5,\ 1+5+9,\ 1+5+9+13, \ldots$, find the k-th term.

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 4.
2. $u_i = 4i - 3, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}[2(1) + (k-1)(4)] = k\,(2k-1)$

Final answer

$a_k = k(2k-1)$

Worked Solutions **Summation**

Q233 **MCQ**

Select the correct option: For the cumulative sequence $2,\ 2+4,\ 2+4+6,\ 2+4+6+8,\ \ldots$**, what is** a_k**:**

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.
2. $u_i = 2i, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}\left[2(2) + (k-1)(2)\right] = k\,(k+1)$

Correct option: A

Final answer

$k\,(k+1)$

Q234 **MCQ**

Select the correct option: For the cumulative sequence $3,\ 3+6,\ 3+6+9,\ 3+6+9+12,\ \ldots$**, what is** a_k**:**

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 3.
2. $u_i = 3i, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}\left[2(3) + (k-1)(3)\right] = \frac{3k(k+1)}{2}$

Correct option: A

Final answer

$$\frac{3k\,(k+1)}{2}$$

Worked Solutions **Summation**

Q235 MCQ

Select the correct option: For the cumulative sequence $4,\ 4+6,\ 4+6+8,\ 4+6+8+10,\ \ldots$, **what is** a_k**:**

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 2.
2. $u_i = 2i + 2, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}[2(4) + (k-1)(2)] = k(k+3)$

Correct option: A

Final answer

$k(k+3)$

Q236 MCQ

Select the correct option: For the cumulative sequence $2,\ 2+6,\ 2+6+10,\ 2+6+10+14,\ \ldots$, **what is** a_k**:**

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 4.
2. $u_i = 4i - 2, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}[2(2) + (k-1)(4)] = 2k^2$

Correct option: A

Final answer

$2k^2$

Worked Solutions **Summation**

Q237 MCQ

Select the correct option: For the cumulative sequence $5,\ 5+8,\ 5+8+11,\ 5+8+11+14,\ \ldots$**, what is** a_k**:**

Solution

1. The inner terms form an arithmetic sequence with first term 5 and common difference 3.
2. $u_i = 3i + 2, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}\left[2(5) + (k-1)(3)\right] = \frac{k(3k+7)}{2}$

Correct option: A

Final answer

$$\frac{k\,(3k+7)}{2}$$

Q238 MCQ

Select the correct option: For the cumulative sequence $3,\ 3+8,\ 3+8+13,\ 3+8+13+18,\ \ldots$**, what is** a_k**:**

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 5.
2. $u_i = 5i - 2, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}\left[2(3) + (k-1)(5)\right] = \frac{k(5k+1)}{2}$

Correct option: A

Final answer

$$\frac{k\,(5k+1)}{2}$$

Worked Solutions **Summation**

Q239 MCQ

Select the correct option: For the cumulative sequence $6,\ 6+8,\ 6+8+10,\ 6+8+10+12,\ \ldots$, **what is** a_k**:**

Solution

1. The inner terms form an arithmetic sequence with first term 6 and common difference 2.
2. $u_i = 2i + 4, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}[2(6) + (k-1)(2)] = k(k+5)$

Correct option: A

Final answer

$k(k+5)$

Q240 MCQ

Select the correct option: For the cumulative sequence $4,\ 4+8,\ 4+8+12,\ 4+8+12+16,\ \ldots$, **what is** a_k**:**

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 4.
2. $u_i = 4i, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}[2(4) + (k-1)(4)] = 2k(k+1)$

Correct option: A

Final answer

$2k(k+1)$

Worked Solutions **Summation**

Q241 Written

For the sequence $3,\ 3+5,\ 3+5+7,\ 3+5+7+9, \ldots$, find the sum from the first term to the n-th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k\,(k+2)$.
2. $S_n = \sum_{k=1}^{n} a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+7)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+7)}{6}$$

Q242 Written

For the sequence $1,\ 1+3,\ 1+3+5,\ 1+3+5+7, \ldots$, find the sum from the first term to the n-th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k^2$.
2. $S_n = \sum_{k=1}^{n} a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+1)}{6}$$

Worked Solutions **Summation**

Q243 **Written**

For the sequence $1,\ 1+5,\ 1+5+9,\ 1+5+9+13, \ldots$, **find the sum from the first term to the** n**-th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k\,(2k-1)$.
2. $S_n = \sum_{k=1}^{n} a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(4n-1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(4n-1)}{6}$$

Q244 **Written**

For the sequence $2,\ 2+4,\ 2+4+6,\ 2+4+6+8, \ldots$, **find the sum from the first term to the** n**-th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k\,(k+1)$.
2. $S_n = \sum_{k=1}^{n} a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(n+2)}{3}$

Final answer

$$S_n = \frac{n(n+1)(n+2)}{3}$$

Worked Solutions **Summation**

Q245 MCQ

Select the correct option: For the cumulative sequence $2,\ 2+5,\ 2+5+8,\ 2+5+8+11,\ \ldots$, **find** a_5:

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 3.
2. $u_i = 3i - 1, \qquad a_k = \sum_{i=1}^{k} u_i$
3. $a_k = \frac{k}{2}[2(2) + (k-1)(3)] = \frac{k(3k+1)}{2}$
4. $a_5 = 40$

Correct option: A

Final answer

40

Worked Solutions **Summation**

Q246 Challenge

The sum of the first n terms of $\{a_n\}$ is $S_n = 2n^2 + 5n$. Find the general term.

Solution

1. $a_1 = S_1 = 7$
2. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
3. $a_n = (2n^2 + 5n) - [2(n-1)^2 + 5(n-1)] = 4n + 3$
4. The formula also gives $a_1 = 7$, so it is valid for every positive integer n.

Final answer

$a_n = 4n + 3$

Worked Solutions **Summation**

Q247 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = n(n-3)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-4)(n-1)$
3. $a_n = 2(n-2)$
4. $a_1 = S_1 = -2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Final answer

$a_n = 2(n-2)$

Worked Solutions **Summation**

Q248 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = n(2n+5)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+3)$
3. $a_n = 4n + 3$
4. $a_1 = S_1 = 7$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Final answer

$a_n = 4n + 3$

Worked Solutions **Summation**

Q249 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = n^2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)^2$
3. $a_n = 2n - 1$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Final answer

$a_n = 2n - 1$

Worked Solutions **Summation**

Q250 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = n(n-4)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-5)(n-1)$
3. $a_n = 2n - 5$
4. $a_1 = S_1 = -3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Final answer

$a_n = 2n - 5$

Worked Solutions **Summation**

Q251 MCQ

Select the correct option: If $S_n = n(n+2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n+1)$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: D

Final answer

$a_n = 2n + 1$

Worked Solutions **Summation**

Q252 MCQ

Select the correct option: If $S_n = 2n(n+2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(n+1)$
3. $a_n = 2(2n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: C

Final answer

$a_n = 2(2n+1)$

Worked Solutions **Summation**

Q253 MCQ

Select the correct option: If $S_n = 3n(n-1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3(n-2)(n-1)$
3. $a_n = 6(n-1)$
4. $a_1 = S_1 = 0$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: B

Final answer

$a_n = 6(n-1)$

Worked Solutions **Summation**

Q254 MCQ

Select the correct option: If $S_n = 2n(2n-1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(2n-3)$
3. $a_n = 2(4n-3)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: A

Final answer

$a_n = 2(4n-3)$

Worked Solutions **Summation**

Q255 MCQ

Select the correct option: If $S_n = 5n(n+1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 5n(n-1)$
3. $a_n = 10n$
4. $a_1 = S_1 = 10$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: D

Final answer

$a_n = 10n$

Worked Solutions **Summation**

Q256 MCQ

Select the correct option: If $S_n = n(2n+3)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+1)$
3. $a_n = 4n+1$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: C

Final answer

$a_n = 4n+1$

Worked Solutions **Summation**

Q257 MCQ

Select the correct option: If $S_n = n(3n+1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(3n-2)$
3. $a_n = 2(3n-1)$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: C

Final answer

$a_n = 2(3n-1)$

Worked Solutions **Summation**

Q258 MCQ

Select the correct option: If $S_n = n\left(n^2 + 2n + 3\right)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)\left(n^2 + 2\right)$
3. $a_n = 3n^2 + n + 2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: C

Final answer

$a_n = 3n^2 + n + 2$

Worked Solutions **Summation**

Q259 **MCQ**

Select the correct option: If $S_n = n(n+1)(2n-1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-3)$
3. $a_n = 2n(3n-2)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: B

Final answer

$a_n = 2n(3n-2)$

Worked Solutions **Summation**

Q260 **MCQ**

Select the correct option: If $S_n = n\left(3n^2 - 2n + 4\right)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)\left(3n^2 - 8n + 9\right)$
3. $a_n = 9n^2 - 13n + 9$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: A

Final answer

$a_n = 9n^2 - 13n + 9$

Worked Solutions **Summation**

Q261 MCQ

Select the correct option: If $S_n = n\left(n^2 - n + 2\right)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)\left(n^2 - 3n + 4\right)$
3. $a_n = 3n^2 - 5n + 4$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: D

Final answer

$a_n = 3n^2 - 5n + 4$

Worked Solutions **Summation**

Q262 MCQ

Select the correct option: If $S_n = n(n+1)(2n+1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-1)$
3. $a_n = 6n^2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: D

Final answer

$a_n = 6n^2$

Worked Solutions **Summation**

Q263 **MCQ**

Select the correct option: If $S_n = n(n+1)(n+2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(n+1)$
3. $a_n = 3n(n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Correct option: B

Final answer

$a_n = 3n(n+1)$

Worked Solutions **Summation**

Q264 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = n^3 + 2n + 6$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^3 - 3n^2 + 5n + 3$
3. $a_n = 3\left(n^2 - n + 1\right)$
4. $a_1 = S_1 = 9$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$a_1 = 9, \quad a_n = 3\left(n^2 - n + 1\right) \ (n \geq 2)$

Worked Solutions **Summation**

Q265 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = n^2 + 2n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^2 - 2$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 2$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$a_1 = 2, \quad a_n = 2n + 1 \ (n \geq 2)$

Worked Solutions **Summation**

Q266 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = (n-1)\left(n^2+n+1\right)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-2)\left(n^2-n+1\right)$
3. $a_n = 3n^2 - 3n + 1$
4. $a_1 = S_1 = 0$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \ (n \geq 2)$

Worked Solutions **Summation**

Q267 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = 5^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 5}{5}$
3. $a_n = 4\,5^{n-1}$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Final answer

$a_n = 4\,5^{n-1}$

Worked Solutions **Summation**

Q268 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = 3^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{3^n - 3}{3}$
3. $a_n = 2\,3^{n-1}$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Final answer

$a_n = 2\,3^{n-1}$

Worked Solutions **Summation**

Q269 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = 2^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n - 2}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Final answer

$a_n = 2^{n-1}$

Worked Solutions **Summation**

Q270 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = 7^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{7^n - 7}{7}$
3. $a_n = 6\,7^{n-1}$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.

Final answer

$a_n = 6\,7^{n-1}$

Worked Solutions **Summation**

Q271 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = 5^n - 2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 10}{5}$
3. $a_n = 4\,5^{n-1}$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$a_1 = 3, \quad a_n = 4\,5^{n-1}\ (n \geq 2)$

Worked Solutions **Summation**

Q272 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n-th term is given by $S_n = 4\,2^n - 3$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2\,2^n - 3$
3. $a_n = 2^{n+1}$
4. $a_1 = S_1 = 5$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$a_1 = 5, \quad a_n = 2^{n+1}\ (n \geq 2)$

Worked Solutions **Summation**

Q273 MCQ

Select the correct option: If $S_n = n(n+4)$, find a_5:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n+3)$
3. $a_n = 2n + 3$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n.
6. $a_5 = 13$

Correct option: B

Final answer

13

Worked Solutions **Summation**

Q274 **Challenge**

Find the general term of the sequence $\{a_n\}$ where S_n is given by $S_n = 3\,3^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3^n - 2n + 3$
3. $a_n = 2\,(3^n - 1)$
4. $a_1 = S_1 = 8$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$a_1 = 8, \quad a_n = 2\,(3^n - 1)\ \ (n \geq 2)$

Worked Solutions **Summation**

Quiz MCQ 1 Concept Quiz · MCQ

Select the correct option: Expand $\sum_{k=1}^{4}(3k-1)$ into a sum of terms:

Solution

1. Substitute $k = 1, 2, 3, 4$.

Correct option: A

Final answer

$2 + 5 + 8 + 11$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: Find the sum $\sum_{k=1}^{n} k(k^2-1)$:

Solution

1. $Expand(k(k^2-1) = k^3 - k)$.

Correct option: A

Final answer

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Worked Solutions **Summation**

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: Find the sum $\sum_{k=1}^{n} k \times 2^{k-1}$:

Solution

1. Write S and rS, subtract, and isolate S.

Correct option: A

Final answer

$(n-1) \times 2^n + 1$

Worked Solutions **Summation**

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: If $S_n = 2^n - 3$, find a_6:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n-6}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = -1$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.
6. $a_6 = 32$

Correct option: C

Final answer

32

Worked Solutions **Summation**

Quiz WRITTEN 5 Concept Quiz · Written

Expand $\sum_{k=1}^{5}(k^2-2)$ into a sum of terms.

Solution

1. Substitute $k=1,2,3,4,5$.

Final answer

$-1+2+7+14+23$

Quiz WRITTEN 6 Concept Quiz · Written

Find the sum $\sum_{k=1}^{n} k^2(k+5)$.

Solution

1. $Expandinto (k^3+5k^2)$.
2. Use the standard power sums and factor.

Final answer

$\frac{n(n+1)(3n^2+23n+10)}{12}$

Worked Solutions **Summation**

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the sum $\sum_{k=1}^{n}(4k-1)\times 3^k$**.**

Solution

1. Show the S-rS alignment (or the inner arithmetic sum for the challenge).
2. $\frac{3(4\,3^n n-3\,3^n+3)}{2}$

Final answer

$$\frac{3\,(4\,3^n n-3\,3^n+3)}{2}$$

Worked Solutions **Summation**

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term of the sequence $\{a_n\}$ when S_n is $2\,2^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2^n - 2n + 3$
3. $a_n = 2^n - 2$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$a_1 = 3, \quad a_n = 2^n - 2\ (n \geq 2)$

Answer key

Use the question number shown on each page.

Q001 $2+4+6+8+10$
Q002 $3+5+7+9+11$
Q003 $3+6+9+12+15$
Q004 $2+6+10+14$
Q005 $-1+1+3+5+7$
Q006 A
Q007 A
Q008 A
Q009 A
Q010 A
Q011 A
Q012 A
Q013 A
Q014 $-2+1+6$
Q015 $5+14+29$
Q016 $1+4+9+16$
Q017 $5+11+21$
Q018 $-1+0+3+8$
Q019 A
Q020 A
Q021 A
Q022 A
Q023 A
Q024 A
Q025 A
Q026 A
Q027 $3+9+27+81$
Q028 $2+2+2$
Q029 $2+4+8+16$
Q030 $3+3+3+3+3+3$
Q031 $3+9+27$
Q032 $8+28+128$
Q033 $-1+1+5+13$
Q034 $5+5+5+5$
Q035 $3+3+3$
Q036 $4+4+4+4+4$
Q037 A
Q038 A
Q039 A
Q040 A
Q041 A
Q042 A
Q043 A
Q044 A
Q045 $\sum_{k=1}^{n}(4k-1)$
Q046 $\sum_{k=1}^{n}(3k+2)$
Q047 $\sum_{k=1}^{n}(2k+1)$
Q048 $\sum_{k=1}^{n}(2k-1)$
Q049 $\sum_{k=1}^{n}(7k-1)$
Q050 $\sum_{k=1}^{n}(4k+5)$
Q051 $\sum_{k=1}^{n}(3k-2)$
Q052 A
Q053 A
Q054 A
Q055 A
Q056 A
Q057 A
Q058 A
Q059 A
Q060 $\sum_{k=6}^{15}k^2$
Q061 $\sum_{k=4}^{13}k^2$
Q062 $\sum_{k=1}^{n}(3^k-2)$
Q063 $\sum_{k=1}^{n}2^{k-1}$
Q064 $\sum_{k=7}^{21}k^2$
Q065 $\sum_{k=12}^{18}k^2$
Q066 $\sum_{k=10}^{20}k^2$
Q067 A
Q068 A
Q069 A
Q070 A
Q071 A
Q072 A
Q073 A
Q074 $\sum_{k=1}^{10}(3k-1)$
Q075 $\sum_{k=1}^{7}8k$
Q076 $\sum_{k=1}^{12}(6k+1)$
Q077 $\sum_{k=1}^{24}(5k-1)$
Q078 $\sum_{k=1}^{n}(3k-2)$
Q079 $-3n$
Q080 $3n$
Q081 $6n$
Q082 $-4n$
Q083 n
Q084 A
Q085 A
Q086 A
Q087 A
Q088 A

Q089 A

Q090 A

Q091 A

Q092 A

Q093 A

Q094 A

Q095 A

Q096 A

Q097 A

Q098 A

Q099 A

Q100 $\frac{n(5n+13)}{2}$

Q101 $n(n-2)$

Q102 $n(2n+1)$

Q103 $\frac{n(3n+7)}{2}$

Q104 $3n^2$

Q105 A

Q106 A

Q107 A

Q108 A

Q109 $\frac{n(n-1)(2n-1)}{6}$

Q110 A

Q111 A

Q112 A

Q113 A

Q114 A

Q115 $\frac{n(n+1)(10n-1)}{6}$

Q116 $\frac{n(n-1)(2n+1)}{2}$

Q117 $\frac{n(n+1)(4n+5)}{6}$

Q118 $\frac{n(n+1)(2n-5)}{6}$

Q119 $\frac{2n(n+1)(n-1)}{3}$

Q120 $\frac{n(n+1)(n+5)}{3}$

Q121 $\frac{n(n+1)(2n+3)}{2}$

Q122 $\frac{n(n+4)(n+5)}{3}$

Q123 16

Q124 35

Q125 280

Q126 184

Q127 100

Q128 245

Q129 $\frac{n(n+1)(n+2)(3n+1)}{12}$

Q130 $\frac{n(n+1)(3n^2-5n-4)}{12}$

Q131 $\frac{n(n+1)(3n^2+11n+4)}{12}$

Q132 $\frac{n(n+1)(n^2-3n-2)}{4}$

Q133 A

Q134 A

Q135 A

Q136 A

Q137 A

Q138 A

Q139 A

Q140 A

Q141 $\frac{n(n+1)(n-1)(n+2)}{4}$

Q142 $\frac{n(n+1)(n^2+n+4)}{4}$

Q143 A

Q144 A

Q145 A

Q146 A

Q147 A

Q148 A

Q149 A

Q150 A

Q151 A

Q152 A

Q153 A

Q154 A

Q155 A

Q156 A

Q157 A

Q158 A

Q159 $3(3^n-1)$

Q160 $4(2^n-1)$

Q161 $2(2^n-1)$

Q162 $\frac{1-(-3)^n}{2}$

Q163 $1-(-2)^n$

Q164 $\frac{5((-5)^n-1)}{6}$

Q165 $1-(-1)^n$

Q166 $2(1-(-3)^n)$

Q167 $\frac{1-(-2)^n}{3}$

Q168 $\frac{3((-3)^n-1)}{4}$

Q169 $\frac{4(4^n-1)}{3}$

Q170 $\frac{4^n-1}{3}$

Q171 $\frac{3^n-1}{2}$

Q172 -5

Q173 39

Q174 62

Q175 60

Q176 105

Q177 $\frac{n}{2(n+2)}$

Q178 $\frac{n}{n+1}$

Q179 $\frac{n}{3(n+3)}$

Q180 A

Q181 A

Q182 A

Q183 A

Q184 A

Q185 A

Q186 A

Q187 A

Q188 A

Q189 $\frac{n}{2(3n+2)}$

Q190 $\frac{n}{2n+1}$

Q191 $\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$

Q192 $\frac{n}{3n+1}$

Q193 $\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$

Q194 A

Q195 A

Q196 A

Q197 A

Q198 A

Q199 A

Q200 A

Q201 A

Q202 $\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$

Q203 $\frac{n}{4(n+1)}$

Q204 $\sqrt{n+2}-\sqrt{2}$

Q205 $\sqrt{n+1}-1$

Q206 $\sqrt{n}$

Q207 $\sqrt{n+1}-1$

Q208 $\frac{\sqrt{2n+1}-1}{2}$

Q209 $\sqrt{n+2}-\sqrt{2}$

Q210 $\frac{1425}{1984}$

Q211 $n\times 3^n$

Q212 $3^n n$

Q213 324

Q214 $\frac{6\,2^{2n}n-5\,2^{2n}+5}{9}$

Q215 14109

Q216 A

Q217 A

Q218 A

Q219 $\frac{(2n+1)\times 3^n-1}{4}$

Q220 A

Q221 A

Q222 A

Q223 A

Q224 A

Q225 $(n-1)\times 2^n+1$

Q226 $(n-1)\times 3^{n+1}+3$

Q227 A

Q228 $\frac{k(k+1)(4k+11)}{6}$

Q229 $a_k=k(k+2)$

Q230 $a_k=k^2$

Q231 $a_k=k(k+1)$

Q232 $a_k=k(2k-1)$

Q233 A

Q234 A

Q235 A

Q236 A

Q237 A

Q238 A

Q239 A

Q240 A

Q241 $S_n=\frac{n(n+1)(2n+7)}{6}$

Q242 $S_n=\frac{n(n+1)(2n+1)}{6}$

Q243 $S_n=\frac{n(n+1)(4n-1)}{6}$

Q244 $S_n=\frac{n(n+1)(n+2)}{3}$

Q245 A

Q246 $a_n=4n+3$

Q247 $a_n=2(n-2)$

Q248 $a_n=4n+3$

Q249 $a_n=2n-1$

Q250 $a_n=2n-5$

Q251 D

Q252 C

Q253 B

Q254 A

Q255 D

Q256 C

Q257 C

Q258 C

Q259 B

Q260 A

Q261 D

Q262 D

Q263 B

Q264 $a_1=9,\quad a_n=3\left(n^2-n+1\right)\ (n\geq 2)$

Q265 $a_1=2,\quad a_n=2n+1\ (n\geq 2)$

Q266 $a_1=0,\quad a_n=3n^2-3n+1\ (n\geq 2)$

Q267 $a_n=4\,5^{n-1}$

Q268 $a_n=2\,3^{n-1}$

Q269 $a_n=2^{n-1}$

Q270 $a_n=6\,7^{n-1}$

Q271 $a_1=3,\quad a_n=4\,5^{n-1}\ (n\geq 2)$

Q272 $a_1=5,\quad a_n=2^{n+1}\ (n\geq 2)$

Q273 B

Q274 $a_1=8,\quad a_n=2(3^n-1)\ (n\geq 2)$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 C

Quiz WRITTEN 5 $-1+2+7+14+23$

Quiz WRITTEN 6 $\frac{n(n+1)(3n^2+23n+10)}{12}$

Quiz WRITTEN 7 $\frac{3(4\,3^n n-3\,3^n+3)}{2}$

Quiz WRITTEN 8 $a_1=3,\quad a_n=2^n-2\ (n\geq 2)$

Concept 4 | Advanced Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Differences

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

odd/even index $\longrightarrow$ separate rules

Eng Eslam Ahmed | 01120009622

Worked Solutions **Advanced Sequences**

Q001 Written

Find the general term of the sequence $\{2, 5, 12, 23, 38, 57, \ldots\}$.

Solution

1. The first differences are $3, 7, 11, 15, 19, \ldots$.
2. Hence $b_k = 4k - 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1}(4k - 1) = 2n^2 - 3n + 3$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$a_n = 2n^2 - 3n + 3$

Worked Solutions **Advanced Sequences**

Q002 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: C

Final answer

$a_n = n^2$

Q003 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k - 2) = \frac{3n^2-7n+8}{2}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Correct option: B

Final answer

$$a_n = \frac{3n^2 - 7n + 8}{2}$$

Worked Solutions **Advanced Sequences**

Q004 **MCQ**

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, then a_n is:

Solution

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1}(6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.

Correct option: A

Final answer

$a_n = 3n^2 - 8n + 12$

Worked Solutions **Advanced Sequences**

Q005 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, find a_5:

Solution

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_5 = 25$

Correct option: B

Final answer

25

Worked Solutions **Advanced Sequences**

Q006 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, find a_6:

Solution

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k - 2) = \frac{3n^2-7n+8}{2}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.
5. $a_6 = 37$

Correct option: A

Final answer

37

Worked Solutions **Advanced Sequences**

Q007 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, find a_6:

Solution

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1}(6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.
5. $a_6 = 72$

Correct option: D

Final answer

72

Worked Solutions **Advanced Sequences**

Q008 **Written**

Find the general term of the sequence $\{2, 10, 24, 44, 70, 102, \ldots\}$.

Solution

1. The first differences are $8, 14, 20, 26, 32, \ldots$.
2. Hence $b_k = 2(3k + 1)$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1}(2(3k + 1)) = n(3n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$a_n = n(3n - 1)$

Worked Solutions **Advanced Sequences**

Q009 **Written**

Find the general term of the sequence $\{1, 6, 15, 28, 45, 66, \dots\}$.

Solution

1. The first differences are $5, 9, 13, 17, 21, \dots$.
2. Hence $b_k = 4k + 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1}(4k + 1) = n(2n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$a_n = n(2n - 1)$

Worked Solutions **Advanced Sequences**

Q010 **MCQ**

Select the correct option: If $a_1 = 3$ and $b_k = 5k + 4$, then a_n is:

Solution

1. The first differences are represented by $b_k = 5k + 4$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1}(5k + 4) = \frac{(n+1)(5n-2)}{2}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)(5n-2)}{2}$$

Q011 **MCQ**

Select the correct option: If $a_1 = 4$ and $b_k = 7k - 3$, then a_n is:

Solution

1. The first differences are represented by $b_k = 7k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 4 + \sum_{k=1}^{n-1}(7k - 3) = \frac{7n^2-13n+14}{2}$.
4. Substituting $n = 1$ gives $a_1 = 4$, matching the given first term.

Correct option: D

Final answer

$$a_n = \frac{7n^2 - 13n + 14}{2}$$

Worked Solutions **Advanced Sequences**

Q012 MCQ

Select the correct option: If $a_1 = 6$ and $b_k = 2(2k+3)$, then a_n is:

Solution

1. The first differences are represented by $b_k = 2(2k+3)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 6 + \sum_{k=1}^{n-1}(2(2k+3)) = 2n(n+2)$.
4. Substituting $n = 1$ gives $a_1 = 6$, matching the given first term.

Correct option: B

Final answer

$a_n = 2n(n+2)$

Worked Solutions **Advanced Sequences**

Q013 **Written**

Find the general term of the sequence $\{5, 8, 9, 8, 5, 0, \ldots\}$.

Solution

1. The first differences are $3, 1, -1, -3, -5, \ldots$.
2. Hence $b_k = 5 - 2k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1}(5 - 2k) = -n(n - 6)$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$a_n = -n(n - 6)$

Worked Solutions **Advanced Sequences**

Q014 Written

Find the general term of the sequence $\{2, 3, 6, 15, 42, 123, \ldots\}$**.**

Solution

1. The first differences are $1, 3, 9, 27, 81, \ldots$.
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1}(3^{k-1}) = \frac{3^n+9}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 9}{6}$$

Worked Solutions **Advanced Sequences**

Q015 **Written**

Find the general term of the sequence $\{1, 2, 5, 14, 41, 122, \ldots\}$.

Solution

1. The first differences are $1, 3, 9, 27, 81, \ldots$.
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1}(3^{k-1}) = \frac{3^n+3}{6}$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 3}{6}$$

Worked Solutions **Advanced Sequences**

Q016 Written

Find the general term of the sequence $\{5, 7, 11, 19, 35, \ldots\}$.

Solution

1. The first differences are $2, 4, 8, 16, \ldots$.
2. Hence $b_k = 2^k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1}(2^k) = 2^n + 3$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$a_n = 2^n + 3$

Worked Solutions **Advanced Sequences**

Q017 Written

Find the general term of the sequence $\{2, 3, 1, 5, -3, 13, \dots\}$.

Solution

1. The first differences are $1, -2, 4, -8, 16, \dots$.
2. Hence $b_k = (-2)^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1}((-2)^{k-1}) = \frac{(-2)^n+14}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions **Advanced Sequences**

Q018 **Written**

Find the general term of the sequence $\{5, 4, 7, -2, 25, -56, \ldots\}$.

Solution

1. The first differences are $-1, 3, -9, 27, -81, \ldots$.
2. Hence $b_k = \frac{(-3)^k}{3}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1}\left(\frac{(-3)^k}{3}\right) = -\frac{(-3)^n - 57}{12}$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = -\frac{(-3)^n - 57}{12}$$

Worked Solutions **Advanced Sequences**

Q019 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = k^2$, then a_n is:

Solution

1. The first differences are represented by $b_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(k^2) = \frac{(n+1)(2n^2-5n+6)}{6}$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)\left(2n^2 - 5n + 6\right)}{6}$$

Q020 Challenge

An oncology team records tissue volumes $1, 3, 7, 13, 21, 31, \dots$ in mm^3. Find the general formula a_n and calculate a_{12} to decide whether the 130 mm^3 threshold is exceeded.

Solution

1. The first differences are represented by $b_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(2k) = n^2 - n + 1$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_{12} = 133$; this is greater than 130, so immediate surgery is required.

Final answer

$a_n = n^2 - n + 1, \quad a_{12} = 133;$ surgery required

Worked Solutions **Advanced Sequences**

Q021 **Written**

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j, find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position $= P_9 + 1 = 82$
3. First value $= 164$

Final answer

164

Q022 **Written**

For groups with $a_j = 1 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j, find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position $= P_9 + 1 = 82$
3. First value $= 163$

Final answer

163

Worked Solutions **Advanced Sequences**

Q023 Written

For groups with $a_j = 1 + (j-1)1$ and $c_j = n$ terms in group j, find the first term of group 10.

Solution

1. $P_9 = 45$ terms precede the group.
2. First position $= P_9 + 1 = 46$
3. First value $= 46$

Final answer

46

Q024 MCQ

Select the correct option: find the first term of group 3:

Solution

1. $P_2 = 3$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 4.

Correct option: B

Final answer

4

Worked Solutions **Advanced Sequences**

Q025 MCQ

Select the correct option: find the first term of group 6:

Solution

1. $P_5 = 25$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 78.

Correct option: C

Final answer

78

Q026 MCQ

Select the correct option: find the first term of group 9:

Solution

1. $P_8 = 36$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 37.

Correct option: D

Final answer

37

Worked Solutions **Advanced Sequences**

Q027 MCQ

Select the correct option: find the first term of group 4:

Solution

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 30.

Correct option: B

Final answer

30

Q028 MCQ

Select the correct option: find the first term of group 7:

Solution

1. $P_6 = 21$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 22.

Correct option: C

Final answer

22

Worked Solutions **Advanced Sequences**

Q029 *MCQ*

Select the correct option: find the first term of group 10:

Solution

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 246.

Correct option: D

Final answer

246

Q030 *MCQ*

Select the correct option: find the first term of group 4:

Solution

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 20.

Correct option: A

Final answer

20

Worked Solutions **Advanced Sequences**

Q031 MCQ

Select the correct option: find the first term of group 7:

Solution

1. $P_6 = 27$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 110.

Correct option: B

Final answer

110

Q032 MCQ

Select the correct option: find the first term of group 10:

Solution

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 164.

Correct option: C

Final answer

164

Worked Solutions **Advanced Sequences**

Q033 *MCQ*

Select the correct option: find the first term of group 5:

Solution

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 58.

Correct option: A

Final answer

58

Q034 *MCQ*

Select the correct option: find the first term of group 8:

Solution

1. $P_7 = 49$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 100.

Correct option: B

Final answer

100

Worked Solutions **Advanced Sequences**

Q035 Written

For groups with $a_j = 1 + (j-1)3$ and $c_j = 3n - 1$ terms in group j, find the first term of group 10.

Solution

1. $P_9 = 126$ terms precede the group.
2. First position $= P_9 + 1 = 127$
3. First value $= 379$

Final answer

379

Q036 MCQ

Select the correct option: find the first term of group 5:

Solution

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 29.

Correct option: D

Final answer

29

Worked Solutions **Advanced Sequences**

Q037 MCQ

Select the correct option: find the first term of group 8:

Solution

1. $P_7 = 28$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 89.

Correct option: A

Final answer

89

Q038 Written

For groups with $a_j = 2 + (j-1)2$ and $c_j = 2n - 1$ terms in group j, derive the sum of the terms in group n.

Solution

1. First value $= 2\left(n^2 - 2n + 2\right)$
2. Last value $= 2n^2$
3. Number of terms $= 2n - 1$
4. $S_n = \frac{c_n}{2}(\text{first} + \text{last}) = 2\,(2n-1)\left(n^2 - n + 1\right)$

Final answer

$2\,(2n-1)\left(n^2 - n + 1\right)$

Worked Solutions **Advanced Sequences**

Q039 **Written**

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value $= 163$
2. Last value $= 199$
3. Number of terms $= 19$
4. $S_{10} = 3439$

Final answer

3439

Q040 **Written**

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value $= 46$
2. Last value $= 55$
3. Number of terms $= 10$
4. $S_{10} = 505$

Final answer

505

Worked Solutions **Advanced Sequences**

Q041 **Written**

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value $= 379$
2. Last value $= 463$
3. Number of terms $= 29$
4. $S_{10} = 12209$

Final answer

12209

Q042 **Written**

In the grouped arithmetic sequence, locate the value 88: state its group and its position within that group.

Solution

1. Sequence position $= (88 - 2)/2 + 1 = 44$
2. $P_6 = 36 < 44 \le P_7 = 49$
3. Therefore the value is in Group 7, position 8.

Final answer

Group 7, position 8

Worked Solutions **Advanced Sequences**

Q043 **Written**

In the grouped arithmetic sequence, locate the value 101: state its group and its position within that group.

Solution

1. Sequence position $= (101 - 1)/2 + 1 = 51$
2. $P_7 = 49 < 51 \leq P_8 = 64$
3. Therefore the value is in Group 8, position 2.

Final answer

Group 8, position 2

Q044 **Written**

In the grouped arithmetic sequence, locate the value 124: state its group and its position within that group.

Solution

1. Sequence position $= (124 - 1)/1 + 1 = 124$
2. $P_{15} = 120 < 124 \leq P_{16} = 136$
3. Therefore the value is in Group 16, position 4.

Final answer

Group 16, position 4

Worked Solutions **Advanced Sequences**

Q045 Written

In the grouped arithmetic sequence, locate the value 100: state its group and its position within that group.

Solution

1. Sequence position $= (100 - 1)/3 + 1 = 34$
2. $P_4 = 26 < 34 \le P_5 = 40$
3. Therefore the value is in Group 5, position 8.

Final answer

Group 5, position 8

Worked Solutions **Advanced Sequences**

Q046 **Challenge**

For the grouped sequence $2, 6 \mid 10, 14, 18 \mid 22, 26, 30, 34 \mid \ldots$, solve: (a) the first term of Group 12; (b) the sum in Group n; (c) the group and position of 202; (d) the total through Group 12.

Solution

1. $c_n = n + 1, \quad P_n = \frac{n(n+3)}{2}$
2. (a) first term $= 310$
3. (b) group sum $= 2\,(n+1)\left(n^2 + 2n - 1\right)$
4. (c) 202 is in Group 9, position 7.
5. (d) total $= 16200$

Final answer

(a) 310; (b) $2\,(n+1)\left(n^2 + 2n - 1\right)$; (c) Group 9, position 7; (d) 16200

Worked Solutions **Advanced Sequences**

Quiz MCQ 1 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k + 1) = \frac{3n^2-n+2}{2}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: If $a_1 = 3$ and $b_k = (-2)^{k-1}$, then a_n is:

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1}((-2)^{k-1}) = \frac{(-2)^n+20}{6}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: C

Final answer

$$a_n = \frac{(-2)^n + 20}{6}$$

Worked Solutions **Advanced Sequences**

Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: find the first term of group 9:

Solution

1. $P_8 = 64$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 195.

Correct option: A

Final answer

195

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: the value 29 is in:

Solution

1. The value is sequence position 15.
2. It lies in Group 5, position 1.

Correct option: B

Final answer

Group 5, position 1

Worked Solutions **Advanced Sequences**

Quiz WRITTEN 5 Concept Quiz · Written

A sequence has $a_1 = 1$ and difference sequence $b_k = 4k - 1$. Find the required general term a_n.

Solution

1. The first differences are represented by $b_k = 4k - 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(4k - 1) = 2n^2 - 3n + 2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$a_n = 2n^2 - 3n + 2$

Quiz WRITTEN 6 Concept Quiz · Written

A sequence has $a_1 = 2$ and difference sequence $b_k = (-2)^{k-1}$. Find the required general term a_n.

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1}((-2)^{k-1}) = \frac{(-2)^n + 14}{6}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions **Advanced Sequences**

Quiz WRITTEN 7 Concept Quiz · Written

For groups with $a_j = 5 + (j-1)3$ and $c_j = n$ terms in group j, find the first term of group 12.

Solution

1. $P_{11} = 66$ terms precede the group.
2. First position $= P_{11} + 1 = 67$
3. First value $= 203$

Final answer

203

Quiz WRITTEN 8 Concept Quiz · Written

In the grouped arithmetic sequence, locate the value 66: state its group and its position within that group.

Solution

1. Sequence position $= (66-2)/4 + 1 = 17$
2. $P_4 = 14 < 17 \leq P_5 = 20$
3. Therefore the value is in Group 5, position 3.

Final answer

Group 5, position 3

Answer key

Use the question number shown on each page.

Q001 $a_n = 2n^2 - 3n + 3$

Q002 C

Q003 B

Q004 A

Q005 B

Q006 A

Q007 D

Q008 $a_n = n(3n-1)$

Q009 $a_n = n(2n-1)$

Q010 A

Q011 D

Q012 B

Q013 $a_n = -n(n-6)$

Q014 $a_n = \dfrac{3^n + 9}{6}$

Q015 $a_n = \dfrac{3^n + 3}{6}$

Q016 $a_n = 2^n + 3$

Q017 $a_n = \dfrac{(-2)^n + 14}{6}$

Q018 $a_n = -\dfrac{(-3)^n - 57}{12}$

Q019 A

Q020 $a_n = n^2 - n + 1, \quad a_{12} = 133;$ surgery required

Q021 164

Q022 163

Q023 46

Q024 B

Q025 C

Q026 D

Q027 B

Q028 C

Q029 D

Q030 A

Q031 B

Q032 C

Q033 A

Q034 B

Q035 379

Q036 D

Q037 A

Q038 $2(2n-1)\left(n^2 - n + 1\right)$

Q039 3439

Q040 505

Q041 12209

Q042 Group 7, position 8

Q043 Group 8, position 2

Q044 Group 16, position 4

Q045 Group 5, position 8

Q046 (a) 310; (b) $2(n+1)\left(n^2 + 2n - 1\right)$; (c) Group 9, position 7; (d) 16200

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a_n = 2n^2 - 3n + 2$

Quiz WRITTEN 6 $a_n = \dfrac{(-2)^n + 14}{6}$

Quiz WRITTEN 7 203

Quiz WRITTEN 8 Group 5, position 3

Concept 5 | Recurrence and Induction

English Student Edition • Focused formulas and guided practice

Formula opener

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1),\ P(k) \Rightarrow P(k+1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k+1) = mq'$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Recurrence and Induction**

Q001 **Written**

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (2)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 26$.
3. $a_4 = 80$.

Final answer

$a_2 = 8,\ a_3 = 26,\ a_4 = 80$

Q002 **Written**

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 2a_n + (3)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -9$.
2. $a_3 = -15$.
3. $a_4 = -27$.

Final answer

$a_2 = -9,\ a_3 = -15,\ a_4 = -27$

Worked Solutions **Recurrence and Induction**

Q003 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 1a_n + (n(n-2))$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -7$.
2. $a_3 = -7$.
3. $a_4 = -4$.

Final answer

$a_2 = -7,\ a_3 = -7,\ a_4 = -4$

Q004 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -3$.
2. $a_3 = -5$.
3. $a_4 = -8$.

Final answer

$a_2 = -3,\ a_3 = -5,\ a_4 = -8$

Worked Solutions **Recurrence and Induction**

Q005 **Written**

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (2n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 4$.
2. $a_3 = 8$.
3. $a_4 = 14$.

Final answer

$a_2 = 4,\ a_3 = 8,\ a_4 = 14$

Q006 **Written**

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3^n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 1$.
2. $a_3 = 10$.
3. $a_4 = 37$.

Final answer

$a_2 = 1,\ a_3 = 10,\ a_4 = 37$

Worked Solutions **Recurrence and Induction**

Q007 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 9$.
2. $a_3 = 27$.
3. $a_4 = 81$.

Correct option: B

Final answer

81

Q008 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 8$.
2. $a_3 = 32$.
3. $a_4 = 128$.

Correct option: D

Final answer

128

Worked Solutions **Recurrence and Induction**

Q009 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (n+1)$. The fourth term is:

Solution

1. $a_2 = 6$.
2. $a_3 = 15$.
3. $a_4 = 34$.

Correct option: A

Final answer

34

Q010 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 3a_n + (2n)$. The fourth term is:

Solution

1. $a_2 = 17$.
2. $a_3 = 55$.
3. $a_4 = 171$.

Correct option: B

Final answer

171

Worked Solutions **Recurrence and Induction**

Q011 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 13$.

Correct option: D

Final answer

13

Q012 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = -2a_n + (5)$. The fourth term is:

Solution

1. $a_2 = -3$.
2. $a_3 = 11$.
3. $a_4 = -17$.

Correct option: A

Final answer

-17

Worked Solutions **Recurrence and Induction**

Q013 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -3a_n + (2)$. The fourth term is:

Solution

1. $a_2 = -1$.
2. $a_3 = 5$.
3. $a_4 = -13$.

Correct option: C

Final answer

-13

Q014 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 18$.

Correct option: D

Final answer

18

Worked Solutions **Recurrence and Induction**

Q015 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (1)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 11$.
3. $a_4 = 23$.

Correct option: C

Final answer

23

Q016 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The fourth term is:

Solution

1. $a_2 = 2$.
2. $a_3 = -1$.
3. $a_4 = -4$.

Correct option: D

Final answer

-4

Worked Solutions **Recurrence and Induction**

Q017 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The fourth term is:

Solution

1. $a_2 = 4$.
2. $a_3 = 6$.
3. $a_4 = 9$.

Correct option: B

Final answer

9

Q018 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 2a_n + (-n)$. The fourth term is:

Solution

1. $a_2 = 1$.
2. $a_3 = 0$.
3. $a_4 = -3$.

Correct option: C

Final answer

-3

Worked Solutions **Recurrence and Induction**

Q019 **Written**

For the sequence $\{a_n\}$ with $a_1 = -1$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 6$.

Final answer

$a_n = 5n - 6$

Q020 **Written**

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 2$.

Final answer

$a_n = 3n - 2$

Worked Solutions **Recurrence and Induction**

Q021 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-2)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -2\,(n-3)$.

Final answer

$a_n = -2\,(n-3)$

Q022 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (6)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 6n - 5$.

Final answer

$a_n = 6n - 5$

Q023 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Final answer

$a_n = 8 - 5n$

Q024 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 5$.

Final answer

$a_n = 3n - 5$

Worked Solutions **Recurrence and Induction**

Q025 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-4)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -4(n-2)$.

Final answer

$a_n = -4(n-2)$

Q026 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3^n$.

Correct option: D

Final answer

$a_n = 3^n$

Worked Solutions **Recurrence and Induction**

Q027 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-1}$.

Correct option: A

Final answer

$a_n = 2^{2n-1}$

Q028 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 5a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4\,5^{n-1}$.

Correct option: A

Final answer

$a_n = 4\,5^{n-1}$

Worked Solutions **Recurrence and Induction**

Q029 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 6$ and recurrence $a_{n+1} = 2a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3\,2^n$.

Correct option: A

Final answer

$a_n = 3\,2^n$

Q030 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2\,3^{n-1}$.

Final answer

$a_n = 2\,3^{n-1}$

Worked Solutions **Recurrence and Induction**

Q031 **Written**

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3\,5^{n-1}$.

Final answer

$a_n = 3\,5^{n-1}$

Q032 **Written**

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4\,3^{n-1}$.

Final answer

$a_n = 4\,3^{n-1}$

Worked Solutions **Recurrence and Induction**

Q033 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-2}$.

Final answer

$a_n = 2^{2n-2}$

Q034 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 4n - 3$.

Correct option: B

Final answer

$a_n = 4n - 3$

Worked Solutions　　**Recurrence and Induction**

Q035 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{2n^3-3n^2+n+24}{6}$.

Correct option: B

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$$

Q036 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = (-4)^{n-1}$.

Correct option: D

Final answer

$$a_n = (-4)^{n-1}$$

Worked Solutions **Recurrence and Induction**

Q037 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (3^n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{3^n+7}{2}$.

Correct option: A

Final answer

$$a_n = \frac{3^n + 7}{2}$$

Q038 Written

For the sequence $\{a_n\}$ with $a_1 = -3$ and recurrence $a_{n+1} = -2a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{3(-2)^n}{2}$.

Final answer

$$a_n = \frac{3\left(-2\right)^n}{2}$$

Worked Solutions **Recurrence and Induction**

Q039 **Written**

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = -5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2\,(-5)^{n-1}$.

Final answer

$a_n = 2\,(-5)^{n-1}$

Q040 **Written**

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = -4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{(-4)^n}{2}$.

Final answer

$$a_n = \frac{(-4)^n}{2}$$

Worked Solutions **Recurrence and Induction**

Q041 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 3n$.

Correct option: B

Final answer

$a_n = 8 - 3n$

Q042 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{n^2 - n + 6}{2}$.

Correct option: C

Final answer

$$a_n = \frac{n^2 - n + 6}{2}$$

Worked Solutions **Recurrence and Induction**

Q043 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Correct option: B

Final answer

$a_n = 8 - 5n$

Q044 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (6)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2(3n - 2)$.

Correct option: C

Final answer

$a_n = 2(3n - 2)$

Worked Solutions **Recurrence and Induction**

Q045 **Written**

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 5$.
2. $a_3 = 13$.
3. $a_4 = 36$.

Final answer

$a_2 = 5,\ a_3 = 13,\ a_4 = 36$

Q046 **Written**

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -2a_n + (5)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 3$.
2. $a_3 = -1$.
3. $a_4 = 7$.

Final answer

$a_2 = 3,\ a_3 = -1,\ a_4 = 7$

Worked Solutions **Recurrence and Induction**

Q047 **Written**

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 22$.
3. $a_4 = 63$.

Final answer

$a_2 = 8,\ a_3 = 22,\ a_4 = 63$

Worked Solutions **Recurrence and Induction**

Q048 **Challenge**

Solve the third-order recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$ with $a_0 = 0, a_1 = 1, a_2 = 6$.

Solution

1. $r^3 - 6r^2 + 11r - 6 = (r-1)(r-2)(r-3)$
2. $a_n = A + B2^n + C3^n$
3. $A = \frac{1}{2},\ B = -2,\ C = \frac{3}{2}$
4. $a_n = -\frac{4\,2^n - 3\,3^n - 1}{2}$

Final answer

$$a_n = -\frac{4\,2^n - 3\,3^n - 1}{2}$$

Worked Solutions **Recurrence and Induction**

Q049 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2n - 3)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(2k - 3) = (n - 2)^2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = (n - 2)^2$

Q050 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (3n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(3k) = \frac{3n^2 - 3n + 2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - 3n + 2}{2}$$

Worked Solutions **Recurrence and Induction**

Q051 **Written**

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (2n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(2k) = n^2 - n + 2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = n^2 - n + 2$

Q052 **Written**

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k^2) = \frac{2n^3-3n^2+n+4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions **Recurrence and Induction**

Q053 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n + 1)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k + 1) = \frac{3n^2-n+2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Q054 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n\,(2n + 1))$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k\,(2k + 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(k\,(2k + 1)) = \frac{4n^3-3n^2-n+12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$$

Worked Solutions **Recurrence and Induction**

Q055 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1}(k^2) = \frac{2n^3-3n^2+n+18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 18}{6}$$

Q056 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k^2) = \frac{2n^3-3n^2+n+4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions **Recurrence and Induction**

Q057 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(k^2 + 1) = \frac{2n^3 - 3n^2 + 7n + 6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{2n^3 - 3n^2 + 7n + 6}{6}$$

Q058 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = a_n + (n(4n-1))$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k-1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1}(k(4k-1)) = \frac{(n+1)(8n^2-23n+30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{(n+1)\left(8n^2 - 23n + 30\right)}{6}$$

Worked Solutions **Recurrence and Induction**

Q059 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (8n)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(8k) = 2\left(2n^2 - 2n + 1\right)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$a_n = 2\left(2n^2 - 2n + 1\right)$

Q060 MCQ

Select the correct option: find a_4 when $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1}(k^2) = \frac{2n^3 - 3n^2 + n + 18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

17

Worked Solutions **Recurrence and Induction**

Q061 MCQ

Select the correct option: find a_5 when $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(3k^2) = \frac{2n^3-3n^2+n+4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

92

Q062 MCQ

Select the correct option: find a_3 when $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(k^2 + 1) = \frac{2n^3-3n^2+7n+6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

9

Worked Solutions **Recurrence and Induction**

Q063 **MCQ**

Select the correct option: find a_6 when $a_1 = 5$ and $a_{n+1} = a_n + (n(4n-1))$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k-1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1}(k(4k-1)) = \frac{(n+1)(8n^2-23n+30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

210

Q064 **MCQ**

Select the correct option: find a_2 when $a_1 = 2$ and $a_{n+1} = a_n + (8n)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(8k) = 2(2n^2 - 2n + 1)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

10

Worked Solutions **Recurrence and Induction**

Q065 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(k^2) = \frac{2n^3-3n^2+n+12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = \dfrac{2n^3 - 3n^2 + n + 12}{6}$

Q066 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = a_n + (2n + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(2k + 1) = n^2$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$a_n = n^2$

Worked Solutions **Recurrence and Induction**

Q067 **Written**

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1}(2^k) = 2^n - 1$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = 2^n - 1$

Q068 **Written**

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (-3^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -3^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1}(-3^k) = -\frac{3^n-7}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3^n - 7}{2}$$

Worked Solutions **Recurrence and Induction**

Q069 **Written**

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 4$ and $a_{n+1} = a_n + (-4^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -4^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1}(-4^k) = -\frac{4^n-16}{3}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = -\dfrac{4^n - 16}{3}$

Worked Solutions **Recurrence and Induction**

Q070 Challenge

A technology startup has a reserve of 400,000 LE, represented by $a_1 = 4$, and burns $0.04n$ hundred-thousand LE each quarter: $a_{n+1} = a_n - 0.04n$. Find an explicit formula for a_n.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -\frac{k}{25}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1}(-\frac{k}{25}) = -\frac{n^2-n-200}{50}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{n^2 - n - 200}{50}$$

Worked Solutions **Recurrence and Induction**

Q071 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 3a_n + (4)$.

Solution

1. $c = -2$ from $c = 3c + (4)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3^n - 2$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = 3^n - 2$

Worked Solutions **Recurrence and Induction**

Q072 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Solution

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3\,5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3\,5^n - 10}{5}$$

Worked Solutions **Recurrence and Induction**

Q073 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = 2a_n + (-1)$.

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n+2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2^n + 2}{2}$$

Worked Solutions **Recurrence and Induction**

Q074 **Written**

Find the general term of $\{a_n\}$ given $a_1 = -3$ and $a_{n+1} = 4a_n + (3)$.

Solution

1. $c = -1$ from $c = 4c + (3)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4^n+2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{4^n + 2}{2}$$

Worked Solutions **Recurrence and Induction**

Q075 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 3$ and $a_{n+1} = 3a_n + (-2)$.

Solution

1. $c = 1$ from $c = 3c + (-2)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2\,3^n + 3}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2\,3^n + 3}{3}$$

Worked Solutions **Recurrence and Induction**

Q076 **Written**

Find the general term of $\{a_n\}$ given $a_1 = -2$ and $a_{n+1} = -3a_n + (4)$.

Solution

1. $c = 1$ from $c = -3c + (4)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = (-3)^n + 1$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = (-3)^n + 1$

Worked Solutions **Recurrence and Induction**

Q077 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = \frac{1}{3}a_n + (2)$.

Solution

1. $c = 3$ from $c = \frac{1}{3}c + (2)$.
2. $a_{n+1} - c = \frac{1}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3\,3^{-n}\,(3^n - 2)$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$a_n = 3\,3^{-n}\,(3^n - 2)$

Worked Solutions **Recurrence and Induction**

Q078 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = \frac{2}{3}a_n + (-\frac{1}{2})$.

Solution

1. $c = -\frac{3}{2}$ from $c = \frac{2}{3}c + (-\frac{1}{2})$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3\,3^{-n}(7\,2^n - 2\,3^n)}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3\,3^{-n}\,(7\,2^n - 2\,3^n)}{4}$$

Worked Solutions **Recurrence and Induction**

Q079 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 4$ and $a_{n+1} = -2a_n + (3)$.

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{3(-2)^n-2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3\left(-2\right)^n - 2}{2}$$

Worked Solutions **Recurrence and Induction**

Q080 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^n - 5}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{2\,(-2)^n - 5}{3}$$

Worked Solutions **Recurrence and Induction**

Q081 **MCQ**

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{2}{3}a_n + (-2)$, the general term is:

Solution

1. $c = -6$ from $c = \frac{2}{3}c + (-2)$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3\,3^{-n}(7\,2^n - 4\,3^n)}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{3\,3^{-n}\,(7\,2^n - 4\,3^n)}{2}$$

Worked Solutions **Recurrence and Induction**

Q082 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = 7a_n + (-4)$, the general term is:

Solution

1. $c = \frac{2}{3}$ from $c = 7c + (-4)$.
2. $a_{n+1} - c = 7(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2(2\,7^n+7)}{21}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{2\,(2\,7^n + 7)}{21}$$

Worked Solutions **Recurrence and Induction**

Q083 **MCQ**

Select the correct option: If $a_1 = 6$ and $a_{n+1} = 2a_n + (-5)$, the general term is:

Solution

1. $c = 5$ from $c = 2c + (-5)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n+10}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = \frac{2^n + 10}{2}$$

Worked Solutions **Recurrence and Induction**

Q084 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = -2a_n + (7)$, the general term is:

Solution

1. $c = \frac{7}{3}$ from $c = -2c + (7)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4(-2)^n - 7}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = -\frac{4\,(-2)^n - 7}{3}$$

Worked Solutions **Recurrence and Induction**

Q085 MCQ

Select the correct option: If $a_1 = 4$ and $a_{n+1} = -5a_n + (-2)$, the general term is:

Solution

1. $c = -\frac{1}{3}$ from $c = -5c + (-2)$.
2. $a_{n+1} - c = -5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{13(-5)^n+5}{15}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{13\,(-5)^n + 5}{15}$$

Q086 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -3a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{4}$ from $c = -3c + (5)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-3)^n - 5}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{(-3)^n - 5}{4}$$

Worked Solutions **Recurrence and Induction**

Q087 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{1}{2}a_n + (7)$, the general term is:

Solution

1. $c = 14$ from $c = \frac{1}{2}c + (7)$.
2. $a_{n+1} - c = \frac{1}{2}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 2\,2^{-n}\,(7\,2^n - 13)$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$a_n = 2\,2^{-n}\,(7\,2^n - 13)$

Worked Solutions **Recurrence and Induction**

Q088 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = 2a_n + (-1)$, the general term is:

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 1$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$a_n = 1$

Worked Solutions **Recurrence and Induction**

Q089 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^n - 5}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{2\,(-2)^n - 5}{3}$$

Worked Solutions **Recurrence and Induction**

Q090 Challenge

A sequence satisfies $a_{n+1} = 4a_n - 9$ with $a_1 = 2$. Find the equilibrium value, the explicit formula, and a_5.

Solution

1. $c = 3$ from $c = 4c + (-9)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2^{2n}-12}{4}$.
5. Substitute $n = 1$ to verify the given first term.
6. $a_5 = -253$

Final answer

$$c = 3, \quad a_n = -\frac{2^{2n} - 12}{4}, \quad a_5 = -253$$

Worked Solutions **Recurrence and Induction**

Q091 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$.
3. For $n = k + 1$, add the next term n to the left side and use the hypothesis.
4. $\frac{k(k+1)}{2} + n = \frac{(k+1)(k+2)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n\,(n+1)}{2}$$

Worked Solutions **Recurrence and Induction**

Q092 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n}(2i-1)^2 = \frac{n(2n-1)(2n+1)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k}(2i-1)^2 = \frac{k(2k-1)(2k+1)}{3}$.
3. For $n = k + 1$, add the next term $(2k+1)^2$ to the left side and use the hypothesis.
4. $\frac{k(2k-1)(2k+1)}{3} + (2k+1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n\,(2n-1)\,(2n+1)}{3}$$

Worked Solutions **Recurrence and Induction**

Q093 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n} i\,(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i\,(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k + 1)\,(k + 2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k + 1)\,(k + 2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n\,(n + 1)\,(n + 2)}{3}$$

Worked Solutions **Recurrence and Induction**

Q094 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} i\,(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i\,(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)\,(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)\,(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$\frac{n\,(n+1)\,(n+2)}{3}$$

Worked Solutions **Recurrence and Induction**

Q095 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3\,4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3\,4^k = \left(2\,2^k - 1\right)\left(2\,2^k + 1\right)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$4^n - 1$

Q096 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 2i + 3 = k(k + 4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k(k + 4) + 2k + 5 = (k + 1)(k + 5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$n(n + 4)$

Worked Solutions **Recurrence and Induction**

Q097 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 2i^2 + 3i$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 2i^2 + 3i = \frac{k(k+1)(4k+11)}{6}$.
3. For $n = k + 1$, add the next term $(k + 1)(2k + 5)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(4k+11)}{6} + (k + 1)(2k + 5) = \frac{(k+1)(k+2)(4k+15)}{6}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$\frac{n(n+1)(4n+11)}{6}$$

Q098 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 3i^2 + 2i$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 3i^2 + 2i = \frac{k(k+1)(2k+3)}{2}$.
3. For $n = k + 1$, add the next term $(k+1)(3k+5)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+3)}{2} + (k+1)(3k+5) = \frac{(k+1)(k+2)(2k+5)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Worked Solutions **Recurrence and Induction**

Q099 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 7 \cdot 2^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 7.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 7 \cdot 2^{i-1} = 7\left(2^k - 1\right)$.
3. For $n = k + 1$, add the next term $7\,2^k$ to the left side and use the hypothesis.
4. $7\left(2^k - 1\right) + 7\,2^k = 7\left(2\,2^k - 1\right)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$7\,(2^n - 1)$

Worked Solutions **Recurrence and Induction**

Q100 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} i\,(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i\,(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n\,(n+1)\,(n+2)}{3}$$

Worked Solutions **Recurrence and Induction**

Q101 **MCQ**

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3\,4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3\,4^k = \left(2\,2^k - 1\right)\left(2\,2^k + 1\right)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$4^n - 1$

Worked Solutions **Recurrence and Induction**

Q102 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^{n} 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 2i + 3 = k(k + 4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k(k + 4) + 2k + 5 = (k + 1)(k + 5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$n(n + 4)$

Worked Solutions **Recurrence and Induction**

Q103 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n}(2i-1)=n^2$.

Solution

1. [1] Base case $n=1$: both sides equal 1.
2. [2] Assume the statement holds for $n=k$: $\sum_{i=1}^{k}(2i-1)=k^2$.
3. For $n=k+1$, add the next term $2k+1$ to the left side and use the hypothesis.
4. $k^2+2k+1=(k+1)^2$, which is the right side at $k+1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

n^2

Q104 Written

Use mathematical induction to prove $\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions **Recurrence and Induction**

Q105 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n} i(i+2) = \frac{n(n+1)(2n+7)}{6}$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} i(i+2) = \frac{k(k+1)(2k+7)}{6}$.
3. For $n = k+1$, add the next term $(k+1)(k+3)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+7)}{6} + (k+1)(k+3) = \frac{(k+1)(k+2)(2k+9)}{6}$, which is the right side at $k+1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(2n+7)}{6}$$

Worked Solutions **Recurrence and Induction**

Q106 **Written**

Use mathematical induction to prove $\sum_{i=1}^{n} 3\,4^{i-1} = 4^n - 1$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^{k} 3\,4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3\,4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3\,4^k = \left(2\,2^k - 1\right)\left(2\,2^k + 1\right)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$4^n - 1$

Worked Solutions **Recurrence and Induction**

Q107 Written

Use mathematical induction to prove $\sum_{i=0}^{n} 10^i = \frac{10\,10^n - 1}{9}$.

Solution

1. [1] Base case $n = 1$: both sides equal 11.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=0}^{k} 10^i = \frac{10\,10^k - 1}{9}$.
3. For $n = k + 1$, add the next term 10^{k+1} to the left side and use the hypothesis.
4. $\frac{10\,10^k - 1}{9} + 10^{k+1} = \frac{100\,10^k - 1}{9}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{10\,10^n - 1}{9}$$

Worked Solutions **Recurrence and Induction**

Q108 **Challenge**

Use mathematical induction to prove $\sum_{i=1}^{n}(i^2+i+1)=\frac{n(n^2+3n+5)}{3}$.

Solution

1. [1] Base case $n=1$: both sides equal 3.
2. [2] Assume the statement holds for $n=k$: $\sum_{i=1}^{k}(i^2+i+1)=\frac{k(k^2+3k+5)}{3}$.
3. For $n=k+1$, add the next term k^2+3k+3 to the left side and use the hypothesis.
4. $\frac{k(k^2+3k+5)}{3}+k^2+3k+3=\frac{(k+1)(k^2+5k+9)}{3}$, which is the right side at $k+1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n\left(n^2+3n+5\right)}{3}$$

Worked Solutions **Recurrence and Induction**

Q109 Written

Prove by mathematical induction that $2^n > 3n$ for $n \geq 4$.

Solution

1. $n = 4:\quad 2^4 = 16 > 12 = 12.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 0)$ after multiplying by the positive base.
4. $2(3k + 0) - (3(k + 1) + 0) = 3\,(k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$2^n > 3n$

Worked Solutions **Recurrence and Induction**

Q110 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 3$.

Solution

1. $n = 3:\quad 3^3 = 27 > 16 = 16.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$3^n > 5n + 1$

Worked Solutions **Recurrence and Induction**

Q111 Written

Prove by mathematical induction that $3^n > 2n$ for $n \geq 1$.

Solution

1. $n = 1$: $3^1 = 3 > 2 = 2.$
2. Assume the inequality holds for $n = k$ with $k \geq 1$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2\,(2k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$3^n > 2n$

Worked Solutions **Recurrence and Induction**

Q112 **Written**

Prove by mathematical induction that $2^n > 1n + 2$ for $n \geq 3$.

Solution

1. $n = 3:\quad 2^3 = 8 > 5 = 5.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$2^n > 1n + 2$

Worked Solutions **Recurrence and Induction**

Q113 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 4$.

Solution

1. $n = 4:\quad 3^4 = 81 > 21 = 21.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$3^n > 5n + 1$

Worked Solutions **Recurrence and Induction**

Q114 **Written**

Prove by mathematical induction that $3^n > 5n + 2$ for $n \geq 4$.

Solution

1. $n = 4:\quad 3^4 = 81 > 22 = 22.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$3^n > 5n + 2$

Worked Solutions **Recurrence and Induction**

Q115 **Written**

Prove by mathematical induction that $4^n > 2n + 3$ for $n \geq 4$.

Solution

1. $n = 4:\quad 4^4 = 256 > 11 = 11.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$4^n > 2n + 3$

Worked Solutions **Recurrence and Induction**

Q116 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:

Solution

1. $n = 3 : \quad 2^3 = 8 > 5 = 5.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: D

Final answer

$2^n > 1n + 2$

Worked Solutions **Recurrence and Induction**

Q117 **MCQ**

Select the correct option: which inequality is proved for $n \geq 2$?:

Solution

1. $n = 2: \quad 3^2 = 9 > 4 = 4.$
2. Assume the inequality holds for $n = k$ with $k \geq 2$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2\,(2k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: A

Final answer

$3^n > 2n$

Worked Solutions **Recurrence and Induction**

Q118 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4: \quad 3^4 = 81 > 21 = 21.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: B

Final answer

$3^n > 5n + 1$

Worked Solutions **Recurrence and Induction**

Q119 **MCQ**

Select the correct option: which inequality is proved for $n \geq 3$?:

Solution

1. $n = 3: \quad 3^3 = 27 > 9 = 9.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(3k + 0)$ after multiplying by the positive base.
4. $3(3k + 0) - (3(k + 1) + 0) = 3\,(2k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: D

Final answer

$3^n > 3n$

Worked Solutions **Recurrence and Induction**

Q120 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4: \quad 3^4 = 81 > 22 = 22.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: A

Final answer

$3^n > 5n + 2$

Worked Solutions **Recurrence and Induction**

Q121 **MCQ**

Select the correct option: which inequality is proved for $n \geq 3$?:

Solution

1. $n = 3: \quad 4^3 = 64 > 10 = 10.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 1)$ after multiplying by the positive base.
4. $4(3k + 1) - (3(k + 1) + 1) = 9k$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: B

Final answer

$4^n > 3n + 1$

Worked Solutions **Recurrence and Induction**

Q122 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4: \quad 4^4 = 256 > 11 = 11.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: D

Final answer

$4^n > 2n + 3$

Worked Solutions **Recurrence and Induction**

Q123 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4: \quad 4^4 = 256 > 17 = 17.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(4k + 1)$ after multiplying by the positive base.
4. $4(4k + 1) - (4(k + 1) + 1) = 12k - 1$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: A

Final answer

$4^n > 4n + 1$

Worked Solutions **Recurrence and Induction**

Q124 **Written**

Prove by mathematical induction that $2^n > 3n + 1$ for $n \geq 4$.

Solution

1. $n = 4:\quad 2^4 = 16 > 13 = 13.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$2^n > 3n + 1$

Worked Solutions **Recurrence and Induction**

Q125 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:

Solution

1. $n = 3: \quad 4^3 = 64 > 9 = 9.$
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 0)$ after multiplying by the positive base.
4. $4(3k + 0) - (3(k + 1) + 0) = 3\,(3k - 1)$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: B

Final answer

$4^n > 3n$

Worked Solutions **Recurrence and Induction**

Q126 **Challenge**

When $h > 0$ and $n \geq 3$, prove $(1+h)^n > 1 + nh^2$.

Solution

1. $n = 3:$ $(1+h)^3 = 1 + 3h + 3h^2 + h^3 > 1 + 3h^2$ because h>0 and $3h + h^3 > 0$.
2. Assume $(1+h)^k > 1 + kh^2$ for $k \geq 3$.
3. Multiplying by the positive factor $1 + h$ gives a lower bound $(1+h)(1+kh^2)$.
4. $(1+h)(1+kh^2) - [1 + (k+1)h^2] = h(1 - h + kh^2)$. The quadratic $1 - h + kh^2$ is positive for $k \geq 3$ because its discriminant is $1 - 4k < 0$.
5. Therefore the inequality holds for every $n \geq 3$ and $h > 0$.

Final answer

$(1+h)^n > 1 + nh^2$

Worked Solutions **Recurrence and Induction**

Q127 **Written**

Prove that $n^3 + (n+1)^3 + (n+2)^3$ is a multiple of 9 for every natural number n, using mathematical induction.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 36 = 9(4)$, so the statement holds.
2. [2] Assume $f(k) = 9m$ for an integer m.
3. [3] The induction difference is $f(k+1) - f(k) = 9\left(k^2 + 3k + 3\right) = 9(k^2 + 3k + 3)$.
4. [4] Therefore $f(k+1) = 9\left[m + k^2 + 3k + 3\right]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$9 \mid [n^3 + (n+1)^3 + (n+2)^3]$

Worked Solutions **Recurrence and Induction**

Q128 **Written**

Prove that $n^3 + 2n$ is a multiple of 3 for every natural number n, using mathematical induction.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m.
3. [3] The induction difference is $f(k+1) - f(k) = 3\left(k^2 + k + 1\right) = 3(k^2 + k + 1)$.
4. [4] Therefore $f(k+1) = 3\left[m + k^2 + k + 1\right]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$3 \mid (n^3 + 2n)$

Q129 Written

Prove by mathematical induction that $n^2 + n$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^{1}(k + 0 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$2 \mid (n^2 + n)$

Worked Solutions **Recurrence and Induction**

Q130 Written

Prove by mathematical induction that $n^2 + 5n + 8$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+2+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$2 \mid (n^2 + 5n + 8)$

Worked Solutions **Recurrence and Induction**

Q131 **Written**

Prove by mathematical induction that $n^2 + 7n + 12$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+3+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$2 \mid (n^2 + 7n + 12)$

Worked Solutions **Recurrence and Induction**

Q132 **Written**

Prove by mathematical induction that $n^2 + 11n + 32$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+5+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$2 \mid (n^2 + 11n + 32)$

Q133 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+0+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

2

Worked Solutions **Recurrence and Induction**

Q134 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 5n + 8$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k + 2 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions **Recurrence and Induction**

Q135 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 7n + 12$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k+3+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions **Recurrence and Induction**

Q136 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 11n + 32$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k + 1) - f(k) = 2\prod_{j=1}^{1}(k + 5 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k + 1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions **Recurrence and Induction**

Q137 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 3n^2 + 2n$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 6 = 6(1)$.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3\prod_{j=1}^{2}(k + 0 + j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: B

Final answer

6

Worked Solutions **Recurrence and Induction**

Q138 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 9n^2 + 26n + 30$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 66 = 6(11)$.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3\prod_{j=1}^{2}(k+2+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

6

Worked Solutions **Recurrence and Induction**

Q139 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 12n^2 + 47n + 60$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 120 = 6(20)$.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^{2}(k+3+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions **Recurrence and Induction**

Q140 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 18n^2 + 107n + 216$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 342 = 6(57)$.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3\prod_{j=1}^{2}(k+5+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions **Recurrence and Induction**

Q141 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n + 2$ for every natural number n?:

Solution

1. [1] Base case: $n = 1, \quad f(1) = 4 = 2(2)$.
2. [2] Assume $f(k) = 2m$ for an integer m.
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2\prod_{j=1}^{1}(k + 0 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions **Recurrence and Induction**

Q142 **Written**

Prove by mathematical induction that $2n^3 - 3n^2 + n$ is a multiple of 6.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 0 = 6(0)$, so the statement holds.
2. [2] Assume $f(k) = 6m$ for an integer m.
3. [3] The induction difference is $f(k+1) - f(k) = 6k^2 = 6(k^2)$.
4. [4] Therefore $f(k+1) = 6\left[m + k^2\right]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$6 \mid (2n^3 - 3n^2 + n)$

Worked Solutions **Recurrence and Induction**

Q143 **Written**

Prove by mathematical induction that $4n^3 - n$ is a multiple of 3.

Solution

1. [1] Base case: $n = 1, \quad f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m.
3. [3] The induction difference is $f(k+1) - f(k) = 3\left(2k+1\right)^2 = 3((2k+1)^2)$.
4. [4] Therefore $f(k+1) = 3\left[m + (2k+1)^2\right]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$3 \mid (4n^3 - n)$

Worked Solutions **Recurrence and Induction**

Q144 Challenge

For any integer $n \geq 1$, prove that $(n+1)^n - 1$ is a multiple of n^2.

Solution

1. Use the binomial theorem on $(n+1)^n = (1+n)^n$.
2. $(1+n)^n - 1 = \binom{n}{1}n + \binom{n}{2}n^2 + \cdots + n^n$.
3. $\binom{n}{1}n = n^2$, and every remaining term also contains n^2.
4. $(n+1)^n - 1 = n^2[1 + \binom{n}{2} + \binom{n}{3}n + \cdots + n^{n-2}]$. The bracket is an integer for every integer $n \geq 1$.
5. Hence the expression is a multiple of n^2.

Final answer

$n^2 \mid [(n+1)^n - 1]$

Worked Solutions **Recurrence and Induction**

Q145 **Written**

For the sequence determined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2-1}{n+1}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 5, \quad a_4 = 6.$
2. The pattern suggests $a_n = n + 2$.

Final answer

$a_2 = 4, \quad a_3 = 5, \quad a_4 = 6; \quad a_n = n + 2$

Q146 **Written**

For the sequence determined by $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = -3, \quad a_3 = -5, \quad a_4 = -7.$
2. The pattern suggests $a_n = 1 - 2n$.

Final answer

$a_2 = -3, \quad a_3 = -5, \quad a_4 = -7; \quad a_n = 1 - 2n$

Worked Solutions **Recurrence and Induction**

Q147 **Written**

For the sequence determined by $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 9, \quad a_4 = 16.$
2. The pattern suggests $a_n = n^2$.

Final answer

$a_2 = 4, \quad a_3 = 9, \quad a_4 = 16; \quad a_n = n^2$

Q148 **Written**

For the sequence determined by $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}.$
2. The pattern suggests $a_n = \frac{2n-1}{n}$.

Final answer

$a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}; \quad a_n = \frac{2n - 1}{n}$

Worked Solutions **Recurrence and Induction**

Q149 **Written**

For a sequence with $a_1 = 3$ and $a_{n+1} = \frac{a_n^2-1}{n+1}$, prove by mathematical induction that $a_n = n + 2$.

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = k + 2$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = \frac{{a_k}^2-1}{k+1}$: $a_{k+1} = k + 3$.
4. [4] This is $k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$a_n = n + 2$

Q150 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2k - 1$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 1$.
4. [4] This is $2k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$a_n = 2n - 1$

Worked Solutions **Recurrence and Induction**

Q151 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 2k$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k+1)$.
4. [4] This is $2(k+1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$a_n = 2n$

Worked Solutions **Recurrence and Induction**

Q152 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 2k + 1$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 3$.
4. [4] This is $2k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$a_n = 2n + 1$

Worked Solutions **Recurrence and Induction**

Q153 **MCQ**

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2(k+1)$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k+2)$.
4. [4] This is $2(k+2)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$a_n = 2(n+1)$

Worked Solutions **Recurrence and Induction**

Q154 **MCQ**

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 3k - 2$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 1$.
4. [4] This is $3k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$a_n = 3n - 2$

Worked Solutions **Recurrence and Induction**

Q155 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 3k - 1$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 2$.
4. [4] This is $3k + 2$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$a_n = 3n - 1$

Worked Solutions **Recurrence and Induction**

Q156 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 3k$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3(k+1)$.
4. [4] This is $3(k+1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$a_n = 3n$

Worked Solutions **Recurrence and Induction**

Q157 **MCQ**

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 3k + 1$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 4$.
4. [4] This is $3k + 4$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$a_n = 3n + 1$

Worked Solutions **Recurrence and Induction**

Q158 **Written**

For a sequence with $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, prove by mathematical induction that $a_n = n^2$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = k^2$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = {a_k}^2 - k^2 a_k + (k+1)^2$: $a_{k+1} = (k+1)^2$.
4. [4] This is $(k+1)^2$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$a_n = n^2$

Worked Solutions **Recurrence and Induction**

Q159 **Written**

For a sequence with $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, prove by mathematical induction that $a_n = 1 - 2n$.

Solution

1. [1] Base case: $a_1 = -1 = -1$.
2. [2] Assume $a_k = 1 - 2k$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = {a_k}^2 + 2ka_k - 2$: $a_{k+1} = -2k - 1$.
4. [4] This is $-2k - 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$a_n = 1 - 2n$

Worked Solutions **Recurrence and Induction**

Q160 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = 2a_n + 0$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2^{k-1}$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = 2a_k + 0$: $a_{k+1} = 2^k$.
4. [4] This is 2^k, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$a_n = 2^{n-1}$

Worked Solutions **Recurrence and Induction**

Q161 **Written**

For a sequence with $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, prove by mathematical induction that $a_n = \frac{2n-1}{n}$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = \frac{2k-1}{k}$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = \frac{a_k - 4}{a_k - 3}$: $a_{k+1} = \frac{2k+1}{k+1}$.
4. [4] This is $\frac{2k+1}{k+1}$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = \frac{2n-1}{n}$$

Worked Solutions **Recurrence and Induction**

Q162 **Challenge**

Three numbers A, B, C in that order form a geometric sequence with common ratio r. If $A, 2B, C$ in that order form an arithmetic sequence, determine the possible values of r.

Solution

1. Let the geometric sequence be A, Ar, Ar^2.
2. Since $A, 2B, C$ is arithmetic, its middle term is the average of the outer terms: $4B = A + C$.
3. Substitute $B = Ar, C = Ar^2$ and cancel the non-zero scale factor A: $r^2 - 4r + 1 = 0$.
4. $r = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$.

Final answer

$r = 2 \pm \sqrt{3}$

Worked Solutions **Recurrence and Induction**

Quiz MCQ 1 Concept Quiz · MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2(2n - 1)$.

Correct option: D

Final answer

$a_n = 2(2n - 1)$

Worked Solutions **Recurrence and Induction**

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -2a_n + (3)$, the general term is:

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-2)^n - 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = -\frac{(-2)^n - 2}{2}$$

Worked Solutions **Recurrence and Induction**

Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4 : \quad 2^4 = 16 > 13 = 13.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Correct option: B

Final answer

$2^n > 3n + 1$

Worked Solutions **Recurrence and Induction**

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2\,(k+1)$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2\,(k+2)$.
4. [4] This is $2\,(k+2)$, the proposed formula with $n = k+1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$a_n = 2\,(n+1)$

Worked Solutions **Recurrence and Induction**

Quiz WRITTEN 5 Concept Quiz · Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 3$.

Final answer

$a_n = 5n - 3$

Worked Solutions **Recurrence and Induction**

Quiz WRITTEN 6 **Concept Quiz · Written**

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Solution

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3\,5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3\,5^n - 10}{5}$$

Worked Solutions **Recurrence and Induction**

Quiz WRITTEN 7 Concept Quiz · Written

Prove by mathematical induction that $2^n > 4n + 1$ for $n \geq 4$.

Solution

1. $n = 4:\quad 2^4 = 16 > 17 = 17.$
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(4k + 1)$ after multiplying by the positive base.
4. $2(4k + 1) - (4(k + 1) + 1) = 4k - 3$ is positive for the allowed k.
5. Hence the inequality holds for k+1 and therefore for every allowed natural number.

Final answer

$2^n > 4n + 1$

Worked Solutions **Recurrence and Induction**

Quiz WRITTEN 8 **Concept Quiz · Written**

For a sequence with $a_1 = 5$ and $a_{n+1} = a_n + 1(2n + 1) + 2$, prove by mathematical induction that $a_n = n^2 + 2n + 2$.

Solution

1. [1] Base case: $a_1 = 5 = 5$.
2. [2] Assume $a_k = k^2 + 2k + 2$ for some positive integer k.
3. [3] Substitute into $a_{k+1} = a_k + 1(2k + 1) + 2$: $a_{k+1} = k^2 + 4k + 5$.
4. [4] This is $k^2 + 4k + 5$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$a_n = n^2 + 2n + 2$

Answer key

Use the question number shown on each page.

Q001 $a_2 = 8,\ a_3 = 26,\ a_4 = 80$

Q002 $a_2 = -9,\ a_3 = -15,\ a_4 = -27$

Q003 $a_2 = -7,\ a_3 = -7,\ a_4 = -4$

Q004 $a_2 = -3,\ a_3 = -5,\ a_4 = -8$

Q005 $a_2 = 4,\ a_3 = 8,\ a_4 = 14$

Q006 $a_2 = 1,\ a_3 = 10,\ a_4 = 37$

Q007 B

Q008 D

Q009 A

Q010 B

Q011 D

Q012 A

Q013 C

Q014 D

Q015 C

Q016 D

Q017 B

Q018 C

Q019 $a_n = 5n - 6$

Q020 $a_n = 3n - 2$

Q021 $a_n = -2(n-3)$

Q022 $a_n = 6n - 5$

Q023 $a_n = 8 - 5n$

Q024 $a_n = 3n - 5$

Q025 $a_n = -4(n-2)$

Q026 D

Q027 A

Q028 A

Q029 A

Q030 $a_n = 2\,3^{n-1}$

Q031 $a_n = 3\,5^{n-1}$

Q032 $a_n = 4\,3^{n-1}$

Q033 $a_n = 2^{2n-2}$

Q034 B

Q035 B

Q036 D

Q037 A

Q038 $a_n = \dfrac{3(-2)^n}{2}$

Q039 $a_n = 2(-5)^{n-1}$

Q040 $a_n = \dfrac{(-4)^n}{2}$

Q041 B

Q042 C

Q043 B

Q044 C

Q045 $a_2 = 5,\ a_3 = 13,\ a_4 = 36$

Q046 $a_2 = 3,\ a_3 = -1,\ a_4 = 7$

Q047 $a_2 = 8,\ a_3 = 22,\ a_4 = 63$

Q048 $a_n = -\dfrac{4\,2^n - 3\,3^n - 1}{2}$

Q049 $a_n = (n-2)^2$

Q050 $a_n = \dfrac{3n^2 - 3n + 2}{2}$

Q051 $a_n = n^2 - n + 2$

Q052 $a_n = \dfrac{2n^3 - 3n^2 + n + 4}{2}$

Q053 $a_n = \dfrac{3n^2 - n + 2}{2}$

Q054 $a_n = \dfrac{4n^3 - 3n^2 - n + 12}{6}$

Q055 C

Q056 A

Q057 C

Q058 D

Q059 C

Q060 A

Q061 D

Q062 A

Q063 B

Q064 A

Q065 $a_n = \dfrac{2n^3 - 3n^2 + n + 12}{6}$

Q066 A

Q067 $a_n = 2^n - 1$

Q068 $a_n = -\dfrac{3^n - 7}{2}$

Q069 $a_n = -\dfrac{4^n - 16}{3}$

Q070 $a_n = -\dfrac{n^2 - n - 200}{50}$

Q071 $a_n = 3^n - 2$

Q072 $a_n = \dfrac{3\,5^n - 10}{5}$

Q073 $a_n = \dfrac{2^n + 2}{2}$

Q074 $a_n = -\dfrac{4^n + 2}{2}$

Q075 $a_n = \dfrac{2\,3^n + 3}{3}$

Q076 $a_n = (-3)^n + 1$

Q077 $a_n = 3\,3^{-n}(3^n - 2)$

Q078 $a_n = \dfrac{3\,3^{-n}(7\,2^n - 2\,3^n)}{4}$

Q079 $a_n = -\dfrac{3(-2)^n - 2}{2}$

Q080 B

Q081 C

Q082 D

Q083 B

Q084 C

Q085 D

Q086 B

Q087 C

Q088 D

Q089 D

Q090 $c = 3, \quad a_n = -\dfrac{2^{2n} - 12}{4}, \quad a_5 = -253$

Q091 $\dfrac{n(n+1)}{2}$

Q092 $\dfrac{n(2n-1)(2n+1)}{3}$

Q093 $\dfrac{n(n+1)(n+2)}{3}$

Q094 A

Q095 B

Q096 C

Q097 B

Q098 C

Q099 A

Q100 C

Q101 A

Q102 B

Q103 n^2

Q104 $\frac{n(n+1)(n+2)}{3}$

Q105 $\frac{n(n+1)(2n+7)}{6}$

Q106 $4^n - 1$

Q107 $\frac{10\,10^n - 1}{9}$

Q108 $\frac{n\left(n^2+3n+5\right)}{3}$

Q109 $2^n > 3n$

Q110 $3^n > 5n + 1$

Q111 $3^n > 2n$

Q112 $2^n > 1n + 2$

Q113 $3^n > 5n + 1$

Q114 $3^n > 5n + 2$

Q115 $4^n > 2n + 3$

Q116 D

Q117 A

Q118 B

Q119 D

Q120 A

Q121 B

Q122 D

Q123 A

Q124 $2^n > 3n + 1$

Q125 B

Q126 $(1+h)^n > 1 + nh^2$

Q127 $9 \mid \left[n^3 + (n+1)^3 + (n+2)^3\right]$

Q128 $3 \mid (n^3 + 2n)$

Q129 $2 \mid (n^2 + n)$

Q130 $2 \mid (n^2 + 5n + 8)$

Q131 $2 \mid (n^2 + 7n + 12)$

Q132 $2 \mid (n^2 + 11n + 32)$

Q133 A

Q134 D

Q135 C

Q136 C

Q137 B

Q138 A

Q139 D

Q140 D

Q141 D

Q142 $6 \mid (2n^3 - 3n^2 + n)$

Q143 $3 \mid (4n^3 - n)$

Q144 $n^2 \mid \left[(n+1)^n - 1\right]$

Q145 $a_2 = 4, \quad a_3 = 5, \quad a_4 = 6; \quad a_n = n + 2$

Q146 $a_2 = -3, \quad a_3 = -5, \quad a_4 = -7; \quad a_n = 1 - 2n$

Q147 $a_2 = 4, \quad a_3 = 9, \quad a_4 = 16; \quad a_n = n^2$

Q148 $a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}; \quad a_n = \frac{2n-1}{n}$

Q149 $a_n = n + 2$

Q150 D

Q151 C

Q152 B

Q153 A

Q154 D

Q155 C

Q156 B

Q157 A

Q158 $a_n = n^2$

Q159 $a_n = 1 - 2n$

Q160 C

Q161 $a_n = \frac{2n-1}{n}$

Q162 $r = 2 \pm \sqrt{3}$

Quiz MCQ 1 D

Quiz MCQ 2 A

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $a_n = 5n - 3$

Quiz WRITTEN 6 $a_n = \frac{3\,5^n - 10}{5}$

Quiz WRITTEN 7 $2^n > 4n + 1$

Quiz WRITTEN 8 $a_n = n^2 + 2n + 2$

Concept 1 | Complex Numbers

English Student Edition • Focused formulas and guided practice

Formula opener

Standard form

$$z = a + bi$$

Conjugate

$$\overline{a + bi} = a - bi$$

Equality

$$a + bi = c + di \Rightarrow a = c,\ b = d$$

Powers

$$i^4 = 1 \qquad i^{4q+r} = i^r$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Complex Numbers**

Q001 Written

State the real and imaginary parts of $3 - 2i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(3,\ -2)$.

Final answer

$(3,\ -2)$

Q002 Written

State the real and imaginary parts of $(4 - \sqrt{5}i)/3$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(4/3, -\sqrt{5}/3)$.

Final answer

$(4/3, -\sqrt{5}/3)$

Worked Solutions **Complex Numbers**

Q003 Written

State the real and imaginary parts of -8.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-8,\ 0)$.

Final answer

$(-8,\ 0)$

Q004 Written

State the real and imaginary parts of $2i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0,\ 2)$.

Final answer

$(0,\ 2)$

Worked Solutions **Complex Numbers**

Q005 Written

Find the real values of x and y satisfying $(2+i)x+(1+i)y=2+3i, x, yreal$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=-1, y=4$.

Final answer

$x=-1, y=4$

Q006 Written

State the real and imaginary parts of $-3+5i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-3,\ 5)$.

Final answer

$(-3,\ 5)$

Worked Solutions **Complex Numbers**

Q007 Written

State the real and imaginary parts of $(-1-\sqrt{3}i)/2$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-1/2, -\sqrt{3}/2)$.

Final answer

$(-1/2, -\sqrt{3}/2)$

Q008 Written

State the real and imaginary parts of 1.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1,\ 0)$.

Final answer

$(1,\ 0)$

Worked Solutions **Complex Numbers**

Q009 Written

State the real and imaginary parts of $3i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0,\ 3)$.

Final answer

$(0,\ 3)$

Q010 Written

Find the real values of x and y satisfying $(x-6)+(x+3y)i=0, x, y real$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=6, y=-2$.

Final answer

$x=6, y=-2$

Worked Solutions **Complex Numbers**

Q011 **Written**

Find the real values of x and y satisfying $(3 + 2i)x + (4 - i)y = 6 - 7i, x, yreal$**.**

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = -2, y = 3$.

Final answer

$x = -2, y = 3$

Q012 **Written**

State the real and imaginary parts of $2 + 5i$**.**

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2,\ 5)$.

Final answer

$(2,\ 5)$

Worked Solutions **Complex Numbers**

Q013 Written

State the real and imaginary parts of $-25 + 13i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-25,\ 13)$.

Final answer

$(-25,\ 13)$

Q014 Written

State the real and imaginary parts of $(2 - \sqrt{5}i)/3$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2/3, -\sqrt{5}/3)$.

Final answer

$(2/3, -\sqrt{5}/3)$

Worked Solutions **Complex Numbers**

Q015 Written

State the real and imaginary parts of $(7 - \sqrt{2}i)/2$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(7/2, -\sqrt{2}/2)$.

Final answer

$(7/2, -\sqrt{2}/2)$

Q016 Written

State the real and imaginary parts of -4.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-4,\ 0)$.

Final answer

$(-4,\ 0)$

Worked Solutions **Complex Numbers**

Q017 Written

State the real and imaginary parts of $1/4$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1/4,\ 0)$.

Final answer

$(1/4,\ 0)$

Q018 Written

State the real and imaginary parts of $\sqrt{3}i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, \sqrt{3})$.

Final answer

$(0, \sqrt{3})$

Worked Solutions **Complex Numbers**

Q019 Written

State the real and imaginary parts of $-i$.

Solution

1. Rewrite the number as (a+bi), including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0,\ -1)$.

Final answer

$(0,\ -1)$

Q020 Written

Find the real values of x and y satisfying $(2x+y)+(3x-y)i=8+7i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x=3, y=2$.

Final answer

$x=3, y=2$

Worked Solutions **Complex Numbers**

Q021 Written

Find the real values of x and y satisfying $(x+1) - 5i = 4 + (2y+1)i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 3, y = -3$.

Final answer

$x = 3, y = -3$

Q022 Written

Find the real values of x and y satisfying $(x-2) + (x+y)i = 0$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 2, y = -2$.

Final answer

$x = 2, y = -2$

Worked Solutions **Complex Numbers**

Q023 Written

Find the real values of x and y satisfying $(3 + 2i)x + (1 - i)y = 7 + 3i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 2, y = 1$.

Final answer

$x = 2, y = 1$

Q024 Written

Find the real values of x and y satisfying $(1 - i)(x + yi) = 2 + i$.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 1/2, y = 3/2$.

Final answer

$x = 1/2, y = 3/2$

Worked Solutions **Complex Numbers**

Q025 **Written**

Solve the complex-number equation and find the exact real values of x and y: $(1+2i)(x+i) = x(y+i)$**.**

Solution

1. Expand: $((1+2i)(x+i) = x - 2 + (1+2x)i)$.
2. Equate coefficients with (xy+xi): $(x - 2 = xy)$ and $(1 + 2x = x)$.
3. Solving gives $x = -1, y = 3$.

Final answer

$x = -1, y = 3$

Q026 **Written**

Calculate $(3-2i) - (2+5i)$ **and give the answer in standard form.**

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 - 7i$.

Final answer

$1 - 7i$

Worked Solutions **Complex Numbers**

Q027 Written

Calculate $(i-2)^2$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $3-4i$.

Final answer

$3-4i$

Q028 Written

Calculate $i^6+i^{27}+i^{17}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -1.

Final answer

-1

Worked Solutions **Complex Numbers**

Q029 Written

Calculate $(4 + 6i) + (-3 + 2i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 + 8i$.

Final answer

$1 + 8i$

Q030 Written

Calculate $(2 + 3i)(1 - 2i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $8 - i$.

Final answer

$8 - i$

Worked Solutions **Complex Numbers**

Q031 Written

Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: 0.

Final answer

0

Q032 Written

Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $4 - i$.

Final answer

$4 - i$

Worked Solutions **Complex Numbers**

Q033 Written

Calculate $(-4+3i)+(5-6i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1-3i$.

Final answer

$1-3i$

Q034 Written

Calculate $(3-3i)-(5-3i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -2.

Final answer

-2

Worked Solutions **Complex Numbers**

Q035 Written

Calculate $(\sqrt{2}+3i)(\sqrt{2}+i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $-1+4\sqrt{2}i$.

Final answer

$-1+4\sqrt{2}i$

Q036 Written

Calculate $(2+3i)+(5+8i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $7+11i$.

Final answer

$7+11i$

Worked Solutions **Complex Numbers**

Q037 Written

Calculate $(2/5 - 1/2i) - (1/3 + 3/7i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $1/15 - 13/14i$.

Final answer

$1/15 - 13/14i$

Q038 Written

Calculate $(2i - 1)^2$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $-3 - 4i$.

Final answer

$-3 - 4i$

Worked Solutions **Complex Numbers**

Q039 **Written**

Calculate $(2/i+i)(2/i-i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -3.

Final answer

-3

Q040 **Written**

Calculate $(2+i)^3+(2-i)^3$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: 4.

Final answer

4

Worked Solutions **Complex Numbers**

Q041 Written

Calculate $i^5 + i^3 + i$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: i.

Final answer

i

Q042 Written

Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: -1.

Final answer

-1

Worked Solutions **Complex Numbers**

Q043 **Written**

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $-i$.

Final answer

$-i$

Q044 **Written**

Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2 i^{40}$.

Solution

1. Use ($i^{40} = 1$) and (($x - i)(x + i) = x^2 + 1$).
2. The equation becomes ($7x(x^2 + 1) = 14x^2$), so ($7x(x - 1)^2 = 0$).
3. Therefore $x = 0$ or $x = 1$.

Final answer

$x = 0$ or $x = 1$

Worked Solutions **Complex Numbers**

Q045 Written

State the complex conjugate of $2 + 3i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $2 - 3i$.

Final answer

$2 - 3i$

Q046 Written

State the complex conjugate of $7 - i\sqrt{11}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $7 + i\sqrt{11}$.

Final answer

$7 + i\sqrt{11}$

Worked Solutions **Complex Numbers**

Q047 **Written**

State the complex conjugate of 6.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 6.

Final answer

6

Q048 **Written**

State the complex conjugate of $-5i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $5i$.

Final answer

$5i$

Worked Solutions **Complex Numbers**

Q049 **Written**

Calculate $(2+i)/(2-i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5 + 4/5i$.

Final answer

$3/5 + 4/5i$

Q050 **Written**

Calculate $(3-i)/(2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1/2 - 3/2i$.

Final answer

$-1/2 - 3/2i$

Worked Solutions **Complex Numbers**

Q051 **Written**

State the complex conjugate of $1 - 4i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $1 + 4i$.

Final answer

$1 + 4i$

Q052 **Written**

State the complex conjugate of $\sqrt{2} + i\sqrt{10}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{2} - i\sqrt{10}$.

Final answer

$\sqrt{2} - i\sqrt{10}$

Worked Solutions **Complex Numbers**

Q053 **Written**

State the complex conjugate of 2.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 2.

Final answer

2

Q054 **Written**

State the complex conjugate of $-8i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8i$.

Final answer

$8i$

Worked Solutions **Complex Numbers**

Q055 Written

Calculate $(3+i)/(1+2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1-i$.

Final answer

$1-i$

Q056 Written

Calculate $(1-i)/i$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1-i$.

Final answer

$-1-i$

Worked Solutions **Complex Numbers**

Q057 Written

State the complex conjugate of $3 + 4i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $3 - 4i$.

Final answer

$3 - 4i$

Q058 Written

State the complex conjugate of $9 + 8i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $9 - 8i$.

Final answer

$9 - 8i$

Worked Solutions **Complex Numbers**

Q059 **Written**

State the complex conjugate of $8 + i\sqrt{3}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8 - i\sqrt{3}$.

Final answer

$8 - i\sqrt{3}$

Q060 **Written**

State the complex conjugate of 5.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 5.

Final answer

5

Worked Solutions **Complex Numbers**

Q061 **Written**

State the complex conjugate of -15.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is -15.

Final answer

-15

Q062 **Written**

State the complex conjugate of $\sqrt{7}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{7}$.

Final answer

$\sqrt{7}$

Worked Solutions **Complex Numbers**

Q063 **Written**

State the complex conjugate of i.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i$.

Final answer

$-i$

Q064 **Written**

State the complex conjugate of $-50i$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $50i$.

Final answer

$50i$

Worked Solutions **Complex Numbers**

Q065 **Written**

State the complex conjugate of $i\sqrt{6}$.

Solution

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i\sqrt{6}$.

Final answer

$-i\sqrt{6}$

Q066 **Written**

Calculate $(1+i)/(1-i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is i.

Final answer

i

Worked Solutions **Complex Numbers**

Q067 **Written**

Calculate $(3 + i)/(2 + i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $7/5 - 1/5i$.

Final answer

$7/5 - 1/5i$

Q068 **Written**

Calculate $(\sqrt{5} + i)/(\sqrt{5} - i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2/3 + \sqrt{5}/3i$.

Final answer

$2/3 + \sqrt{5}/3i$

Worked Solutions **Complex Numbers**

Q069 Written

Calculate $5i/(1 + 2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2 + i$.

Final answer

$2 + i$

Q070 Written

Calculate $5i/(4 + 3i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5 + 4/5i$.

Final answer

$3/5 + 4/5i$

Worked Solutions **Complex Numbers**

Q071 **Written**

Calculate $(1 - 3i)/(3 - 2i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $9/13 - 7/13i$.

Final answer

$9/13 - 7/13i$

Q072 **Written**

Calculate $1/i$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-i$.

Final answer

$-i$

Worked Solutions **Complex Numbers**

Q073 **Written**

Calculate $(2+i)/(3i)$ and give the answer in the form a+bi.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1/3 - 2/3i$.

Final answer

$1/3 - 2/3i$

Q074 **Written**

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)}\right)^{2027}$.

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The base simplifies exactly to (displaystyle
rac-2-144i85).
3. Hence the exact result is $((-2-144i)/85)^{2027}$; no expansion of the 2027th power is required.

Final answer

$((-2-144i)/85)^{2027}$

Worked Solutions **Complex Numbers**

Q075 **Written**

Express the square roots of $-6intermsofi$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm i\sqrt{6}$.

Final answer

$\pm i\sqrt{6}$

Q076 **Written**

Express $\sqrt{-12}\sqrt{-6}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-6\sqrt{2}$.

Final answer

$-6\sqrt{2}$

Worked Solutions **Complex Numbers**

Q077 Written

Express $\sqrt{8}/\sqrt{-18}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-(2/3)i$.

Final answer

$-(2/3)i$

Q078 Written

Express $\sqrt{-7}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $i\sqrt{7}$.

Final answer

$i\sqrt{7}$

Worked Solutions **Complex Numbers**

Q079 **Written**

Express $\sqrt{-27}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $3i\sqrt{3}$.

Final answer

$3i\sqrt{3}$

Q080 **Written**

Express $-\sqrt{-12}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2i\sqrt{3}$.

Final answer

$-2i\sqrt{3}$

Worked Solutions **Complex Numbers**

Q081 Written

Express the square roots of -12 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm 2i\sqrt{3}$.

Final answer

$\pm 2i\sqrt{3}$

Q082 Written

Express $\sqrt{-3}\sqrt{-6}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-3\sqrt{2}$.

Final answer

$-3\sqrt{2}$

Worked Solutions **Complex Numbers**

Q083 **Written**

Express $\sqrt{27}/\sqrt{-3}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-3i$.

Final answer

$-3i$

Q084 **Written**

Express $(3+\sqrt{-2})^2$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $7+6\sqrt{2}i$.

Final answer

$7+6\sqrt{2}i$

Worked Solutions **Complex Numbers**

Q085 Written

Express $\sqrt{-2}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $i\sqrt{2}$.

Final answer

$i\sqrt{2}$

Q086 Written

Express $\sqrt{-9}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $3i$.

Final answer

$3i$

Worked Solutions **Complex Numbers**

Q087 **Written**

Express $\sqrt{-45}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $3i\sqrt{5}$.

Final answer

$3i\sqrt{5}$

Q088 **Written**

Express $-\sqrt{-24}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2i\sqrt{6}$.

Final answer

$-2i\sqrt{6}$

Worked Solutions **Complex Numbers**

Q089 **Written**

Express $-\sqrt{-36}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-6i$.

Final answer

$-6i$

Q090 **Written**

Express the square roots of -2 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm i\sqrt{2}$.

Final answer

$\pm i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q091 Written

Express the square roots of -18 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm 3i\sqrt{2}$.

Final answer

$\pm 3i\sqrt{2}$

Q092 Written

Express the square roots of -16 in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $\pm 4i$.

Final answer

$\pm 4i$

Worked Solutions **Complex Numbers**

Q093 Written

Express $\sqrt{-2}\,\sqrt{-6}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2\sqrt{3}$.

Final answer

$-2\sqrt{3}$

Q094 Written

Express $\sqrt{-28}\,\sqrt{-35}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-14\sqrt{5}$.

Final answer

$-14\sqrt{5}$

Worked Solutions **Complex Numbers**

Q095 **Written**

Express $\sqrt{8}/\sqrt{-2}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2i$.

Final answer

$-2i$

Q096 **Written**

Express $\sqrt{27}/\sqrt{-12}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-(3/2)i$.

Final answer

$-(3/2)i$

Worked Solutions **Complex Numbers**

Q097 Written

Express $\sqrt{-32}+\sqrt{-98}+\sqrt{-8}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $13i\sqrt{2}$.

Final answer

$13i\sqrt{2}$

Q098 Written

Express $\sqrt{-98}-\sqrt{-72}+\sqrt{-50}$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $6i\sqrt{2}$.

Final answer

$6i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q099 **Written**

Express $(\sqrt{3}+\sqrt{-5})^2$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore $-2+2i\sqrt{15}$.

Final answer

$-2+2i\sqrt{15}$

Q100 **Written**

Express $(2+\sqrt{-3})(2-\sqrt{-3})$ in terms of i.

Solution

1. Use (sqrt-a=isqrt a) for (a>0), then simplify the positive radical.
2. Therefore 7.

Final answer

7

Worked Solutions **Complex Numbers**

Q101 Written

Solve over the complex numbers: $(2x-8)^2 = -10$.

Solution

1. Take square roots: $(2x - 8 = pm$ isqrt10).
2. Divide by 2 to obtain $x = 4 \pm (\sqrt{10}/2)i$.

Final answer

$x = 4 \pm (\sqrt{10}/2)i$

Q102 MCQ

For $2+3i$**, select the ordered pair (real part, imaginary part):**

Solution

1. Simplify $2+3i$ using the relevant family rule.
2. The correct value is $(2,\ 3)$.

Correct option: C

Final answer

$(2,\ 3)$

Worked Solutions **Complex Numbers**

Q103 MCQ

For $-4 + 5i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-4 + 5i$ using the relevant family rule.
2. The correct value is $(-4,\ 5)$.

Correct option: D

Final answer

$(-4,\ 5)$

Q104 MCQ

For $7 - 2i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $7 - 2i$ using the relevant family rule.
2. The correct value is $(7,\ -2)$.

Correct option: D

Final answer

$(7,\ -2)$

Worked Solutions **Complex Numbers**

Q105 MCQ

For $(1 - \sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $(1 - \sqrt{2}i)/3$ using the relevant family rule.
2. The correct value is $\left(\frac{1}{3},\ -\frac{\sqrt{2}}{3}\right)$.

Correct option: B

Final answer

$\left(\frac{1}{3},\ -\frac{\sqrt{2}}{3}\right)$

Q106 MCQ

For -6, select the ordered pair (real part, imaginary part):

Solution

1. Simplify -6 using the relevant family rule.
2. The correct value is $(-6,\ 0)$.

Correct option: D

Final answer

$(-6,\ 0)$

Worked Solutions **Complex Numbers**

Q107 MCQ

For $4i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $4i$ using the relevant family rule.
2. The correct value is $(0,\ 4)$.

Correct option: A

Final answer

$(0,\ 4)$

Q108 MCQ

For $-i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-i$ using the relevant family rule.
2. The correct value is $(0,\ -1)$.

Correct option: B

Final answer

$(0,\ -1)$

Worked Solutions **Complex Numbers**

Q109 *MCQ*

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{5}i$ using the relevant family rule.
2. The correct value is $\left(0,\ \sqrt{5}\right)$.

Correct option: C

Final answer

$\left(0,\ \sqrt{5}\right)$

Q110 *MCQ*

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $3/2 + \sqrt{3}i$ using the relevant family rule.
2. The correct value is $\left(\frac{3}{2},\ \sqrt{3}\right)$.

Correct option: B

Final answer

$\left(\frac{3}{2},\ \sqrt{3}\right)$

Worked Solutions **Complex Numbers**

Q111 MCQ

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-1/4 - 2i$ using the relevant family rule.
2. The correct value is $\left(-\frac{1}{4},\ -2\right)$.

Correct option: D

Final answer

$\left(-\frac{1}{4},\ -2\right)$

Q112 MCQ

For $5/3$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $5/3$ using the relevant family rule.
2. The correct value is $\left(\frac{5}{3},\ 0\right)$.

Correct option: D

Final answer

$\left(\frac{5}{3},\ 0\right)$

Worked Solutions **Complex Numbers**

Q113 MCQ

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{7}i$ using the relevant family rule.
2. The correct value is $\left(0,\ -\sqrt{7}\right)$.

Correct option: C

Final answer

$\left(0,\ -\sqrt{7}\right)$

Q114 MCQ

For $-9+\sqrt{2}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-9+\sqrt{2}i$ using the relevant family rule.
2. The correct value is $\left(-9,\ \sqrt{2}\right)$.

Correct option: D

Final answer

$\left(-9,\ \sqrt{2}\right)$

Worked Solutions **Complex Numbers**

Q115 *MCQ*

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $2/5 - 3/4i$ using the relevant family rule.
2. The correct value is $\left(\frac{2}{5},\ -\frac{3}{4}\right)$.

Correct option: B

Final answer

$\left(\frac{2}{5},\ -\frac{3}{4}\right)$

Q116 *MCQ*

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{11}$ using the relevant family rule.
2. The correct value is $\left(-\sqrt{11},\ 0\right)$.

Correct option: B

Final answer

$\left(-\sqrt{11},\ 0\right)$

Worked Solutions **Complex Numbers**

Q117 *MCQ*

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{13}i$ using the relevant family rule.
2. The correct value is $\left(0,\ \sqrt{13}\right)$.

Correct option: C

Final answer

$\left(0,\ \sqrt{13}\right)$

Q118 *MCQ*

For $(2+i)x+(1-i)y=3+3i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(2,\ -1)$.

Correct option: D

Final answer

$(2,\ -1)$

Worked Solutions **Complex Numbers**

Q119 MCQ

For $(1+2i)x+(3-i)y=5-4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1,\ 2)$.

Correct option: A

Final answer

$(-1,\ 2)$

Q120 MCQ

For $(3-2i)x+(1+i)y=6+i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1,\ 3)$.

Correct option: B

Final answer

$(1,\ 3)$

Worked Solutions **Complex Numbers**

Q121 *MCQ*

For $(2-i)x+(4+2i)y=4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-2,\ 1)$.

Correct option: B

Final answer

$(-2,\ 1)$

Q122 *MCQ*

For $(1+i)x+(2+3i)y=1$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(3,\ -1)$.

Correct option: C

Final answer

$(3,\ -1)$

Worked Solutions **Complex Numbers**

Q123 MCQ

For $(4+i)x+(1-2i)y=2+5i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1,\ -2)$.

Correct option: D

Final answer

$(1,\ -2)$

Q124 MCQ

For $(2+3i)x+(3+i)y=-8-5i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1,\ -2)$.

Correct option: D

Final answer

$(-1,\ -2)$

Worked Solutions **Complex Numbers**

Q125 *MCQ*

For $(1-i)x+(2-i)y=6-4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(2,\ 2)$.

Correct option: C

Final answer

$(2,\ 2)$

Q126 *MCQ*

For $(3+4i)-(1-2i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2=-1$), reducing powers of i modulo 4.
2. The simplified value is $2+6i$.

Correct option: C

Final answer

$2+6i$

Worked Solutions **Complex Numbers**

Q127 MCQ

For $(-2+i)+(5-3i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2=-1)$, reducing powers of i modulo 4.
2. The simplified value is $3-2i$.

Correct option: C

Final answer

$3-2i$

Q128 MCQ

For $(1+2i)(3-i)$, select the simplified standard form:

Solution

1. Expand and use $(i^2=-1)$, reducing powers of i modulo 4.
2. The simplified value is $5+5i$.

Correct option: D

Final answer

$5+5i$

Worked Solutions **Complex Numbers**

Q129 *MCQ*

For $(\sqrt{3}+i)(\sqrt{3}-i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 4.

Correct option: A

Final answer

4

Q130 *MCQ*

For $(2-i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2-i)^2$.

Correct option: C

Final answer

$(2-i)^2$

Worked Solutions **Complex Numbers**

Q131 MCQ

For $(1 + 3i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-8 + 6i$.

Correct option: A

Final answer

$-8 + 6i$

Q132 MCQ

For i^{37}, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is i.

Correct option: B

Final answer

i

Worked Solutions **Complex Numbers**

Q133 MCQ

For $i^{22} + i^{9}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: B

Final answer

$-1 + i$

Q134 MCQ

For $i^{15} - i^{6}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $1 - i$.

Correct option: B

Final answer

$1 - i$

Worked Solutions **Complex Numbers**

Q135 MCQ

For $(4 - 2i) + (1 + 5i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5 + 3i$.

Correct option: C

Final answer

$5 + 3i$

Q136 MCQ

For $(2 + i)(2 - i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 5.

Correct option: A

Final answer

5

Worked Solutions **Complex Numbers**

Q137 MCQ

For $(\sqrt{2}+2i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $\left(\sqrt{2}+2i\right)^2$.

Correct option: C

Final answer

$\left(\sqrt{2}+2i\right)^2$

Q138 MCQ

For $i^8 + i^{11} + i^{14}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-i$.

Correct option: C

Final answer

$-i$

Worked Solutions **Complex Numbers**

Q139 MCQ

For $(3-i)^3$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(3-i)^3$.

Correct option: D

Final answer

$(3-i)^3$

Q140 MCQ

For $(1-2i)(-2+i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5i$.

Correct option: D

Final answer

$5i$

Worked Solutions **Complex Numbers**

Q141 MCQ

For $i^5 + i^{10}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: D

Final answer

$-1 + i$

Q142 MCQ

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $2 + 6i$.

Correct option: D

Final answer

$2 + 6i$

Worked Solutions **Complex Numbers**

Q143 MCQ

For $(2-3i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2-3i)^2$.

Correct option: C

Final answer

$(2-3i)^2$

Q144 MCQ

For the expression State the complex conjugate of $3-2i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $3+2i$.

Correct option: C

Final answer

$3+2i$

Worked Solutions **Complex Numbers**

Q145 *MCQ*

For the expression State the complex conjugate of $-4+\sqrt{3}i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-4-\sqrt{3}i$.

Correct option: A

Final answer

$-4-\sqrt{3}i$

Q146 *MCQ*

For the expression State the complex conjugate of $7i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-7i$.

Correct option: C

Final answer

$-7i$

Worked Solutions **Complex Numbers**

Q147 MCQ

For the expression State the complex conjugate of $-\sqrt{5}$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\sqrt{5}$.

Correct option: C

Final answer

$-\sqrt{5}$

Q148 MCQ

For the expression State the complex conjugate of $1/2 - 3i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} + 3i$.

Correct option: B

Final answer

$\frac{1}{2} + 3i$

Worked Solutions **Complex Numbers**

Q149 MCQ

For $(2+i)/(1-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2}+\frac{3i}{2}$.

Correct option: A

Final answer

$\frac{1}{2}+\frac{3i}{2}$

Q150 MCQ

For $(4-3i)/(2+i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $1-2i$.

Correct option: A

Final answer

$1-2i$

Worked Solutions **Complex Numbers**

Q151 MCQ

For $(1+2i)/(3-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{10}+\frac{7i}{10}$.

Correct option: B

Final answer

$\frac{1}{10}+\frac{7i}{10}$

Q152 MCQ

For $5i/(2-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $i\,(2+i)$.

Correct option: C

Final answer

$i\,(2+i)$

Worked Solutions **Complex Numbers**

Q153 MCQ

For $(3 - i)/(1 + 3i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D

Final answer

$-i$

Q154 MCQ

For $1/i$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D

Final answer

$-i$

Worked Solutions **Complex Numbers**

Q155 *MCQ*

For $(2+3i)/(2-3i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13}+\frac{12i}{13}$.

Correct option: B

Final answer

$-\frac{5}{13}+\frac{12i}{13}$

Q156 *MCQ*

For $(\sqrt{2}+i)/(\sqrt{2}-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{2}+i}{\sqrt{2}-i}$.

Correct option: B

Final answer

$\frac{\sqrt{2}+i}{\sqrt{2}-i}$

Worked Solutions **Complex Numbers**

Q157 MCQ

For $(1-4i)/(3+2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13}-\frac{14i}{13}$.

Correct option: D

Final answer

$-\frac{5}{13}-\frac{14i}{13}$

Q158 MCQ

For $7i/(1-i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{7}{2}+\frac{7i}{2}$.

Correct option: C

Final answer

$-\frac{7}{2}+\frac{7i}{2}$

Worked Solutions **Complex Numbers**

Q159 MCQ

For the expression State the complex conjugate of $-2 + 5i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-2 - 5i$.

Correct option: B

Final answer

$-2 - 5i$

Q160 MCQ

For the expression State the complex conjugate of $6 - \sqrt{7}i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $6 + \sqrt{7}i$.

Correct option: D

Final answer

$6 + \sqrt{7}i$

Worked Solutions **Complex Numbers**

Q161 *MCQ*

For the expression State the complex conjugate of $-i\sqrt{2}$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{2}i$.

Correct option: B

Final answer

$\sqrt{2}i$

Q162 *MCQ*

For the expression State the complex conjugate of 9:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is 9.

Correct option: C

Final answer

9

Worked Solutions **Complex Numbers**

Q163 MCQ

For the expression State the complex conjugate of $-3/4 + 2i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{3}{4} - 2i$.

Correct option: A

Final answer

$-\frac{3}{4} - 2i$

Q164 MCQ

For $(5 + 2i)/(1 - 2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{5} + \frac{12i}{5}$.

Correct option: D

Final answer

$\frac{1}{5} + \frac{12i}{5}$

Worked Solutions **Complex Numbers**

Q165 MCQ

For $(2-i)/(4+i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{7}{17}-\frac{6i}{17}$.

Correct option: A

Final answer

$\frac{7}{17}-\frac{6i}{17}$

Q166 MCQ

For $3i/(2+3i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{9}{13}+\frac{6i}{13}$.

Correct option: A

Final answer

$\frac{9}{13}+\frac{6i}{13}$

Worked Solutions **Complex Numbers**

Q167 MCQ

For $(\sqrt{3}-i)/(\sqrt{3}+i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\dfrac{\sqrt{3}-i}{\sqrt{3}+i}$.

Correct option: C

Final answer

$$\frac{\sqrt{3}-i}{\sqrt{3}+i}$$

Q168 MCQ

For $(1+i)/(2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\dfrac{1}{2}-\dfrac{i}{2}$.

Correct option: A

Final answer

$$\frac{1}{2}-\frac{i}{2}$$

Worked Solutions **Complex Numbers**

Q169 MCQ

For the expression State the complex conjugate of $4 - 7i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $4 + 7i$.

Correct option: D

Final answer

$4 + 7i$

Q170 MCQ

For the expression State the complex conjugate of $\sqrt{10} + 2i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{10} - 2i$.

Correct option: A

Final answer

$\sqrt{10} - 2i$

Worked Solutions **Complex Numbers**

Q171 MCQ

For the expression State the complex conjugate of $-8i$:, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $8i$.

Correct option: B

Final answer

$8i$

Q172 MCQ

For $(3 + 2i)/(3 - 2i)$, select the value in the form a+bi:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{5}{13} + \frac{12i}{13}$.

Correct option: B

Final answer

$$\frac{5}{13} + \frac{12i}{13}$$

Worked Solutions **Complex Numbers**

Q173 MCQ

For $\sqrt{-5}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{5}$.

Correct option: B

Final answer

$i\sqrt{5}$

Q174 MCQ

For $\sqrt{-20}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{5}$.

Correct option: A

Final answer

$2i\sqrt{5}$

Worked Solutions **Complex Numbers**

Q175 MCQ

For $-\sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i\sqrt{2}$.

Correct option: A

Final answer

$-3i\sqrt{2}$

Q176 MCQ

For $squarerootsof-8$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{2}$.

Correct option: C

Final answer

$\pm 2i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q177 MCQ

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 3.

Correct option: A

Final answer

3

Q178 MCQ

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -4.

Correct option: B

Final answer

-4

Worked Solutions **Complex Numbers**

Q179 *MCQ*

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: C

Final answer

$-2i$

Q180 *MCQ*

For $(2+\sqrt{-3})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $1+4i\sqrt{3}$.

Correct option: D

Final answer

$1+4i\sqrt{3}$

Worked Solutions **Complex Numbers**

Q181 MCQ

For $\sqrt{-72}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: D

Final answer

$6i\sqrt{2}$

Q182 MCQ

For $squarerootsof - 50$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 5i\sqrt{2}$.

Correct option: D

Final answer

$\pm 5i\sqrt{2}$

Q183 MCQ

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -6.

Correct option: A

Final answer

-6

Q184 MCQ

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: D

Final answer

$-2i$

Worked Solutions **Complex Numbers**

Q185 MCQ

For $\sqrt{-32}+\sqrt{-8}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: B

Final answer

$6i\sqrt{2}$

Q186 MCQ

For $\sqrt{-98}-\sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $4i\sqrt{2}$.

Correct option: B

Final answer

$4i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q187 MCQ

For $(\sqrt{5}+\sqrt{-2})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $3+2i\sqrt{10}$.

Correct option: A

Final answer

$3+2i\sqrt{10}$

Q188 MCQ

For $(3+\sqrt{-5})(3-\sqrt{-5})$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 14.

Correct option: D

Final answer

14

Worked Solutions **Complex Numbers**

Q189 MCQ

For $\sqrt{-7}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{7}$.

Correct option: C

Final answer

$i\sqrt{7}$

Q190 MCQ

For $\sqrt{-28}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{7}$.

Correct option: A

Final answer

$2i\sqrt{7}$

Worked Solutions **Complex Numbers**

Q191 MCQ

For $squarerootsof - 12$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{3}$.

Correct option: A

Final answer

$\pm 2i\sqrt{3}$

Q192 MCQ

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i$.

Correct option: A

Final answer

$-3i$

Worked Solutions **Complex Numbers**

Q193 MCQ

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -6.

Correct option: D

Final answer

-6

Q194 MCQ

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-4i$.

Correct option: C

Final answer

$-4i$

Worked Solutions **Complex Numbers**

Q195 **MCQ**

For $\sqrt{-8} + \sqrt{-18}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{2}$.

Correct option: C

Final answer

$5i\sqrt{2}$

Q196 **MCQ**

For $(1 + \sqrt{-2})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-1 + 2i\sqrt{2}$.

Correct option: A

Final answer

$-1 + 2i\sqrt{2}$

Worked Solutions **Complex Numbers**

Q197 MCQ

For $(4+\sqrt{-3})(4-\sqrt{-3})$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 19.

Correct option: B

Final answer

19

Q198 MCQ

For $\sqrt{-75}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{3}$.

Correct option: D

Final answer

$5i\sqrt{3}$

Q199 MCQ

Solve for the real values x and y: $(1+3i)(x+i)=x(y+i)$**:**

Solution

1. Expand and equate real and imaginary parts.
2. The system gives $(x=-1/2)$ and $(y=7)$.

Correct option: B

Final answer

$\left(-\frac{1}{2},\ 7\right)$

Q200 MCQ

Given x is real, solve: $(4x-4i)(x^2+xi)=10x^2i^{32}$**:**

Solution

1. Use $(i^{32}=1)$ and $((x-i)(x+i)=x^2+1)$.
2. The real equation is $(4x(x^2+1)=10x^2)$, giving $x=0,\ 1/2$, or 2.

Correct option: D

Final answer

$\left\{0,\ 2,\ \frac{1}{2}\right\}$

Worked Solutions **Complex Numbers**

Q201 MCQ

Simplify exactly: $(i(8-6i)/((1+2i)(3-i)))^{2027}$:

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The exact base simplifies to ((7+i)/5). Keep the 2027th power unexpanded.

Correct option: B

Final answer

$((7+i)/5)^{2027}$

Q202 MCQ

Solve over the complex numbers: $(3x-6)^2=-8$:

Solution

1. Take square roots: ($3x-6=pm$ 2isqrt2).
2. Divide by 3 to obtain ($x=2pm$
rac23isqrt2).

Correct option: B

Final answer

$x=2\pm(2/3)i\sqrt{2}$

Worked Solutions **Complex Numbers**

Quiz MCQ 1 Concept Quiz · MCQ

For 4-3i, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $4 - 3i$ using the relevant family rule.
2. The correct value is $(4,\ -3)$.

Correct option: A

Final answer

$(4,\ -3)$

Quiz MCQ 2 Concept Quiz · MCQ

For (1+2i)(2-i), select the simplified value:

Solution

1. Simplify $(1 + 2i)(2 - i)$ using the relevant family rule.
2. The correct value is $4 + 3i$.

Correct option: A

Final answer

$4 + 3i$

Worked Solutions **Complex Numbers**

Quiz MCQ 3 Concept Quiz · MCQ

For (3+i)/(1-i), select the value in the form a+bi:

Solution

1. Simplify $(3+i)/(1-i)$ using the relevant family rule.
2. The correct value is $1+2i$.

Correct option: D

Final answer

$1+2i$

Quiz MCQ 4 Concept Quiz · MCQ

For sqrt(-27), select the expression in terms of i:

Solution

1. Simplify $\sqrt{-27}$ using the relevant family rule.
2. The correct value is $3i\sqrt{3}$.

Correct option: A

Final answer

$3i\sqrt{3}$

Worked Solutions **Complex Numbers**

Quiz WRITTEN 5 Concept Quiz · Written

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $(5/2, -\sqrt{5}/2)$

Final answer

$(5/2, -\sqrt{5}/2)$

Quiz WRITTEN 6 Concept Quiz · Written

Solve for real x,y: $(2+i)x + (1-i)y = 3+i$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 4/3, y = 1/3$

Final answer

$x = 4/3, y = 1/3$

Worked Solutions **Complex Numbers**

Quiz WRITTEN 7 **Concept Quiz · Written**

Calculate $i^{18} + i^7 - i^3$ in standard form.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: -1

Final answer

-1

Quiz WRITTEN 8 **Concept Quiz · Written**

Solve over the complex numbers: $(x-3)^2 = -12$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 3 \pm 2i\sqrt{3}$

Final answer

$x = 3 \pm 2i\sqrt{3}$

Answer key

Use the question number shown on each page.

Q001 $(3,\ -2)$

Q002 $(4/3, -\sqrt{5}/3)$

Q003 $(-8,\ 0)$

Q004 $(0,\ 2)$

Q005 $x = -1, y = 4$

Q006 $(-3,\ 5)$

Q007 $(-1/2, -\sqrt{3}/2)$

Q008 $(1,\ 0)$

Q009 $(0,\ 3)$

Q010 $x = 6, y = -2$

Q011 $x = -2, y = 3$

Q012 $(2,\ 5)$

Q013 $(-25,\ 13)$

Q014 $(2/3, -\sqrt{5}/3)$

Q015 $(7/2, -\sqrt{2}/2)$

Q016 $(-4,\ 0)$

Q017 $(1/4,\ 0)$

Q018 $(0, \sqrt{3})$

Q019 $(0,\ -1)$

Q020 $x = 3, y = 2$

Q021 $x = 3, y = -3$

Q022 $x = 2, y = -2$

Q023 $x = 2, y = 1$

Q024 $x = 1/2, y = 3/2$

Q025 $x = -1, y = 3$

Q026 $1 - 7i$

Q027 $3 - 4i$

Q028 -1

Q029 $1 + 8i$

Q030 $8 - i$

Q031 0

Q032 $4 - i$

Q033 $1 - 3i$

Q034 -2

Q035 $-1 + 4\sqrt{2}i$

Q036 $7 + 11i$

Q037 $1/15 - 13/14i$

Q038 $-3 - 4i$

Q039 -3

Q040 4

Q041 i

Q042 -1

Q043 $-i$

Q044 $x = 0$ or $x = 1$

Q045 $2 - 3i$

Q046 $7 + i\sqrt{11}$

Q047 6

Q048 $5i$

Q049 $3/5 + 4/5i$

Q050 $-1/2 - 3/2i$

Q051 $1 + 4i$

Q052 $\sqrt{2} - i\sqrt{10}$

Q053 2

Q054 $8i$

Q055 $1 - i$

Q056 $-1 - i$

Q057 $3 - 4i$

Q058 $9 - 8i$

Q059 $8 - i\sqrt{3}$

Q060 5

Q061 -15

Q062 $\sqrt{7}$

Q063 $-i$

Q064 $50i$

Q065 $-i\sqrt{6}$

Q066 i

Q067 $7/5 - 1/5i$

Q068 $2/3 + \sqrt{5}/3i$

Q069 $2 + i$

Q070 $3/5 + 4/5i$

Q071 $9/13 - 7/13i$

Q072 $-i$

Q073 $1/3 - 2/3i$

Q074 $((-2 - 144i)/85)^{2027}$

Q075 $\pm i\sqrt{6}$

Q076 $-6\sqrt{2}$

Q077 $-(2/3)i$

Q078 $i\sqrt{7}$

Q079 $3i\sqrt{3}$

Q080 $-2i\sqrt{3}$

Q081 $\pm 2i\sqrt{3}$

Q082 $-3\sqrt{2}$

Q083 $-3i$

Q084 $7 + 6\sqrt{2}i$

Q085 $i\sqrt{2}$

Q086 $3i$

Q087 $3i\sqrt{5}$

Q088 $-2i\sqrt{6}$

Q089 $-6i$

Q090 $\pm i\sqrt{2}$

Q091 $\pm 3i\sqrt{2}$

Q092 $\pm 4i$

Q093 $-2\sqrt{3}$

Q094 $-14\sqrt{5}$

Q095 $-2i$

Q096 $-(3/2)i$

Q097 $13i\sqrt{2}$

Q098 $6i\sqrt{2}$

Q099 $-2 + 2i\sqrt{15}$

Q100 7

Q101 $x = 4 \pm (\sqrt{10}/2)i$

Q102 C

Q103 D

Q104 D

Q105 B
Q106 D
Q107 A
Q108 B
Q109 C
Q110 B
Q111 D
Q112 D
Q113 C
Q114 D
Q115 B
Q116 B
Q117 C
Q118 D
Q119 A
Q120 B
Q121 B
Q122 C
Q123 D
Q124 D
Q125 C
Q126 C
Q127 C
Q128 D
Q129 A
Q130 C
Q131 A
Q132 B
Q133 B
Q134 B
Q135 C
Q136 A
Q137 C
Q138 C
Q139 D
Q140 D
Q141 D
Q142 D
Q143 C
Q144 C
Q145 A
Q146 C
Q147 C
Q148 B
Q149 A
Q150 A
Q151 B
Q152 C
Q153 D
Q154 D
Q155 B
Q156 B
Q157 D
Q158 C
Q159 B
Q160 D
Q161 B
Q162 C
Q163 A
Q164 D
Q165 A
Q166 A
Q167 C
Q168 A
Q169 D
Q170 A
Q171 B
Q172 B
Q173 B
Q174 A
Q175 A
Q176 C
Q177 A
Q178 B
Q179 C
Q180 D
Q181 D
Q182 D
Q183 A
Q184 D
Q185 B
Q186 B
Q187 A
Q188 D
Q189 C
Q190 A
Q191 A
Q192 A
Q193 D
Q194 C
Q195 C
Q196 A
Q197 B
Q198 D
Q199 B
Q200 D
Q201 B
Q202 B
Quiz MCQ 1 A
Quiz MCQ 2 A
Quiz MCQ 3 D
Quiz MCQ 4 A
Quiz WRITTEN 5 $(5/2, -\sqrt{5}/2)$
Quiz WRITTEN 6 $x = 4/3, y = 1/3$
Quiz WRITTEN 7 -1
Quiz WRITTEN 8 $x = 3 \pm 2i\sqrt{3}$

Concept 2 | Quadratic Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum/product

$$\alpha + \beta = -\frac{b}{a}, \ \alpha\beta = \frac{c}{a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Quadratic Equations**

Q001 **Written**

Solve $x^2 + 8 = 0$.

Solution

1. $x^2 = -8$
2. $x = \pm\sqrt{-8} = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
3. Therefore the final answer is $x = \pm 2i\sqrt{2}$.

Final answer

$x = \pm 2i\sqrt{2}$

Q002 **Written**

Solve $x^2 + 4 = 0$.

Solution

1. $x^2 = -4$
2. $x = \pm 2i$
3. Therefore the final answer is $x = \pm 2i$.

Final answer

$x = \pm 2i$

Worked Solutions **Quadratic Equations**

Q003 Written

Solve $x^2 + 49 = 0$.

Solution

1. $x^2 = -49$
2. Therefore the final answer is $x = \pm 7i$.

Final answer

$x = \pm 7i$

Q004 Written

Solve $9x^2 - 4 = 0$.

Solution

1. $(3x - 2)(3x + 2) = 0$
2. Therefore the final answer is $x = \pm\frac{2}{3}$.

Final answer

$x = \pm\frac{2}{3}$

Worked Solutions **Quadratic Equations**

Q005 Written

Solve $x^2 - 12 = 0$.

Solution

1. $x = \pm\sqrt{12}$
2. Therefore the final answer is $x = \pm 2\sqrt{3}$.

Final answer

$x = \pm 2\sqrt{3}$

Q006 Written

Solve $4x^2 + 3 = 0$.

Solution

1. $x^2 = -3/4$
2. Therefore the final answer is $x = \pm\frac{i\sqrt{3}}{2}$.

Final answer

$x = \pm\frac{i\sqrt{3}}{2}$

Worked Solutions **Quadratic Equations**

Q007 MCQ

Solve $x^2 + 8 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 2i\sqrt{2}$.

Correct option: A

Final answer

$x = \pm 2i\sqrt{2}$

Q008 MCQ

Solve $x^2 + 4 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 2i$.

Correct option: A

Final answer

$x = \pm 2i$

Worked Solutions **Quadratic Equations**

Q009 *MCQ*

Solve $x^2 + 49 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 7i$.

Correct option: C

Final answer

$x = \pm 7i$

Q010 *MCQ*

Solve $9x^2 - 4 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm\frac{2}{3}$.

Correct option: C

Final answer

$x = \pm\frac{2}{3}$

Worked Solutions **Quadratic Equations**

Q011 MCQ

Solve $x^2 - 12 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm 2\sqrt{3}$.

Correct option: B

Final answer

$x = \pm 2\sqrt{3}$

Q012 MCQ

Solve $4x^2 + 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm \frac{i\sqrt{3}}{2}$.

Correct option: B

Final answer

$x = \pm \frac{i\sqrt{3}}{2}$

Worked Solutions **Quadratic Equations**

Q013 **Written**

Solve $(x-3)^2 = -8$.

Solution

1. $x - 3 = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
2. Therefore the final answer is $x = 3 \pm 2i\sqrt{2}$.

Final answer

$x = 3 \pm 2i\sqrt{2}$

Q014 **Written**

Solve $(x-2)^2 = 5$.

Solution

1. $x - 2 = \pm\sqrt{5}$
2. Therefore the final answer is $x = 2 \pm \sqrt{5}$.

Final answer

$x = 2 \pm \sqrt{5}$

Worked Solutions **Quadratic Equations**

Q015 Written

Solve $(x+6)^2 = -6$.

Solution

1. $x + 6 = \pm i\sqrt{6}$
2. Therefore the final answer is $x = -6 \pm i\sqrt{6}$.

Final answer

$x = -6 \pm i\sqrt{6}$

Q016 Written

Solve $(x+4)^2 = -18$.

Solution

1. $x + 4 = \pm i\sqrt{18} = \pm 3i\sqrt{2}$
2. Therefore the final answer is $x = -4 \pm 3i\sqrt{2}$.

Final answer

$x = -4 \pm 3i\sqrt{2}$

Worked Solutions **Quadratic Equations**

Q017 Written

Solve $(2x-3)^2 = 7$.

Solution

1. $2x - 3 = \pm\sqrt{7}$
2. Therefore the final answer is $x = \dfrac{3 \pm \sqrt{7}}{2}$.

Final answer

$x = \dfrac{3 \pm \sqrt{7}}{2}$

Q018 MCQ

Solve $(x-3)^2 = -8$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 3 \pm 2i\sqrt{2}$.

Correct option: A

Final answer

$x = 3 \pm 2i\sqrt{2}$

Worked Solutions **Quadratic Equations**

Q019 MCQ

Solve $(x-2)^2 = 5$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 2 \pm \sqrt{5}$.

Correct option: B

Final answer

$x = 2 \pm \sqrt{5}$

Q020 MCQ

Solve $(x+6)^2 = -6$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = -6 \pm i\sqrt{6}$.

Correct option: C

Final answer

$x = -6 \pm i\sqrt{6}$

Worked Solutions **Quadratic Equations**

Q021 *MCQ*

Solve $(x+4)^2 = -18$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = -4 \pm 3i\sqrt{2}$.

Correct option: C

Final answer

$x = -4 \pm 3i\sqrt{2}$

Q022 *MCQ*

Solve $(2x-3)^2 = 7$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{3 \pm \sqrt{7}}{2}$.

Correct option: D

Final answer

$x = \dfrac{3 \pm \sqrt{7}}{2}$

Worked Solutions **Quadratic Equations**

Q023 Written

Solve $2x^2 - 7x - 9 = 0$.

Solution

1. $(x + 1)(2x - 9) = 0$
2. $x = -1 or x = 9/2$
3. Therefore the final answer is $x = -1, \dfrac{9}{2}$.

Final answer

$x = -1, \dfrac{9}{2}$

Q024 Written

Solve $2x^2 + 5x - 3 = 0$.

Solution

1. $(2x - 1)(x + 3) = 0$
2. Therefore the final answer is $x = \dfrac{1}{2}, -3$.

Final answer

$x = \dfrac{1}{2}, -3$

Worked Solutions **Quadratic Equations**

Q025 **Written**

Solve $2x^2 + 7x = 0$.

Solution

1. $x(2x + 7) = 0$
2. Therefore the final answer is $x = 0, -\frac{7}{2}$.

Final answer

$x = 0, -\frac{7}{2}$

Q026 **Written**

Solve $2x^2 + 5x - 12 = 0$.

Solution

1. $(2x - 3)(x + 4) = 0$
2. Therefore the final answer is $x = \frac{3}{2}, -4$.

Final answer

$x = \frac{3}{2}, -4$

Worked Solutions **Quadratic Equations**

Q027 Written

Solve $x^2 - 8x + 15 = 0$.

Solution

1. $(x-3)(x-5) = 0$
2. Therefore the final answer is $x = 3,\ 5$.

Final answer

$x = 3,\ 5$

Q028 MCQ

Solve $2x^2 - 7x - 9 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = -1,\ \frac{9}{2}$.

Correct option: A

Final answer

$x = -1,\ \frac{9}{2}$

Worked Solutions **Quadratic Equations**

Q029 MCQ

Solve $2x^2 + 5x - 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{1}{2}, -3$.

Correct option: D

Final answer

$x = \frac{1}{2}, -3$

Q030 MCQ

Solve $2x^2 + 7x = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 0, -\frac{7}{2}$.

Correct option: D

Final answer

$x = 0, -\frac{7}{2}$

Worked Solutions **Quadratic Equations**

Q031 MCQ

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \frac{3}{2}, -4$.

Correct option: A

Final answer

$x = \frac{3}{2}, -4$

Q032 MCQ

Solve $x^2 - 8x + 15 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 3,\ 5$.

Correct option: A

Final answer

$x = 3,\ 5$

Worked Solutions **Quadratic Equations**

Q033 **Written**

Solve $3x^2 - 4x - 1 = 0$.

Solution

1. $D = 16 + 12 = 28$
2. $x = (4 \pm 2\sqrt{7})/6 = (2 \pm \sqrt{7})/3$
3. Therefore the final answer is $x = \dfrac{2 \pm \sqrt{7}}{3}$.

Final answer

$x = \dfrac{2 \pm \sqrt{7}}{3}$

Q034 **MCQ**

Solve $3x^2 - 4x - 1 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{2 \pm \sqrt{7}}{3}$.

Correct option: B

Final answer

$x = \dfrac{2 \pm \sqrt{7}}{3}$

Worked Solutions **Quadratic Equations**

Q035 Written

Solve $2x^2 + 2\sqrt{6}\,x + 11 = 0.$

Solution

1. $D = (2\sqrt{6})^2 - 4(2)(11) = -64$
2. $x = (-2\sqrt{6} \pm \sqrt{(-64)})/4 = (-\sqrt{6} \pm 4i)/2$
3. Therefore the final answer is $x = \dfrac{-\sqrt{6} \pm 4i}{2}$.

Final answer

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Q036 Written

Solve $2x^2 + 2\sqrt{3}\,x + 5 = 0.$

Solution

1. $D = 12 - 40 = -28$
2. $x = (-2\sqrt{3} \pm i\sqrt{28})/4 = (-\sqrt{3} \pm i\sqrt{7})/2$
3. Therefore the final answer is $x = \dfrac{-\sqrt{3} \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Worked Solutions **Quadratic Equations**

Q037 Written

Solve $x^2 + x + 1 = 0$.

Solution

1. $D = 1 - 4 = -3$
2. $x = (-1 \pm i\sqrt{3})/2$
3. Therefore the final answer is $x = \dfrac{-1 \pm i\sqrt{3}}{2}$.

Final answer

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Q038 Written

Solve $3x^2 - 5x + 3 = 0$.

Solution

1. $D = 25 - 36 = -11$
2. $x = (5 \pm i\sqrt{11})/6$
3. Therefore the final answer is $x = \dfrac{5 \pm i\sqrt{11}}{6}$.

Final answer

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Worked Solutions **Quadratic Equations**

Q039 **Written**

Solve $x^2 - 2\sqrt{3}\,x + 5 = 0$.

Solution

1. $D = 12 - 20 = -8$
2. $x = (2\sqrt{3} \pm i\sqrt{8})/2 = \sqrt{3} \pm i\sqrt{2}$
3. Therefore the final answer is $x = \sqrt{3} \pm i\sqrt{2}$.

Final answer

$x = \sqrt{3} \pm i\sqrt{2}$

Q040 **Written**

Solve $3x^2 + 2\sqrt{2}\,x + 1 = 0$.

Solution

1. $D = 8 - 12 = -4$
2. $x = (-2\sqrt{2} \pm 2i)/6 = (-\sqrt{2} \pm i)/3$
3. Therefore the final answer is $x = \dfrac{-\sqrt{2} \pm i}{3}$.

Final answer

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Worked Solutions **Quadratic Equations**

Q041 Written

Solve $x^2 - x + 2 = 0$.

Solution

1. $D = 1 - 8 = -7$
2. $x = (1 \pm i\sqrt{7})/2$
3. Therefore the final answer is $x = \dfrac{1 \pm i\sqrt{7}}{2}$.

Final answer

$x = \dfrac{1 \pm i\sqrt{7}}{2}$

Q042 Written

Solve $3x^2 - \sqrt{5}\,x + 1 = 0$.

Solution

1. $D = 5 - 12 = -7$
2. $x = (\sqrt{5} \pm i\sqrt{7})/6$
3. Therefore the final answer is $x = \dfrac{\sqrt{5} \pm i\sqrt{7}}{6}$.

Final answer

$x = \dfrac{\sqrt{5} \pm i\sqrt{7}}{6}$

Worked Solutions **Quadratic Equations**

Q043 Written

Solve $6x^2 - 12x + 15 = 0$.

Solution

1. $D = 144 - 360 = -216$
2. $x = (12 \pm i6\sqrt{6})/12 = 1 \pm i\sqrt{6}/2$
3. Therefore the final answer is $x = 1 \pm \dfrac{i\sqrt{6}}{2}$.

Final answer

$x = 1 \pm \dfrac{i\sqrt{6}}{2}$

Q044 MCQ

Solve $2x^2 + 2\sqrt{6}\,x + 11 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-\sqrt{6} \pm 4i}{2}$.

Correct option: B

Final answer

$x = \dfrac{-\sqrt{6} \pm 4i}{2}$

Worked Solutions **Quadratic Equations**

Q045 MCQ

Solve $2x^2 + 2\sqrt{3}\,x + 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-\sqrt{3} \pm i\sqrt{7}}{2}$.

Correct option: C

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Q046 MCQ

Solve $x^2 + x + 1 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-1 \pm i\sqrt{3}}{2}$.

Correct option: D

Final answer

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Worked Solutions **Quadratic Equations**

Q047 MCQ

Solve $3x^2 - 5x + 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{5 \pm i\sqrt{11}}{6}$.

Correct option: B

Final answer

$x = \dfrac{5 \pm i\sqrt{11}}{6}$

Q048 MCQ

Solve $x^2 - 2\sqrt{3}\,x + 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \sqrt{3} \pm i\sqrt{2}$.

Correct option: D

Final answer

$x = \sqrt{3} \pm i\sqrt{2}$

Worked Solutions **Quadratic Equations**

Q049 **MCQ**

Solve $3x^2 + 2\sqrt{2}\,x + 1 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-\sqrt{2} \pm i}{3}$.

Correct option: B

Final answer

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Q050 **MCQ**

Solve $x^2 - x + 2 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{1 \pm i\sqrt{7}}{2}$.

Correct option: C

Final answer

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Worked Solutions **Quadratic Equations**

Q051 MCQ

Solve $3x^2 - \sqrt{5}\,x + 1 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{\sqrt{5} \pm i\sqrt{7}}{6}$.

Correct option: C

Final answer

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Q052 MCQ

Solve $6x^2 - 12x + 15 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = 1 \pm \dfrac{i\sqrt{6}}{2}$.

Correct option: B

Final answer

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Worked Solutions **Quadratic Equations**

Q053 **Written**

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.

Solution

1. $D = 4^2 - 4(3)(7) = -68 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q054 **Written**

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.

Solution

1. $D = (-m)^2 - 4(m^2 - 3m - 9) = -3m^2 + 12m + 36$
2. $D >= 0- > m^2 - 4m - 12 <= 0- > (m+2)(m-6) <= 0$
3. Therefore the final answer is $-2 \leq m \leq 6$.

Final answer

$-2 \leq m \leq 6$

Worked Solutions **Quadratic Equations**

Q055 **Written**

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = k^2 - 4(3 - k) = (k - 2)(k + 6)$
2. $UseD > 0, D = 0, D < 0intervals$
3. Therefore the final answer is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

---

Q056 **Written**

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.

Solution

1. $D = 25 - 48 = -23 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q057 **Written**

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.

Solution

1. $D = 49 + 32 = 81 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q058 **Written**

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.

Solution

1. $D = 100 - 100 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q059 Written

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.

Solution

1. $D = 25 - 12 = 13 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q060 Written

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.

Solution

1. $D = 144 - 144 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q061 **Written**

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.

Solution

1. $D = 64 - 80 = -16 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q062 **Written**

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.

Solution

1. $D = 16 - 16 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q063 Written

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

Solution

1. $D = 9 - 20 = -11 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q064 Written

Determine the nature of the solutions of $x^2 - x - 2 = 0$.

Solution

1. $D = 1 + 8 = 9 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Worked Solutions **Quadratic Equations**

Q065 MCQ

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q066 MCQ

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-2 \leq m \leq 6$.

Correct option: D

Final answer

$-2 \leq m \leq 6$

Worked Solutions Quadratic Equations

Q067 MCQ

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k < -6$ or $k > 2:$ distinct real; $k = -6, 2:$ repeated real; $-6 < k < 2:$ distinct imaginary.

Correct option: C

Final answer

$k < -6$ or $k > 2:$ distinct real; $k = -6, 2:$ repeated real; $-6 < k < 2:$ distinct imaginary

Q068 MCQ

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: C

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q069 MCQ

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions **Quadratic Equations**

Q070 MCQ

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is One repeated real solution.

Correct option: A

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q071 *MCQ*

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct real solutions.

Correct option: C

Final answer

Two distinct real solutions

Worked Solutions **Quadratic Equations**

Q072 MCQ

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is One repeated real solution.

Correct option: C

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q073 *MCQ*

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: B

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q074 MCQ

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is One repeated real solution.

Correct option: A

Final answer

One repeated real solution

Worked Solutions **Quadratic Equations**

Q075 *MCQ*

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions **Quadratic Equations**

Q076 MCQ

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions **Quadratic Equations**

Q077 Written

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

Solution

1. $D = 4m^2 - 4(m + 2)$
2. $D < 0 -> m^2 - m - 2 < 0 -> (m - 2)(m + 1) < 0$
3. Therefore the final answer is $-1 < m < 2$.

Final answer

$-1 < m < 2$

Q078 Written

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 - 4(2a - 3) = (a - 2)(a - 6)$
2. Classify by sign of D
3. Therefore the final answer is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Final answer

$a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions **Quadratic Equations**

Q079 **Written**

Find the values of a for which $x^2 - (a+1)x + 4 = 0$ has a repeated solution.

Solution

1. $D = (a+1)^2 - 16 = 0$
2. $a + 1 = \pm 4$
3. Therefore the final answer is $a = 3$ or $a = -5$.

Final answer

$a = 3$ or $a = -5$

Q080 **Written**

Find the range of m for which $x^2 + (m+6)x - 2m = 0$ has two distinct imaginary solutions.

Solution

1. $D = (m+6)^2 + 8m = m^2 + 20m + 36 = (m+2)(m+18)$
2. $D < 0$
3. Therefore the final answer is $-18 < m < -2$.

Final answer

$-18 < m < -2$

Worked Solutions **Quadratic Equations**

Q081 Written

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

Solution

1. $D = 4m^2 - 4(2m^2 - 3m) = 4m(3 - m)$
2. $D >= 0$
3. Therefore the final answer is $0 \leq m \leq 3$.

Final answer

$0 \leq m \leq 3$

Q082 MCQ

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-1 < m < 2$.

Correct option: C

Final answer

$-1 < m < 2$

Q083 MCQ

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Correct option: C

Final answer

$a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions **Quadratic Equations**

Q084 MCQ

Find the values of a for which $x^2 - (a+1)x + 4 = 0$ has a repeated solution Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a = 3$ or $a = -5$.

Correct option: A

Final answer

$a = 3$ or $a = -5$

Q085 MCQ

Find the range of m for which $x^2 + (m+6)x - 2m = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-18 < m < -2$.

Correct option: D

Final answer

$-18 < m < -2$

Worked Solutions **Quadratic Equations**

Q086 MCQ

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $0 \le m \le 3$.

Correct option: C

Final answer

$0 \le m \le 3$

Q087 Written

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.

Solution

1. $UseS = -b/a = 6/3 = 2andP = c/a = -5/3$
2. Therefore the final answer is $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$.

Final answer

$\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$

Worked Solutions **Quadratic Equations**

Q088 Written

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.

Solution

1. $S = -2/4 = -1/2; P = -7/4$
2. Therefore the final answer is $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$.

Final answer

$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$

Q089 Written

Find the sum and product of the roots of $x^2 + x + 2 = 0$.

Solution

1. $S = -1; P = 2$
2. Therefore the final answer is $S = -1, \quad P = 2$.

Final answer

$S = -1, \quad P = 2$

Q090 Written

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.

Solution

1. $S = 7/3; P = -2$
2. Therefore the final answer is $S = \frac{7}{3}, \quad P = -2$.

Final answer

$S = \frac{7}{3}, \quad P = -2$

Q091 MCQ

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$.

Correct option: C

Final answer

$\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$

Worked Solutions **Quadratic Equations**

Q092 *MCQ*

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$.

Correct option: A

Final answer

$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$

Q093 *MCQ*

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $S = -1, \quad P = 2$.

Correct option: B

Final answer

$S = -1, \quad P = 2$

Worked Solutions **Quadratic Equations**

Q094 MCQ

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $S = \frac{7}{3}, \quad P = -2$.

Correct option: A

Final answer

$S = \frac{7}{3}, \quad P = -2$

Q095 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 3, P = -3/2$
2. $S^2 - 2P = 9 + 3 = 12$
3. Therefore the final answer is 12.

Final answer

12

Worked Solutions **Quadratic Equations**

Q096 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = -3/2$
2. $S^3 - 3PS = 27 + 27/2 = 81/2$
3. Therefore the final answer is $\frac{81}{2}$.

Final answer
$\frac{81}{2}$

Q097 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3/2$
2. $S^2 - 2P = 4 - 3 = 1$
3. Therefore the final answer is 1.

Final answer
1

Worked Solutions **Quadratic Equations**

Q098 **Written**

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3/2$
2. $S^3 - 3PS = 8 - 9 = -1$
3. Therefore the final answer is -1.

Final answer

-1

Q099 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3$
2. $S^2 - 2P = 4 - 6 = -2$
3. Therefore the final answer is -2.

Final answer

-2

Worked Solutions **Quadratic Equations**

Q100 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3$
2. $S^3 - 3PS = 8 - 18 = -10$
3. Therefore the final answer is -10.

Final answer

-10

Q101 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = -2, P = -1/3$
2. $S^2 - 2P = 4 + 2/3 = 14/3$
3. Therefore the final answer is $\frac{14}{3}$.

Final answer

$\frac{14}{3}$

Worked Solutions **Quadratic Equations**

Q102 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = -2, P = -1/3$
2. $S^3 - 3PS = -8 - 2 = -10$
3. Therefore the final answer is -10.

Final answer

-10

Q103 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 12.

Correct option: A

Final answer

12

Worked Solutions **Quadratic Equations**

Q104 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{81}{2}$.

Correct option: B

Final answer

$\frac{81}{2}$

Q105 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 1.

Correct option: A

Final answer

1

Worked Solutions **Quadratic Equations**

Q106 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -1.

Correct option: B

Final answer

-1

Q107 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -2.

Correct option: D

Final answer

-2

Worked Solutions **Quadratic Equations**

Q108 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -10.

Correct option: B

Final answer

-10

Q109 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{14}{3}$.

Correct option: B

Final answer

$\frac{14}{3}$

Worked Solutions **Quadratic Equations**

Q110 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -10.

Correct option: D

Final answer

-10

Q111 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

Solution

1. $2(\alpha + \beta)/(\alpha\beta) = 2(3)/(-3/2) = -4$
2. Therefore the final answer is -4.

Final answer

-4

Worked Solutions **Quadratic Equations**

Q112 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

Solution

1. $(\alpha^2 + \beta^2)/(\alpha\beta) = 1/(3/2) = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$\frac{2}{3}$

Q113 Written

Find the sum and product of the roots of $5x^2 - 3 = 0$.

Solution

1. $S = 0; P = -3/5$
2. Therefore the final answer is $S = 0, \quad P = -\frac{3}{5}$.

Final answer

$S = 0, \quad P = -\frac{3}{5}$

Worked Solutions **Quadratic Equations**

Q114 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$\frac{2}{3}$

Q115 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = (-2)/(-1/3) = 6$
2. Therefore the final answer is 6.

Final answer

6

Worked Solutions **Quadratic Equations**

Q116 **MCQ**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is -4.

Correct option: C

Final answer

-4

Q117 **MCQ**

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D

Final answer

$\frac{2}{3}$

Worked Solutions **Quadratic Equations**

Q118 MCQ

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $S = 0, \quad P = -\frac{3}{5}$.

Correct option: D

Final answer

$S = 0, \quad P = -\frac{3}{5}$

Q119 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D

Final answer

$\frac{2}{3}$

Worked Solutions **Quadratic Equations**

Q120 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 6.

Correct option: D

Final answer

6

Q121 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

Solution

1. $P + S + 1 = -3/2 + 3 + 1 = 5/2$
2. Therefore the final answer is $\frac{5}{2}$.

Final answer

$\frac{5}{2}$

Worked Solutions **Quadratic Equations**

Q122 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

Solution

1. $P - S + 1 = 3/2 - 2 + 1 = 1/2$
2. Therefore the final answer is $\frac{1}{2}$.

Final answer
$\frac{1}{2}$

Q123 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

Solution

1. $P + 2S + 4 = 3 + 4 + 4 = 11$
2. Therefore the final answer is 11.

Final answer
11

Worked Solutions **Quadratic Equations**

Q124 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

Solution

1. $P - 2S + 4 = -1/3 + 4 + 4 = 23/3$
2. Therefore the final answer is $\frac{23}{3}$.

Final answer

$\frac{23}{3}$

Q125 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{5}{2}$.

Correct option: A

Final answer

$\frac{5}{2}$

Worked Solutions Quadratic Equations

Q126 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{1}{2}$.

Correct option: C

Final answer

$\frac{1}{2}$

Q127 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 11.

Correct option: D

Final answer

11

Worked Solutions **Quadratic Equations**

Q128 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{23}{3}$.

Correct option: D

Final answer

$\frac{23}{3}$

Q129 Written

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

Solution

1. $Let roots \alpha, 3\alpha. 4\alpha = 8 -> \alpha = 2.$
2. $k = \alpha(3\alpha) = 6$
3. Therefore the final answer is $k = 6, \quad x = 2, 6$.

Final answer

$k = 6, \quad x = 2, 6$

Worked Solutions **Quadratic Equations**

Q130 Written

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

Solution

1. $Roots \alpha, 2\alpha; 3\alpha = -6 -> \alpha = -2; m = 2\alpha^2 = 8$
2. Therefore the final answer is $m = 8, \quad x = -2, -4$.

Final answer

$m = 8, \quad x = -2, -4$

Q131 Written

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

Solution

1. $Roots \alpha, 2\alpha; 2\alpha^2 = 1/8 -> \alpha = \pm 1/4$.
2. $m = 8(3\alpha) = \pm 6$.
3. Therefore the final answer is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Final answer

$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Worked Solutions **Quadratic Equations**

Q132 MCQ

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k = 6, \quad x = 2, 6$.

Correct option: A

Final answer

$k = 6, \quad x = 2, 6$

Q133 MCQ

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m = 8, \quad x = -2, -4$.

Correct option: A

Final answer

$m = 8, \quad x = -2, -4$

Worked Solutions **Quadratic Equations**

Q134 MCQ

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Correct option: D

Final answer

$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Worked Solutions **Quadratic Equations**

Q135 Written

If the difference between the two roots of $x^2 - (m-1)x - 6 = 0$ is 5, find m and the roots.

Solution

1. $Let roots be \alpha and \alpha + 5.$
2. $\alpha(\alpha+5) = -6- > \alpha = -2 or - 3.$
3. $S = 2\alpha + 5 = m - 1 gives m = 2 or 0; obtain the paired roots.$
4. Therefore the final answer is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Final answer

$(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Q136 Written

If the difference between the roots of $x^2 - (m+1)x + 2 = 0$ is 1, find m and the roots.

Solution

1. $Let roots \alpha, \alpha + 1. \alpha(\alpha+1) = 2- > \alpha = 1 or - 2.$
2. $m + 1 = 2\alpha + 1- > m = 2 or - 4.$
3. Therefore the final answer is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Final answer

$(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Worked Solutions **Quadratic Equations**

Q137 Written

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

Solution

1. $Roots\alpha, \alpha + 4; 2\alpha + 4 = -6- > \alpha = -5; m = \alpha(\alpha + 4) = 5$
2. Therefore the final answer is $m = 5, \quad x = -5, -1$.

Final answer

$m = 5, \quad x = -5, -1$

Q138 Written

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

Solution

1. $Roots\alpha, \alpha + 1; \alpha(\alpha + 1) = 3/4- > \alpha = 1/2or - 3/2.$
2. $-m/4 = 2\alpha + 1- > m = -8or8.$
3. Therefore the final answer is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Final answer

$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

Worked Solutions **Quadratic Equations**

Q139 MCQ

If the difference between the two roots of $x^2 - (m-1)x - 6 = 0$ is 5, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Correct option: D

Final answer

$(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Worked Solutions **Quadratic Equations**

Q140 MCQ

If the difference between the roots of $x^2 - (m+1)x + 2 = 0$ is 1, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Correct option: A

Final answer

$(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Worked Solutions **Quadratic Equations**

Q141 **MCQ**

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m = 5, \quad x = -5, -1$.

Correct option: B

Final answer

$m = 5, \quad x = -5, -1$

Worked Solutions **Quadratic Equations**

Q142 MCQ

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Correct option: B

Final answer

$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

Worked Solutions **Quadratic Equations**

Q143 Written

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots.

Solution

1. $Letroots2t, 3t; 5t = 2-> t = 2/5; m = 6t^2 = 24/25$
2. Therefore the final answer is $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$.

Final answer

$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$

Q144 MCQ

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$.

Correct option: A

Final answer

$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$

Worked Solutions **Quadratic Equations**

Q145 **Written**

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

Solution

1. $Roots\alpha, \alpha^2; \alpha + \alpha^2 = 6- > \alpha = 2or - 3.$
2. $k = \alpha^3 = 8or - 27.$
3. Therefore the final answer is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Final answer

$(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

Worked Solutions **Quadratic Equations**

Q146 *MCQ*

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Correct option: B

Final answer

$(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

Worked Solutions **Quadratic Equations**

Q147 Written

Factorise $2x^2 - 3x + 2$ over the complex numbers.

Solution

1. $D = 9 - 16 = -7; roots(3 \pm i\sqrt{7})/4$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$.

Final answer

$$2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$$

Q148 Written

Factorise $x^2 + 2x + 5$ over the complex numbers.

Solution

1. $Roots - 1 \pm 2i$
2. Therefore the final answer is $(x + 1 - 2i)(x + 1 + 2i)$.

Final answer

$$(x + 1 - 2i)(x + 1 + 2i)$$

Worked Solutions **Quadratic Equations**

Q149 Written

Factorise $2x^2 - 6x + 1$ over the complex numbers.

Solution

1. $D = 28; roots(3 \pm \sqrt{7})/2$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$.

Final answer

$$2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$$

Q150 Written

Factorise $x^2 + 4x + 5$ over the complex numbers.

Solution

1. $D = 16 - 20 = -4; roots - 2 \pm i$
2. Therefore the final answer is $(x + 2 - i)(x + 2 + i)$.

Final answer

$(x + 2 - i)(x + 2 + i)$

Worked Solutions **Quadratic Equations**

Q151 Written

Factorise $x^2 + 4x + 2$ over the complex numbers.

Solution

1. $Roots - 2 \pm \sqrt{2}$
2. Therefore the final answer is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Final answer

$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$

Q152 Written

Factorise $x^2 - 4x + 7$ over the complex numbers.

Solution

1. $D = 16 - 28 = -12; roots 2 \pm i\sqrt{3}$
2. Therefore the final answer is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Final answer

$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

Worked Solutions **Quadratic Equations**

Q153 Written

Factorise $2x^2 - x + 1$ over the complex numbers.

Solution

1. $D = 1 - 8 = -7; roots(1 \pm i\sqrt{7})/4$
2. Therefore the final answer is $2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$.

Final answer

$2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$

Q154 Written

Factorise $3x^2 - 5x + 3$ over the complex numbers.

Solution

1. $D = 25 - 36 = -11; roots(5 \pm i\sqrt{11})/6$
2. Therefore the final answer is $3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$.

Final answer

$3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$

Worked Solutions **Quadratic Equations**

Q155 Written

Factorise $2x^2 - 2x - 1$ over the complex numbers.

Solution

1. $D = 4 + 8 = 12; roots (1 \pm \sqrt{3})/2$
2. Therefore the final answer is $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$.

Final answer

$2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$

Worked Solutions **Quadratic Equations**

Q156 MCQ

Factorise $2x^2 - 3x + 2$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Correct option: C

Final answer

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Worked Solutions **Quadratic Equations**

Q157 MCQ

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x + 1 - 2i)(x + 1 + 2i)$.

Correct option: A

Final answer

$(x + 1 - 2i)(x + 1 + 2i)$

Worked Solutions **Quadratic Equations**

Q158 *MCQ*

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$.

Correct option: D

Final answer

$2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$

Worked Solutions **Quadratic Equations**

Q159 MCQ

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x + 2 - i)(x + 2 + i)$.

Correct option: D

Final answer

$(x + 2 - i)(x + 2 + i)$

Q160 MCQ

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Correct option: B

Final answer

$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$

Worked Solutions **Quadratic Equations**

Q161 MCQ

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Correct option: A

Final answer

$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

Worked Solutions **Quadratic Equations**

Q162 MCQ

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.

Correct option: C

Final answer

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Worked Solutions **Quadratic Equations**

Q163 MCQ

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $3\left(x - \dfrac{5+i\sqrt{11}}{6}\right)\left(x - \dfrac{5-i\sqrt{11}}{6}\right)$.

Correct option: A

Final answer

$3\left(x - \dfrac{5+i\sqrt{11}}{6}\right)\left(x - \dfrac{5-i\sqrt{11}}{6}\right)$

Worked Solutions **Quadratic Equations**

Q164 MCQ

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2\left(x - \dfrac{1+\sqrt{3}}{2}\right)\left(x - \dfrac{1-\sqrt{3}}{2}\right)$.

Correct option: B

Final answer

$$2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$$

Worked Solutions **Quadratic Equations**

Q165 Written

Factorise $x^4 + 2x^2 - 3$ over the complex numbers.

Solution

1. $x^4 + 2x^2 - 3 = (x^2 - 1)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$.

Final answer

$(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$

Q166 Written

Factorise $x^4 - x^2 - 6$ over the complex numbers.

Solution

1. $Let y = x^2 : y^2 - y - 6 = (y-3)(y+2)$
2. $Factor x^2 - 3 and x^2 + 2$
3. Therefore the final answer is $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$.

Final answer

$(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$

Worked Solutions **Quadratic Equations**

Q167 **Written**

Factorise $x^4 - 9$ over the complex numbers.

Solution

1. $x^4 - 9 = (x^2 - 3)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$

Q168 **Written**

Factorise $x^4 - 3x^2 + 2$ over the complex numbers.

Solution

1. $Let y = x^2 : y^2 - 3y + 2 = (y - 1)(y - 2)$
2. $Factor x^2 - 1 and x^2 - 2$
3. Therefore the final answer is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Final answer

$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$

Worked Solutions **Quadratic Equations**

Q169 MCQ

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$.

Correct option: B

Final answer

$(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$

Q170 MCQ

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$.

Correct option: B

Final answer

$(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$

Worked Solutions **Quadratic Equations**

Q171 MCQ

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{3})$.

Correct option: D

Final answer
$(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{3})$

Q172 MCQ

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$.

Correct option: C

Final answer
$(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$

Worked Solutions **Quadratic Equations**

Q173 Written

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

Solution

1. $S = 1, P = 1/2$
2. $x^2 - x + 1/2 = 0; multiply by 2$
3. Therefore the final answer is $2x^2 - 2x + 1 = 0$.

Final answer

$2x^2 - 2x + 1 = 0$

Q174 Written

Find two numbers whose sum is -1 and product is 2.

Solution

1. $They are roots of x^2 + x + 2 = 0$
2. $D = -7$
3. Therefore the final answer is $\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$.

Final answer

$\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$

Worked Solutions **Quadratic Equations**

Q175 **Written**

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

Solution

1. $S = 6, P = 13$
2. $x^2 - Sx + P = 0$
3. Therefore the final answer is $x^2 - 6x + 13 = 0$.

Final answer

$x^2 - 6x + 13 = 0$

Q176 **Written**

Find two numbers whose sum and product are both 1.

Solution

1. $Roots of x^2 - x + 1 = 0$
2. $D = -3$
3. Therefore the final answer is $\dfrac{1 + i\sqrt{3}}{2}, \dfrac{1 - i\sqrt{3}}{2}$.

Final answer

$\dfrac{1 + i\sqrt{3}}{2}, \dfrac{1 - i\sqrt{3}}{2}$

Worked Solutions **Quadratic Equations**

Q177 **Written**

Create a quadratic equation with roots -1 and 2.

Solution

1. $S = 1, P = -2$
2. Therefore the final answer is $x^2 - x - 2 = 0$.

Final answer

$x^2 - x - 2 = 0$

Q178 **Written**

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.

Solution

1. $S = 2, P = 17$
2. Therefore the final answer is $x^2 - 2x + 17 = 0$.

Final answer

$x^2 - 2x + 17 = 0$

Worked Solutions **Quadratic Equations**

Q179 Written

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

Solution

1. $S = 10/3, P = 2$
2. $x^2 - (10/3)x + 2 = 0; multiplyby3$
3. Therefore the final answer is $3x^2 - 10x + 6 = 0$.

Final answer

$3x^2 - 10x + 6 = 0$

Q180 Written

Find two numbers whose sum is 2 and product is 5.

Solution

1. $Rootsof x^2 - 2x + 5 = 0$
2. $D = -16$
3. Therefore the final answer is $1 + 2i,\ 1 - 2i$.

Final answer

$1 + 2i,\ 1 - 2i$

Worked Solutions **Quadratic Equations**

Q181 Written

Find two numbers whose sum is 4 and product is 2.

Solution

1. $Rootsof x^2 - 4x + 2 = 0$
2. Therefore the final answer is $2 + \sqrt{2},\ 2 - \sqrt{2}$.

Final answer

$2 + \sqrt{2},\ 2 - \sqrt{2}$

Q182 MCQ

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2x^2 - 2x + 1 = 0$.

Correct option: B

Final answer

$2x^2 - 2x + 1 = 0$

Worked Solutions **Quadratic Equations**

Q183 MCQ

Find two numbers whose sum is -1 and product is 2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\dfrac{-1+i\sqrt{7}}{2}, \dfrac{-1-i\sqrt{7}}{2}$.

Correct option: B

Final answer

$\dfrac{-1+i\sqrt{7}}{2}, \dfrac{-1-i\sqrt{7}}{2}$

Q184 MCQ

Create a quadratic equation with roots $3+2i$ and $3-2i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 6x + 13 = 0$.

Correct option: B

Final answer

$x^2 - 6x + 13 = 0$

Worked Solutions **Quadratic Equations**

Q185 *MCQ*

Find two numbers whose sum and product are both 1 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$.

Correct option: A

Final answer

$\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

Q186 *MCQ*

Create a quadratic equation with roots -1 and 2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - x - 2 = 0$.

Correct option: A

Final answer

$x^2 - x - 2 = 0$

Worked Solutions **Quadratic Equations**

Q187 MCQ

Create a quadratic equation with roots $1+4i$ and $1-4i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2-2x+17=0$.

Correct option: D

Final answer

$x^2-2x+17=0$

Q188 MCQ

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $3x^2-10x+6=0$.

Correct option: A

Final answer

$3x^2-10x+6=0$

Worked Solutions **Quadratic Equations**

Q189 MCQ

Find two numbers whose sum is 2 and product is 5 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $1+2i,\ 1-2i$.

Correct option: B

Final answer

$1+2i,\ 1-2i$

Q190 MCQ

Find two numbers whose sum is 4 and product is 2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2+\sqrt{2},\ 2-\sqrt{2}$.

Correct option: B

Final answer

$2+\sqrt{2},\ 2-\sqrt{2}$

Worked Solutions **Quadratic Equations**

Q191 Written

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

Solution

1. $S = -2, P = -4$
2. $Newsum = S + 4 = 2; newproduct = P + 2S + 4 = -4$
3. $Equationx^2 - 2x - 4 = 0$
4. Therefore the final answer is $x^2 - 2x - 4 = 0$.

Final answer

$x^2 - 2x - 4 = 0$

Q192 MCQ

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 2x - 4 = 0$.

Correct option: C

Final answer

$x^2 - 2x - 4 = 0$

Worked Solutions **Quadratic Equations**

Q193 **Written**

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2.

Solution

1. $S = -1, P = -3$
2. $Newsum S^2 - 2P = 1 + 6 = 7; product P^2 = 9$
3. Therefore the final answer is $x^2 - 7x + 9 = 0$.

Final answer

$x^2 - 7x + 9 = 0$

Q194 **Written**

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2.

Solution

1. $S = 2, P = 7$
2. $Newsum = S^2 - 2P = -10; product = P^2 = 49$
3. Therefore the final answer is $x^2 + 10x + 49 = 0$.

Final answer

$x^2 + 10x + 49 = 0$

Worked Solutions **Quadratic Equations**

Q195 MCQ

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 7x + 9 = 0$.

Correct option: D

Final answer

$x^2 - 7x + 9 = 0$

Q196 MCQ

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 + 10x + 49 = 0$.

Correct option: C

Final answer

$x^2 + 10x + 49 = 0$

Worked Solutions **Quadratic Equations**

Q197 **Written**

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

Solution

1. $S = -3/2, P = 2$
2. $Newsum = S/P = -3/4; product = 1/P = 1/2$
3. $x^2 + (3/4)x + 1/2 = 0; multiplyby4$
4. Therefore the final answer is $4x^2 + 3x + 2 = 0$.

Final answer

$4x^2 + 3x + 2 = 0$

Q198 **MCQ**

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $4x^2 + 3x + 2 = 0$.

Correct option: A

Final answer

$4x^2 + 3x + 2 = 0$

Worked Solutions **Quadratic Equations**

Q199 **Written**

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

Solution

1. $S = -2, P = 3$
2. $Newsum = 4S = -8$
3. $Newproduct = 3(\alpha^2 + \beta^2) + 10P = 3(S^2 - 2P) + 10P = 24$
4. Therefore the final answer is $x^2 + 8x + 24 = 0$.

Final answer

$x^2 + 8x + 24 = 0$

Q200 **MCQ**

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 + 8x + 24 = 0$.

Correct option: D

Final answer

$x^2 + 8x + 24 = 0$

Worked Solutions **Quadratic Equations**

Q201 **Written**

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m.

Solution

1. $D > 0- > (m-2)(m+1) > 0- > m < -1 or m > 2$
2. $S = -2m > 0- > m < 0$
3. $P = m + 2 > 0- > m > -2$
4. $Intersection : -2 < m < -1$
5. Therefore the final answer is $-2 < m < -1$.

Final answer

$-2 < m < -1$

Worked Solutions **Quadratic Equations**

Q202 **Written**

If $x^2 + 2(m-1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m.

Solution

1. $D = 4(4-m)(m+1) > 0- > -1 < m < 4$
2. $S = 2(1-m) > 0- > m < 1$
3. $P = (2m+1)(m-3) > 0- > m < -1/2 or m > 3$
4. $Intersection : -1 < m < -1/2$
5. Therefore the final answer is $-1 < m < -\frac{1}{2}$.

Final answer

$-1 < m < -\frac{1}{2}$

Worked Solutions **Quadratic Equations**

Q203 MCQ

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-2 < m < -1$.

Correct option: A

Final answer

$-2 < m < -1$

Q204 MCQ

If $x^2 + 2(m-1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-1 < m < -\frac{1}{2}$.

Correct option: B

Final answer

$-1 < m < -\frac{1}{2}$

Worked Solutions **Quadratic Equations**

Q205 **Written**

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m.

Solution

1. $D > 0 : 4m^2 - 4(m+6) > 0 -> (m-3)(m+2) > 0$
2. $S = 2m < 0 -> m < 0$
3. $P = m + 6 > 0 -> m > -6$
4. $Intersection : -6 < m < -2$
5. Therefore the final answer is $-6 < m < -2$.

Final answer

$-6 < m < -2$

Worked Solutions **Quadratic Equations**

Q206 **Written**

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k.

Solution

1. $D = (k - 10)(k + 2) > 0 - > k < -2 or k > 10$
2. $S = -k < 0 - > k > 0$
3. $P = 2k + 5 > 0 - > k > -5/2$
4. $Intersection k > 10$
5. Therefore the final answer is $k > 10$.

Final answer

$k > 10$

Worked Solutions **Quadratic Equations**

Q207 MCQ

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-6 < m < -2$.

Correct option: C

Final answer

$-6 < m < -2$

Q208 MCQ

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k > 10$.

Correct option: B

Final answer

$k > 10$

Worked Solutions **Quadratic Equations**

Q209 Written

If $x^2 - (m-3)x - m - 4 = 0$ has roots of different signs, find the range of m.

Solution

1. $Oppositesignsiff P = -m - 4 < 0$
2. Therefore the final answer is $m > -4$.

Final answer

$m > -4$

Q210 Written

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k.

Solution

1. $P = -k - 3 < 0$
2. Therefore the final answer is $k > -3$.

Final answer

$k > -3$

Worked Solutions **Quadratic Equations**

Q211 MCQ

If $x^2 - (m-3)x - m - 4 = 0$ has roots of different signs, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m > -4$.

Correct option: A

Final answer

$m > -4$

Q212 MCQ

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $k > -3$.

Correct option: D

Final answer

$k > -3$

Worked Solutions **Quadratic Equations**

Q213 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 + 4a = a(a+4)$
2. Classify sign intervals at -4 and 0
3. Therefore the final answer is $a < -4 \text{ or } a > 0: \text{ distinct real; } a = -4, 0: \text{ repeated real; } -4 < a < 0: \text{ distinct imaginary.}$

Final answer

$a < -4 \text{ or } a > 0: \text{ distinct real; } a = -4, 0: \text{ repeated real; } -4 < a < 0: \text{ distinct imaginary}$

Worked Solutions **Quadratic Equations**

Q214 Challenge

For $x^2 - (k+2)x + k^2 = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = (k+2)^2 - 4k^2 = -3k^2 + 4k + 4 = -3(k-2)(k+2/3)$
2. Classify D sign
3. Therefore the final answer is $-\frac{2}{3} < k < 2:$ distinct real; $k = -\frac{2}{3}, 2:$ repeated; $k < -\frac{2}{3}$ or $k > 2:$ distinct imaginary.

Final answer

$-\frac{2}{3} < k < 2:$ distinct real; $k = -\frac{2}{3}, 2:$ repeated; $k < -\frac{2}{3}$ or $k > 2:$ distinct imaginary

Worked Solutions **Quadratic Equations**

Q215 Challenge

For $x^2 - 2(a-1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = 4(a-1)^2 - 4(a^2 - 3a + 4) = 4(a-3)$
2. Classify by sign of a-3
3. Therefore the final answer is $a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct imaginary.

Final answer

$a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct imaginary

Worked Solutions **Quadratic Equations**

Q216 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

Solution

1. $\alpha\beta(\alpha + \beta) = PS = (3/2)(3) = 9/2$
2. Therefore the final answer is $\frac{9}{2}$.

Final answer

$\frac{9}{2}$

Worked Solutions **Quadratic Equations**

Q217 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

Solution

1. $S^2 - 4P = 9 - 6 = 3$
2. Therefore the final answer is 3.

Final answer

3

Worked Solutions **Quadratic Equations**

Q218 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = 3/2$
2. $S^3 - 3PS = 27 - 27/2 = 27/2$
3. Therefore the final answer is $\frac{27}{2}$.

Final answer

$$\frac{27}{2}$$

Worked Solutions **Quadratic Equations**

Q219 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

Solution

1. $(\alpha^3 + \beta^3)/(\alpha\beta) = (27/2)/(3/2) = 9$
2. Therefore the final answer is 9.

Final answer

9

Worked Solutions **Quadratic Equations**

Q220 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

Solution

1. $Subtractequations : (a-b)(x-c) = 0, socommonrootr = c.$
2. $Substitutec : c(c+a+b) = 0; c \neq 0- > a+b+c = 0.$
3. $Theotherrootsarebc/c = bandca/c = a.Theirequationisx^2 - (a+b)x + ab = x^2 + cx + ab.$
4. Therefore the final answer is Common root c; other roots a and b..

Final answer

Common root c; other roots a and b.

Worked Solutions **Quadratic Equations**

Q221 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

Solution

1. $Let y = x^2 : 6y^2 - 7y - 3 = (2y - 3)(3y + 1)$
2. $x^2 = 3/2 or - 1/3; hence x = \pm\sqrt{6}/2 or \pm i\sqrt{3}/3$
3. $For real displacement x, only \pm\sqrt{6}/2 are admissible$
4. Therefore the final answer is $x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$.

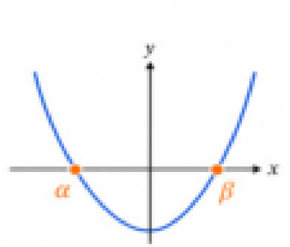
Final answer

$x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$

Worked Solutions **Quadratic Equations**

Q222 Challenge

In the opposite figure, α, β are the two roots. If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.



Solution

1. $New sum = S/P; new product = 1/P$
2. $Equation x^2 - (S/P)x + 1/P = 0; multiply by P$
3. Therefore the final answer is For $P = \alpha\beta \neq 0: \quad Px^2 - Sx + 1 = 0.$

Final answer

For $P = \alpha\beta \neq 0: \quad Px^2 - Sx + 1 = 0$

Worked Solutions **Quadratic Equations**

Q223 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

Solution

1. $D = (m-5)(m+3) > 0- > m < -3 or m > 5$
2. $S = -(m+1) > 0- > m < -1$
3. $P = m + 4 > 0- > m > -4$
4. Intersection (-4,-3)
5. Therefore the final answer is $-4 < m < -3$.

Final answer

$-4 < m < -3$

Q224 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

Solution

1. $D > 0- > m < -3orm > 5$
2. $S < 0- > m > -1$
3. $P > 0- > m > -4$
4. $Intersectionm > 5$
5. Therefore the final answer is $m > 5$.

Final answer

$m > 5$

Worked Solutions **Quadratic Equations**

Q225 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

Solution

1. $Oppositesignsiff P = m + 4 < 0$
2. Therefore the final answer is $m < -4$.

Final answer

$m < -4$

Worked Solutions **Quadratic Equations**

Q226 **Challenge**

A model has $R = 3x^2 + 2\sqrt{3}\,x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

Solution

1. $Set R = 0. D = 12 - 24 = -12.$
2. $x = (-2\sqrt{3} \pm i2\sqrt{3})/6 = (-\sqrt{3} \pm i\sqrt{3})/3.$
3. Because x is real, the complex pair is not admissible; no real equilibrium exists.
4. Therefore the final answer is $x = \dfrac{-\sqrt{3} \pm i\sqrt{3}}{3};$ no real equilibrium.

Final answer

$x = \dfrac{-\sqrt{3} \pm i\sqrt{3}}{3};$ no real equilibrium

Worked Solutions **Quadratic Equations**

Q227 **Challenge**

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Correct option: B

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions **Quadratic Equations**

Q228 **Challenge**

For $x^2 - (k+2)x + k^2 = 0$, determine the nature of the solutions as k varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-\frac{2}{3} < k < 2 :$ distinct real; $k = -\frac{2}{3}, 2 :$ repeated; $k < -\frac{2}{3}$ or $k > 2 :$ distinct imaginary.

Correct option: B

Final answer

$-\frac{2}{3} < k < 2 :$ distinct real; $k = -\frac{2}{3}, 2 :$ repeated; $k < -\frac{2}{3}$ or $k > 2 :$ distinct imaginary

Worked Solutions **Quadratic Equations**

Q229 Challenge

For $x^2 - 2(a-1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct imaginary.

Correct option: B

Final answer

$a > 3:$ distinct real; $a = 3:$ repeated real; $a < 3:$ distinct imaginary

Worked Solutions **Quadratic Equations**

Q230 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{9}{2}$.

Correct option: B

Final answer

$\frac{9}{2}$

Worked Solutions **Quadratic Equations**

Q231 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 3.

Correct option: D

Final answer

3

Worked Solutions **Quadratic Equations**

Q232 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $\frac{27}{2}$.

Correct option: B

Final answer

$\frac{27}{2}$

Worked Solutions **Quadratic Equations**

Q233 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 9.

Correct option: A

Final answer

9

Worked Solutions **Quadratic Equations**

Q234 **Challenge**

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Common root c; other roots a and b..

Correct option: C

Final answer

Common root c; other roots a and b.

Worked Solutions **Quadratic Equations**

Q235 **Challenge**

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$.

Correct option: A

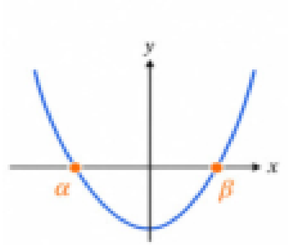
Final answer

$x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$

Worked Solutions **Quadratic Equations**

Q236 Challenge

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$ Select the correct answer:



Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is For $P = \alpha\beta \neq 0: \quad Px^2 - Sx + 1 = 0$.

Correct option: A

Final answer

For $P = \alpha\beta \neq 0: \quad Px^2 - Sx + 1 = 0$

Worked Solutions **Quadratic Equations**

Q237 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $-4 < m < -3$.

Correct option: C

Final answer

$-4 < m < -3$

Worked Solutions **Quadratic Equations**

Q238 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m > 5$.

Correct option: D

Final answer

$m > 5$

Worked Solutions **Quadratic Equations**

Q239 Challenge

For $x^2 + (m+1)x + m + 4 = 0$, find the range of m for two roots of opposite signs Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $m < -4$.

Correct option: D

Final answer

$m < -4$

Worked Solutions **Quadratic Equations**

Q240 **Challenge**

A model has $R = 3x^2 + 2\sqrt{3}\,x + 2$. Determine the values of x for which $R = 0$, and interpret the result Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x = \dfrac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Correct option: D

Final answer

$x = \dfrac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Worked Solutions **Quadratic Equations**

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $x^2 - 5x + 6 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $2, 3$.

Correct option: B

Final answer

$2, 3$

Worked Solutions **Quadratic Equations**

Quiz MCQ 2 **Concept Quiz · MCQ**

For $x^2 - 4x + 7 = 0$, determine the nature of the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is Two distinct non-real complex roots.

Correct option: A

Final answer

Two distinct non-real complex roots

Worked Solutions **Quadratic Equations**

Quiz MCQ 3 Concept Quiz · MCQ

If alpha,beta are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is 7.

Correct option: C

Final answer

7

Quiz MCQ 4 Concept Quiz · MCQ

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the source demand.
2. The correct value is $x^2 - 4x + 5 = 0$.

Correct option: A

Final answer

$x^2 - 4x + 5 = 0$

Worked Solutions **Quadratic Equations**

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $2x^2 + 3x - 2 = 0$.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $x = \frac{1}{2}, -2$

Final answer

$x = \frac{1}{2}, -2$

Quiz WRITTEN 6 **Concept Quiz · Written**

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $-1 < m < 2$

Final answer

$-1 < m < 2$

Worked Solutions **Quadratic Equations**

Quiz WRITTEN 7 Concept Quiz · Written

If alpha,beta are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: 28

Final answer

28

Quiz WRITTEN 8 Concept Quiz · Written

If alpha,beta are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $x^2 + 6x + 25 = 0$

Final answer

$x^2 + 6x + 25 = 0$

Answer key

Use the question number shown on each page.

Q001 $x = \pm 2i\sqrt{2}$

Q002 $x = \pm 2i$

Q003 $x = \pm 7i$

Q004 $x = \pm \frac{2}{3}$

Q005 $x = \pm 2\sqrt{3}$

Q006 $x = \pm \frac{i\sqrt{3}}{2}$

Q007 A

Q008 A

Q009 C

Q010 C

Q011 B

Q012 B

Q013 $x = 3 \pm 2i\sqrt{2}$

Q014 $x = 2 \pm \sqrt{5}$

Q015 $x = -6 \pm i\sqrt{6}$

Q016 $x = -4 \pm 3i\sqrt{2}$

Q017 $x = \frac{3 \pm \sqrt{7}}{2}$

Q018 A

Q019 B

Q020 C

Q021 C

Q022 D

Q023 $x = -1,\ \frac{9}{2}$

Q024 $x = \frac{1}{2},\ -3$

Q025 $x = 0,\ -\frac{7}{2}$

Q026 $x = \frac{3}{2},\ -4$

Q027 $x = 3,\ 5$

Q028 A

Q029 D

Q030 D

Q031 A

Q032 A

Q033 $x = \frac{2 \pm \sqrt{7}}{3}$

Q034 B

Q035 $x = \frac{-\sqrt{6} \pm 4i}{2}$

Q036 $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

Q037 $x = \frac{-1 \pm i\sqrt{3}}{2}$

Q038 $x = \frac{5 \pm i\sqrt{11}}{6}$

Q039 $x = \sqrt{3} \pm i\sqrt{2}$

Q040 $x = \frac{-\sqrt{2} \pm i}{3}$

Q041 $x = \frac{1 \pm i\sqrt{7}}{2}$

Q042 $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

Q043 $x = 1 \pm \frac{i\sqrt{6}}{2}$

Q044 B

Q045 C

Q046 D

Q047 B

Q048 D

Q049 B

Q050 C

Q051 C

Q052 B

Q053 Two distinct imaginary (non-real complex) solutions

Q054 $-2 \le m \le 6$

Q055 $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Two distinct imaginary (non-real complex) solutions

Q057 Two distinct real solutions

Q058 One repeated real solution

Q059 Two distinct real solutions

Q060 One repeated real solution

Q061 Two distinct imaginary (non-real complex) solutions

Q062 One repeated real solution

Q063 Two distinct imaginary (non-real complex) solutions

Q064 Two distinct real solutions

Q065 A

Q066 D

Q067 C

Q068 C

Q069 A

Q070 A

Q071 C

Q072 C

Q073 B

Q074 A

Q075 A

Q076 A

Q077 $-1 < m < 2$

Q078 $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Q079 $a = 3$ or $a = -5$

Q080 $-18 < m < -2$

Q081 $0 \le m \le 3$

Q082 C

Q083 C

Q084 A

Q085 D

Q086 C

Q087 $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$

Q088 $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$

Q089 $S = -1, \quad P = 2$

Q090 $S = \frac{7}{3}, \quad P = -2$

Q091 C

Q092 A

Q093 B

Q094 A

Q095 12

Q096 $\frac{81}{2}$

Q097 1

Q098 -1

Q099 -2

Q100 -10

Q101 $\frac{14}{3}$

Q102 -10

Q103 A

Q104 B

Q105 A

Q106 B

Q107 D

Q108 B

Q109 B

Q110 D

Q111 -4

Q112 $\frac{2}{3}$

Q113 $S = 0, \quad P = -\frac{3}{5}$

Q114 $\frac{2}{3}$

Q115 6

Q116 C

Q117 D

Q118 D

Q119 D

Q120 D

Q121 $\frac{5}{2}$

Q122 $\frac{1}{2}$

Q123 11

Q124 $\frac{23}{3}$

Q125 A

Q126 C

Q127 D

Q128 D

Q129 $k = 6, \quad x = 2, 6$

Q130 $m = 8, \quad x = -2, -4$

Q131 $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Q132 A

Q133 A

Q134 D

Q135 $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Q136 $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Q137 $m = 5, \quad x = -5, -1$

Q138 $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

Q139 D

Q140 A

Q141 B

Q142 B

Q143 $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$

Q144 A

Q145 $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

Q146 B

Q147 $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$

Q148 $(x + 1 - 2i)(x + 1 + 2i)$

Q149 $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$

Q150 $(x + 2 - i)(x + 2 + i)$

Q151 $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$

Q152 $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

Q153 $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$

Q154 $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

Q155 $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$

Q156 C

Q157 A

Q158 D

Q159 D

Q160 B

Q161 A

Q162 C

Q163 A

Q164 B

Q165 $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$

Q166 $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$

Q167 $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$

Q168 $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$

Q169 B

Q170 B

Q171 D

Q172 C

Q173 $2x^2 - 2x + 1 = 0$

Q174 $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$

Q175 $x^2 - 6x + 13 = 0$

Q176 $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$

Q177 $x^2 - x - 2 = 0$

Q178 $x^2 - 2x + 17 = 0$

Q179 $3x^2 - 10x + 6 = 0$

Q180 $1 + 2i, \ 1 - 2i$

Q181 $2 + \sqrt{2}, \ 2 - \sqrt{2}$

Q182 B

Q183 B

Q184 B

Q185 A

Q186 A

Q187 D

Q188 A

Q189 B

Q190 B

Q191 $x^2 - 2x - 4 = 0$

Q192 C

Q193 $x^2 - 7x + 9 = 0$

Q194 $x^2 + 10x + 49 = 0$

Q195 D

Q196 C

Q197 $4x^2 + 3x + 2 = 0$

Q198 A

Q199 $x^2 + 8x + 24 = 0$

Q200 D

Q201 $-2 < m < -1$

Q202 $-1 < m < -\frac{1}{2}$

Q203 A

Q204 B

Q205 $-6 < m < -2$

Q206 $k > 10$

Q207 C

Q208 B

Q209 $m > -4$

Q210 $k > -3$

Q211 A

Q212 D

Q213 $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Q214 $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Q215 $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Q216 $\frac{9}{2}$

Q217 3

Q218 $\frac{27}{2}$

Q219 9

Q220 Common root c; other roots a and b.

Q221 $x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$

Q222 For $P = \alpha\beta \neq 0$: $\quad Px^2 - Sx + 1 = 0$

Q223 $-4 < m < -3$

Q224 $m > 5$

Q225 $m < -4$

Q226 $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Q227 B

Q228 B

Q229 B

Q230 B

Q231 D

Q232 B

Q233 A

Q234 C

Q235 A

Q236 A

Q237 C

Q238 D

Q239 D

Q240 D

Quiz MCQ 1 B

Quiz MCQ 2 A

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $x = \frac{1}{2}, -2$

Quiz WRITTEN 6 $-1 < m < 2$

Quiz WRITTEN 7 28

Quiz WRITTEN 8 $x^2 + 6x + 25 = 0$

Concept 3 | Polynomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Remainder

$P(a) = R$ when divisor $= x - a$

Factor

$P(a) = 0 \Rightarrow (x - a) \mid P(x)$

Roots

$P(x) = a \prod (x - r_i)$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Polynomial Theorems**

Q001 Written

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.

Solution

1. $Usex = -2$
2. $P(-2) = 2$
3. Therefore the final answer is 2.

Final answer

2

Q002 Written

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.

Solution

1. $Usex = 1$
2. $P(1) = 6$
3. Therefore the final answer is 6.

Final answer

6

Worked Solutions **Polynomial Theorems**

Q003 MCQ

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 2.

Correct option: C

Final answer

2

Q004 MCQ

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 6.

Correct option: C

Final answer

6

Worked Solutions **Polynomial Theorems**

Q005 **Written**

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.

Solution

1. $Usex = 2/3$
2. $EvaluateP(2/3) = 8$
3. Therefore the final answer is 8.

Final answer

8

Q006 **Written**

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.

Solution

1. $Usex = 1/2$
2. $P(1/2) = -1/8$
3. Therefore the final answer is $-\frac{1}{8}$.

Final answer

$-\frac{1}{8}$

Worked Solutions　　**Polynomial Theorems**

Q007 **Written**

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.

Solution

1. $Usex = 2/3$
2. $P(2/3) = 0$
3. Therefore the final answer is 0.

Final answer

0

Q008 **Written**

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.

Solution

1. $Usex = -1/2$
2. $P(-1/2) = 4$
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions **Polynomial Theorems**

Q009 MCQ

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 8.

Correct option: D

Final answer

8

Q010 MCQ

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $-\frac{1}{8}$.

Correct option: A

Final answer

$-\frac{1}{8}$

Worked Solutions **Polynomial Theorems**

Q011 MCQ

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 0.

Correct option: D

Final answer

0

Q012 MCQ

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 4.

Correct option: A

Final answer

4

Worked Solutions **Polynomial Theorems**

Q013 **Written**

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a.

Solution

1. $P(-3) = 4$
2. $9a - 35 = 4, hence a = 13/3$
3. Therefore the final answer is $a = 13/3$.

Final answer

$a = 13/3$

Q014 **Written**

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a.

Solution

1. $P(3) = 1$
2. $9a + 37 = 1, hence a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$a = -4$

Worked Solutions **Polynomial Theorems**

Q015 **Written**

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2, find a.

Solution

1. $P(-1) = -2$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$a = -4$

Q016 **Written**

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a.

Solution

1. $P(-3) = 8$
2. $a = -16$
3. Therefore the final answer is $a = -16$.

Final answer

$a = -16$

Worked Solutions **Polynomial Theorems**

Q017 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = 13/3$.

Correct option: B

Final answer

$a = 13/3$

Q018 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -4$.

Correct option: B

Final answer

$a = -4$

Worked Solutions **Polynomial Theorems**

Q019 MCQ

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -4$.

Correct option: C

Final answer

$a = -4$

Q020 MCQ

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -16$.

Correct option: A

Final answer

$a = -16$

Worked Solutions **Polynomial Theorems**

Q021 Written

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a.

Solution

1. $P(2) = 0$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$a = -4$

Q022 MCQ

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -4$.

Correct option: A

Final answer

$a = -4$

Worked Solutions **Polynomial Theorems**

Q023 **Written**

A polynomial $P(x)$ leaves remainders 3 on division by $x-1$ and -5 on division by $x+3$. Find the remainder on division by x^2+2x-3.

Solution

1. $LetR = ax + b$
2. $a + b = 3and - 3a + b = -5$
3. $R = 2x + 1$
4. Therefore the final answer is $2x + 1$.

Final answer

$2x + 1$

Q024 **Written**

A polynomial $P(x)$ leaves remainders 3 on division by $x-2$ and -7 on division by $x+3$. Find the remainder on division by x^2+x-6.

Solution

1. $LetR = ax + b$
2. $2a + b = 3and - 3a + b = -7$
3. $R = -10x + 13$
4. Therefore the final answer is $-10x + 13$.

Final answer

$-10x + 13$

Worked Solutions **Polynomial Theorems**

Q025 **Written**

A polynomial $P(x)$ leaves remainders 5 on division by $x-1$ and -1 on division by $x+2$. Find the remainder on division by $(x-1)(x+2)$.

Solution

1. $Let R = ax + b$
2. $a + b = 5 and - 2a + b = -1$
3. $R = 2x + 3$
4. Therefore the final answer is $2x + 3$.

Final answer

$2x + 3$

Q026 **Written**

A polynomial $P(x)$ leaves remainders 4 on division by $x-2$ and -11 on division by $x+3$. Find the remainder on division by x^2+x-6.

Solution

1. $Let R = ax + b$
2. $2a + b = 4 and - 3a + b = -11$
3. $R = -15x + 19$
4. Therefore the final answer is $-15x + 19$.

Final answer

$-15x + 19$

Worked Solutions **Polynomial Theorems**

Q027 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x-1$ and -5 on division by $x+3$. Find the remainder on division by x^2+2x-3 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $2x+1$.

Correct option: A

Final answer

$2x+1$

Q028 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x-2$ and -7 on division by $x+3$. Find the remainder on division by x^2+x-6 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $-10x+13$.

Correct option: A

Final answer

$-10x+13$

Worked Solutions **Polynomial Theorems**

Q029 MCQ

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $2x + 3$.

Correct option: A

Final answer

$2x + 3$

Q030 MCQ

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $-15x + 19$.

Correct option: A

Final answer

$-15x + 19$

Worked Solutions **Polynomial Theorems**

Q031 Written

A polynomial $P(x)$ leaves remainder $7x+6$ on division by x^2-x-6, and $4x-3$ on division by x^2+2x-3. Find the remainder on division by x^2+x-2.

Solution

1. $Let R = ax + b$
2. $Usesharedroots - 2and1 : -2a + b = -8anda + b = 1$
3. $R = 3x - 2$
4. Therefore the final answer is $3x - 2$.

Final answer

$3x - 2$

Q032 Written

A polynomial $P(x)$ leaves remainder $3x+2$ on division by x^2-x-2, and $7x-6$ on division by x^2+2x-3. Find the remainder on division by x^2-1.

Solution

1. $Let R = ax + b$
2. $At - 1and1 : -a + b = -1anda + b = 1$
3. $R = x$
4. Therefore the final answer is x.

Final answer

x

Worked Solutions **Polynomial Theorems**

Q033 Written

A polynomial $P(x)$ leaves remainder $2x-2$ on division by x^2-2x-3, and $-x+7$ on division by x^2-4. Find the remainder on division by x^2-x-2.

Solution

1. Use shared roots -1 and 2
2. $-a+b=-4and2a+b=5, soR=3x-1$
3. Therefore the final answer is $3x-1$.

Final answer

$3x-1$

Q034 Written

A polynomial $P(x)$ leaves remainder $-x+4$ on division by x^2-3x+2, and $3x$ on division by x^2-4x+3. Find the remainder on division by x^2-5x+6.

Solution

1. Use shared roots 2 and 3
2. $2a+b=2and3a+b=9, soR=7x-12$
3. Therefore the final answer is $7x-12$.

Final answer

$7x-12$

Worked Solutions **Polynomial Theorems**

Q035 MCQ

A polynomial $P(x)$ leaves remainder $7x+6$ on division by x^2-x-6, and $4x-3$ on division by x^2+2x-3. Find the remainder on division by x^2+x-2 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $3x-2$.

Correct option: B

Final answer

$3x-2$

Q036 MCQ

A polynomial $P(x)$ leaves remainder $3x+2$ on division by x^2-x-2, and $7x-6$ on division by x^2+2x-3. Find the remainder on division by x^2-1 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is x.

Correct option: B

Final answer

x

Worked Solutions **Polynomial Theorems**

Q037 MCQ

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $3x - 1$.

Correct option: A

Final answer

$3x - 1$

Q038 MCQ

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $7x - 12$.

Correct option: D

Final answer

$7x - 12$

Worked Solutions **Polynomial Theorems**

Q039 Written

A polynomial $P(x)$ leaves remainder $x-5$ on division by $(x+1)^2$, and 6 on division by $x-2$. Find the remainder on division by $(x+1)^2(x-2)$.

Solution

1. $Let R = ax^2 + bx + c$
2. $R(-1) = -6, R'(-1) = 1, R(2) = 6$
3. $Solve to get R = x^2 + 3x - 4$
4. Therefore the final answer is $x^2 + 3x - 4$.

Final answer

$x^2 + 3x - 4$

Q040 MCQ

A polynomial $P(x)$ leaves remainder $x-5$ on division by $(x+1)^2$, and 6 on division by $x-2$. Find the remainder on division by $(x+1)^2(x-2)$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x^2 + 3x - 4$.

Correct option: A

Final answer

$x^2 + 3x - 4$

Worked Solutions **Polynomial Theorems**

Q041 Written

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

Solution

1. Use -2 and coefficients 1,0,5,11
2. Bottom row 1,-2,9,-7
3. Therefore the final answer is $Q = x^2 - 2x + 9, R = -7$.

Final answer

$Q = x^2 - 2x + 9, R = -7$

Q042 MCQ

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 - 2x + 9, R = -7$.

Correct option: A

Final answer

$Q = x^2 - 2x + 9, R = -7$

Worked Solutions **Polynomial Theorems**

Q043 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,0,5,-6
2. Bottom row 1,1,6,0
3. Therefore the final answer is $Q = x^2 + x + 6, R = 0$.

Final answer

$Q = x^2 + x + 6, R = 0$

Q044 Written

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

Solution

1. Use 3 and coefficients 3,-5,-3,-1
2. Bottom row 3,4,9,26
3. Therefore the final answer is $Q = 3x^2 + 4x + 9, R = 26$.

Final answer

$Q = 3x^2 + 4x + 9, R = 26$

Worked Solutions **Polynomial Theorems**

Q045 Written

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

Solution

1. Use 2 and coefficients 1,0,-5,6
2. Bottom row 1,2,-1,4
3. Therefore the final answer is $Q = x^2 + 2x - 1, R = 4$.

Final answer

$Q = x^2 + 2x - 1, R = 4$

Q046 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,2,-1,-3
2. Bottom row 1,3,2,-1
3. Therefore the final answer is $Q = x^2 + 3x + 2, R = -1$.

Final answer

$Q = x^2 + 3x + 2, R = -1$

Worked Solutions **Polynomial Theorems**

Q047 Written

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 4,-2,2
2. Bottom row 4,2,4
3. Therefore the final answer is $Q = 4x + 2, R = 4$.

Final answer

$Q = 4x + 2, R = 4$

Q048 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 + x + 6, R = 0$.

Correct option: B

Final answer

$Q = x^2 + x + 6, R = 0$

Worked Solutions **Polynomial Theorems**

Q049 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = 3x^2 + 4x + 9, R = 26$.

Correct option: D

Final answer

$Q = 3x^2 + 4x + 9, R = 26$

Q050 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 + 2x - 1, R = 4$.

Correct option: D

Final answer

$Q = x^2 + 2x - 1, R = 4$

Worked Solutions **Polynomial Theorems**

Q051 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 + 3x + 2, R = -1$.

Correct option: A

Final answer

$Q = x^2 + 3x + 2, R = -1$

Q052 MCQ

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = 4x + 2, R = 4$.

Correct option: B

Final answer

$Q = 4x + 2, R = 4$

Worked Solutions **Polynomial Theorems**

Q053 **Written**

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

Solution

1. Use -2 and coefficients 3,5,0,-2,-12
2. Bottom row 3,-1,2,-6,0
3. Therefore the final answer is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Final answer

$Q = 3x^3 - x^2 + 2x - 6, R = 0$

Q054 **MCQ**

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Correct option: D

Final answer

$Q = 3x^3 - x^2 + 2x - 6, R = 0$

Worked Solutions **Polynomial Theorems**

Q055 **Written**

Factorise $x^3 + 2x^2 - x - 2$.

Solution

1. $P(1) = 0; dividebyx - 1$
2. $Quotientx^2 + 3x + 2 = (x + 1)(x + 2)$
3. Therefore the final answer is $(x - 1)(x + 1)(x + 2)$.

Final answer

$(x - 1)(x + 1)(x + 2)$

Q056 **Written**

Factorise $x^3 - 2x^2 - 5x + 6$.

Solution

1. $P(1) = 0; quotientx^2 - x - 6 = (x - 3)(x + 2)$
2. Completed factorisation
3. Therefore the final answer is $(x - 1)(x - 3)(x + 2)$.

Final answer

$(x - 1)(x - 3)(x + 2)$

Worked Solutions **Polynomial Theorems**

Q057 **Written**

Factorise $x^3 - x^2 - 16x - 20$.

Solution

1. $P(-2) = 0; dividebyx + 2twice$
2. Remaining x-5
3. Therefore the final answer is $(x + 2)^2(x - 5)$.

Final answer

$(x + 2)^2(x - 5)$

Q058 **Written**

Factorise $2x^3 - x^2 - 5x - 2$.

Solution

1. $P(2) = 0; quotient2x^2 + 3x + 1$
2. Therefore the final answer is $(x - 2)(2x + 1)(x + 1)$.

Final answer

$(x - 2)(2x + 1)(x + 1)$

Worked Solutions **Polynomial Theorems**

Q059 **Written**

Factorise $x^3 - 7x - 6$.

Solution

1. Test divisor 3 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 3)(x + 1)(x + 2)$.

Final answer

$(x - 3)(x + 1)(x + 2)$

Q060 **Written**

Factorise $x^3 + 4x^2 + x - 6$.

Solution

1. Test divisor 1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 1)(x + 2)(x + 3)$.

Final answer

$(x - 1)(x + 2)(x + 3)$

Worked Solutions **Polynomial Theorems**

Q061 Written

Factorise $x^3 + 6x^2 + 11x + 6$.

Solution

1. Test divisor -1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x + 1)(x + 2)(x + 3)$.

Final answer

$(x + 1)(x + 2)(x + 3)$

Q062 Written

Factorise $x^3 - 3x + 2$.

Solution

1. Test divisor 1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 1)^2(x + 2)$.

Final answer

$(x - 1)^2(x + 2)$

Worked Solutions **Polynomial Theorems**

Q063 Written

Factorise $x^3 + x^2 - 8x - 12$.

Solution

1. Test divisor -2 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x-3)(x+2)^2$.

Final answer

$(x-3)(x+2)^2$

Q064 Written

Factorise $x^3 - 3x^2 - 9x - 5$.

Solution

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x-5)(x+1)^2$.

Final answer

$(x-5)(x+1)^2$

Worked Solutions **Polynomial Theorems**

Q065 **Written**

Factorise $2x^3 - 3x^2 - 3x + 2$**.**

Solution

1. Test divisor 2 and divide
2. $Factorquotient 2x^2 + x - 1$
3. Therefore the final answer is $(x-2)(x+1)(2x-1)$.

Final answer

$(x-2)(x+1)(2x-1)$

Q066 **Written**

Factorise $3x^3 + x^2 - 8x + 4$**.**

Solution

1. Test divisor 1 and divide
2. $Factorquotient 3x^2 + 4x - 4$
3. Therefore the final answer is $(x-1)(x+2)(3x-2)$.

Final answer

$(x-1)(x+2)(3x-2)$

Worked Solutions **Polynomial Theorems**

Q067 Written

Factorise $2x^3 + 7x^2 + 8x + 3$.

Solution

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x+1)^2(2x+3)$.

Final answer

$(x+1)^2(2x+3)$

Q068 MCQ

Factorise $x^3 + 2x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)(x+1)(x+2)$.

Correct option: D

Final answer

$(x-1)(x+1)(x+2)$

Worked Solutions **Polynomial Theorems**

Q069 **MCQ**

Factorise $x^3 - 2x^2 - 5x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)(x-3)(x+2)$.

Correct option: D

Final answer

$(x-1)(x-3)(x+2)$

Q070 **MCQ**

Factorise $x^3 - x^2 - 16x - 20$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x+2)^2(x-5)$.

Correct option: C

Final answer

$(x+2)^2(x-5)$

Worked Solutions **Polynomial Theorems**

Q071 MCQ

Factorise $2x^3 - x^2 - 5x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)(2x+1)(x+1)$.

Correct option: B

Final answer

$(x-2)(2x+1)(x+1)$

Q072 MCQ

Factorise $x^3 - 7x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-3)(x+1)(x+2)$.

Correct option: B

Final answer

$(x-3)(x+1)(x+2)$

Worked Solutions **Polynomial Theorems**

Q073 MCQ

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x + 2)(x + 3)$.

Correct option: C

Final answer

$(x - 1)(x + 2)(x + 3)$

Q074 MCQ

Factorise $x^3 + 6x^2 + 11x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x + 1)(x + 2)(x + 3)$.

Correct option: B

Final answer

$(x + 1)(x + 2)(x + 3)$

Worked Solutions **Polynomial Theorems**

Q075 MCQ

Factorise $x^3 - 3x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)^2(x+2)$.

Correct option: D

Final answer

$(x-1)^2(x+2)$

Q076 MCQ

Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-3)(x+2)^2$.

Correct option: D

Final answer

$(x-3)(x+2)^2$

Worked Solutions **Polynomial Theorems**

Q077 MCQ

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-5)(x+1)^2$.

Correct option: D

Final answer

$(x-5)(x+1)^2$

Q078 MCQ

Factorise $2x^3 - 3x^2 - 3x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)(x+1)(2x-1)$.

Correct option: A

Final answer

$(x-2)(x+1)(2x-1)$

Worked Solutions **Polynomial Theorems**

Q079 MCQ

Factorise $3x^3 + x^2 - 8x + 4$ **Select the correct answer:**

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)(x+2)(3x-2)$.

Correct option: C

Final answer

$(x-1)(x+2)(3x-2)$

Q080 MCQ

Factorise $2x^3 + 7x^2 + 8x + 3$ **Select the correct answer:**

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x+1)^2(2x+3)$.

Correct option: D

Final answer

$(x+1)^2(2x+3)$

Worked Solutions **Polynomial Theorems**

Q081 Written

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$.

Solution

1. $P(2) = 0; dividebyx - 2twice$
2. $Remainingquadraticx^2 - 2x - 4$
3. Therefore the final answer is $(x-2)^2(x^2-2x-4)$.

Final answer

$(x-2)^2(x^2-2x-4)$

Q082 Written

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$.

Solution

1. Test roots and divide successively
2. Four linear factors
3. Therefore the final answer is $(x-3)(x-2)(x+1)(x+7)$.

Final answer

$(x-3)(x-2)(x+1)(x+7)$

Worked Solutions **Polynomial Theorems**

Q083 **Written**

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$.

Solution

1. $P(1) = 0; dividetwice$
2. $Remainingquadratic x^2 - 2x + 3$
3. Therefore the final answer is $(x-1)^2(x^2-2x+3)$.

Final answer

$(x-1)^2(x^2-2x+3)$

Q084 **Written**

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$.

Solution

1. Test roots 1,2,3,4 successively
2. Four linear factors
3. Therefore the final answer is $(x-4)(x-3)(x-2)(x-1)$.

Final answer

$(x-4)(x-3)(x-2)(x-1)$

Worked Solutions **Polynomial Theorems**

Q085 Written

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$.

Solution

1. Test roots 1 and -2 successively
2. Completed factorisation
3. Therefore the final answer is $(x-5)(x-1)^2(x+2)$.

Final answer

$(x-5)(x-1)^2(x+2)$

Q086 Written

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$.

Solution

1. Test roots 2,-2,4,-3/2
2. Completed factorisation
3. Therefore the final answer is $(x-4)(x-2)(x+2)(2x+3)$.

Final answer

$(x-4)(x-2)(x+2)(2x+3)$

Worked Solutions **Polynomial Theorems**

Q087 Written

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$.

Solution

1. Test roots 1,-2
2. Quadratic quotient remains
3. Therefore the final answer is $(x-1)(x+2)(x^2-5x+3)$.

Final answer

$(x-1)(x+2)(x^2-5x+3)$

Q088 Written

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$.

Solution

1. Test roots 3,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x-3)(x+1)(x^2-2x+2)$.

Final answer

$(x-3)(x+1)(x^2-2x+2)$

Worked Solutions **Polynomial Theorems**

Q089 Written

Factorise $x^4 - x^3 + x^2 + x - 2$.

Solution

1. Test roots 1,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x-1)(x+1)(x^2-x+2)$.

Final answer

$(x-1)(x+1)(x^2-x+2)$

Q090 MCQ

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)^2(x^2-2x-4)$.

Correct option: D

Final answer

$(x-2)^2(x^2-2x-4)$

Q091 MCQ

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-3)(x-2)(x+1)(x+7)$.

Correct option: A

Final answer

$(x-3)(x-2)(x+1)(x+7)$

Q092 MCQ

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)^2(x^2-2x+3)$.

Correct option: D

Final answer

$(x-1)^2(x^2-2x+3)$

Worked Solutions **Polynomial Theorems**

Q093 *MCQ*

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-4)(x-3)(x-2)(x-1)$.

Correct option: D

Final answer

$(x-4)(x-3)(x-2)(x-1)$

Q094 *MCQ*

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-5)(x-1)^2(x+2)$.

Correct option: C

Final answer

$(x-5)(x-1)^2(x+2)$

Worked Solutions **Polynomial Theorems**

Q095 MCQ

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ **Select the correct answer:**

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-4)(x-2)(x+2)(2x+3)$.

Correct option: D

Final answer

$(x-4)(x-2)(x+2)(2x+3)$

Q096 MCQ

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ **Select the correct answer:**

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)(x+2)(x^2-5x+3)$.

Correct option: A

Final answer

$(x-1)(x+2)(x^2-5x+3)$

Worked Solutions **Polynomial Theorems**

Q097 *MCQ*

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-3)(x+1)(x^2-2x+2)$.

Correct option: B

Final answer

$(x-3)(x+1)(x^2-2x+2)$

Q098 *MCQ*

Factorise $x^4 - x^3 + x^2 + x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-1)(x+1)(x^2-x+2)$.

Correct option: C

Final answer

$(x-1)(x+1)(x^2-x+2)$

Worked Solutions **Polynomial Theorems**

Q099 Written

Factorise $2x^3 - 3x^2 - x - 2$.

Solution

1. $P(2) = 0; quotient 2x^2 + x + 1 has discriminant - 7$
2. Therefore the final answer is $(x-2)(2x^2+x+1)$.

Final answer

$(x-2)(2x^2+x+1)$

Q100 Written

Factorise $2x^3 - 7x^2 + 3x + 6$.

Solution

1. Test divisor 2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x-2)(2x^2-3x-3)$.

Final answer

$(x-2)(2x^2-3x-3)$

Worked Solutions **Polynomial Theorems**

Q101 Written

Factorise $x^3 + 4x^2 + 2x - 4$.

Solution

1. Test divisor -2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x+2)(x^2+2x-2)$.

Final answer

$(x+2)(x^2+2x-2)$

Q102 Written

Factorise $x^3 - 3x^2 + 2$.

Solution

1. Test divisor 1 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x-1)(x^2-2x-2)$.

Final answer

$(x-1)(x^2-2x-2)$

Worked Solutions **Polynomial Theorems**

Q103 MCQ

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)(2x^2+x+1)$.

Correct option: C

Final answer

$(x-2)(2x^2+x+1)$

Q104 MCQ

Factorise $2x^3 - 7x^2 + 3x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x-2)(2x^2-3x-3)$.

Correct option: B

Final answer

$(x-2)(2x^2-3x-3)$

Worked Solutions **Polynomial Theorems**

Q105 MCQ

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x + 2)(x^2 + 2x - 2)$.

Correct option: B

Final answer

$(x + 2)(x^2 + 2x - 2)$

Q106 MCQ

Factorise $x^3 - 3x^2 + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x^2 - 2x - 2)$.

Correct option: D

Final answer

$(x - 1)(x^2 - 2x - 2)$

Worked Solutions **Polynomial Theorems**

Q107 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.

Solution

1. $Substitute x = -1$
2. $(-1)^{57} + 57 = 56$
3. Therefore the final answer is 56.

Final answer

56

Worked Solutions **Polynomial Theorems**

Q108 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a, then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

Solution

1. Use the divisor zeros: $P(-1/2) = 0$ and $P(-2) = 0$.
2. Hence $a = 10$ for $2x + 1$, and $a = -11$ for $x + 2$.
3. Each linear factor corresponds to a polynomial root.

Final answer

$a = 10 \quad \text{for } 2x + 1$

$a = -11 \quad \text{for } x + 2$

linear factors $\leftrightarrow$ roots

Worked Solutions **Polynomial Theorems**

Q109 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Solution

1. $a + b = 0 and - 3a + b = -4$
2. $R = x - 1$
3. Therefore the final answer is $x - 1$.

Final answer

$x - 1$

Worked Solutions **Polynomial Theorems**

Q110 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

Solution

1. $P = (2x-1)^2(x^2-3)$
2. $Remainingrootsare + / - \sqrt(3)$
3. Therefore the final answer is $x = \sqrt{3}, -\sqrt{3}$.

Final answer

$x = \sqrt{3}, -\sqrt{3}$

Worked Solutions **Polynomial Theorems**

Q111 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b.

Solution

1. $P(1) = 0 gives a + b = -10$
2. $P(-2) = 0 gives 2a - b = -26$
3. $a = -12, b = 2$
4. Therefore the final answer is $a = -12, b = 2$.

Final answer

$a = -12, b = 2$

Worked Solutions **Polynomial Theorems**

Q112 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$.

Solution

1. $P = (x-1)(x+2)(x^2 - 4x - 6)$
2. $Solvequadratic : x = 2 \pm \sqrt{(10)}$
3. $All fourrootssatisfy P = 0$
4. Therefore the final answer is $x = 1, -2, 2 \pm \sqrt{10}$.

Final answer

$x = 1, -2, 2 \pm \sqrt{10}$

Worked Solutions **Polynomial Theorems**

Q113 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 56.

Correct option: B

Final answer

56

Worked Solutions **Polynomial Theorems**

Q114 **Challenge**

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a, then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.

$$a = 10 \quad \text{for } 2x + 1$$

2. The correct value is $a = -11 \quad \text{for } x + 2$.

$$\text{linear factors} \leftrightarrow \text{roots}$$

Correct option: D

Final answer

$a = 10 \quad \text{for } 2x + 1$

$a = -11 \quad \text{for } x + 2$

linear factors $\leftrightarrow$ roots

Worked Solutions **Polynomial Theorems**

Q115 **Challenge**

A polynomial $P(x)$ is divisible by $x-1$ and leaves remainder -4 on division by $x+3$. Find the remainder on division by x^2+2x-3 Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x-1$.

Correct option: C

Final answer

$x-1$

Worked Solutions **Polynomial Theorems**

Q116 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x = \sqrt{3}, -\sqrt{3}$.

Correct option: A

Final answer

$x = \sqrt{3}, -\sqrt{3}$

Worked Solutions **Polynomial Theorems**

Q117 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -12, b = 2$.

Correct option: C

Final answer

$a = -12, b = 2$

Worked Solutions **Polynomial Theorems**

Q118 **Challenge**

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x = 1, -2, 2 \pm \sqrt{10}$.

Correct option: A

Final answer

$x = 1, -2, 2 \pm \sqrt{10}$

Worked Solutions **Polynomial Theorems**

Quiz MCQ 1 **Concept Quiz · MCQ**

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 9.

Correct option: C

Final answer

9

Quiz MCQ 2 **Concept Quiz · MCQ**

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $P(-1) = 0$.

Correct option: C

Final answer

$P(-1) = 0$

Worked Solutions **Polynomial Theorems**

Quiz MCQ 3 Concept Quiz · MCQ

The remainder on division by a quadratic divisor has the form Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $ax + b$.

Correct option: B

Final answer

$ax + b$

Quiz MCQ 4 Concept Quiz · MCQ

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Syntheticdivision$.

Correct option: C

Final answer

$Syntheticdivision$

Worked Solutions **Polynomial Theorems**

Quiz WRITTEN 5 Concept Quiz · Written

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: 4

Final answer

4

Quiz WRITTEN 6 Concept Quiz · Written

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $R = 3x$

Final answer

$R = 3x$

Worked Solutions **Polynomial Theorems**

Quiz WRITTEN 7 Concept Quiz · Written

Factorise $x^3 - 6x^2 + 11x - 6$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $(x-1)(x-2)(x-3)$

Final answer

$(x-1)(x-2)(x-3)$

Quiz WRITTEN 8 Concept Quiz · Written

If $x-1$ and $x+2$ are factors of $x^4 - 3x^3 + ax^2 + bx + 12$, find a, b.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $a = -12, b = 2$

Final answer

$a = -12, b = 2$

Answer key

Use the question number shown on each page.

Q001 2
Q002 6
Q003 C
Q004 C
Q005 8
Q006 $-\frac{1}{8}$
Q007 0
Q008 4
Q009 D
Q010 A
Q011 D
Q012 A
Q013 $a = 13/3$
Q014 $a = -4$
Q015 $a = -4$
Q016 $a = -16$
Q017 B
Q018 B
Q019 C
Q020 A
Q021 $a = -4$
Q022 A
Q023 $2x + 1$
Q024 $-10x + 13$
Q025 $2x + 3$
Q026 $-15x + 19$
Q027 A
Q028 A
Q029 A
Q030 A
Q031 $3x - 2$
Q032 x
Q033 $3x - 1$
Q034 $7x - 12$
Q035 B
Q036 B
Q037 A
Q038 D
Q039 $x^2 + 3x - 4$
Q040 A
Q041 $Q = x^2 - 2x + 9, R = -7$
Q042 A
Q043 $Q = x^2 + x + 6, R = 0$
Q044 $Q = 3x^2 + 4x + 9, R = 26$
Q045 $Q = x^2 + 2x - 1, R = 4$
Q046 $Q = x^2 + 3x + 2, R = -1$
Q047 $Q = 4x + 2, R = 4$
Q048 B
Q049 D
Q050 D
Q051 A
Q052 B
Q053 $Q = 3x^3 - x^2 + 2x - 6, R = 0$
Q054 D
Q055 $(x-1)(x+1)(x+2)$
Q056 $(x-1)(x-3)(x+2)$
Q057 $(x+2)^2(x-5)$
Q058 $(x-2)(2x+1)(x+1)$
Q059 $(x-3)(x+1)(x+2)$
Q060 $(x-1)(x+2)(x+3)$
Q061 $(x+1)(x+2)(x+3)$
Q062 $(x-1)^2(x+2)$
Q063 $(x-3)(x+2)^2$
Q064 $(x-5)(x+1)^2$
Q065 $(x-2)(x+1)(2x-1)$
Q066 $(x-1)(x+2)(3x-2)$
Q067 $(x+1)^2(2x+3)$
Q068 D
Q069 D
Q070 C
Q071 B
Q072 B
Q073 C
Q074 B
Q075 D
Q076 D
Q077 D
Q078 A
Q079 C
Q080 D
Q081 $(x-2)^2(x^2 - 2x - 4)$
Q082 $(x-3)(x-2)(x+1)(x+7)$
Q083 $(x-1)^2(x^2 - 2x + 3)$
Q084 $(x-4)(x-3)(x-2)(x-1)$
Q085 $(x-5)(x-1)^2(x+2)$
Q086 $(x-4)(x-2)(x+2)(2x+3)$
Q087 $(x-1)(x+2)(x^2 - 5x + 3)$
Q088 $(x-3)(x+1)(x^2 - 2x + 2)$
Q089 $(x-1)(x+1)(x^2 - x + 2)$
Q090 D
Q091 A
Q092 D
Q093 D
Q094 C
Q095 D
Q096 A
Q097 B
Q098 C
Q099 $(x-2)(2x^2 + x + 1)$
Q100 $(x-2)(2x^2 - 3x - 3)$
Q101 $(x+2)(x^2 + 2x - 2)$
Q102 $(x-1)(x^2 - 2x - 2)$
Q103 C
Q104 B

Q105 B

Q106 D

Q107 56

$a = 10$ for $2x + 1$

Q108 $a = -11$ for $x + 2$

linear factors ↔ roots

Q109 $x - 1$

Q110 $x = \sqrt{3}, -\sqrt{3}$

Q111 $a = -12, b = 2$

Q112 $x = 1, -2, 2 \pm \sqrt{10}$

Q113 B

Q114 D

Q115 C

Q116 A

Q117 C

Q118 A

Quiz MCQ 1 C

Quiz MCQ 2 C

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 4

Quiz WRITTEN 6 $R = 3x$

Quiz WRITTEN 7 $(x - 1)(x - 2)(x - 3)$

Quiz WRITTEN 8 $a = -12, b = 2$

Concept 4 | Higher-Degree Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Factor theorem

$P(a) = 0 \Rightarrow x - a$ is a factor

Degree

$\deg(PQ) = \deg P + \deg Q$

Conjugate roots

$a + bi \Rightarrow a - bi$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Higher-Degree Equations**

Q001 **Written**

$x^3 - 8 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$.

Final answer

$x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$

Q002 **Written**

$x^3 = -1$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, (1 - \sqrt{3}\,i)/2, (1 + \sqrt{3}\,i)/2$.

Final answer

$x = -1, (1 - \sqrt{3}\,i)/2, (1 + \sqrt{3}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q003 **Written**

$x^3 = 27$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$.

Final answer

$x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$

Q004 **Written**

$x^3 = -125$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$.

Final answer

$x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q005 **Written**

$8x^3 = 1$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1/2, (-1 - \sqrt{3}\,i)/4, (-1 + \sqrt{3}\,i)/4$.

Final answer

$x = 1/2, (-1 - \sqrt{3}\,i)/4, (-1 + \sqrt{3}\,i)/4$

Q006 **MCQ**

$x^3 - 8 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$.

Correct option: B

Final answer

$x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$

Worked Solutions **Higher-Degree Equations**

Q007 MCQ

$x^3 = -1$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, (1 - \sqrt{3}\,i)/2, (1 + \sqrt{3}\,i)/2$.

Correct option: D

Final answer

$x = -1, (1 - \sqrt{3}\,i)/2, (1 + \sqrt{3}\,i)/2$

Q008 MCQ

$x^3 = 27$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$.

Correct option: B

Final answer

$x = 3, (-3 - 3\sqrt{3}\,i)/2, (-3 + 3\sqrt{3}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q009 MCQ

$x^3 = -125$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$.

Correct option: A

Final answer

$x = -5, (5 - 5\sqrt{3}\,i)/2, (5 + 5\sqrt{3}\,i)/2$

Q010 MCQ

$8x^3 = 1$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 1/2, (-1 - \sqrt{3}\,i)/4, (-1 + \sqrt{3}\,i)/4$.

Correct option: C

Final answer

$x = 1/2, (-1 - \sqrt{3}\,i)/4, (-1 + \sqrt{3}\,i)/4$

Worked Solutions **Higher-Degree Equations**

Q011 **Written**

$x^3 - 2x^2 - 5x + 6 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 1, 3$.

Final answer

$x = -2, 1, 3$

Q012 **Written**

$3x^3 + x^2 - 8x + 4 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 2/3, 1$.

Final answer

$x = -2, 2/3, 1$

Worked Solutions **Higher-Degree Equations**

Q013 **Written**

$2x^3 - 3x^2 - x - 2 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$.

Final answer

$x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$

Q014 **Written**

$x^3 - x^2 - 16x - 20 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{repeated}), 5$.

Final answer

$x = -2(\text{repeated}), 5$

Worked Solutions **Higher-Degree Equations**

Q015 **Written**

$x^3 + 6x^2 + 11x + 6 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, -1$.

Final answer

$x = -3, -2, -1$

Q016 **Written**

$x^3 + 3x^2 - 4x - 12 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, 2$.

Final answer

$x = -3, -2, 2$

Worked Solutions **Higher-Degree Equations**

Q017 Written

$3x^3 - 2x^2 - 3x + 2 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 2/3, 1$.

Final answer

$x = -1, 2/3, 1$

Q018 Written

$x^3 - x^2 - 4 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}\,i)/2, (-1 + \sqrt{7}\,i)/2$.

Final answer

$x = 2, (-1 - \sqrt{7}\,i)/2, (-1 + \sqrt{7}\,i)/2$

Worked Solutions

Higher-Degree Equations

Q019 Written

$x^3 + 4x^2 + 2x - 4 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Final answer

$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$

Q020 Written

$x^3 - 4x^2 + 9x - 10 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, 1 - 2i, 1 + 2i$.

Final answer

$x = 2, 1 - 2i, 1 + 2i$

Worked Solutions **Higher-Degree Equations**

Q021 **Written**

$x^3 + x^2 - 8x - 12 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{repeated}), 3$.

Final answer

$x = -2(\text{repeated}), 3$

Q022 **Written**

$2x^3 + 7x^2 + 8x + 3 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1(\text{repeated}), -3/2$.

Final answer

$x = -1(\text{repeated}), -3/2$

Worked Solutions **Higher-Degree Equations**

Q023 Written

$x^3 + 6x^2 + 12x + 8 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{triple})$.

Final answer

$x = -2(\text{triple})$

Q024 MCQ

$x^3 - 2x^2 - 5x + 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, 1, 3$.

Correct option: C

Final answer

$x = -2, 1, 3$

Worked Solutions **Higher-Degree Equations**

Q025 *MCQ*

$3x^3 + x^2 - 8x + 4 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, 2/3, 1$.

Correct option: D

Final answer

$x = -2, 2/3, 1$

Q026 *MCQ*

$2x^3 - 3x^2 - x - 2 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$.

Correct option: D

Final answer

$x = 2, (-1 - \sqrt{7}\,i)/4, (-1 + \sqrt{7}\,i)/4$

Worked Solutions **Higher-Degree Equations**

Q027 MCQ

$x^3 - x^2 - 16x - 20 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2(\text{repeated}), 5$.

Correct option: B

Final answer

$x = -2(\text{repeated}), 5$

Q028 MCQ

$x^3 + 6x^2 + 11x + 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, -2, -1$.

Correct option: C

Final answer

$x = -3, -2, -1$

Worked Solutions **Higher-Degree Equations**

Q029 MCQ

$x^3 + 3x^2 - 4x - 12 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, -2, 2$.

Correct option: D

Final answer

$x = -3, -2, 2$

Q030 MCQ

$3x^3 - 2x^2 - 3x + 2 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, 2/3, 1$.

Correct option: D

Final answer

$x = -1, 2/3, 1$

Worked Solutions **Higher-Degree Equations**

Q031 MCQ

$x^3 - x^2 - 4 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}\,i)/2, (-1 + \sqrt{7}\,i)/2$.

Correct option: A

Final answer

$x = 2, (-1 - \sqrt{7}\,i)/2, (-1 + \sqrt{7}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q032 MCQ

$x^3 + 4x^2 + 2x - 4 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Correct option: D

Final answer

$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$

Worked Solutions **Higher-Degree Equations**

Q033 MCQ

$x^3 - 4x^2 + 9x - 10 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, 1 - 2i, 1 + 2i$.

Correct option: D

Final answer

$x = 2, 1 - 2i, 1 + 2i$

Q034 MCQ

$x^3 + x^2 - 8x - 12 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2(\text{repeated}), 3$.

Correct option: D

Final answer

$x = -2(\text{repeated}), 3$

Worked Solutions **Higher-Degree Equations**

Q035 MCQ

$2x^3 + 7x^2 + 8x + 3 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1(\text{repeated}), -3/2$.

Correct option: B

Final answer

$x = -1(\text{repeated}), -3/2$

Q036 MCQ

$x^3 + 6x^2 + 12x + 8 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2(\text{triple})$.

Correct option: D

Final answer

$x = -2(\text{triple})$

Worked Solutions **Higher-Degree Equations**

Q037 **Written**

$x^4 - 7x^2 - 18 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$.

Final answer

$x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$

Q038 **Written**

$x^4 - 5x^2 + 4 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1, 1, 2$.

Final answer

$x = -2, -1, 1, 2$

Worked Solutions **Higher-Degree Equations**

Q039 **Written**

$x^4 - 5x^2 - 36 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -2\,i, 2\,i$.

Final answer

$x = -3, 3, -2\,i, 2\,i$

Q040 **Written**

$x^4 - 6x^2 - 27 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$.

Final answer

$x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$

Worked Solutions **Higher-Degree Equations**

Q041 Written

$x^4 - x^2 - 2 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -\sqrt{2}, \sqrt{2}, -i, i$.

Final answer

$x = -\sqrt{2}, \sqrt{2}, -i, i$

Q042 MCQ

$x^4 - 7x^2 - 18 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$.

Correct option: D

Final answer

$x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$

Worked Solutions **Higher-Degree Equations**

Q043 MCQ

$x^4 - 5x^2 + 4 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -2, -1, 1, 2$.

Correct option: D

Final answer

$x = -2, -1, 1, 2$

Q044 MCQ

$x^4 - 5x^2 - 36 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, 3, -2\,i, 2\,i$.

Correct option: C

Final answer

$x = -3, 3, -2\,i, 2\,i$

Worked Solutions **Higher-Degree Equations**

Q045 MCQ

$x^4 - 6x^2 - 27 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$.

Correct option: C

Final answer

$x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$

Q046 MCQ

$x^4 - x^2 - 2 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -\sqrt{2}, \sqrt{2}, -i,\ i$.

Correct option: B

Final answer

$x = -\sqrt{2}, \sqrt{2}, -i,\ i$

Worked Solutions **Higher-Degree Equations**

Q047 Written

$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$.

Final answer

$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$

Q048 Written

$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3, 4$.

Final answer

$x = 1, 2, 3, 4$

Worked Solutions **Higher-Degree Equations**

Q049 Written

$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3(\text{repeated})$.

Final answer

$x = 1, 2, 3(\text{repeated})$

Q050 Written

$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 3, 1 - i, 1 + i$.

Final answer

$x = -1, 3, 1 - i, 1 + i$

Worked Solutions **Higher-Degree Equations**

Q051 MCQ

$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$.

Correct option: B

Final answer

$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$

Worked Solutions **Higher-Degree Equations**

Q052 MCQ

$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 1, 2, 3, 4$.

Correct option: A

Final answer

$x = 1, 2, 3, 4$

Q053 MCQ

$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 1, 2, 3(\text{repeated})$.

Correct option: C

Final answer

$x = 1, 2, 3(\text{repeated})$

Worked Solutions **Higher-Degree Equations**

Q054 MCQ

$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, 3, 1 - i, 1 + i$.

Correct option: C

Final answer

$x = -1, 3, 1 - i, 1 + i$

Q055 Written

$x^3 + 4x^2 + ax + b = 0$ **has roots -3 and 1; find a,b and the other root.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 1, b = -6;\ \text{other roots:}\ -2$.

Final answer

$a = 1, b = -6;\ \text{other roots:}\ -2$

Worked Solutions **Higher-Degree Equations**

Q056 **Written**

$x^3 - ax^2 - x + b = 0$ **has roots -1 and 2; find a,b and the other root.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 2, b = 2;$ other roots: 1.

Final answer

$a = 2, b = 2;$ other roots: 1

Q057 **Written**

$x^3 + ax^2 + bx - 8 = 0$ **has roots 1 and 2; find a,b and the other root.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -7, b = 14;$ other roots: 4.

Final answer

$a = -7, b = 14;$ other roots: 4

Worked Solutions **Higher-Degree Equations**

Q058 MCQ

$x^3 + 4x^2 + ax + b = 0$ **has roots -3 and 1; find a,b and the other root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 1, b = -6;$ other roots: -2.

Correct option: C

Final answer

$a = 1, b = -6;$ other roots: -2

Worked Solutions **Higher-Degree Equations**

Q059 **MCQ**

$x^3 - ax^2 - x + b = 0$ **has roots -1 and 2; find a,b and the other root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 2, b = 2;$ other roots: 1.

Correct option: B

Final answer

$a = 2, b = 2;$ other roots: 1

Worked Solutions **Higher-Degree Equations**

Q060 MCQ

$x^3 + ax^2 + bx - 8 = 0$ **has roots 1 and 2; find a,b and the other root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -7, b = 14;$ other roots: 4.

Correct option: C

Final answer

$a = -7, b = 14;$ other roots: 4

Worked Solutions **Higher-Degree Equations**

Q061 **Written**

$x^3 + ax^2 + bx + 5 = 0$ has root 2+i; find real a,b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -3, b = 1;$ other roots: $-1, 2 - i$.

Final answer

$a = -3, b = 1;$ other roots: $-1, 2 - i$

Q062 **Written**

$x^3 + ax^2 + bx - 4 = 0$ has root 1+i; find a,b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 6;$ other roots: $1 - i, 2$.

Final answer

$a = -4, b = 6;$ other roots: $1 - i, 2$

Worked Solutions **Higher-Degree Equations**

Q063 Written

$x^3 + ax^2 + bx - 20 = 0$ **has root** $1 - 3\,i$**; find real a,b and the other roots.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 14;$ other roots: $1 + 3i, 2$.

Final answer

$a = -4, b = 14;$ other roots: $1 + 3i, 2$

Q064 Written

$x^3 + ax^2 + bx + 10 = 0$ **has root** $1 + 2\,i$**; find real a,b and the other roots.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 0, b = 1;$ other roots: $1 - 2i, -2$.

Final answer

$a = 0, b = 1;$ other roots: $1 - 2i, -2$

Worked Solutions **Higher-Degree Equations**

Q065 MCQ

$x^3 + ax^2 + bx + 5 = 0$ **has root 2+i; find real a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -3, b = 1;$ other roots: $-1, 2 - i$.

Correct option: A

Final answer

$a = -3, b = 1;$ other roots: $-1, 2 - i$

Q066 MCQ

$x^3 + ax^2 + bx - 4 = 0$ **has root 1+i; find a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -4, b = 6;$ other roots: $1 - i, 2$.

Correct option: B

Final answer

$a = -4, b = 6;$ other roots: $1 - i, 2$

Worked Solutions **Higher-Degree Equations**

Q067 **MCQ**

$x^3 + ax^2 + bx - 20 = 0$ **has root** $1 - 3\,i$**; find real a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -4, b = 14;$ other roots: $1 + 3i, 2$.

Correct option: D

Final answer

$a = -4, b = 14;$ other roots: $1 + 3i, 2$

Q068 **MCQ**

$x^3 + ax^2 + bx + 10 = 0$ **has root** $1 + 2\,i$**; find real a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 0, b = 1;$ other roots: $1 - 2i, -2$.

Correct option: B

Final answer

$a = 0, b = 1;$ other roots: $1 - 2i, -2$

Worked Solutions **Higher-Degree Equations**

Q069 Written

$x^3 - kx^2 + (3k-1)x - 24 = 0$ **has root 2; find k and the other roots.**

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $k = 9;$ other roots: $3, 4$.

Final answer

$k = 9;$ other roots: $3, 4$

Q070 MCQ

$x^3 - kx^2 + (3k-1)x - 24 = 0$ **has root 2; find k and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $k = 9;$ other roots: $3, 4$.

Correct option: B

Final answer

$k = 9;$ other roots: $3, 4$

Worked Solutions **Higher-Degree Equations**

Q071 Written

For an imaginary cube root of unity w, evaluate w^9.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q072 Written

For an imaginary cube root of unity w, evaluate w^{2010}.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Worked Solutions **Higher-Degree Equations**

Q073 **Written**

For an imaginary cube root of unity w, evaluate w^3.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q074 **MCQ**

For an imaginary cube root of unity w, evaluate w^9 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: A

Final answer

1

Worked Solutions **Higher-Degree Equations**

Q075 MCQ

For an imaginary cube root of unity w, evaluate w^{2010} Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: B

Final answer

1

Q076 MCQ

For an imaginary cube root of unity w, evaluate w^{3} Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: D

Final answer

1

Worked Solutions **Higher-Degree Equations**

Q077 Written

For an imaginary cube root of unity w, evaluate $w^4 + w^5$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1.

Final answer

-1

Q078 Written

For an imaginary cube root of unity w, evaluate $w^2 + w$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1.

Final answer

-1

Worked Solutions **Higher-Degree Equations**

Q079 Written

For an imaginary cube root of unity w, evaluate $w^2 + w^4 + w^6$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q080 MCQ

For an imaginary cube root of unity w, evaluate $w^4 + w^5$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1.

Correct option: A

Final answer

-1

Worked Solutions **Higher-Degree Equations**

Q081 MCQ

For an imaginary cube root of unity w, evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1.

Correct option: B

Final answer

-1

Q082 MCQ

For an imaginary cube root of unity w, evaluate $w^2 + w^4 + w^6$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 0.

Correct option: D

Final answer

0

Worked Solutions **Higher-Degree Equations**

Q083 Written

For an imaginary cube root of unity w, evaluate $w + 1/w$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1.

Final answer

-1

Q084 Written

For an imaginary cube root of unity w, evaluate $(1 + w)(1 + w^2)$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Worked Solutions **Higher-Degree Equations**

Q085 Written

For an imaginary cube root of unity w, evaluate $1/w + 1/w^2 + 1$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q086 MCQ

For an imaginary cube root of unity w, evaluate $w + 1/w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1.

Correct option: D

Final answer

-1

Worked Solutions **Higher-Degree Equations**

Q087 MCQ

For an imaginary cube root of unity w, evaluate $(1+w)(1+w^2)$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 1.

Correct option: C

Final answer

1

Q088 MCQ

For an imaginary cube root of unity w, evaluate $1/w+1/w^2+1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 0.

Correct option: D

Final answer

0

Worked Solutions **Higher-Degree Equations**

Q089 **Written**

$x^3 - 2x^2 + (a-3)x + a = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -4$ or $9/4$.

Final answer

$a = -4$ or $9/4$

Q090 **Written**

$x^3 - x^2 + (a-2)x + a = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -3$ or 1.

Final answer

$a = -3$ or 1

Worked Solutions **Higher-Degree Equations**

Q091 **Written**

$x^3 - 7x^2 + (a+6)x - a = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 5$ or 9.

Final answer

$a = 5$ or 9

Q092 **Written**

$x^3 - (a+1)x^2 + (3a+5)x - 2a - 5 = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -6, -2,$ or 10.

Final answer

$a = -6, -2,$ or 10

Worked Solutions **Higher-Degree Equations**

Q093 Written

$x^3 + (2a-1)x^2 - 3(a-2)x + a - 6 = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose $(D = 0)$, then retain the real parameter values.
3. Therefore the final answer is $a = -7, -3,$ or 2.

Final answer

$a = -7, -3,$ or 2

Q094 MCQ

$x^3 - 2x^2 + (a-3)x + a = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -4$ or $9/4$.

Correct option: D

Final answer

$a = -4$ or $9/4$

Worked Solutions **Higher-Degree Equations**

Q095 MCQ

$x^3 - x^2 + (a-2)x + a = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -3$ or 1.

Correct option: D

Final answer

$a = -3$ or 1

Q096 MCQ

$x^3 - 7x^2 + (a+6)x - a = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 5$ or 9.

Correct option: B

Final answer

$a = 5$ or 9

Worked Solutions **Higher-Degree Equations**

Q097 MCQ

$x^3 - (a+1)x^2 + (3a+5)x - 2a - 5 = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -6, -2,$ or 10.

Correct option: A

Final answer

$a = -6, -2,$ or 10

Q098 MCQ

$x^3 + (2a-1)x^2 - 3(a-2)x + a - 6 = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -7, -3,$ or 2.

Correct option: A

Final answer

$a = -7, -3,$ or 2

Worked Solutions **Higher-Degree Equations**

Q099 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}\,i$; find real a,b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$.

Final answer

$a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$

Worked Solutions **Higher-Degree Equations**

Q100 **Challenge**

For an imaginary cube root of unity w, evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions **Higher-Degree Equations**

Q101 Challenge

$x^3 + (a-1)x^2 + (a-3)x - 2a + 3 = 0$ **Find all real parameter values a producing a repeated root.**

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 2/3, 2,\ \text{or } 6$.

Final answer

$a = 2/3, 2,\ \text{or } 6$

Worked Solutions **Higher-Degree Equations**

Q102 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

Solution

1. Set inflow equal to outflow: $(x^4 + x^2 + x = x^3 + 2)$.
2. Move all terms to one side.
3. Therefore the final answer is $x^4 - x^3 + x^2 + x - 2 = 0$.

Final answer

$x^4 - x^3 + x^2 + x - 2 = 0$

Worked Solutions **Higher-Degree Equations**

Q103 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels.

Solution

1. Factor $(x^4 - x^3 + x^2 + x - 2)$ and solve the remaining quadratic.
2. Apply the stated physical restriction (xge0) only after listing the algebraic roots.
3. Therefore the final answer is $x = -1, 1, (1 - \sqrt{7}\,i)/2, (1 + \sqrt{7}\,i)/2$.

Final answer

$x = -1, 1, (1 - \sqrt{7}\,i)/2, (1 + \sqrt{7}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q104 Challenge

Classify the stable pressure solutions as real or complex.

Solution

1. Separate the real roots from the non-real conjugate pair.

$$\text{Both types occur: real } x = -1, 1$$

2. Therefore the final answer is non-real conjugates $x = (1 \pm \sqrt{7}\, i)/2$.

$$\text{if } x \geq 0: \quad x = 1$$

Final answer

$$\text{Both types occur: real } x = -1, 1$$

non-real conjugates $x = (1 \pm \sqrt{7}\, i)/2$

$$\text{if } x \geq 0: \quad x = 1$$

Worked Solutions **Higher-Degree Equations**

Q105 **Challenge**

$x^3 + ax + b = 0$ **has root** $1 + \sqrt{2}\,i$**; find real a,b and the other roots Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$.

Correct option: C

Final answer

$a = -1, b = 6;$ other roots: $1 - \sqrt{2}i, -2$

Worked Solutions **Higher-Degree Equations**

Q106 **Challenge**

For an imaginary cube root of unity w, evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is 4.

Correct option: D

Final answer

4

Worked Solutions **Higher-Degree Equations**

Q107 **Challenge**

$x^3 + (a-1)x^2 + (a-3)x - 2a + 3 = 0$ **Find all real parameter values a producing a repeated root Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $a = 2/3, 2,$ or 6.

Correct option: D

Final answer

$a = 2/3, 2,$ or 6

Worked Solutions **Higher-Degree Equations**

Q108 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x^4 - x^3 + x^2 + x - 2 = 0$.

Correct option: A

Final answer

$x^4 - x^3 + x^2 + x - 2 = 0$

Worked Solutions **Higher-Degree Equations**

Q109 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ for the stable pressure levels Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = -1, 1, (1 - \sqrt{7}\,i)/2, (1 + \sqrt{7}\,i)/2$.

Correct option: D

Final answer

$x = -1, 1, (1 - \sqrt{7}\,i)/2, (1 + \sqrt{7}\,i)/2$

Worked Solutions **Higher-Degree Equations**

Q110 Challenge

Classify the stable pressure solutions as real or complex Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.

$$\text{Both types occur: real } x = -1, 1$$

2. The correct value is non-real conjugates $x = (1 \pm \sqrt{7}\, i)/2$.

$$\text{if } x \geq 0 : \quad x = 1$$

Correct option: B

Final answer

Both types occur: real $x = -1, 1$

non-real conjugates $x = (1 \pm \sqrt{7}\, i)/2$

$$\text{if } x \geq 0 : \quad x = 1$$

Worked Solutions **Higher-Degree Equations**

Quiz MCQ 1 **Concept Quiz · MCQ**

$Solve x^3 - 8 = 0$. **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$.

Correct option: D

Final answer

$x = 2, -1 - \sqrt{3}\,i, -1 + \sqrt{3}\,i$

Quiz MCQ 2 **Concept Quiz · MCQ**

$A real cubic has root 1 + 2\,i. Which root must also occur$? **Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $1 - 2\,i$.

Correct option: D

Final answer

$1 - 2\,i$

Quiz MCQ 3 Concept Quiz · MCQ

For an imaginary cube root of unity w, evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is -1.

Correct option: B

Final answer

-1

Quiz MCQ 4 Concept Quiz · MCQ

For the remaining quadratic factor, a repeated root requires Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the source demand.
2. The correct value is $D = 0$.

Correct option: B

Final answer

$D = 0$

Worked Solutions **Higher-Degree Equations**

Quiz WRITTEN 5 **Concept Quiz · Written**

$Solve x^4 - 5x^2 + 4 = 0.$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $x = -2, -1, 1, 2$

Final answer

$x = -2, -1, 1, 2$

Quiz WRITTEN 6 **Concept Quiz · Written**

$For x^3 + ax^2 + bx + 10 = 0$ **with root** $1 + 2\,i$**, find a,b and the other roots.**

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = 0, b = 1;$ other roots: $1 - 2i, -2$

Final answer

$a = 0, b = 1;$ other roots: $1 - 2i, -2$

Worked Solutions **Higher-Degree Equations**

Quiz WRITTEN 7 **Concept Quiz · Written**

For an imaginary cube root of unity w, evaluate $(1 + w)(1 + w^2)$.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: 1

Final answer

1

Quiz WRITTEN 8 **Concept Quiz · Written**

$Findallrealaforx^3 - 2x^2 + (a - 3)x + a = 0tohavearepeatedroot.$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = -4$ or $9/4$

Final answer

$a = -4$ or $9/4$

Answer key

Use the question number shown on each page.

Q001 $x = 2, -1-\sqrt{3}\,i, -1+\sqrt{3}\,i$

Q002 $x = -1, (1-\sqrt{3}\,i)/2, (1+\sqrt{3}\,i)/2$

Q003 $x = 3, (-3-3\sqrt{3}\,i)/2, (-3+3\sqrt{3}\,i)/2$

Q004 $x = -5, (5-5\sqrt{3}\,i)/2, (5+5\sqrt{3}\,i)/2$

Q005 $x = 1/2, (-1-\sqrt{3}\,i)/4, (-1+\sqrt{3}\,i)/4$

Q006 B

Q007 D

Q008 B

Q009 A

Q010 C

Q011 $x = -2, 1, 3$

Q012 $x = -2, 2/3, 1$

Q013 $x = 2, (-1-\sqrt{7}\,i)/4, (-1+\sqrt{7}\,i)/4$

Q014 $x = -2(\text{repeated}), 5$

Q015 $x = -3, -2, -1$

Q016 $x = -3, -2, 2$

Q017 $x = -1, 2/3, 1$

Q018 $x = 2, (-1-\sqrt{7}\,i)/2, (-1+\sqrt{7}\,i)/2$

Q019 $x = -2, -1-\sqrt{3}, -1+\sqrt{3}$

Q020 $x = 2, 1-2\,i, 1+2\,i$

Q021 $x = -2(\text{repeated}), 3$

Q022 $x = -1(\text{repeated}), -3/2$

Q023 $x = -2(\text{triple})$

Q024 C

Q025 D

Q026 D

Q027 B

Q028 C

Q029 D

Q030 D

Q031 A

Q032 D

Q033 D

Q034 D

Q035 B

Q036 D

Q037 $x = -3, 3, -\sqrt{2}\,i, \sqrt{2}\,i$

Q038 $x = -2, -1, 1, 2$

Q039 $x = -3, 3, -2\,i, 2\,i$

Q040 $x = -3, 3, -\sqrt{3}\,i, \sqrt{3}\,i$

Q041 $x = -\sqrt{2}, \sqrt{2}, -i, i$

Q042 D

Q043 D

Q044 C

Q045 C

Q046 B

Q047 $x = 2(\text{repeated}), 1-\sqrt{5}, 1+\sqrt{5}$

Q048 $x = 1, 2, 3, 4$

Q049 $x = 1, 2, 3(\text{repeated})$

Q050 $x = -1, 3, 1-i, 1+i$

Q051 B

Q052 A

Q053 C

Q054 C

Q055 $a = 1, b = -6$; other roots: -2

Q056 $a = 2, b = 2$; other roots: 1

Q057 $a = -7, b = 14$; other roots: 4

Q058 C

Q059 B

Q060 C

Q061 $a = -3, b = 1$; other roots: $-1, 2-i$

Q062 $a = -4, b = 6$; other roots: $1-i, 2$

Q063 $a = -4, b = 14$; other roots: $1+3i, 2$

Q064 $a = 0, b = 1$; other roots: $1-2i, -2$

Q065 A

Q066 B

Q067 D

Q068 B

Q069 $k = 9$; other roots: $3, 4$

Q070 B

Q071 1

Q072 1

Q073 1

Q074 A

Q075 B

Q076 D

Q077 -1

Q078 -1

Q079 0

Q080 A

Q081 B

Q082 D

Q083 -1

Q084 1

Q085 0

Q086 D

Q087 C

Q088 D

Q089 $a = -4$ or $9/4$

Q090 $a = -3$ or 1

Q091 $a = 5$ or 9

Q092 $a = -6, -2,$ or 10

Q093 $a = -7, -3,$ or 2

Q094 D

Q095 D

Q096 B

Q097 A

Q098 A

Q099 $a = -1, b = 6$; other roots: $1-\sqrt{2}i, -2$

Q100 4

Q101 $a = 2/3, 2,$ or 6

Q102 $x^4 - x^3 + x^2 + x - 2 = 0$

Q103 $x = -1, 1, (1-\sqrt{7}\,i)/2, (1+\sqrt{7}\,i)/2$

Q104 Both types occur: real $x = -1, 1$
non-real conjugates $x = (1 \pm \sqrt{7}\,i)/2$
if $x \geq 0$: $x = 1$

Q105 C

Q106 D

Q107 D

Q108 A

Q109 D

Q110 B

Quiz MCQ 1 D

Quiz MCQ 2 D

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $x = -2, -1, 1, 2$

Quiz WRITTEN 6 $a = 0, b = 1$; other roots: $1 - 2i, -2$

Quiz WRITTEN 7 1

Quiz WRITTEN 8 $a = -4$ or $9/4$

Concept 1 | Coordinate Geometry Basics

English Student Edition • Focused formulas and guided practice

Formula opener

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

Gradient

$$m = \frac{y_2-y_1}{x_2-x_1}$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Coordinate Geometry Basics**

Q001 Written

A (-4), B (8): distance AB

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Q002 Written

A (-5), B (7): distance AB

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Worked Solutions **Coordinate Geometry Basics**

Q003 Written

For two points A (-2) and B (6), find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Q004 Written

For two points A (4) and B (10), find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 6.
3. Final answer: 6.

Final answer

6

Worked Solutions **Coordinate Geometry Basics**

Q005 MCQ

Find a (-4), B (8): distance AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C

Final answer

12

Q006 MCQ

Find a (-5), B (7): distance AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C

Final answer

12

Worked Solutions **Coordinate Geometry Basics**

Q007 MCQ

For two points A (-2) and B (6), find the distance AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: C

Final answer

8

Q008 MCQ

For two points A (4) and B (10), find the distance AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 6.

Correct option: C

Final answer

6

Worked Solutions **Coordinate Geometry Basics**

Q009 Written

A (-4), B (8): internal point C in ratio 2:1

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q010 Written

A (-5), B (7): internal point C in ratio 2:1

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 3.
3. Final answer: 3.

Final answer

3

Worked Solutions **Coordinate Geometry Basics**

Q011 Written

For two points A (-2) and B (6), find the coordinate of point C dividing segment AB internally in the ratio 3:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q012 Written

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Worked Solutions **Coordinate Geometry Basics**

Q013 MCQ

Find a (-4), B (8): internal point C in ratio 2:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B

Final answer

4

Q014 MCQ

Find a (-5), B (7): internal point C in ratio 2:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 3.

Correct option: C

Final answer

3

Worked Solutions **Coordinate Geometry Basics**

Q015 MCQ

For two points A (-2) and B (6), find the coordinate of point C dividing segment AB internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B

Final answer

4

Q016 MCQ

For two points A (4) and B (10), find the coordinate of point C dividing segment AB internally in the ratio 2:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: B

Final answer

8

Worked Solutions **Coordinate Geometry Basics**

Q017 Written

A (-4), B (8): external point D in ratio 2:3

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -28.
3. Final answer: -28.

Final answer

-28

Q018 Written

A (-5), B (7): external point D in ratio 2:5

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -13.
3. Final answer: -13.

Final answer

-13

Worked Solutions **Coordinate Geometry Basics**

Q019 Written

For two points A (-2) and B (6), find the coordinate of point D dividing segment AB externally in the ratio 5:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 18.
3. Final answer: 18.

Final answer

18

Q020 Written

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions **Coordinate Geometry Basics**

Q021 MCQ

Find a (-4), B (8): external point D in ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -28.

Correct option: A

Final answer

-28

Q022 MCQ

Find a (-5), B (7): external point D in ratio 2:5:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -13.

Correct option: B

Final answer

-13

Worked Solutions **Coordinate Geometry Basics**

Q023 MCQ

For two points A (-2) and B (6), find the coordinate of point D dividing segment AB externally in the ratio 5:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 18.

Correct option: A

Final answer

18

Q024 MCQ

For two points A (4) and B (10), find the coordinate of point D dividing segment AB externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: B

Final answer

1

Worked Solutions **Coordinate Geometry Basics**

Q025 Written

A (-4), B (8): midpoint M

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q026 Written

A (-5), B (7): midpoint M

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions **Coordinate Geometry Basics**

Q027 **Written**

For two points A (-2) and B (6), find the coordinate of midpoint M of segment AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q028 **Written**

For two points A (4) and B (10), find the coordinate of midpoint M of segment AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 7.
3. Final answer: 7.

Final answer

7

Worked Solutions **Coordinate Geometry Basics**

Q029 MCQ

Find a (-4), B (8): midpoint M:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: C

Final answer

2

Q030 MCQ

Find a (-5), B (7): midpoint M:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: A

Final answer

1

Worked Solutions **Coordinate Geometry Basics**

Q031 MCQ

For two points A (-2) and B (6), find the coordinate of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: D

Final answer

2

Q032 MCQ

For two points A (4) and B (10), find the coordinate of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 7.

Correct option: A

Final answer

7

Q033 Challenge

Pods A (-8) and B (-3): fibre-optic tether length

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is 5.

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Q034 Challenge

Pods A (-8), B (-3): internal sensor point in ratio 1:4

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -7.

Final answer

-7

Q035 Challenge

Pods A (-8), B (-3): external outpost in ratio 3:4

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -23.

Final answer

-23

Q036 Challenge

Pods A (-8), B (-3): midpoint supply hub

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $-11/2$.

Final answer

$-11/2$

Worked Solutions **Coordinate Geometry Basics**

Q037 Challenge

Find pods A (-8) and B (-3): fibre-optic tether length:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: A

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Q038 Challenge

Find pods A (-8), B (-3): internal sensor point in ratio 1:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -7.

Correct option: C

Final answer

-7

Worked Solutions **Coordinate Geometry Basics**

Q039 Challenge

Find pods A (-8), B (-3): external outpost in ratio 3:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -23.

Correct option: B

Final answer

-23

Worked Solutions **Coordinate Geometry Basics**

Q040 Challenge

Find pods A (-8), B (-3): midpoint supply hub:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $-11/2$.

Correct option: C

Final answer

$-11/2$

Worked Solutions **Coordinate Geometry Basics**

Q041 Written

Point a has coordinates (2,3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is I.

Final answer

I

Q042 Written

Point b has coordinates (2,-3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is IV.

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q043 Written

Point c has coordinates (-2,3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is II.

Final answer

II

Q044 Written

Point d has coordinates (-2,-3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is III.

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q045 Written

Point a has coordinates (-4,2): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is II.

Final answer

II

Q046 Written

Point b has coordinates (1,2): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is I.

Final answer

I

Q047 Written

Point c has coordinates (3,-2): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is IV.

Final answer

IV

Q048 Written

Point d has coordinates (-3,-1): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is III.

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q049 Written

Point a has coordinates (-3,1): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is II.

Final answer

II

Q050 Written

Point b has coordinates (2,-8): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is IV.

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q051 Written

Point c has coordinates (4,4): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is I.

Final answer

I

Q052 Written

Point d has coordinates (-6,-4): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is III.

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q053 Written

Point e has coordinates (-2,6): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is II.

Final answer

II

Q054 Written

Point f has coordinates (3,-3): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is IV.

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q055 Written

Point g has coordinates (-5,-7): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is III.

Final answer

III

Q056 Written

Point h has coordinates (1,4): determine its quadrant

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is I.

Final answer

I

Worked Solutions **Coordinate Geometry Basics**

Q057 MCQ

Point a has coordinates (2,3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is I.

Correct option: C

Final answer

I

Q058 MCQ

Point b has coordinates (2,-3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is IV.

Correct option: B

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q059 MCQ

Point c has coordinates (-2,3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: A

Final answer

II

Q060 MCQ

Point d has coordinates (-2,-3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is III.

Correct option: C

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q061 MCQ

Point a has coordinates (-4,2): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: A

Final answer

II

Q062 MCQ

Point b has coordinates (1,2): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is I.

Correct option: C

Final answer

I

Worked Solutions **Coordinate Geometry Basics**

Q063 MCQ

Point c has coordinates (3,-2): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is IV.

Correct option: B

Final answer

IV

Q064 MCQ

Point d has coordinates (-3,-1): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is III.

Correct option: A

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q065 MCQ

Point a has coordinates (-3,1): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: B

Final answer

II

Q066 MCQ

Point b has coordinates (2,-8): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is IV.

Correct option: A

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q067 MCQ

Point c has coordinates (4,4): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is I.

Correct option: A

Final answer

I

Q068 MCQ

Point d has coordinates (-6,-4): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is III.

Correct option: C

Final answer

III

Worked Solutions **Coordinate Geometry Basics**

Q069 MCQ

Point e has coordinates (-2,6): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: B

Final answer

II

Q070 MCQ

Point f has coordinates (3,-3): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is IV.

Correct option: C

Final answer

IV

Worked Solutions **Coordinate Geometry Basics**

Q071 MCQ

Point g has coordinates (-5,-7): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is III.

Correct option: C

Final answer

III

Q072 MCQ

Point h has coordinates (1,4): determine its quadrant:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is I.

Correct option: D

Final answer

I

Worked Solutions **Coordinate Geometry Basics**

Q073 **Written**

For point A $(-3,\ 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3,\ -2)$.

Final answer

$(-3,\ -2)$

Q074 **Written**

For point A $(-1,\ 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1,\ -2)$.

Final answer

$(-1,\ -2)$

Worked Solutions **Coordinate Geometry Basics**

Q075 **Written**

For point A $(-4,\quad -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-4,\quad 2)$.

Final answer

$(-4,\quad 2)$

Q076 **Written**

For point A $(1,\quad -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1,\quad 2)$.

Final answer

$(1,\quad 2)$

Worked Solutions **Coordinate Geometry Basics**

Q077 Written

For point A $(-3,\ 4)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3,\ -4)$.

Final answer

$(-3,\ -4)$

Q078 MCQ

For point A $(-3,\ 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3,\ -2)$.

Correct option: C

Final answer

$(-3,\ -2)$

Worked Solutions **Coordinate Geometry Basics**

Q079 MCQ

For point A $(-1,\ 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1,\ -2)$.

Correct option: D

Final answer

$(-1,\ -2)$

Q080 MCQ

For point A $(-4,\ -2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-4,\ 2)$.

Correct option: C

Final answer

$(-4,\ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q081 MCQ

For point A $(1,\quad -2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1,\quad 2)$.

Correct option: D

Final answer

$(1,\quad 2)$

Q082 MCQ

For point A $(-3,\quad 4)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3,\quad -4)$.

Correct option: C

Final answer

$(-3,\quad -4)$

Worked Solutions **Coordinate Geometry Basics**

Q083 **Written**

For point A $(-3,\ 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3,\ 2)$.

Final answer

$(3,\ 2)$

Q084 **Written**

For point A $(-1,\ 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1,\ 2)$.

Final answer

$(1,\ 2)$

Q085 Written

For point A $(-4,\quad -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4,\quad -2)$.

Final answer

$(4,\quad -2)$

Q086 Written

For point A $(1,\quad -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1,\quad -2)$.

Final answer

$(-1,\quad -2)$

Worked Solutions **Coordinate Geometry Basics**

Q087 Written

For point A $(-3,\ \ 4)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3,\ \ 4)$.

Final answer

$(3,\ \ 4)$

Q088 MCQ

For point A $(-3,\ \ 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ \ 2)$.

Correct option: C

Final answer

$(3,\ \ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q089 MCQ

For point A $(-1,\ \ 2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1,\ \ 2)$.

Correct option: C

Final answer

$(1,\ \ 2)$

Q090 MCQ

For point A $(-4,\ \ -2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4,\ \ -2)$.

Correct option: A

Final answer

$(4,\ \ -2)$

Worked Solutions **Coordinate Geometry Basics**

Q091 MCQ

For point A $(1,\ -2)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1,\ -2)$.

Correct option: A

Final answer

$(-1,\ -2)$

Q092 MCQ

For point A $(-3,\ 4)$, find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ 4)$.

Correct option: A

Final answer

$(3,\ 4)$

Worked Solutions **Coordinate Geometry Basics**

Q093 Written

For point A $(-3,\ 2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3,\ -2)$.

Final answer

$(3,\ -2)$

Q094 Written

For point A $(-1,\ 2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1,\ -2)$.

Final answer

$(1,\ -2)$

Worked Solutions **Coordinate Geometry Basics**

Q095 **Written**

For point A $(-4, \ -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4, \ 2)$.

Final answer

$(4, \ 2)$

Q096 **Written**

For point A $(1, \ -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, \ 2)$.

Final answer

$(-1, \ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q097 Written

For point A $(-3,\ \ 4)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3,\ \ -4)$.

Final answer

$(3,\ \ -4)$

Q098 MCQ

For point A $(-3,\ \ 2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ \ -2)$.

Correct option: B

Final answer

$(3,\ \ -2)$

Worked Solutions **Coordinate Geometry Basics**

Q099 MCQ

For point A $(-1,\ 2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1,\ -2)$.

Correct option: B

Final answer

$(1,\ -2)$

Q100 MCQ

For point A $(-4,\ -2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4,\ 2)$.

Correct option: B

Final answer

$(4,\ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q101 MCQ

For point A $(1,\ -2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1,\ 2)$.

Correct option: A

Final answer

$(-1,\ 2)$

Q102 MCQ

For point A $(-3,\ 4)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ -4)$.

Correct option: D

Final answer

$(3,\ -4)$

Worked Solutions **Coordinate Geometry Basics**

Q103 Challenge

A robotic welding arm is programmed at point A $(-3, \ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(-3, \ -4)$.

Final answer

$(-3, \ -4)$

Worked Solutions **Coordinate Geometry Basics**

Q104 **Challenge**

A robotic welding arm is programmed at point A $(-3,\ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3,\ 4)$.

Final answer

$(3,\ 4)$

Worked Solutions **Coordinate Geometry Basics**

Q105 **Challenge**

A robotic welding arm is programmed at point A $(-3,\ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3,\ -4)$.

Final answer

$(3,\ -4)$

Worked Solutions **Coordinate Geometry Basics**

Q106 **Challenge**

A robotic welding arm is programmed at point A $(-3,\ \ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3,\ \ -4)$.

Correct option: A

Final answer

$(-3,\ \ -4)$

Worked Solutions **Coordinate Geometry Basics**

Q107 Challenge

A robotic welding arm is programmed at point A $(-3,\ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ 4)$.

Correct option: B

Final answer

$(3,\ 4)$

Worked Solutions **Coordinate Geometry Basics**

Q108 Challenge

A robotic welding arm is programmed at point A $(-3,\ 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ -4)$.

Correct option: C

Final answer

$(3,\ -4)$

Worked Solutions **Coordinate Geometry Basics**

Q109 Written

Distance between A $(2,\ \ 6)$ and B $(5,\ \ 10)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

5

Q110 Written

Distance between A $(4,\ \ 3)$ and B $(1,\ \ 2)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{10}$.

Final answer

$\sqrt{10}$

Worked Solutions **Coordinate Geometry Basics**

Q111 **Written**

Distance between A($\sqrt{3}$,-1) and B $(0,\ \ 1)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{7}$.

Final answer

$\sqrt{7}$

Q112 **Written**

Distance between A $(1,\ \ 2)$ **and B** $(11,\ \ 7)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $5\sqrt{5}$.

Final answer

$5\sqrt{5}$

Worked Solutions **Coordinate Geometry Basics**

Q113 Written

Distance between A $(3, \ -1)$ and B $(-1, \ 3)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $4\sqrt{2}$.

Final answer

$4\sqrt{2}$

Q114 MCQ

Find distance between A $(2, \ 6)$ and B $(5, \ 10)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Q115 *MCQ*

Find distance between A $(4,\ \ 3)$ **and B** $(1,\ \ 2)$**:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A

Final answer

$\sqrt{10}$

Q116 *MCQ*

Find distance between A($\sqrt{3}$,-1) and B $(0,\ \ 1)$**:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{7}$.

Correct option: C

Final answer

$\sqrt{7}$

Worked Solutions

Coordinate Geometry Basics

Q117 MCQ

Find distance between A $(1,\quad 2)$ and B $(11,\quad 7)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $5\sqrt{5}$.

Correct option: B

Final answer

$5\sqrt{5}$

Q118 MCQ

Find distance between A $(3,\quad -1)$ and B $(-1,\quad 3)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $4\sqrt{2}$.

Correct option: A

Final answer

$4\sqrt{2}$

Worked Solutions **Coordinate Geometry Basics**

Q119 Written

A $(3, \ -2)$, **B** $(x, \ 1)$, **distance 5: find x**

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, \ 7$.

Final answer

$x = -1, \ 7$

Q120 Written

A $(-1, \ 3)$, **B** $(x, \ 1)$, **distance** $\sqrt{5}$**: find x**

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -2, \ 0$.

Final answer

$x = -2, \ 0$

Worked Solutions **Coordinate Geometry Basics**

Q121 Written

A $(3, \quad -1)$, B $(x, \quad 2)$, distance 5: find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1,\ 7$.

Final answer

$x = -1,\ 7$

Q122 Written

A $(1, \quad -3)$, B $(-2, \quad y)$, distance $\sqrt{13}$: find y

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $y = -5,\ -1$.

Final answer

$y = -5,\ -1$

Worked Solutions **Coordinate Geometry Basics**

Q123 Written

A $(-4,\ \ -2)$, B $(x,\ \ 3)$, distance 5sqrt(5): find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -14,\ 6$.

Final answer

$x = -14,\ 6$

Q124 MCQ

A $(3,\ \ -2)$, B $(x,\ \ 1)$, distance 5: find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1,\ 7$.

Correct option: B

Final answer

$x = -1,\ 7$

Worked Solutions **Coordinate Geometry Basics**

Q125 MCQ

A $(-1,\ \ 3)$, B $(x,\ \ 1)$, distance $\sqrt{5}$: find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2,\ 0$.

Correct option: C

Final answer

$x = -2,\ 0$

Q126 MCQ

A $(3,\ \ -1)$, B $(x,\ \ 2)$, distance 5: find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1,\ 7$.

Correct option: B

Final answer

$x = -1,\ 7$

Worked Solutions **Coordinate Geometry Basics**

Q127 MCQ

A $(1, \quad -3)$**, B** $(-2, \quad y)$**, distance** $\sqrt{13}$**: find y:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $y = -5, \ -1$.

Correct option: A

Final answer

$y = -5, \ -1$

Q128 MCQ

A $(-4, \quad -2)$**, B** $(x, \quad 3)$**, distance 5sqrt(5): find x:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -14, \ 6$.

Correct option: D

Final answer

$x = -14, \ 6$

Worked Solutions **Coordinate Geometry Basics**

Q129 Written

Distance from origin to A $(2, \quad -4)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $2\sqrt{5}$.

Final answer

$2\sqrt{5}$

Q130 Written

Distance from origin to A $(-3, \quad 4)$

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Q131 MCQ

Find distance from origin to A $(2,\ -4)$**:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: C

Final answer

$2\sqrt{5}$

Q132 MCQ

Find distance from origin to A $(-3,\ 4)$**:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C

Final answer

5

Q133 Challenge

Distance between (x,7) and (3x-1,-5) is 13: find x

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $x = -2,\ 3$.

Final answer

$x = -2,\ 3$

Worked Solutions **Coordinate Geometry Basics**

Q134 Challenge

Distance between (x,7) and (3x-1,-5) is 13: find x:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2,\ 3$.

Correct option: C

Final answer

$x = -2,\ 3$

Worked Solutions **Coordinate Geometry Basics**

Q135 **Written**

Point on the x-axis equidistant from A $(-2, \ 1)$ and B $(3, \ 4)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2, \ 0)$.

Final answer

$(2, \ 0)$

Q136 **Written**

Point on the y-axis equidistant from A $(4, \ 2)$ and B $(1, \ -1)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0, \ 3)$.

Final answer

$(0, \ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q137 Written

Point on the x-axis equidistant from A $(-1,\ 2)$ and B $(4,\ 3)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2,\ 0)$.

Final answer

$(2,\ 0)$

Q138 Written

Point on the y-axis equidistant from A $(-5,\ 2)$ and B $(3,\ 5)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0,\ 5/6)$.

Final answer

$(0,\ 5/6)$

Worked Solutions **Coordinate Geometry Basics**

Q139 **Written**

Point on the x-axis equidistant from A $(-1, \ 2)$ and B $(3, \ 8)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(17/2, \ 0)$.

Final answer

$(17/2, \ 0)$

Q140 **MCQ**

Find point on the x-axis equidistant from A $(-2, \ 1)$ and B $(3, \ 4)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, \ 0)$.

Correct option: C

Final answer

$(2, \ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q141 MCQ

Find point on the y-axis equidistant from A $(4,\ 2)$ and B $(1,\ -1)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ 3)$.

Correct option: A

Final answer

$(0,\ 3)$

Q142 MCQ

Find point on the x-axis equidistant from A $(-1,\ 2)$ and B $(4,\ 3)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ 0)$.

Correct option: C

Final answer

$(2,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q143 MCQ

Find point on the y-axis equidistant from A $(-5,\ 2)$ and B $(3,\ 5)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ 5/6)$.

Correct option: C

Final answer

$(0,\ 5/6)$

Q144 MCQ

Find point on the x-axis equidistant from A $(-1,\ 2)$ and B $(3,\ 8)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(17/2,\ 0)$.

Correct option: B

Final answer

$(17/2,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q145 Written

Point on $y = 2x - 1$ equidistant from A $(1,\ 4)$ and B $(3,\ 1)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(13/8,\ 9/4)$.

Final answer

$(13/8,\ 9/4)$

Q146 Written

Point on $y = 3x + 1$ equidistant from A $(4,\ -5)$ and B $(2,\ -3)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(-4,\ -11)$.

Final answer

$(-4,\ -11)$

Worked Solutions **Coordinate Geometry Basics**

Q147 Written

Point on $y = 2x - 3$ equidistant from A $(1, \ 4)$ and B $(3, \ -2)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2, \ 1)$.

Final answer

$(2, \ 1)$

Q148 Written

Point on $y = x - 1$ equidistant from A $(-1, \ 5)$ and B $(7, \ -1)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(3, \ 2)$.

Final answer

$(3, \ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q149 Written

Point on $y = -x + 1$ equidistant from A $(1,\ 2)$ and B $(-3,\ 6)$

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(-2,\ 3)$.

Final answer

$(-2,\ 3)$

Q150 MCQ

Find point on $y = 2x - 1$ equidistant from A $(1,\ 4)$ and B $(3,\ 1)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(13/8,\ 9/4)$.

Correct option: B

Final answer

$(13/8,\ 9/4)$

Worked Solutions **Coordinate Geometry Basics**

Q151 MCQ

Find point on $y = 3x + 1$ equidistant from A $(4, \ -5)$ and B $(2, \ -3)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-4, \ -11)$.

Correct option: C

Final answer

$(-4, \ -11)$

Q152 MCQ

Find point on $y = 2x - 3$ equidistant from A $(1, \ 4)$ and B $(3, \ -2)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, \ 1)$.

Correct option: A

Final answer

$(2, \ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q153 *MCQ*

Find point on $y = x - 1$ equidistant from A $(-1, \ 5)$ and B $(7, \ -1)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, \ 2)$.

Correct option: C

Final answer

$(3, \ 2)$

Q154 *MCQ*

Find point on $y = -x + 1$ equidistant from A $(1, \ 2)$ and B $(-3, \ 6)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2, \ 3)$.

Correct option: B

Final answer

$(-2, \ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q155 Challenge

A $(2,\ 1)$, B $(4,\ 5)$, C $(3,\ y)$ with CA=CB and D $(x,\ 0)$ with DA=DB: find CD

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $3\sqrt{5}$.

Final answer

$3\sqrt{5}$

Worked Solutions **Coordinate Geometry Basics**

Q156 **Challenge**

A $(2,\ 1)$**, B** $(4,\ 5)$**, C** $(3,\ y)$ **with CA=CB and D** $(x,\ 0)$ **with DA=DB: find CD:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $3\sqrt{5}$.

Correct option: A

Final answer

$3\sqrt{5}$

Worked Solutions **Coordinate Geometry Basics**

Q157 Written

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(4/3,\ 11/3)$.

Final answer

$(4/3,\ 11/3)$

Worked Solutions **Coordinate Geometry Basics**

Q158 Written

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-15/4,\ 13/4)$.

Final answer

$(-15/4,\ 13/4)$

Worked Solutions **Coordinate Geometry Basics**

Q159 Written

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of point P dividing segment AB internally in the ratio 4:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2,\ -9/5)$.

Final answer

$(2,\ -9/5)$

Q160 Written

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3,\ 3)$.

Final answer

$(3,\ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q161 Written

Given A $(-4,\ \ 5)$, B $(1,\ \ -5)$, and C $(3,\ \ 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2/3,\ \ -5/3)$.

Final answer

$(-2/3,\ \ -5/3)$

Worked Solutions **Coordinate Geometry Basics**

Q162 MCQ

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3,\ 11/3)$.

Correct option: C

Final answer

$(4/3,\ 11/3)$

Worked Solutions **Coordinate Geometry Basics**

Q163 MCQ

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-15/4,\ 13/4)$.

Correct option: C

Final answer

$(-15/4,\ 13/4)$

Worked Solutions **Coordinate Geometry Basics**

Q164 MCQ

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of point P dividing segment AB internally in the ratio 4:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ -9/5)$.

Correct option: C

Final answer

$(2,\ -9/5)$

Worked Solutions **Coordinate Geometry Basics**

Q165 MCQ

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ 3)$.

Correct option: C

Final answer

$(3,\ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q166 MCQ

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2/3,\ -5/3)$.

Correct option: A

Final answer

$(-2/3,\ -5/3)$

Worked Solutions **Coordinate Geometry Basics**

Q167 Written

Given A $(-2, \ -3)$, B $(3, \ 7)$, and C $(5, \ 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, \ 19/2)$.

Final answer

$(2, \ 19/2)$

Worked Solutions **Coordinate Geometry Basics**

Q168 Written

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(11,\ -18)$.

Final answer

$(11,\ -18)$

Worked Solutions **Coordinate Geometry Basics**

Q169 Written

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(10,\ -15)$.

Final answer

$(10,\ -15)$

Worked Solutions Coordinate Geometry Basics

Q170 Written

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-8,\ -11)$.

Final answer

$(-8,\ -11)$

Worked Solutions **Coordinate Geometry Basics**

Q171 Written

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0,\ -15/2)$.

Final answer

$(0,\ -15/2)$

Worked Solutions **Coordinate Geometry Basics**

Q172 MCQ

Given A $(-2, \ -3)$, B $(3, \ 7)$, and C $(5, \ 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, \ 19/2)$.

Correct option: B

Final answer

$(2, \ 19/2)$

Worked Solutions **Coordinate Geometry Basics**

Q173 MCQ

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(11,\ -18)$.

Correct option: C

Final answer

$(11,\ -18)$

Worked Solutions **Coordinate Geometry Basics**

Q174 MCQ

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(10,\ -15)$.

Correct option: B

Final answer

$(10,\ -15)$

Worked Solutions **Coordinate Geometry Basics**

Q175 *MCQ*

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8,\ -11)$.

Correct option: B

Final answer

$(-8,\ -11)$

Q176 MCQ

Given A $(-4,\ \ 5)$, B $(1,\ \ -5)$, and C $(3,\ \ 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ \ -15/2)$.

Correct option: C

Final answer

$(0,\ \ -15/2)$

Worked Solutions **Coordinate Geometry Basics**

Q177 Written

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3/2,\ -1/2)$.

Final answer

$(3/2,\ -1/2)$

Worked Solutions **Coordinate Geometry Basics**

Q178 **Written**

Given A $(-2, \ -3)$, B $(3, \ 7)$, and C $(5, \ 2)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, \ 2)$.

Final answer

$(2, \ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q179 Written

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-1,\ 0)$.

Final answer

$(-1,\ 0)$

Q180 Written

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2,\ 1)$.

Final answer

$(-2,\ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q181 Written

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0,\ 3)$.

Final answer

$(0,\ 3)$

Worked Solutions **Coordinate Geometry Basics**

Q182 Written

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2/3,\ 1)$.

Final answer

$(2/3,\ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q183 Written

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2,\ -1)$.

Final answer

$(2,\ -1)$

Worked Solutions **Coordinate Geometry Basics**

Q184 Written

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(5/3,\ 5/3)$.

Final answer

$(5/3,\ 5/3)$

Worked Solutions **Coordinate Geometry Basics**

Q185 Written

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of midpoint M of segment BC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2,\ -5/2)$.

Final answer

$(2,\ -5/2)$

Worked Solutions **Coordinate Geometry Basics**

Q186 Written

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0,\ 0)$.

Final answer

$(0,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q187 MCQ

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3/2,\ -1/2)$.

Correct option: A

Final answer

$(3/2,\ -1/2)$

Worked Solutions **Coordinate Geometry Basics**

Q188 MCQ

Given A $(-2,\ -3)$, B $(3,\ 7)$, and C $(5,\ 2)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ 2)$.

Correct option: C

Final answer

$(2,\ 2)$

Worked Solutions **Coordinate Geometry Basics**

Q189 MCQ

Given A $(-3,\ 4)$, B $(1,\ -4)$, and C $(-4,\ 3)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1,\ 0)$.

Correct option: B

Final answer

$(-1,\ 0)$

Worked Solutions **Coordinate Geometry Basics**

Q190 MCQ

Given A $(-3,\ \ 4)$, B $(1,\ \ -4)$, and C $(-4,\ \ 3)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2,\ \ 1)$.

Correct option: A

Final answer

$(-2,\ \ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q191 MCQ

Given A $(2,\ \ 3)$, B $(2,\ \ -3)$, and C $(-2,\ \ 3)$, find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ \ 3)$.

Correct option: A

Final answer

$(0,\ \ 3)$

Q192 MCQ

Given A $(2,\ 3)$, B $(2,\ -3)$, and C $(-2,\ 3)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2/3,\ 1)$.

Correct option: D

Final answer

$(2/3,\ 1)$

Worked Solutions **Coordinate Geometry Basics**

Q193 MCQ

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ -1)$.

Correct option: B

Final answer

$(2,\ -1)$

Worked Solutions **Coordinate Geometry Basics**

Q194 MCQ

Given A $(-2,\ 1)$, B $(6,\ -3)$, and C $(1,\ 7)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(5/3,\ 5/3)$.

Correct option: A

Final answer

$(5/3,\ 5/3)$

Worked Solutions **Coordinate Geometry Basics**

Q195 MCQ

Given A $(-4,\ 5)$, B $(1,\ -5)$, and C $(3,\ 0)$, find the coordinates of midpoint M of segment BC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2,\ -5/2)$.

Correct option: B

Final answer

$(2,\ -5/2)$

Worked Solutions **Coordinate Geometry Basics**

Q196 MCQ

Given A $(-4, \quad 5)$, B $(1, \quad -5)$, and C $(3, \quad 0)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, \quad 0)$.

Correct option: C

Final answer

$(0, \quad 0)$

Worked Solutions **Coordinate Geometry Basics**

Q197 Challenge

A $(-2,\ \ 3)$, B $(6,\ \ 1)$, C $(0,\ \ y)$, CA=CB: find the centroid G

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(4/3,\ \ -2/3)$.

Final answer

$(4/3,\ \ -2/3)$

Q198 Challenge

A $(-2,\ 3)$, B $(6,\ 1)$, C $(0,\ y)$, CA=CB: find the centroid G:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3,\ -2/3)$.

Correct option: C

Final answer

$(4/3,\ -2/3)$

Worked Solutions **Coordinate Geometry Basics**

Q199 **Written**

Point with A $(-3,\ \ 2)$ symmetric about centre (2, 6)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7,\ \ 10)$.

Final answer

$(7,\ \ 10)$

Q200 **Written**

Point with A $(1,\ \ -2)$ symmetric about centre (2, 3)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(3,\ \ 8)$.

Final answer

$(3,\ \ 8)$

Worked Solutions **Coordinate Geometry Basics**

Q201 Written

Point with A $(3,\ 2)$ symmetric about centre (-1, 5)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-5,\ 8)$.

Final answer

$(-5,\ 8)$

Q202 Written

Point with A $(-3,\ 1)$ symmetric about centre (2, 0)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7,\ -1)$.

Final answer

$(7,\ -1)$

Worked Solutions **Coordinate Geometry Basics**

Q203 Written

Point with A $(7,\quad 4)$ symmetric about centre (-3, 5)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-13,\quad 6)$.

Final answer

$(-13,\quad 6)$

Q204 Written

Point with A $(4,\quad -8)$ symmetric about centre (-3, 2)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-10,\quad 12)$.

Final answer

$(-10,\quad 12)$

Worked Solutions **Coordinate Geometry Basics**

Q205 Written

Point with A $(0,\ -4)$ symmetric about centre (-4, 1)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-8,\ 6)$.

Final answer

$(-8,\ 6)$

Q206 Written

Point with A $(-4,\ 2)$ symmetric about centre (-1, -3)

Solution

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(2,\ -8)$.

Final answer

$(2,\ -8)$

Worked Solutions **Coordinate Geometry Basics**

Q207 MCQ

Find point with A $(-3,\ 2)$ symmetric about centre (2, 6):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(7,\ 10)$.

Correct option: C

Final answer

$(7,\ 10)$

Q208 MCQ

Find point with A $(1,\ -2)$ symmetric about centre (2, 3):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3,\ 8)$.

Correct option: B

Final answer

$(3,\ 8)$

Worked Solutions **Coordinate Geometry Basics**

Q209 MCQ

Find point with A $(3,\ \ 2)$ symmetric about centre (-1, 5):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-5,\ \ 8)$.

Correct option: A

Final answer

$(-5,\ \ 8)$

Q210 MCQ

Find point with A $(-3,\ \ 1)$ symmetric about centre (2, 0):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(7,\ \ -1)$.

Correct option: C

Final answer

$(7,\ \ -1)$

Worked Solutions **Coordinate Geometry Basics**

Q211 MCQ

Find point with A $(7, \ 4)$ symmetric about centre (-3, 5):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-13, \ 6)$.

Correct option: C

Final answer

$(-13, \ 6)$

Q212 MCQ

Find point with A $(4, \ -8)$ symmetric about centre (-3, 2):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-10, \ 12)$.

Correct option: C

Final answer

$(-10, \ 12)$

Worked Solutions **Coordinate Geometry Basics**

Q213 MCQ

Find point with A $(0, \ -4)$ symmetric about centre (-4, 1):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8, \ 6)$.

Correct option: B

Final answer

$(-8, \ 6)$

Q214 MCQ

Find point with A $(-4, \ 2)$ symmetric about centre (-1, -3):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, \ -8)$.

Correct option: A

Final answer

$(2, \ -8)$

Worked Solutions **Coordinate Geometry Basics**

Q215 Challenge

B $(m,\ 5)$, C $(4,\ n)$ symmetric about A $(1,\ -1)$: find m+n

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -9.

Final answer

-9

Worked Solutions Coordinate Geometry Basics

Q216 Challenge

B $(m, \ 5)$**, C** $(4, \ n)$ **symmetric about A** $(1, \ -1)$**: find m+n:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -9.

Correct option: C

Final answer

-9

Worked Solutions **Coordinate Geometry Basics**

Quiz MCQ 1 **Concept Quiz · MCQ**

Point A $(-3,\ 4)$ lies in which quadrant?:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is II.

Correct option: A

Final answer

II

Quiz MCQ 2 **Concept Quiz · MCQ**

Find the distance between A $(1,\ 2)$ and B $(4,\ 6)$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: B

Final answer

5

Worked Solutions **Coordinate Geometry Basics**

Quiz MCQ 3 Concept Quiz · MCQ

Point P divides the segment from A $(-2,\ 3)$ to B $(6,\ 1)$ internally in ratio 1:3. Find P:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0,\ 5/2)$.

Correct option: C

Final answer

$(0,\ 5/2)$

Quiz MCQ 4 Concept Quiz · MCQ

Point B is symmetric to A $(2,\ -1)$ about P $(-3,\ 4)$. Find B:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8,\ 9)$.

Correct option: C

Final answer

$(-8,\ 9)$

Worked Solutions **Coordinate Geometry Basics**

Quiz WRITTEN 5 Concept Quiz · Written

Find the point on $y = 2x + 1$ equidistant from A $(0,\ \ 2)$ and B $(4,\ \ 0)$.

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(1,\ \ 3)$.

Final answer

$(1,\ \ 3)$

Quiz WRITTEN 6 Concept Quiz · Written

For points A (-5) and B (7), find the point dividing AB externally in ratio 2:5.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -13.
3. Final answer: -13.

Final answer

-13

Worked Solutions **Coordinate Geometry Basics**

Quiz WRITTEN 7 **Concept Quiz · Written**

If the distance between A $(3,\ -1)$ and B $(x,\ 2)$ is 5, find x.

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1,\ 7$.

Final answer

$x = -1,\ 7$

Quiz WRITTEN 8 **Concept Quiz · Written**

Find the centroid of A $(-2,\ 1)$, B $(6,\ -3)$, C $(1,\ 7)$.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(5/3,\ 5/3)$.

Final answer

$(5/3,\ 5/3)$

Answer key

Use the question number shown on each page.

Q001 12
Q002 12
Q003 8
Q004 6
Q005 C
Q006 C
Q007 C
Q008 C
Q009 4
Q010 3
Q011 4
Q012 8
Q013 B
Q014 C
Q015 B
Q016 B
Q017 -28
Q018 -13
Q019 18
Q020 1
Q021 A
Q022 B
Q023 A
Q024 B
Q025 2
Q026 1
Q027 2
Q028 7
Q029 C
Q030 A
Q031 D
Q032 A
Q033 5
Q034 -7
Q035 -23
Q036 $-11/2$
Q037 A
Q038 C
Q039 B
Q040 C
Q041 I
Q042 IV
Q043 II
Q044 III
Q045 II
Q046 I
Q047 IV
Q048 III
Q049 II
Q050 IV
Q051 I
Q052 III
Q053 II
Q054 IV
Q055 III
Q056 I
Q057 C
Q058 B
Q059 A
Q060 C
Q061 A
Q062 C
Q063 B
Q064 A
Q065 B
Q066 A
Q067 A
Q068 C
Q069 B
Q070 C
Q071 C
Q072 D
Q073 $(-3,\ -2)$
Q074 $(-1,\ -2)$
Q075 $(-4,\ 2)$
Q076 $(1,\ 2)$
Q077 $(-3,\ -4)$
Q078 C
Q079 D
Q080 C
Q081 D
Q082 C
Q083 $(3,\ 2)$
Q084 $(1,\ 2)$
Q085 $(4,\ -2)$
Q086 $(-1,\ -2)$
Q087 $(3,\ 4)$
Q088 C
Q089 C
Q090 A
Q091 A
Q092 A
Q093 $(3,\ -2)$
Q094 $(1,\ -2)$
Q095 $(4,\ 2)$
Q096 $(-1,\ 2)$
Q097 $(3,\ -4)$
Q098 B
Q099 B
Q100 B
Q101 A
Q102 D
Q103 $(-3,\ -4)$
Q104 $(3,\ 4)$

Q105 $(3,\ -4)$
Q106 A
Q107 B
Q108 C
Q109 5
Q110 $\sqrt{10}$
Q111 $\sqrt{7}$
Q112 $5\sqrt{5}$
Q113 $4\sqrt{2}$
Q114 C
Q115 A
Q116 C
Q117 B
Q118 A
Q119 $x = -1,\ 7$
Q120 $x = -2,\ 0$
Q121 $x = -1,\ 7$
Q122 $y = -5,\ -1$
Q123 $x = -14,\ 6$
Q124 B
Q125 C
Q126 B
Q127 A
Q128 D
Q129 $2\sqrt{5}$
Q130 5
Q131 C
Q132 C
Q133 $x = -2,\ 3$
Q134 C
Q135 $(2,\ 0)$
Q136 $(0,\ 3)$
Q137 $(2,\ 0)$
Q138 $(0,\ 5/6)$
Q139 $(17/2,\ 0)$
Q140 C
Q141 A
Q142 C
Q143 C
Q144 B
Q145 $(13/8,\ 9/4)$
Q146 $(-4,\ -11)$
Q147 $(2,\ 1)$
Q148 $(3,\ 2)$
Q149 $(-2,\ 3)$
Q150 B
Q151 C
Q152 A
Q153 C
Q154 B
Q155 $3\sqrt{5}$
Q156 A
Q157 $(4/3,\ 11/3)$
Q158 $(-15/4,\ 13/4)$
Q159 $(2,\ -9/5)$
Q160 $(3,\ 3)$
Q161 $(-2/3,\ -5/3)$
Q162 C
Q163 C
Q164 C
Q165 C
Q166 A
Q167 $(2,\ 19/2)$
Q168 $(11,\ -18)$
Q169 $(10,\ -15)$
Q170 $(-8,\ -11)$
Q171 $(0,\ -15/2)$
Q172 B
Q173 C
Q174 B
Q175 B
Q176 C
Q177 $(3/2,\ -1/2)$
Q178 $(2,\ 2)$
Q179 $(-1,\ 0)$
Q180 $(-2,\ 1)$
Q181 $(0,\ 3)$
Q182 $(2/3,\ 1)$
Q183 $(2,\ -1)$
Q184 $(5/3,\ 5/3)$
Q185 $(2,\ -5/2)$
Q186 $(0,\ 0)$
Q187 A
Q188 C
Q189 B
Q190 A
Q191 A
Q192 D
Q193 B
Q194 A
Q195 B
Q196 C
Q197 $(4/3,\ -2/3)$
Q198 C
Q199 $(7,\ 10)$
Q200 $(3,\ 8)$
Q201 $(-5,\ 8)$
Q202 $(7,\ -1)$
Q203 $(-13,\ 6)$
Q204 $(-10,\ 12)$
Q205 $(-8,\ 6)$
Q206 $(2,\ -8)$
Q207 C
Q208 B
Q209 A
Q210 C
Q211 C
Q212 C
Q213 B
Q214 A
Q215 -9
Q216 C
Quiz MCQ 1 A
Quiz MCQ 2 B
Quiz MCQ 3 C
Quiz MCQ 4 C
Quiz WRITTEN 5 $(1,\ 3)$
Quiz WRITTEN 6 -13
Quiz WRITTEN 7 $x = -1,\ 7$
Quiz WRITTEN 8 $(5/3,\ 5/3)$

Concept 2 | Triangles and Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Line

$$y - y_1 = m(x - x_1)$$

Perpendicular

$$m_1 m_2 = -1$$

Area

$$A = \frac{1}{2}bh$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Triangles and Lines**

Q001 **Written**

Determine the shape of triangle ABC with A $(3,\ \ 2)$, B $(-1,\ \ 0)$, C $\left(1+\sqrt{3},\ \ 1-2\sqrt{3}\right)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer

equilateral

Worked Solutions **Triangles and Lines**

Q002 **Written**

Determine the shape of triangle ABC with A $(2,\ 2)$, B $(2,\ 0)$, C $\left(2+\sqrt{3},\ 1\right)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer

equilateral

Worked Solutions **Triangles and Lines**

Q003 MCQ

Determine the shape of triangle ABC with A $(3,\ \ 2)$**, B** $(-1,\ \ 0)$**, C** $\left(1+\sqrt{3},\ \ 1-2\sqrt{3}\right)$**:**

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *equilateral*.

Correct option: D

Final answer

equilateral

Worked Solutions **Triangles and Lines**

Q004 MCQ

Determine the shape of triangle ABC with A $(2,\ \ 2)$, B $(2,\ \ 0)$, C $\left(2+\sqrt{3},\ \ 1\right)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $equilateral$.

Correct option: C

Final answer

$equilateral$

Q005 Written

Determine the shape of triangle ABC with A $(-1,\ \ 2)$, B $(1,\ \ 1)$, C $(0,\ \ 4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right-angled;\ right\ angle\ at\ A$.

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ A$

Worked Solutions **Triangles and Lines**

Q006 Written

Determine the shape of triangle ABC with A $(0,\ 2)$, B $(-1,\ -2)$, C $(4,\ 1)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ A$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ A$

Q007 MCQ

Determine the shape of triangle ABC with A $(-1,\ 2)$, B $(1,\ 1)$, C $(0,\ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right - angled;\ right\ angle\ at\ A$.

Correct option: B

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ A$

Worked Solutions **Triangles and Lines**

Q008 MCQ

Determine the shape of triangle ABC with A $(0,\ \ 2)$, B $(-1,\ \ -2)$, C $(4,\ \ 1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right - angled;\ right\ angle\ at\ A$.

Correct option: D

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ A$

Q009 Written

Determine the shape of triangle ABC with A $(-3,\ \ 4)$, B $(1,\ \ 7)$, C $(3,\ \ -4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $scalene\ right - angled;\ right\ angle\ at\ A$.

Final answer

$scalene\ right - angled;\ right\ angle\ at\ A$

Worked Solutions **Triangles and Lines**

Q010 **MCQ**

Determine the shape of triangle ABC with A $(-3,\ \ 4)$, B $(1,\ \ 7)$, C $(3,\ \ -4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $scalene\ right - angled;\ right\ angle\ at\ A$.

Correct option: D

Final answer

$scalene\ right - angled;\ right\ angle\ at\ A$

Q011 **Written**

Determine the shape of triangle ABC with A $(4,\ \ 9)$, B $(0,\ \ 6)$, C $(3,\ \ 2)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ B$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ B$

Worked Solutions **Triangles and Lines**

Q012 **Written**

Determine the shape of triangle ABC with A $(1,\ \ 2)$, B $(5,\ \ 8)$, C $(11,\ \ 4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ B$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ B$

Q013 **Written**

Determine the shape of triangle ABC with A $(-1,\ \ 1)$, B $(4,\ \ 2)$, C $(1,\ \ 4)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: $isosceles\ right - angled;\ right\ angle\ at\ C$.

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ C$

Worked Solutions **Triangles and Lines**

Q014 MCQ

Determine the shape of triangle ABC with A $(4,\ 9)$, B $(0,\ 6)$, C $(3,\ 2)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right - angled;\ right\ angle\ at\ B$.

Correct option: B

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ B$

Q015 MCQ

Determine the shape of triangle ABC with A $(1,\ 2)$, B $(5,\ 8)$, C $(11,\ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right - angled;\ right\ angle\ at\ B$.

Correct option: C

Final answer

$isosceles\ right - angled;\ right\ angle\ at\ B$

Worked Solutions **Triangles and Lines**

Q016 MCQ

Determine the shape of triangle ABC with A $(-1,\ \ 1)$, B $(4,\ \ 2)$, C $(1,\ \ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $isosceles\ right-angled;\ right\ angle\ at\ C$.

Correct option: B

Final answer

$isosceles\ right-angled;\ right\ angle\ at\ C$

Worked Solutions **Triangles and Lines**

Q017 Challenge

If A $(1,\ 2)$, B $(2k,\ k)$, C $(4,\ 5)$ and ABC is isosceles and right-angled at B, find k.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $k = 2$.

Final answer

$k = 2$

Worked Solutions **Triangles and Lines**

Q018 Challenge

If A $(1,\ 2)$, B $(2k,\ k)$, C $(4,\ 5)$ and ABC is isosceles and right-angled at B, find k:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $k = 2$.

Correct option: C

Final answer

$k = 2$

Worked Solutions **Triangles and Lines**

Q019 Written

Find the equation of the line through (3,-4) parallel to the x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -4$.

Final answer

$y = -4$

Q020 Written

Find the equation of the line through (-3,8) parallel to the y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$x = -3$

Worked Solutions **Triangles and Lines**

Q021 **Written**

Find the equation of the line through (2,6) parallel to the x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$y = 6$

Q022 **Written**

Find the equation of the line through (3,5) perpendicular to the x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$x = 3$

Worked Solutions **Triangles and Lines**

Q023 Written

Find the equation of the line through (-4,5) parallel to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 5$.

Final answer

$y = 5$

Q024 Written

Find the equation of the line through (2,1) parallel to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 1$.

Final answer

$y = 1$

Worked Solutions **Triangles and Lines**

Q025 **Written**

Find the equation of the line through (-4,-7) parallel to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -7$.

Final answer

$y = -7$

Q026 **Written**

Find the equation of the line through (1,-2) perpendicular to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 1$.

Final answer

$x = 1$

Worked Solutions **Triangles and Lines**

Q027 **Written**

Find the equation of the line through (-3,-4) perpendicular to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$x = -3$

Q028 **Written**

Find the equation of the line through (3,-5) perpendicular to x-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$x = 3$

Worked Solutions **Triangles and Lines**

Q029 **Written**

Find the equation of the line through (2,-7) parallel to y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 2$.

Final answer

$x = 2$

Q030 **Written**

Find the equation of the line through (4,3) parallel to y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$x = 4$

Worked Solutions **Triangles and Lines**

Q031 Written

Find the equation of the line through (-6,-2) parallel to y-axis.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -6$.

Final answer

$x = -6$

Q032 MCQ

Find the equation of the line through (3,-4) parallel to the x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -4$.

Correct option: B

Final answer

$y = -4$

Worked Solutions **Triangles and Lines**

Q033 MCQ

Find the equation of the line through (-3,8) parallel to the y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: C

Final answer

$x = -3$

Q034 MCQ

Find the equation of the line through (2,6) parallel to the x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: D

Final answer

$y = 6$

Worked Solutions **Triangles and Lines**

Q035 MCQ

Find the equation of the line through (3,5) perpendicular to the x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D

Final answer

$x = 3$

Q036 MCQ

Find the equation of the line through (-4,5) parallel to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 5$.

Correct option: B

Final answer

$y = 5$

Worked Solutions **Triangles and Lines**

Q037 MCQ

Find the equation of the line through (2,1) parallel to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 1$.

Correct option: D

Final answer

$y = 1$

Q038 MCQ

Find the equation of the line through (-4,-7) parallel to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -7$.

Correct option: B

Final answer

$y = -7$

Worked Solutions **Triangles and Lines**

Q039 MCQ

Find the equation of the line through (1,-2) perpendicular to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 1$.

Correct option: C

Final answer

$x = 1$

Q040 MCQ

Find the equation of the line through (-3,-4) perpendicular to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: D

Final answer

$x = -3$

Worked Solutions **Triangles and Lines**

Q041 MCQ

Find the equation of the line through (3,-5) perpendicular to x-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D

Final answer

$x = 3$

Q042 MCQ

Find the equation of the line through (2,-7) parallel to y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 2$.

Correct option: B

Final answer

$x = 2$

Worked Solutions **Triangles and Lines**

Q043 MCQ

Find the equation of the line through (4,3) parallel to y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: D

Final answer

$x = 4$

Q044 MCQ

Find the equation of the line through (-6,-2) parallel to y-axis:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -6$.

Correct option: A

Final answer

$x = -6$

Worked Solutions **Triangles and Lines**

Q045 Written

Find the equation of the line through (-3,1) and (-3,2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$x = -3$

Q046 Written

Find the equation of the line through (-2,3) and (-5,3).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q047 Written

Find the equation of the line through (1,-1) and (4,0).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$.

Final answer

$y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$

Q048 Written

Find the equation of the line through (-2,-2) and (2,-2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$y = -2$

Worked Solutions **Triangles and Lines**

Q049 Written

Find the equation of the line through (-4,2) and (-4,8).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -4$.

Final answer

$x = -4$

Q050 Written

Find the equation of the line through (-2,-6) and (-4,1).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $7x + 2y + 26 = 0$.

Final answer

$7x + 2y + 26 = 0$

Worked Solutions **Triangles and Lines**

Q051 Written

Find the equation of the line through (4,0) and (4,3).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$x = 4$

Q052 Written

Find the equation of the line through (-1,2) and (-1,6).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -1$.

Final answer

$x = -1$

Worked Solutions **Triangles and Lines**

Q053 Written

Find the equation of the line through (-5,-3) and (-5,2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -5$.

Final answer

$x = -5$

Q054 Written

Find the equation of the line through (-2,3) and (-3,3).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q055 **Written**

Find the equation of the line through (4,-2) and (1,-2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$y = -2$

Q056 **Written**

Find the equation of the line through (-3,6) and (2,6).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$y = 6$

Worked Solutions **Triangles and Lines**

Q057 Written

Find the equation of the line through (1,2) and (3,10).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 4x - 2$.

Final answer

$y = 4x - 2$

Q058 Written

Find the equation of the line through (-1,-8) and (4,2).

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x - 6$.

Final answer

$y = 2x - 6$

Worked Solutions **Triangles and Lines**

Q059 MCQ

Find the equation of the line through (-3,1) and (-3,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: B

Final answer

$x = -3$

Q060 MCQ

Find the equation of the line through (-2,3) and (-5,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: C

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q061 MCQ

Find the equation of the line through (1,-1) and (4,0):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$.

Correct option: D

Final answer

$y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$

Q062 MCQ

Find the equation of the line through (-2,-2) and (2,-2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: A

Final answer

$y = -2$

Worked Solutions **Triangles and Lines**

Q063 MCQ

Find the equation of the line through (-4,2) and (-4,8):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -4$.

Correct option: B

Final answer

$x = -4$

Q064 MCQ

Find the equation of the line through (-2,-6) and (-4,1):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $7x + 2y + 26 = 0$.

Correct option: D

Final answer

$7x + 2y + 26 = 0$

Worked Solutions **Triangles and Lines**

Q065 MCQ

Find the equation of the line through (4,0) and (4,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: C

Final answer

$x = 4$

Q066 MCQ

Find the equation of the line through (-1,2) and (-1,6):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -1$.

Correct option: A

Final answer

$x = -1$

Worked Solutions **Triangles and Lines**

Q067 MCQ

Find the equation of the line through (-5,-3) and (-5,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -5$.

Correct option: C

Final answer

$x = -5$

Q068 MCQ

Find the equation of the line through (-2,3) and (-3,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: D

Final answer

$y = 3$

Worked Solutions **Triangles and Lines**

Q069 MCQ

Find the equation of the line through (4,-2) and (1,-2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: D

Final answer

$y = -2$

Q070 MCQ

Find the equation of the line through (-3,6) and (2,6):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: C

Final answer

$y = 6$

Worked Solutions **Triangles and Lines**

Q071 MCQ

Find the equation of the line through (1,2) and (3,10):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 4x - 2$.

Correct option: D

Final answer

$y = 4x - 2$

Q072 MCQ

Find the equation of the line through (-1,-8) and (4,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x - 6$.

Correct option: D

Final answer

$y = 2x - 6$

Worked Solutions **Triangles and Lines**

Q073 **Written**

Find the equation of the line through (2,-1) with gradient 3.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3x - 7$.

Final answer

$y = 3x - 7$

Q074 **Written**

Find the equation of the line through (-3,4) with gradient 2.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x + 10$.

Final answer

$y = 2x + 10$

Worked Solutions **Triangles and Lines**

Q075 Written

Find the equation of the line through (-1,5) with gradient -1.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -x + 4$.

Final answer

$y = -x + 4$

Q076 Written

Find the equation of the line through (-3,-8) with gradient 1.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x - 5$.

Final answer

$y = x - 5$

Worked Solutions **Triangles and Lines**

Q077 Written

Find the equation of the line through (2,-1) with gradient -2.

Solution

1. Recognise $x = k$ or $y = k$; otherwise calculate m.
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2x + 3$.

Final answer

$y = -2x + 3$

Q078 MCQ

Find the equation of the line through (2,-1) with gradient 3:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: D

Final answer

$y = 3x - 7$

Worked Solutions **Triangles and Lines**

Q079 MCQ

Find the equation of the line through (-3,4) with gradient 2:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x + 10$.

Correct option: C

Final answer

$y = 2x + 10$

Q080 MCQ

Find the equation of the line through (-1,5) with gradient -1:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -x + 4$.

Correct option: D

Final answer

$y = -x + 4$

Worked Solutions **Triangles and Lines**

Q081 MCQ

Find the equation of the line through (-3,-8) with gradient 1:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x - 5$.

Correct option: B

Final answer

$y = x - 5$

Q082 MCQ

Find the equation of the line through (2,-1) with gradient -2:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2x + 3$.

Correct option: D

Final answer

$y = -2x + 3$

Worked Solutions **Triangles and Lines**

Q083 **Challenge**

Sensors A $(-4,\ 2)$ and B $(9,\ 2)$ have the same height. Find the straight-line boundary through A and B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $y = 2$.

Final answer

$y = 2$

Worked Solutions **Triangles and Lines**

Q084 Challenge

Sensors A $(-4,\ 2)$ and B $(9,\ 2)$ have the same height. Find the straight-line boundary through A and B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2$.

Correct option: D

Final answer

$y = 2$

Worked Solutions **Triangles and Lines**

Q085 Written

A $(2,\ \ 3)$, B $(4,\ \ -1)$, and C $(a-1,\ \ 2-a)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$a = 7$

Q086 Written

A $(2,\ \ 5)$, B $(4,\ \ 9)$, and C $(-1,\ \ a)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = -1$.

Final answer

$a = -1$

Worked Solutions **Triangles and Lines**

Q087 Written

A $(1, \quad -1)$, B $(-2, \quad -3)$, and C $(4, \quad a)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 1$.

Final answer

$a = 1$

Q088 Written

A $(1, \quad 2)$, B $(3, \quad 1)$, and C $(a, \quad -1)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$a = 7$

Worked Solutions **Triangles and Lines**

Q089 Written

A $(-3, \quad -9)$, B $(5, \quad 7)$, and C $(a+3, \quad 3a-2)$ are collinear. Find a.

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 5$.

Final answer

$a = 5$

Q090 MCQ

A $(2, \quad 3)$, B $(4, \quad -1)$, and C $(a-1, \quad 2-a)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 7$.

Correct option: D

Final answer

$a = 7$

Worked Solutions **Triangles and Lines**

Q091 MCQ

A $(2, \quad 5)$, B $(4, \quad 9)$, and C $(-1, \quad a)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1$.

Correct option: D

Final answer

$a = -1$

Q092 MCQ

A $(1, \quad -1)$, B $(-2, \quad -3)$, and C $(4, \quad a)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1$.

Correct option: C

Final answer

$a = 1$

Worked Solutions **Triangles and Lines**

Q093 MCQ

A $(1,\ \ 2)$, B $(3,\ \ 1)$, and C $(a,\ \ -1)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 7$.

Correct option: D

Final answer

$a = 7$

Q094 MCQ

A $(-3,\ \ -9)$, B $(5,\ \ 7)$, and C $(a+3,\ \ 3a-2)$ are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 5$.

Correct option: D

Final answer

$a = 5$

Worked Solutions **Triangles and Lines**

Q095 Written

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through (-1,3).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 2 = 0$.

Final answer

$x + y - 2 = 0$

Q096 Written

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + 3y - 23 = 0$.

Final answer

$x + 3y - 23 = 0$

Worked Solutions **Triangles and Lines**

Q097 **Written**

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through (-1,2).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 1 = 0$.

Final answer

$x + y - 1 = 0$

Q098 **Written**

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through (-3,-1).

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $2x - 5y + 1 = 0$.

Final answer

$2x - 5y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q099 MCQ

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through (-1,3):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 2 = 0$.

Correct option: B

Final answer

$x + y - 2 = 0$

Q100 MCQ

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through (5,6):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 23 = 0$.

Correct option: A

Final answer

$x + 3y - 23 = 0$

Worked Solutions **Triangles and Lines**

Q101 MCQ

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through (-1,2):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: B

Final answer

$x + y - 1 = 0$

Q102 MCQ

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through (-3,-1):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 5y + 1 = 0$.

Correct option: D

Final answer

$2x - 5y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q103 **Challenge**

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $3x + 13y = 0$.

Final answer

$3x + 13y = 0$

Worked Solutions **Triangles and Lines**

Q104 **Challenge**

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 13y = 0$.

Correct option: C

Final answer

$3x + 13y = 0$

Worked Solutions **Triangles and Lines**

Q105 Written

Find the gradient of line $y = 2x + 3$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$m = 2$

Q106 Written

Find the gradient of line $y = -3x + 2$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$m = -3$

Worked Solutions **Triangles and Lines**

Q107 Written

Find the gradient of line $x - 3y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1/3$.

Final answer

$m = 1/3$

Q108 Written

Find the gradient of line $2x - y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$m = 2$

Worked Solutions **Triangles and Lines**

Q109 Written

Find the gradient of line $x + 3y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$m = -1/3$

Q110 Written

Find the gradient of line $y = 3x + 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$m = 3$

Worked Solutions **Triangles and Lines**

Q111 Written

Find the gradient of line $y = 4x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 4$.

Final answer

$m = 4$

Q112 Written

Find the gradient of line $y = -4x - 7$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$m = -4$

Worked Solutions **Triangles and Lines**

Q113 Written

Find the gradient of line $6x - 2y = 5$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$m = 3$

Q114 Written

Find the gradient of line $8x + 2y - 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$m = -4$

Worked Solutions **Triangles and Lines**

Q115 Written

Find the gradient of line $x + 4y - 4 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/4$.

Final answer

$m = -1/4$

Q116 Written

Find the gradient of line $y = 3x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$m = 3$

Worked Solutions **Triangles and Lines**

Q117 **Written**

Find the gradient of line $y = -3x + 4$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$m = -3$

Q118 **Written**

Find the gradient of line $y = x - 3$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$m = 1$

Worked Solutions **Triangles and Lines**

Q119 Written

Find the gradient of line $6x + 2y + 5 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$m = -3$

Q120 Written

Find the gradient of line $2x - 2y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$m = 1$

Worked Solutions **Triangles and Lines**

Q121 **Written**

Find the gradient of line $x + 3y - 2 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$m = -1/3$

Q122 **Written**

Find the gradient of line $y = 2x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$m = 2$

Worked Solutions **Triangles and Lines**

Q123 Written

Find the gradient of line $y = -x + 4$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$m = -1$

Q124 Written

Find the gradient of line $y = 3x/2 + 6$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3/2$.

Final answer

$m = 3/2$

Worked Solutions **Triangles and Lines**

Q125 Written

Find the gradient of line $x + y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$m = -1$

Q126 Written

Find the gradient of line $2x + 3y + 8 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -2/3$.

Final answer

$m = -2/3$

Worked Solutions **Triangles and Lines**

Q127 Written

Find the gradient of line $3x + 6y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/2$.

Final answer

$m = -1/2$

Q128 MCQ

Find the gradient of line $y = 2x + 3$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D

Final answer

$m = 2$

Worked Solutions **Triangles and Lines**

Q129 MCQ

Find the gradient of line $y = -3x + 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C

Final answer

$m = -3$

Q130 MCQ

Find the gradient of line $x - 3y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1/3$.

Correct option: A

Final answer

$m = 1/3$

Worked Solutions **Triangles and Lines**

Q131 MCQ

Find the gradient of line $2x - y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: B

Final answer

$m = 2$

Q132 MCQ

Find the gradient of line $x + 3y + 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A

Final answer

$m = -1/3$

Worked Solutions **Triangles and Lines**

Q133 MCQ

Find the gradient of line $y = 3x + 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A

Final answer

$m = 3$

Q134 MCQ

Find the gradient of line $y = 4x - 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 4$.

Correct option: B

Final answer

$m = 4$

Worked Solutions **Triangles and Lines**

Q135 MCQ

Find the gradient of line $y = -4x - 7$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: D

Final answer

$m = -4$

Q136 MCQ

Find the gradient of line $6x - 2y = 5$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A

Final answer

$m = 3$

Worked Solutions **Triangles and Lines**

Q137 MCQ

Find the gradient of line $8x + 2y - 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: C

Final answer

$m = -4$

Q138 MCQ

Find the gradient of line $x + 4y - 4 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/4$.

Correct option: C

Final answer

$m = -1/4$

Worked Solutions **Triangles and Lines**

Q139 MCQ

Find the gradient of line $y = 3x - 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: D

Final answer

$m = 3$

Q140 MCQ

Find the gradient of line $y = -3x + 4$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: A

Final answer

$m = -3$

Worked Solutions **Triangles and Lines**

Q141 MCQ

Find the gradient of line $y = x - 3$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: D

Final answer

$m = 1$

Q142 MCQ

Find the gradient of line $6x + 2y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C

Final answer

$m = -3$

Worked Solutions **Triangles and Lines**

Q143 MCQ

Find the gradient of line $2x - 2y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: C

Final answer

$m = 1$

Q144 MCQ

Find the gradient of line $x + 3y - 2 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A

Final answer

$m = -1/3$

Worked Solutions **Triangles and Lines**

Q145 MCQ

Find the gradient of line $y = 2x - 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D

Final answer

$m = 2$

Q146 MCQ

Find the gradient of line $y = -x + 4$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: A

Final answer

$m = -1$

Worked Solutions **Triangles and Lines**

Q147 MCQ

Find the gradient of line $y = 3x/2 + 6$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3/2$.

Correct option: B

Final answer

$m = 3/2$

Q148 MCQ

Find the gradient of line $x + y + 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: D

Final answer

$m = -1$

Worked Solutions **Triangles and Lines**

Q149 MCQ

Find the gradient of line $2x + 3y + 8 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -2/3$.

Correct option: B

Final answer

$m = -2/3$

Q150 MCQ

Find the gradient of line $3x + 6y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/2$.

Correct option: B

Final answer

$m = -1/2$

Worked Solutions **Triangles and Lines**

Q151 Written

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 14 = 0$.

Final answer

$3x - 2y - 14 = 0$

Q152 Written

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + y + 1 = 0$.

Final answer

$2x + y + 1 = 0$

Worked Solutions **Triangles and Lines**

Q153 **Written**

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x - y - 4 = 0$.

Final answer

$2x - y - 4 = 0$

Q154 **Written**

Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y - 4 = 0$.

Final answer

$2x + 3y - 4 = 0$

Worked Solutions **Triangles and Lines**

Q155 Written

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x + 2y - 7 = 0$.

Final answer

$3x + 2y - 7 = 0$

Q156 MCQ

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 14 = 0$.

Correct option: C

Final answer

$3x - 2y - 14 = 0$

Worked Solutions **Triangles and Lines**

Q157 MCQ

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + y + 1 = 0$.

Correct option: D

Final answer

$2x + y + 1 = 0$

Q158 MCQ

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y - 4 = 0$.

Correct option: C

Final answer

$2x - y - 4 = 0$

Worked Solutions **Triangles and Lines**

Q159 MCQ

Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y - 4 = 0$.

Correct option: D

Final answer

$2x + 3y - 4 = 0$

Q160 MCQ

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 2y - 7 = 0$.

Correct option: D

Final answer

$3x + 2y - 7 = 0$

Worked Solutions **Triangles and Lines**

Q161 **Written**

Find the line through (-2,5) perpendicular to $3x + 5y + 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $5x - 3y + 25 = 0$.

Final answer

$5x - 3y + 25 = 0$

Q162 **Written**

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 4 = 0$.

Final answer

$3x - 2y - 4 = 0$

Worked Solutions **Triangles and Lines**

Q163 Written

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $x - 2y - 12 = 0$.

Final answer

$x - 2y - 12 = 0$

Q164 Written

Find the equation of the line through (-1,3) perpendicular to $5x - 2y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 5y - 13 = 0$.

Final answer

$2x + 5y - 13 = 0$

Worked Solutions **Triangles and Lines**

Q165 Written

Find the equation of the line through (-2,-5) perpendicular to $3x - 2y - 7 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y + 19 = 0$.

Final answer

$2x + 3y + 19 = 0$

Q166 MCQ

Find the line through (-2,5) perpendicular to $3x + 5y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $5x - 3y + 25 = 0$.

Correct option: D

Final answer

$5x - 3y + 25 = 0$

Worked Solutions **Triangles and Lines**

Q167 MCQ

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 4 = 0$.

Correct option: D

Final answer

$3x - 2y - 4 = 0$

Q168 MCQ

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x - 2y - 12 = 0$.

Correct option: B

Final answer

$x - 2y - 12 = 0$

Worked Solutions **Triangles and Lines**

Q169 MCQ

Find the equation of the line through (-1,3) perpendicular to $5x - 2y + 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 5y - 13 = 0$.

Correct option: A

Final answer

$2x + 5y - 13 = 0$

Q170 MCQ

Find the equation of the line through (-2,-5) perpendicular to $3x - 2y - 7 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y + 19 = 0$.

Correct option: D

Final answer

$2x + 3y + 19 = 0$

Worked Solutions **Triangles and Lines**

Q171 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $m/k = 2\,(k = 3,\quad m = 6)$.

Final answer

$m/k = 2\,(k = 3,\quad m = 6)$

Worked Solutions **Triangles and Lines**

Q172 **Challenge**

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m/k = 2\,(k = 3,\quad m = 6)$.

Correct option: D

Final answer

$m/k = 2\,(k = 3,\quad m = 6)$

Worked Solutions **Triangles and Lines**

Q173 Written

For A $(5, \quad 11)$, B $(1, \quad -1)$, find the equation of the perpendicular bisector of AB.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $x + 3y - 18 = 0$.

Final answer

$x + 3y - 18 = 0$

Q174 Written

For A $(-1, \quad 2)$, B $(3, \quad -4)$, find the equation of the perpendicular bisector of AB.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - 3y - 5 = 0$.

Final answer

$2x - 3y - 5 = 0$

Worked Solutions **Triangles and Lines**

Q175 **Written**

Find the perpendicular bisector of segment A $(-2,\ 3)$, B $(2,\ 1)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 2 = 0$.

Final answer

$2x - y + 2 = 0$

Q176 **Written**

Find the perpendicular bisector of segment A $(-1,\ 2)$, B $(5,\ -2)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $3x - 2y - 6 = 0$.

Final answer

$3x - 2y - 6 = 0$

Worked Solutions **Triangles and Lines**

Q177 Written

Find the perpendicular bisector of segment A $(0,\ \ 6)$, B $(4,\ \ 4)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 1 = 0$.

Final answer

$2x - y + 1 = 0$

Q178 MCQ

For A $(5,\ \ 11)$, B $(1,\ \ -1)$, find the equation of the perpendicular bisector of AB:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 18 = 0$.

Correct option: A

Final answer

$x + 3y - 18 = 0$

Worked Solutions **Triangles and Lines**

Q179 MCQ

For A $(-1,\ \ 2)$, B $(3,\ \ -4)$, find the equation of the perpendicular bisector of AB:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 3y - 5 = 0$.

Correct option: C

Final answer

$2x - 3y - 5 = 0$

Q180 MCQ

Find the perpendicular bisector of segment A $(-2,\ \ 3)$, B $(2,\ \ 1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y + 2 = 0$.

Correct option: C

Final answer

$2x - y + 2 = 0$

Worked Solutions **Triangles and Lines**

Q181 *MCQ*

Find the perpendicular bisector of segment A $(-1,\ \ 2)$, B $(5,\ \ -2)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x-2y-6=0$.

Correct option: D

Final answer

$3x-2y-6=0$

Q182 *MCQ*

Find the perpendicular bisector of segment A $(0,\ \ 6)$, B $(4,\ \ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x-y+1=0$.

Correct option: A

Final answer

$2x-y+1=0$

Worked Solutions **Triangles and Lines**

Q183 Written

Find the point B symmetric to A $(-4,\ 3)$ with respect to $3x - y + 5 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (2,\ 1)$.

Final answer

$B = (2,\ 1)$

Q184 Written

Find the point B symmetric to A $(0,\ 4)$ with respect to $2x - y - 1 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (4,\ 2)$.

Final answer

$B = (4,\ 2)$

Worked Solutions **Triangles and Lines**

Q185 **Written**

Find the point B symmetric to A $(1,\quad 2)$ with respect to $x + 2y - 10 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (3,\quad 6)$.

Final answer

$B = (3,\quad 6)$

Q186 **Written**

Find the point B symmetric to A $(4,\quad -1)$ with respect to $4x - 2y - 3 = 0$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (-2,\quad 2)$.

Final answer

$B = (-2,\quad 2)$

Worked Solutions **Triangles and Lines**

Q187 MCQ

Find the point B symmetric to A $(-4,\ \ 3)$ with respect to $3x - y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (2,\ \ 1)$.

Correct option: A

Final answer

$B = (2,\ \ 1)$

Q188 MCQ

Find the point B symmetric to A $(0,\ \ 4)$ with respect to $2x - y - 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (4,\ \ 2)$.

Correct option: A

Final answer

$B = (4,\ \ 2)$

Worked Solutions **Triangles and Lines**

Q189 *MCQ*

Find the point B symmetric to A $(1,\ \ 2)$ with respect to $x + 2y - 10 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (3,\ \ 6)$.

Correct option: C

Final answer

$B = (3,\ \ 6)$

Q190 *MCQ*

Find the point B symmetric to A $(4,\ \ -1)$ with respect to $4x - 2y - 3 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-2,\ \ 2)$.

Correct option: C

Final answer

$B = (-2,\ \ 2)$

Worked Solutions **Triangles and Lines**

Q191 Challenge

Reflect A $(5,\ \ 0)$ across the targeting line $y = 2x$. Find the receiver point B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $B = (-3,\ \ 4)$.

Final answer

$B = (-3,\ \ 4)$

Worked Solutions **Triangles and Lines**

Q192 **Challenge**

Reflect A $(5,\ \ 0)$ across the targeting line $y = 2x$. Find the receiver point B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-3,\ \ 4)$.

Correct option: C

Final answer

$B = (-3,\ \ 4)$

Worked Solutions **Triangles and Lines**

Q193 **Written**

Find the distance from (-2,1) to $3x - 2y + 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $3/\sqrt{13} = 3\sqrt{13}/13$.

Final answer

$3/\sqrt{13} = 3\sqrt{13}/13$

Q194 **Written**

Find the distance from (3,-2) to $x + 2y - 4 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{5}$.

Final answer

$\sqrt{5}$

Worked Solutions **Triangles and Lines**

Q195 Written

Find the perpendicular distance from (-3,2) to $2x - 3y + 6 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $6/\sqrt{13} = 6\sqrt{13}/13$.

Final answer

$6/\sqrt{13} = 6\sqrt{13}/13$

Q196 Written

Find the perpendicular distance from (-3,7) to $12x - 5y = 7$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 6.

Final answer

6

Worked Solutions **Triangles and Lines**

Q197 **Written**

Find the perpendicular distance from (-8,1) to $y = 2x - 3$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $4\sqrt{5}$.

Final answer

$4\sqrt{5}$

Q198 **Written**

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 1.

Final answer

1

Worked Solutions **Triangles and Lines**

Q199 Written

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{13}$.

Final answer

$\sqrt{13}$

Q200 Written

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 10.

Final answer

10

Worked Solutions **Triangles and Lines**

Q201 Written

Find the perpendicular distance from (1,2) to $y = 3x + 1$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/5$.

Final answer

$\sqrt{10}/5$

Q202 MCQ

Find the distance from (-2,1) to $3x - 2y + 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3/\sqrt{13} = 3\sqrt{13}/13$.

Correct option: A

Final answer

$3/\sqrt{13} = 3\sqrt{13}/13$

Worked Solutions **Triangles and Lines**

Q203 MCQ

Find the distance from (3,-2) to $x + 2y - 4 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{5}$.

Correct option: D

Final answer

$\sqrt{5}$

Q204 MCQ

Find the perpendicular distance from (-3,2) to $2x - 3y + 6 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $6/\sqrt{13} = 6\sqrt{13}/13$.

Correct option: D

Final answer

$6/\sqrt{13} = 6\sqrt{13}/13$

Worked Solutions **Triangles and Lines**

Q205 MCQ

Find the perpendicular distance from (-3,7) to $12x - 5y = 7$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 6.

Correct option: B

Final answer

6

Q206 MCQ

Find the perpendicular distance from (-8,1) to $y = 2x - 3$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $4\sqrt{5}$.

Correct option: D

Final answer

$4\sqrt{5}$

Worked Solutions **Triangles and Lines**

Q207 MCQ

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 1.

Correct option: D

Final answer

1

Q208 MCQ

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{13}$.

Correct option: A

Final answer

$\sqrt{13}$

Worked Solutions **Triangles and Lines**

Q209 MCQ

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: B

Final answer

10

Q210 MCQ

Find the perpendicular distance from (1,2) to $y = 3x + 1$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/5$.

Correct option: D

Final answer

$\sqrt{10}/5$

Worked Solutions **Triangles and Lines**

Q211 **Written**

Find the perpendicular length from the origin to $3x - y = 5$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/2$.

Final answer

$\sqrt{10}/2$

Q212 **Written**

Find the perpendicular length from (-1,5) to $y = 3x - 2$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}$.

Final answer

$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q213 MCQ

Find the perpendicular length from the origin to $3x - y = 5$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/2$.

Correct option: D

Final answer

$\sqrt{10}/2$

Q214 MCQ

Find the perpendicular length from (-1,5) to $y = 3x - 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: B

Final answer

$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q215 **Challenge**

Find the shortest distance from P $(-1,\quad 5)$ to the radiation line $y = x/3 + 2$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $\sqrt{10}$.

Final answer

$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q216 Challenge

Find the shortest distance from P $(-1, \;\; 5)$ to the radiation line $y = x/3 + 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A

Final answer

$\sqrt{10}$

Worked Solutions **Triangles and Lines**

Q217 Written

Find the area of triangle ABC with A $(1,\quad 1)$, B $(3,\quad 7)$, C $(-3,\quad -1)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 10.

Final answer

10

Q218 Written

Find the area for A $(1,\quad 5)$, B $(-3,\quad -3)$, C $(3,\quad -6)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 30.

Final answer

30

Worked Solutions **Triangles and Lines**

Q219 Written

Find the area for A $(-2, \quad 1)$, B $(2, \quad -1)$, C $(0, \quad 4)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 8.

Final answer

8

Q220 Written

Find the area for A $(0, \quad -6)$, B $(2, \quad -2)$, C $(1, \quad 5)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 9.

Final answer

9

Worked Solutions **Triangles and Lines**

Q221 **Written**

Find the area for A $(-2,\ \ 4)$, **B** $(-3,\ \ -5)$, **C** $(5,\ \ -1)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 34.

Final answer

34

Q222 **Written**

Find the area for A $(2,\ \ 5)$, **B** $(0,\ \ 1)$, **C** $(4,\ \ 2)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 7.

Final answer

7

Worked Solutions **Triangles and Lines**

Q223 MCQ

Find the area of triangle ABC with A $(1,\ 1)$, B $(3,\ 7)$, C $(-3,\ -1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: D

Final answer

10

Q224 MCQ

Find the area for A $(1,\ 5)$, B $(-3,\ -3)$, C $(3,\ -6)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 30.

Correct option: C

Final answer

30

Worked Solutions **Triangles and Lines**

Q225 MCQ

Find the area for A $(-2, \quad 1)$**, B** $(2, \quad -1)$**, C** $(0, \quad 4)$**:**

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 8.

Correct option: A

Final answer

8

Q226 MCQ

Find the area for A $(0, \quad -6)$**, B** $(2, \quad -2)$**, C** $(1, \quad 5)$**:**

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 9.

Correct option: B

Final answer

9

Worked Solutions **Triangles and Lines**

Q227 MCQ

Find the area for A $(-2,\ 4)$, **B** $(-3,\ -5)$, **C** $(5,\ -1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 34.

Correct option: B

Final answer

34

Q228 MCQ

Find the area for A $(2,\ 5)$, **B** $(0,\ 1)$, **C** $(4,\ 2)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 7.

Correct option: D

Final answer

7

Worked Solutions **Triangles and Lines**

Q229 Challenge

Find the area of the wound triangle A $(-2, \quad 1)$, B $(1, \quad 5)$, C $(-21, \quad -21)$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 5.

Final answer

5

Worked Solutions **Triangles and Lines**

Q230 Challenge

Find the area of the wound triangle A $(-2,\ \ 1)$, B $(1,\ \ 5)$, C $(-21,\ \ -21)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 5.

Correct option: D

Final answer

5

Worked Solutions **Triangles and Lines**

Q231 **Written**

The line $(k-4)x+(k+2)y+k+5=0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1/2,\quad -3/2)$.

Final answer

$(1/2,\quad -3/2)$

Q232 **Written**

The line $(2k+1)x-(k+2)y+7k+8=0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-2,\quad 3)$.

Final answer

$(-2,\quad 3)$

Worked Solutions **Triangles and Lines**

Q233 Written

The line $(k+2)x + ky - 6 = 0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(3, \quad -3)$.

Final answer

$(3, \quad -3)$

Q234 Written

The line $(k+3)x - (2k+1)y + k - 2 = 0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1, \quad 1)$.

Final answer

$(1, \quad 1)$

Worked Solutions **Triangles and Lines**

Q235 Written

The line $(k+3)x+(3k+2)y+7=0$ passes through a fixed point for every k. Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-3,\ \ 1)$.

Final answer

$(-3,\ \ 1)$

Q236 MCQ

The line $(k-4)x+(k+2)y+k+5=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(1/2,\ \ -3/2)$.

Correct option: D

Final answer

$(1/2,\ \ -3/2)$

Worked Solutions **Triangles and Lines**

Q237 MCQ

The line $(2k+1)x-(k+2)y+7k+8=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-2,\ \ 3)$.

Correct option: B

Final answer

$(-2,\ \ 3)$

Q238 MCQ

The line $(k+2)x+ky-6=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(3,\ \ -3)$.

Correct option: D

Final answer

$(3,\ \ -3)$

Worked Solutions **Triangles and Lines**

Q239 MCQ

The line $(k+3)x-(2k+1)y+k-2=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(1,\quad 1)$.

Correct option: D

Final answer

$(1,\quad 1)$

Q240 MCQ

The line $(k+3)x+(3k+2)y+7=0$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-3,\quad 1)$.

Correct option: A

Final answer

$(-3,\quad 1)$

Worked Solutions **Triangles and Lines**

Q241 Challenge

The line $-(k-2)x+(k-3)y=-3k+5$ passes through a fixed point for every k. Find it.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $(4,\quad 1)$.

Final answer

$(4,\quad 1)$

Worked Solutions **Triangles and Lines**

Q242 **Challenge**

The line $-(k-2)x+(k-3)y=-3k+5$ passes through a fixed point for every k. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(4, \quad 1)$.

Correct option: A

Final answer

$(4, \quad 1)$

Worked Solutions **Triangles and Lines**

Q243 Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4,\ -3,\ or 6$.

Final answer

$a = -4,\ -3,\ or 6$

Q244 Written

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -1,\ -1/2,\ or 1$.

Final answer

$a = -1,\ -1/2,\ or 1$

Worked Solutions **Triangles and Lines**

Q245 Written

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 11/10,\ 3/2,\ or - 9/2$.

Final answer

$a = 11/10,\ 3/2,\ or - 9/2$

Q246 Written

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 1,\ -3/2,\ or 2$.

Final answer

$a = 1,\ -3/2,\ or 2$

Worked Solutions **Triangles and Lines**

Q247 MCQ

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -4,\ -3,\ or 6$.

Correct option: A

Final answer

$a = -4,\ -3,\ or 6$

Q248 MCQ

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1,\ -1/2,\ or 1$.

Correct option: A

Final answer

$a = -1,\ -1/2,\ or 1$

Worked Solutions **Triangles and Lines**

Q249 MCQ

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 11/10,\ 3/2,\ or - 9/2$.

Correct option: C

Final answer

$a = 11/10,\ 3/2,\ or - 9/2$

Q250 MCQ

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1,\ -3/2,\ or 2$.

Correct option: D

Final answer

$a = 1,\ -3/2,\ or 2$

Worked Solutions **Triangles and Lines**

Q251 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 15.

Final answer

15

Worked Solutions **Triangles and Lines**

Q252 **Challenge**

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 15.

Correct option: A

Final answer

15

Worked Solutions **Triangles and Lines**

Quiz MCQ 1 **Concept Quiz · MCQ**

Determine the shape of triangle A $(0,\ \ 0)$, B $(3,\ \ 0)$, C $(0,\ \ 4)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $scalene\ right-angled$.

Correct option: C

Final answer

$scalene\ right-angled$

Quiz MCQ 2 **Concept Quiz · MCQ**

Find the equation of the line through (2,-1) with gradient 3:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: B

Final answer

$y = 3x - 7$

Worked Solutions **Triangles and Lines**

Quiz MCQ 3 **Concept Quiz · MCQ**

Find the perpendicular distance from (1,2) to $3x + 4y - 10 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $1/5$.

Correct option: D

Final answer

$1/5$

Quiz MCQ 4 **Concept Quiz · MCQ**

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(3, \quad -3)$.

Correct option: C

Final answer

$(3, \quad -3)$

Worked Solutions **Triangles and Lines**

Quiz WRITTEN 5 Concept Quiz · Written

Find the area of triangle A $(0,\ 0)$, B $(4,\ 0)$, C $(0,\ 3)$.

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2}$ base $\times$ height.
3. Area: 6.

Final answer

6

Quiz WRITTEN 6 Concept Quiz · Written

Find the perpendicular bisector of the segment joining A $(1,\ 2)$ and B $(5,\ 4)$.

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $y = -2x + 9$.

Final answer

$y = -2x + 9$

Worked Solutions **Triangles and Lines**

Quiz WRITTEN 7 Concept Quiz · Written

Find the equation of the line through (1,2) perpendicular to $y = 3x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $y = -x/3 + 7/3$.

Final answer

$y = -x/3 + 7/3$

Quiz WRITTEN 8 Concept Quiz · Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a.

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4,\ -3,\ or 6$.

Final answer

$a = -4,\ -3,\ or 6$

Answer key

Use the question number shown on each page.

Q001 *equilateral*
Q002 *equilateral*
Q003 D
Q004 C
Q005 $isosceles\ right-angled;\ right\ angle\ at\ A$
Q006 $isosceles\ right-angled;\ right\ angle\ at\ A$
Q007 B
Q008 D
Q009 $scalene\ right-angled;\ right\ angle\ at\ A$
Q010 D
Q011 $isosceles\ right-angled;\ right\ angle\ at\ B$
Q012 $isosceles\ right-angled;\ right\ angle\ at\ B$
Q013 $isosceles\ right-angled;\ right\ angle\ at\ C$
Q014 B
Q015 C
Q016 B
Q017 $k = 2$
Q018 C
Q019 $y = -4$
Q020 $x = -3$
Q021 $y = 6$
Q022 $x = 3$
Q023 $y = 5$
Q024 $y = 1$
Q025 $y = -7$
Q026 $x = 1$
Q027 $x = -3$
Q028 $x = 3$
Q029 $x = 2$
Q030 $x = 4$
Q031 $x = -6$
Q032 B
Q033 C
Q034 D
Q035 D
Q036 B
Q037 D
Q038 B
Q039 C
Q040 D
Q041 D
Q042 B
Q043 D
Q044 A
Q045 $x = -3$
Q046 $y = 3$
Q047 $y = x/3 - 4/3(\text{equivalently} x - 3y - 4 = 0)$
Q048 $y = -2$
Q049 $x = -4$
Q050 $7x + 2y + 26 = 0$
Q051 $x = 4$
Q052 $x = -1$
Q053 $x = -5$
Q054 $y = 3$
Q055 $y = -2$
Q056 $y = 6$
Q057 $y = 4x - 2$
Q058 $y = 2x - 6$
Q059 B
Q060 C
Q061 D
Q062 A
Q063 B
Q064 D
Q065 C
Q066 A
Q067 C
Q068 D
Q069 D
Q070 C
Q071 D
Q072 D
Q073 $y = 3x - 7$
Q074 $y = 2x + 10$
Q075 $y = -x + 4$
Q076 $y = x - 5$
Q077 $y = -2x + 3$
Q078 D
Q079 C
Q080 D
Q081 B
Q082 D
Q083 $y = 2$
Q084 D
Q085 $a = 7$
Q086 $a = -1$
Q087 $a = 1$
Q088 $a = 7$
Q089 $a = 5$
Q090 D
Q091 D
Q092 C
Q093 D
Q094 D
Q095 $x + y - 2 = 0$
Q096 $x + 3y - 23 = 0$
Q097 $x + y - 1 = 0$
Q098 $2x - 5y + 1 = 0$
Q099 B
Q100 A
Q101 B
Q102 D
Q103 $3x + 13y = 0$
Q104 C

Q105 $m = 2$
Q106 $m = -3$
Q107 $m = 1/3$
Q108 $m = 2$
Q109 $m = -1/3$
Q110 $m = 3$
Q111 $m = 4$
Q112 $m = -4$
Q113 $m = 3$
Q114 $m = -4$
Q115 $m = -1/4$
Q116 $m = 3$
Q117 $m = -3$
Q118 $m = 1$
Q119 $m = -3$
Q120 $m = 1$
Q121 $m = -1/3$
Q122 $m = 2$
Q123 $m = -1$
Q124 $m = 3/2$
Q125 $m = -1$
Q126 $m = -2/3$
Q127 $m = -1/2$
Q128 D
Q129 C
Q130 A
Q131 B
Q132 A
Q133 A
Q134 B
Q135 D
Q136 A
Q137 C
Q138 C
Q139 D
Q140 A
Q141 D
Q142 C
Q143 C
Q144 A
Q145 D
Q146 A
Q147 B
Q148 D
Q149 B
Q150 B
Q151 $3x - 2y - 14 = 0$
Q152 $2x + y + 1 = 0$
Q153 $2x - y - 4 = 0$
Q154 $2x + 3y - 4 = 0$
Q155 $3x + 2y - 7 = 0$
Q156 C
Q157 D
Q158 C
Q159 D
Q160 D
Q161 $5x - 3y + 25 = 0$
Q162 $3x - 2y - 4 = 0$
Q163 $x - 2y - 12 = 0$
Q164 $2x + 5y - 13 = 0$
Q165 $2x + 3y + 19 = 0$
Q166 D
Q167 D
Q168 B
Q169 A
Q170 D
Q171 $m/k = 2\,(k = 3,\ \ m = 6)$
Q172 D
Q173 $x + 3y - 18 = 0$
Q174 $2x - 3y - 5 = 0$
Q175 $2x - y + 2 = 0$
Q176 $3x - 2y - 6 = 0$
Q177 $2x - y + 1 = 0$
Q178 A
Q179 C
Q180 C
Q181 D
Q182 A
Q183 $B = (2,\ \ 1)$
Q184 $B = (4,\ \ 2)$
Q185 $B = (3,\ \ 6)$
Q186 $B = (-2,\ \ 2)$
Q187 A
Q188 A
Q189 C
Q190 C
Q191 $B = (-3,\ \ 4)$
Q192 C
Q193 $3/\sqrt{13} = 3\sqrt{13}/13$
Q194 $\sqrt{5}$
Q195 $6/\sqrt{13} = 6\sqrt{13}/13$
Q196 6
Q197 $4\sqrt{5}$
Q198 1
Q199 $\sqrt{13}$
Q200 10
Q201 $\sqrt{10}/5$
Q202 A
Q203 D
Q204 D
Q205 B
Q206 D
Q207 D
Q208 A
Q209 B
Q210 D
Q211 $\sqrt{10}/2$
Q212 $\sqrt{10}$
Q213 D
Q214 B
Q215 $\sqrt{10}$
Q216 A
Q217 10
Q218 30
Q219 8
Q220 9
Q221 34
Q222 7
Q223 D
Q224 C
Q225 A
Q226 B
Q227 B
Q228 D
Q229 5
Q230 D
Q231 $(1/2,\ \ -3/2)$
Q232 $(-2,\ \ 3)$

Q233 $(3,\ -3)$

Q234 $(1,\ 1)$

Q235 $(-3,\ 1)$

Q236 D

Q237 B

Q238 D

Q239 D

Q240 A

Q241 $(4,\ 1)$

Q242 A

Q243 $a = -4,\ -3,\ or 6$

Q244 $a = -1,\ -1/2,\ or 1$

Q245 $a = 11/10,\ 3/2,\ or - 9/2$

Q246 $a = 1,\ -3/2,\ or 2$

Q247 A

Q248 A

Q249 C

Q250 D

Q251 15

Q252 A

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 D

Quiz MCQ 4 C

Quiz WRITTEN 5 6

Quiz WRITTEN 6 $y = -2x + 9$

Quiz WRITTEN 7 $y = -x/3 + 7/3$

Quiz WRITTEN 8 $a = -4,\ -3,\ or 6$

Concept 3 | Circles

English Student Edition • Focused formulas and guided practice

Formula opener

Circle

$$(x-a)^2 + (y-b)^2 = r^2$$

General form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Tangent

$$xx_1 + yy_1 = r^2$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Circles**

Q001 **Written**

Find the equation of the circle with centre (3,-2) and radius 5.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x-3)^2 + (y+2)^2 = 25$.

Final answer

$(x-3)^2 + (y+2)^2 = 25$

Q002 **Written**

Find the equation of the circle with centre (-3,1) and radius 2.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x+3)^2 + (y-1)^2 = 4$.

Final answer

$(x+3)^2 + (y-1)^2 = 4$

Worked Solutions **Circles**

Q003 **Written**

Find the equation of the circle with centre the origin and radius 4.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $x^2+y^2=16$.

Final answer

$x^2+y^2=16$

Q004 **Written**

Find the equation of the circle with centre (3,4), radius 6.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $(x-3)^2+(y-4)^2=36$.

Final answer

$(x-3)^2+(y-4)^2=36$

Worked Solutions **Circles**

Q005 Written

Find the equation of the circle with centre (-2,6), radius 2sqrt(3).

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $(x+2)^2+(y-6)^2=12$.

Final answer

$(x+2)^2+(y-6)^2=12$

Q006 Written

Find the equation of the circle with centre the origin, radius 5.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2+(y-k)^2=r^2$ and compare coefficients when needed.
3. Circle: $x^2+y^2=25$.

Final answer

$x^2+y^2=25$

Worked Solutions **Circles**

Q007 MCQ

Find the equation of the circle with centre (3,-2) and radius 5:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-3)^2 + (y+2)^2 = 25$.

Correct option: A

Final answer

$(x-3)^2 + (y+2)^2 = 25$

Q008 MCQ

Find the equation of the circle with centre (-3,1) and radius 2:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+3)^2 + (y-1)^2 = 4$.

Correct option: A

Final answer

$(x+3)^2 + (y-1)^2 = 4$

Worked Solutions **Circles**

Q009 *MCQ*

Find the equation of the circle with centre the origin and radius 4:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 16$.

Correct option: D

Final answer

$x^2 + y^2 = 16$

Q010 *MCQ*

Find the equation of the circle with centre (3,4), radius 6:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y - 4)^2 = 36$.

Correct option: B

Final answer

$(x - 3)^2 + (y - 4)^2 = 36$

Worked Solutions **Circles**

Q011 MCQ

Find the equation of the circle with centre (-2,6), radius 2sqrt(3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+2)^2 + (y-6)^2 = 12$.

Correct option: B

Final answer

$(x+2)^2 + (y-6)^2 = 12$

Q012 MCQ

Find the equation of the circle with centre the origin, radius 5:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 25$.

Correct option: A

Final answer

$x^2 + y^2 = 25$

Worked Solutions **Circles**

Q013 **Written**

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle ; centre $(3, -1)$, radius 2.

Final answer

circle ; centre $(3, -1)$, radius 2

Q014 **Written**

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle ; centre $(2, 3)$, radius 3.

Final answer

circle ; centre $(2, 3)$, radius 3

Worked Solutions **Circles**

Q015 Written

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $\text{circle}\,;\ \text{centre}\ (2,\ -4)\,,\ \text{radius}\,4$.

Final answer

$\text{circle}\,;\ \text{centre}\ (2,\ -4)\,,\ \text{radius}\,4$

Q016 Written

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $\text{circle}\,;\ \text{centre}\ (2,\ 0)\,,\ \text{radius}\,2$.

Final answer

$\text{circle}\,;\ \text{centre}\ (2,\ 0)\,,\ \text{radius}\,2$

Worked Solutions **Circles**

Q017 Written

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(-1, 2)$, radius 5.

Final answer

circle; centre $(-1, 2)$, radius 5

Q018 Written

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(-2, 3)$, radius $\sqrt{13}$.

Final answer

circle; centre $(-2, 3)$, radius $\sqrt{13}$

Worked Solutions **Circles**

Q019 MCQ

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{circle}\,;\,\text{centre }(3,\,-1)\,,\ \text{radius}\,2$.

Correct option: A

Final answer

$\text{circle}\,;\,\text{centre }(3,\,-1)\,,\ \text{radius}\,2$

Worked Solutions **Circles**

Q020 *MCQ*

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle ; centre $(2,\ 3)$, radius 3.

Correct option: C

Final answer

circle ; centre $(2,\ 3)$, radius 3

Worked Solutions **Circles**

Q021 MCQ

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(2, -4)$, radius 4.

Correct option: D

Final answer

circle; centre $(2, -4)$, radius 4

Worked Solutions **Circles**

Q022 *MCQ*

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{circle}\,;\ \text{centre}\ (2,\ 0)\,,\ \text{radius}\,2$.

Correct option: C

Final answer

$\text{circle}\,;\ \text{centre}\ (2,\ 0)\,,\ \text{radius}\,2$

Worked Solutions **Circles**

Q023 **MCQ**

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{circle; centre } (-1,\ 2),\ \text{radius } 5$.

Correct option: C

Final answer

$\text{circle; centre } (-1,\ 2),\ \text{radius } 5$

Worked Solutions **Circles**

Q024 *MCQ*

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{circle}; \text{ centre } (-2,\ 3),\ \text{radius}\sqrt{13}$.

Correct option: C

Final answer

$\text{circle}; \text{ centre } (-2,\ 3),\ \text{radius}\sqrt{13}$

Worked Solutions **Circles**

Q025 Challenge

A circle passes through A $(2,\ 5)$, B $(8,\ 1)$, and has its centre on $x + y = 8$. Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: centre $(5,\ 3)$, radius$^2 = 13; (x-5)^2 + (y-3)^2 = 13$.

Final answer

centre $(5,\ 3)$, radius$^2 = 13; (x-5)^2 + (y-3)^2 = 13$

Worked Solutions **Circles**

Q026 **Challenge**

A circle passes through A $(2, 5)$, B $(8, 1)$, and has its centre on $x + y = 8$. Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is centre $(5, 3)$, radius$^2 = 13; (x-5)^2 + (y-3)^2 = 13$.

Correct option: B

Final answer

centre $(5, 3)$, radius$^2 = 13; (x-5)^2 + (y-3)^2 = 13$

Worked Solutions **Circles**

Q027 **Written**

Find the equation of the circle with centre (2,3) passing through (4,6).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-2)^2+(y-3)^2=13$.

Final answer

$(x-2)^2+(y-3)^2=13$

Q028 **Written**

Find the equation of the circle with centre (-3,1) passing through (4,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x+3)^2+(y-1)^2=58$.

Final answer

$(x+3)^2+(y-1)^2=58$

Worked Solutions **Circles**

Q029 Written

Find the equation of the circle with centre (1,-3) through (-3,0).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-1)^2+(y+3)^2=25$.

Final answer

$(x-1)^2+(y+3)^2=25$

Q030 Written

Find the equation of the circle with centre (-4,1) through (3,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x+4)^2+(y-1)^2=58$.

Final answer

$(x+4)^2+(y-1)^2=58$

Worked Solutions **Circles**

Q031 **Written**

Find the equation of the circle with centre (2,-1) through the origin.

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-2)^2+(y+1)^2=5$.

Final answer

$(x-2)^2+(y+1)^2=5$

Q032 **Written**

Find the equation of the circle with centre the origin through (4,-3).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2+y^2=25$.

Final answer

$x^2+y^2=25$

Worked Solutions **Circles**

Q033 *MCQ*

Find the equation of the circle with centre (2,3) passing through (4,6):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2+(y-3)^2=13$.

Correct option: C

Final answer

$(x-2)^2+(y-3)^2=13$

Q034 *MCQ*

Find the equation of the circle with centre (-3,1) passing through (4,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+3)^2+(y-1)^2=58$.

Correct option: A

Final answer

$(x+3)^2+(y-1)^2=58$

Worked Solutions **Circles**

Q035 MCQ

Find the equation of the circle with centre (1,-3) through (-3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-1)^2+(y+3)^2=25$.

Correct option: D

Final answer

$(x-1)^2+(y+3)^2=25$

Q036 MCQ

Find the equation of the circle with centre (-4,1) through (3,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+4)^2+(y-1)^2=58$.

Correct option: A

Final answer

$(x+4)^2+(y-1)^2=58$

Worked Solutions **Circles**

Q037 MCQ

Find the equation of the circle with centre (2,-1) through the origin:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2+(y+1)^2=5$.

Correct option: B

Final answer

$(x-2)^2+(y+1)^2=5$

Q038 MCQ

Find the equation of the circle with centre the origin through (4,-3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2+y^2=25$.

Correct option: A

Final answer

$x^2+y^2=25$

Worked Solutions **Circles**

Q039 **Written**

Find the equation of the circle with diameter endpoints A $(1, 3)$, B $(3, -5)$.

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-2)^2+(y+1)^2=17$.

Final answer

$(x-2)^2+(y+1)^2=17$

Q040 **Written**

Find the equation of the circle with diameter endpoints (3,4), (5,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-4)^2+(y-1)^2=10$.

Final answer

$(x-4)^2+(y-1)^2=10$

Worked Solutions **Circles**

Q041 **Written**

Find the equation of the circle with diameter endpoints (4,7), (-2,-1).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x-1)^2+(y-3)^2=25$.

Final answer

$(x-1)^2+(y-3)^2=25$

Q042 **Written**

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x+1)^2+(y+2)^2=25$.

Final answer

$(x+1)^2+(y+2)^2=25$

Worked Solutions **Circles**

Q043 **Written**

Find the equation of the circle with diameter endpoints (-3,0), (-1,2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x+2)^2 + (y-1)^2 = 2$.

Final answer

$(x+2)^2 + (y-1)^2 = 2$

Q044 **Written**

Find the equation of the circle with diameter endpoints (-3,6), (3,-2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + (y-2)^2 = 25$.

Final answer

$x^2 + (y-2)^2 = 25$

Worked Solutions **Circles**

Q045 *MCQ*

Find the equation of the circle with diameter endpoints A $(1, 3)$, B $(3, -5)$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2+(y+1)^2=17$.

Correct option: A

Final answer

$(x-2)^2+(y+1)^2=17$

Q046 *MCQ*

Find the equation of the circle with diameter endpoints (3,4), (5,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-4)^2+(y-1)^2=10$.

Correct option: C

Final answer

$(x-4)^2+(y-1)^2=10$

Worked Solutions **Circles**

Q047 MCQ

Find the equation of the circle with diameter endpoints (4,7), (-2,-1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-1)^2+(y-3)^2=25$.

Correct option: C

Final answer

$(x-1)^2+(y-3)^2=25$

Q048 MCQ

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2+(y+2)^2=25$.

Correct option: A

Final answer

$(x+1)^2+(y+2)^2=25$

Worked Solutions **Circles**

Q049 MCQ

Find the equation of the circle with diameter endpoints (-3,0), (-1,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 1)^2 = 2$.

Correct option: B

Final answer

$(x + 2)^2 + (y - 1)^2 = 2$

Q050 MCQ

Find the equation of the circle with diameter endpoints (-3,6), (3,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + (y - 2)^2 = 25$.

Correct option: C

Final answer

$x^2 + (y - 2)^2 = 25$

Worked Solutions **Circles**

Q051 Written

Find the equation of the circle passing through (1,1), (2,-1), and (3,2).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 5x - y + 4 = 0$.

Final answer

$x^2 + y^2 - 5x - y + 4 = 0$

Q052 Written

Find the equation of the circle passing through (0,1), (-1,2), and (3,1).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Final answer

$x^2 + y^2 - 3x - 7y + 6 = 0$

Worked Solutions **Circles**

Q053 Written

Find the equation of the circle with through (0,0), (-1,1), (2,4).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x - 4y = 0$.

Final answer

$x^2 + y^2 - 2x - 4y = 0$

Q054 Written

Find the equation of the circle with through (-4,6), (0,8), (3,-1).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 6y - 16 = 0$.

Final answer

$x^2 + y^2 - 6y - 16 = 0$

Worked Solutions **Circles**

Q055 Written

Find the equation of the circle with through (1,1), (5,-1), (-3,-7).

Solution

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Final answer

$x^2 + y^2 - 2x + 8y - 8 = 0$

Q056 MCQ

Find the equation of the circle passing through (1,1), (2,-1), and (3,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 5x - y + 4 = 0$.

Correct option: A

Final answer

$x^2 + y^2 - 5x - y + 4 = 0$

Worked Solutions **Circles**

Q057 MCQ

Find the equation of the circle passing through (0,1), (-1,2), and (3,1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Correct option: B

Final answer

$x^2 + y^2 - 3x - 7y + 6 = 0$

Q058 MCQ

Find the equation of the circle with through (0,0), (-1,1), (2,4):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x - 4y = 0$.

Correct option: D

Final answer

$x^2 + y^2 - 2x - 4y = 0$

Worked Solutions **Circles**

Q059 MCQ

Find the equation of the circle with through (-4,6), (0,8), (3,-1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6y - 16 = 0$.

Correct option: D

Final answer

$x^2 + y^2 - 6y - 16 = 0$

Q060 MCQ

Find the equation of the circle with through (1,1), (5,-1), (-3,-7):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Correct option: D

Final answer

$x^2 + y^2 - 2x + 8y - 8 = 0$

Q061 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 - 6x - 6y + 8 = 0$; centre $(3,\ 3)$, radius $\sqrt{10}$.

Final answer

$x^2 + y^2 - 6x - 6y + 8 = 0$; centre $(3,\ 3)$, radius $\sqrt{10}$

Worked Solutions **Circles**

Q062 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6x - 6y + 8 = 0$; centre $(3,\ 3)$, radius $\sqrt{10}$.

Correct option: A

Final answer

$x^2 + y^2 - 6x - 6y + 8 = 0$; centre $(3,\ 3)$, radius $\sqrt{10}$

Worked Solutions **Circles**

Q063 **Written**

Find the circle with centre (4,-2) tangent to the x-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-4)^2+(y+2)^2=4$.

Final answer

$(x-4)^2+(y+2)^2=4$

Q064 **Written**

Find the circle with centre (-2,3) tangent to the y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+2)^2+(y-3)^2=4$.

Final answer

$(x+2)^2+(y-3)^2=4$

Worked Solutions **Circles**

Q065 Written

Find the circle with centre (-1,4) tangent to x-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2+(y-4)^2=16$.

Final answer

$(x+1)^2+(y-4)^2=16$

Q066 Written

Find the circle with centre (3,-4) tangent to x-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-3)^2+(y+4)^2=16$.

Final answer

$(x-3)^2+(y+4)^2=16$

Worked Solutions **Circles**

Q067 Written

Find the circle with centre (5,-4) tangent to x-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-5)^2+(y+4)^2=16$.

Final answer

$(x-5)^2+(y+4)^2=16$

Q068 Written

Find the circle with centre (-1,6) tangent to y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2+(y-6)^2=1$.

Final answer

$(x+1)^2+(y-6)^2=1$

Worked Solutions **Circles**

Q069 Written

Find the circle with centre (2,-5) tangent to y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-2)^2+(y+5)^2=4$.

Final answer

$(x-2)^2+(y+5)^2=4$

Q070 Written

Find the circle with centre (-3,-4) tangent to y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+3)^2+(y+4)^2=9$.

Final answer

$(x+3)^2+(y+4)^2=9$

Worked Solutions **Circles**

Q071 MCQ

Find the circle with centre (4,-2) tangent to the x-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-4)^2+(y+2)^2=4$.

Correct option: B

Final answer

$(x-4)^2+(y+2)^2=4$

Worked Solutions **Circles**

Q072 MCQ

Find the circle with centre (-2,3) tangent to the y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+2)^2 + (y-3)^2 = 4$.

Correct option: B

Final answer

$(x+2)^2 + (y-3)^2 = 4$

Worked Solutions **Circles**

Q073 *MCQ*

Find the circle with centre (-1,4) tangent to x-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2 + (y-4)^2 = 16$.

Correct option: B

Final answer

$(x+1)^2 + (y-4)^2 = 16$

Worked Solutions **Circles**

Q074 MCQ

Find the circle with centre (3,-4) tangent to x-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-3)^2+(y+4)^2=16$.

Correct option: D

Final answer

$(x-3)^2+(y+4)^2=16$

Worked Solutions **Circles**

Q075 MCQ

Find the circle with centre (5,-4) tangent to x-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-5)^2+(y+4)^2=16$.

Correct option: D

Final answer

$(x-5)^2+(y+4)^2=16$

Worked Solutions **Circles**

Q076 MCQ

Find the circle with centre (-1,6) tangent to y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2+(y-6)^2=1$.

Correct option: D

Final answer

$(x+1)^2+(y-6)^2=1$

Worked Solutions **Circles**

Q077 MCQ

Find the circle with centre (2,-5) tangent to y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2+(y+5)^2=4$.

Correct option: B

Final answer

$(x-2)^2+(y+5)^2=4$

Worked Solutions **Circles**

Q078 MCQ

Find the circle with centre (-3,-4) tangent to y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+3)^2 + (y+4)^2 = 9$.

Correct option: C

Final answer

$(x+3)^2 + (y+4)^2 = 9$

Worked Solutions **Circles**

Q079 **Written**

Find the circles through (-2,1) tangent to both the x-axis and y-axis.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$.

Final answer

$(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$

Q080 **Written**

Find the circles through (1,2) tangent to both axes.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$.

Final answer

$(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$

Worked Solutions **Circles**

Q081 Written

Find the circle with through (-1,-2) tangent to both axes.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$.

Final answer

$(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Q082 Written

Find the circle with through (-2,-1) tangent to both axes.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$.

Final answer

$(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Worked Solutions **Circles**

Q083 **Written**

Find the circle with through (8,-1) tangent to both axes.

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$.

Final answer

$(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$

Worked Solutions **Circles**

Q084 MCQ

Find the circles through (-2,1) tangent to both the x-axis and y-axis:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$.

Correct option: C

Final answer

$(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$

Worked Solutions **Circles**

Q085 MCQ

Find the circles through (1,2) tangent to both axes:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$.

Correct option: D

Final answer

$(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$

Worked Solutions **Circles**

Q086 MCQ

Find the circle with through (-1,-2) tangent to both axes:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2 + (y+1)^2 = 1$ or $(x+5)^2 + (y+5)^2 = 25$.

Correct option: C

Final answer

$(x+1)^2 + (y+1)^2 = 1$ or $(x+5)^2 + (y+5)^2 = 25$

Worked Solutions **Circles**

Q087 MCQ

Find the circle with through (-2,-1) tangent to both axes:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$.

Correct option: C

Final answer

$(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Worked Solutions **Circles**

Q088 MCQ

Find the circle with through (8,-1) tangent to both axes:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$.

Correct option: B

Final answer

$(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$

Worked Solutions **Circles**

Q089 **Written**

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3).

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-1)^2 + (y+1)^2 = 25$.

Final answer

$(x-1)^2 + (y+1)^2 = 25$

Q090 **Written**

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2).

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x+5/2)^2 + (y-5/2)^2 = 25/2$.

Final answer

$(x+5/2)^2 + (y-5/2)^2 = 25/2$

Worked Solutions **Circles**

Q091 Written

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4).

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-2)^2 + (y-3)^2 = 2$.

Final answer

$(x-2)^2 + (y-3)^2 = 2$

Q092 Written

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3).

Solution

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x-4)^2 + (y-2)^2 = 26$.

Final answer

$(x-4)^2 + (y-2)^2 = 26$

Worked Solutions **Circles**

Q093 MCQ

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-1)^2 + (y+1)^2 = 25$.

Correct option: D

Final answer

$(x-1)^2 + (y+1)^2 = 25$

Worked Solutions **Circles**

Q094 MCQ

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$.

Correct option: C

Final answer

$(x + 5/2)^2 + (y - 5/2)^2 = 25/2$

Worked Solutions **Circles**

Q095 MCQ

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-2)^2 + (y-3)^2 = 2$.

Correct option: C

Final answer

$(x-2)^2 + (y-3)^2 = 2$

Worked Solutions **Circles**

Q096 MCQ

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-4)^2 + (y-2)^2 = 26$.

Correct option: D

Final answer

$(x-4)^2 + (y-2)^2 = 26$

Worked Solutions **Circles**

Q097 **Challenge**

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x+1)^2+y^2=5$; centre $(-1,\,0)$, radius $\sqrt{5}$.

Final answer

$(x+1)^2+y^2=5$; centre $(-1,\,0)$, radius $\sqrt{5}$

Worked Solutions **Circles**

Q098 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$.

Correct option: C

Final answer

$(x+1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$

Worked Solutions **Circles**

Q099 Written

Find the range of k for $x^2 + y^2 - 2\mathbf{kx} + 2\mathbf{ky} + 4k + 6 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1 \text{ or } k > 3$.

Final answer

$k < -1 \text{ or } k > 3$

Q100 Written

Find the range of k for $x^2 + y^2 + 2\mathbf{kx} - 4\mathbf{ky} + 4k^2 + 6 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -\sqrt{6} \text{ or } k > \sqrt{6}$.

Final answer

$k < -\sqrt{6} \text{ or } k > \sqrt{6}$

Worked Solutions **Circles**

Q101 Written

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < 5/4$.

Final answer

$k < 5/4$

Q102 Written

Find the range of k for $x^2 + y^2 + 6\,\mathbf{kx} - 2\,\mathbf{ky} + 28k + 6 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1/5$ or $k > 3$.

Final answer

$k < -1/5$ or $k > 3$

Worked Solutions **Circles**

Q103 Written

Find the range of k for $x^2 + y^2 + 2\mathbf{kx} - 2(k+3)y + 4k^2 + 1 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $-1 < k < 4$.

Final answer

$-1 < k < 4$

Worked Solutions **Circles**

Q104 MCQ

Find the range of k for $x^2 + y^2 - 2\,\mathbf{kx} + 2\,\mathbf{ky} + 4k + 6 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1$ or $k > 3$.

Correct option: C

Final answer

$k < -1$ or $k > 3$

Worked Solutions **Circles**

Q105 MCQ

Find the range of k for $x^2 + y^2 + 2\mathbf{kx} - 4\mathbf{ky} + 4k^2 + 6 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -\sqrt{6}$ or $k > \sqrt{6}$.

Correct option: B

Final answer

$k < -\sqrt{6}$ or $k > \sqrt{6}$

Worked Solutions **Circles**

Q106 MCQ

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < 5/4$.

Correct option: C

Final answer

$k < 5/4$

Worked Solutions **Circles**

Q107 MCQ

Find the range of k for $x^2 + y^2 + 6\,\mathbf{kx} - 2\,\mathbf{ky} + 28k + 6 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1/5$ or $k > 3$.

Correct option: B

Final answer

$k < -1/5$ or $k > 3$

Worked Solutions **Circles**

Q108 MCQ

Find the range of k for $x^2 + y^2 + 2\,\mathbf{kx} - 2(k+3)y + 4k^2 + 1 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-1 < k < 4$.

Correct option: D

Final answer

$-1 < k < 4$

Worked Solutions **Circles**

Q109 **Challenge**

An acoustic-safety circle is $x^2 + y^2 + 6x - 2\mathbf{ky} + 2k^2 - 4k + 9 = 0$. Find the exact k-range giving a positive radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < k < 4$.

Final answer

$0 < k < 4$

Worked Solutions **Circles**

Q110 **Challenge**

An acoustic-safety circle is $x^2 + y^2 + 6x - 2\mathbf{ky} + 2k^2 - 4k + 9 = 0$. Find the exact k-range giving a positive radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < k < 4$.

Correct option: B

Final answer

$0 < k < 4$

Worked Solutions **Circles**

Q111 **Written**

Find the coordinates of the intersections of $(x+2)^2 + y^2 = 10$ and $y = -2x + 1$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(1,\ -1)$ and $(-1,\ 3)$.

Final answer

$(1,\ -1)$ and $(-1,\ 3)$

Q112 **Written**

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(0,\ 2)$ and $(2,\ 0)$.

Final answer

$(0,\ 2)$ and $(2,\ 0)$

Worked Solutions **Circles**

Q113 **Written**

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(3, -1)$.

Final answer

$(3, -1)$

Q114 **Written**

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2, 1)$.

Final answer

$(-2, 1)$

Worked Solutions **Circles**

Q115 Written

Find the coordinates of the intersections of $(x-1)^2+y^2=10$ with $x+3y-1=0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2,\ 1)$ and $(4,\ -1)$.

Final answer

$(-2,\ 1)$ and $(4,\ -1)$

Worked Solutions **Circles**

Q116 MCQ

Find the coordinates of the intersections of $(x+2)^2 + y^2 = 10$ and $y = -2x + 1$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(1, -1)$ and $(-1, 3)$.

Correct option: C

Final answer

$(1, -1)$ and $(-1, 3)$

Worked Solutions **Circles**

Q117 *MCQ*

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is (0, 2) and (2, 0).

Correct option: D

Final answer

(0, 2) and (2, 0)

Worked Solutions **Circles**

Q118 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(3, -1)$.

Correct option: C

Final answer

$(3, -1)$

Worked Solutions **Circles**

Q119 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$.

Correct option: D

Final answer

$(-2, 1)$

Worked Solutions **Circles**

Q120 MCQ

Find the coordinates of the intersections of $(x-1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$ and $(4, -1)$.

Correct option: B

Final answer

$(-2, 1)$ and $(4, -1)$

Worked Solutions **Circles**

Q121 **Written**

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $0\,\text{points}$.

Final answer

$0\,\text{points}$

Q122 **Written**

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $2\,\text{points}\ :\ (1,\,-3)$ and $(3,\,1)$.

Final answer

$2\,\text{points}\ :\ (1,\,-3)$ and $(3,\,1)$

Worked Solutions **Circles**

Q123 Written

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $2\,\text{points}\ :\ \left(-1+\sqrt{6}/2,\ 1+\sqrt{6}/2\right)$ and $\left(-1-\sqrt{6}/2,\ 1-\sqrt{6}/2\right)$.

Final answer

$2\,\text{points}\ :\ \left(-1+\sqrt{6}/2,\ 1+\sqrt{6}/2\right)$ and $\left(-1-\sqrt{6}/2,\ 1-\sqrt{6}/2\right)$

Q124 Written

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $0\,\text{points}$.

Final answer

$0\,\text{points}$

Worked Solutions **Circles**

Q125 **Written**

Find the number of intersections of $(x+1)^2+(y-2)^2=5$ with $x+2y+2=0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $1\,\text{point} \;:\; (-2, 0)$.

Final answer

$1\,\text{point} \;:\; (-2, 0)$

Worked Solutions **Circles**

Q126 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0 points.

Correct option: D

Final answer

0 points

Worked Solutions **Circles**

Q127 MCQ

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\,\text{points}\ :\ (1,\,-3)$ and $(3,\,1)$.

Correct option: A

Final answer

$2\,\text{points}\ :\ (1,\,-3)$ and $(3,\,1)$

Worked Solutions **Circles**

Q128 MCQ

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\,\text{points}\ :\ \left(-1 + \sqrt{6}/2,\ 1 + \sqrt{6}/2\right)$ and $\left(-1 - \sqrt{6}/2,\ 1 - \sqrt{6}/2\right)$.

Correct option: C

Final answer

$2\,\text{points}\ :\ \left(-1 + \sqrt{6}/2,\ 1 + \sqrt{6}/2\right)$ and $\left(-1 - \sqrt{6}/2,\ 1 - \sqrt{6}/2\right)$

Worked Solutions **Circles**

Q129 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0\,\text{points}$.

Correct option: C

Final answer

$0\,\text{points}$

Worked Solutions **Circles**

Q130 MCQ

Find the number of intersections of $(x+1)^2 + (y-2)^2 = 5$ with $x + 2y + 2 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $1\,\text{point}\ :\ (-2,\,0)$.

Correct option: D

Final answer

$1\,\text{point}\ :\ (-2,\,0)$

Worked Solutions **Circles**

Q131 Written

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 < k < 5$.

Final answer

$-5 < k < 5$

Q132 Written

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{5}$.

Final answer

$r = \sqrt{5}$

Worked Solutions **Circles**

Q133 **Written**

If $x^2 + y^2 = 4$ and $y =$ mx $+ 4$ are tangent, find m.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm\sqrt{3}$.

Final answer

$m = \pm\sqrt{3}$

Q134 **Written**

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $0 < r < \sqrt{10}$.

Final answer

$0 < r < \sqrt{10}$

Worked Solutions **Circles**

Q135 Written

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm 4$.

Final answer

$m = \pm 4$

Q136 Written

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $n < -10 \text{ or } n > 10$.

Final answer

$n < -10 \text{ or } n > 10$

Worked Solutions **Circles**

Q137 Written

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 <= m <= 5$.

Final answer

$-5 <= m <= 5$

Q138 Written

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{2}$.

Final answer

$r = \sqrt{2}$

Worked Solutions **Circles**

Q139 **Written**

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r > \sqrt{17}$.

Final answer

$r > \sqrt{17}$

Worked Solutions **Circles**

Q140 MCQ

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 < k < 5$.

Correct option: D

Final answer

$-5 < k < 5$

Worked Solutions **Circles**

Q141 MCQ

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{5}$.

Correct option: B

Final answer

$r = \sqrt{5}$

Worked Solutions **Circles**

Q142 MCQ

If $x^2 + y^2 = 4$ and $y = \mathbf{mx} + 4$ are tangent, find m:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm\sqrt{3}$.

Correct option: B

Final answer

$m = \pm\sqrt{3}$

Worked Solutions **Circles**

Q143 MCQ

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < \sqrt{10}$.

Correct option: B

Final answer

$0 < r < \sqrt{10}$

Worked Solutions **Circles**

Q144 MCQ

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm 4$.

Correct option: C

Final answer

$m = \pm 4$

Worked Solutions **Circles**

Q145 MCQ

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $n < -10$ or $n > 10$.

Correct option: A

Final answer

$n < -10$ or $n > 10$

Worked Solutions **Circles**

Q146 MCQ

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 <= m <= 5$.

Correct option: C

Final answer

$-5 <= m <= 5$

Worked Solutions **Circles**

Q147 MCQ

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{2}$.

Correct option: B

Final answer

$r = \sqrt{2}$

Worked Solutions **Circles**

Q148 MCQ

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r > \sqrt{17}$.

Correct option: D

Final answer

$r > \sqrt{17}$

Worked Solutions **Circles**

Q149 **Challenge**

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < r < 2$.

Final answer

$0 < r < 2$

Worked Solutions **Circles**

Q150 **Challenge**

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < 2$.

Correct option: D

Final answer

$0 < r < 2$

Worked Solutions **Circles**

Q151 Written

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $\sqrt{14}$.

Final answer

$\sqrt{14}$

Q152 Written

Find the chord length cut off by $(x-2)^2 + (y+1)^2 = 9$ on $x + 2y - 5 = 0$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 4.

Final answer

4

Worked Solutions **Circles**

Q153 **Written**

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $2\sqrt{5}$.

Final answer

$2\sqrt{5}$

Q154 **Written**

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $3\sqrt{2}$.

Final answer

$3\sqrt{2}$

Worked Solutions **Circles**

Q155 Written

Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 2.

Final answer

2

Q156 MCQ

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\sqrt{14}$.

Correct option: A

Final answer

$\sqrt{14}$

Worked Solutions **Circles**

Q157 MCQ

Find the chord length cut off by $(x-2)^2+(y+1)^2=9$ on $x+2y-5=0$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 4.

Correct option: C

Final answer

4

Q158 MCQ

Find the chord length cut off by $x^2+y^2=10$ on $2x-y=5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: D

Final answer

$2\sqrt{5}$

Worked Solutions **Circles**

Q159 *MCQ*

Find the chord length cut off by $(x-2)^2+(y-1)^2=5$ on $x+y=4$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3\sqrt{2}$.

Correct option: B

Final answer

$3\sqrt{2}$

Q160 *MCQ*

Find the chord length cut off by $x^2+y^2=3$ on $x-y=2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2.

Correct option: D

Final answer

2

Worked Solutions **Circles**

Q161 Written

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$k = \pm 2$

Q162 Written

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is sqrt(2).

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm\sqrt{31}$.

Final answer

$k = \pm\sqrt{31}$

Worked Solutions **Circles**

Q163 Written

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is 4sqrt(2).

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$k = \pm 2$

Q164 Written

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is 3sqrt(2).

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = 0 \text{ or } k = -2$.

Final answer

$k = 0 \text{ or } k = -2$

Worked Solutions **Circles**

Q165 MCQ

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: C

Final answer

$k = \pm 2$

Q166 MCQ

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is sqrt(2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm\sqrt{31}$.

Correct option: D

Final answer

$k = \pm\sqrt{31}$

Worked Solutions **Circles**

Q167 MCQ

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is 4sqrt(2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: A

Final answer

$k = \pm 2$

Worked Solutions **Circles**

Q168 MCQ

For $(x-1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is 3sqrt(2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = 0 \,\text{or}\, k = -2$.

Correct option: B

Final answer

$k = 0 \,\text{or}\, k = -2$

Worked Solutions **Circles**

Q169 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $k = -1$ or $k = 7$.

Final answer

$k = -1$ or $k = 7$

Worked Solutions **Circles**

Q170 **Challenge**

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = -1$ or $k = 7$.

Correct option: A

Final answer

$k = -1$ or $k = 7$

Worked Solutions **Circles**

Q171 Written

Find the tangent to $x^2 + y^2 = 25$ at (4,3).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $4x + 3y = 25$.

Final answer

$4x + 3y = 25$

Q172 Written

Find the tangent to $x^2 + y^2 = 34$ at (5,3).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $5x + 3y = 34$.

Final answer

$5x + 3y = 34$

Worked Solutions **Circles**

Q173 **Written**

Find the tangents $x^2 + y^2 = 10$ at (3,-1).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $3x - y = 10$.

Final answer

$3x - y = 10$

Q174 **Written**

Find the tangents $x^2 + y^2 = 29$ at (-5,-2).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $-5x - 2y = 29$.

Final answer

$-5x - 2y = 29$

Worked Solutions **Circles**

Q175 **Written**

Find the tangents $x^2 + y^2 = 5$ at (-2,1).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-2x + y = 5$.

Final answer

$-2x + y = 5$

Worked Solutions **Circles**

Q176 MCQ

Find the tangent to $x^2 + y^2 = 25$ at (4,3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $4x + 3y = 25$.

Correct option: A

Final answer

$4x + 3y = 25$

Worked Solutions **Circles**

Q177 MCQ

Find the tangent to $x^2 + y^2 = 34$ at (5,3):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $5x + 3y = 34$.

Correct option: C

Final answer

$5x + 3y = 34$

Worked Solutions **Circles**

Q178 MCQ

Find the tangents $x^2 + y^2 = 10$ at (3,-1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3x - y = 10$.

Correct option: A

Final answer

$3x - y = 10$

Worked Solutions **Circles**

Q179 MCQ

Find the tangents $x^2 + y^2 = 29$ at (-5,-2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5x - 2y = 29$.

Correct option: C

Final answer

$-5x - 2y = 29$

Worked Solutions **Circles**

Q180 MCQ

Find the tangents $x^2 + y^2 = 5$ at (-2,1):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$.

Correct option: A

Final answer

$-2x + y = 5$

Worked Solutions **Circles**

Q181 Written

Find the tangents from (1,3) to $x^2 + y^2 = 5$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $2x + y = 5$ and $-x + 2y = 5$.

Final answer

$2x + y = 5$ and $-x + 2y = 5$

Q182 Written

Find the tangents from (-2,4) to $x^2 + y^2 = 10$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x-h)(x_1-h)+(y-k)(y_1-k)=r^2$ and rearrange.
3. Tangent: $-3x + y = 10$ and $x + 3y = 10$.

Final answer

$-3x + y = 10$ and $x + 3y = 10$

Worked Solutions **Circles**

Q183 **Written**

Find the tangents from (-1,3) to $x^2 + y^2 = 5$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-2x + y = 5$ and $x + 2y = 5$.

Final answer

$-2x + y = 5$ and $x + 2y = 5$

Q184 **Written**

Find the tangents from (2,1) to $x^2 + y^2 = 1$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $y = 1$ and $4x - 3y = 5$.

Final answer

$y = 1$ and $4x - 3y = 5$

Worked Solutions **Circles**

Q185 **Written**

Find the tangents from (2,6) to $x^2 + y^2 = 20$.

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-x + 2y = 10$ and $2x + y = 10$.

Final answer

$-x + 2y = 10$ and $2x + y = 10$

Worked Solutions **Circles**

Q186 MCQ

Find the tangents from (1,3) to $x^2 + y^2 = 5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x + y = 5$ and $-x + 2y = 5$.

Correct option: D

Final answer

$2x + y = 5$ and $-x + 2y = 5$

Worked Solutions **Circles**

Q187 MCQ

Find the tangents from (-2,4) to $x^2 + y^2 = 10$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-3x + y = 10$ and $x + 3y = 10$.

Correct option: A

Final answer

$-3x + y = 10$ and $x + 3y = 10$

Worked Solutions **Circles**

Q188 MCQ

Find the tangents from (-1,3) to $x^2 + y^2 = 5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$ and $x + 2y = 5$.

Correct option: D

Final answer

$-2x + y = 5$ and $x + 2y = 5$

Worked Solutions **Circles**

Q189 MCQ

Find the tangents from (2,1) to $x^2 + y^2 = 1$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $y = 1$ and $4x - 3y = 5$.

Correct option: B

Final answer

$y = 1$ and $4x - 3y = 5$

Worked Solutions **Circles**

Q190 MCQ

Find the tangents from (2,6) to $x^2 + y^2 = 20$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ and $2x + y = 10$.

Correct option: C

Final answer

$-x + 2y = 10$ and $2x + y = 10$

Worked Solutions **Circles**

Q191 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Final answer

$-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

Worked Solutions **Circles**

Q192 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Correct option: B

Final answer

$-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

Worked Solutions **Circles**

Q193 Written

Classify $x^2 + y^2 = 4$ and $(x-4)^2 + y^2 = 9$.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: intersect at 2 points.

Final answer

intersect at 2 points

Q194 Written

Classify $x^2 + y^2 = 9$ and $(x-3)^2 + (y-4)^2 = 4$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: B : touch externally.

Final answer

B : touch externally

Worked Solutions **Circles**

Q195 Written

Classify $(x-1)^2+y^2=5^2$ and $x^2+y^2+2x-6y=3$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $E\,:\ \text{intersect at}\,2\,\text{points}$.

Final answer

$E\,:\ \text{intersect at}\,2\,\text{points}$

Q196 Written

Classify $x^2+y^2=4$ and $(x-1)^2+y^2=1$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $C\,:\ \text{touch internally}$.

Final answer

$C\,:\ \text{touch internally}$

Worked Solutions **Circles**

Q197 **Written**

Classify $x^2 + y^2 = 18$ and $(x+2)^2 + (y-2)^2 = 2$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $C:$ touch internally.

Final answer

$C:$ touch internally

Q198 **Written**

Classify $(x+3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $E:$ intersect at 2 points.

Final answer

$E:$ intersect at 2 points

Worked Solutions **Circles**

Q199 Written

Classify $x^2 + (y-3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E.

Solution

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: D : one is inside the other.

Final answer

D : one is inside the other

Worked Solutions **Circles**

Q200 *MCQ*

For the expression Classify $x^2 + y^2 = 4$ and $(x-4)^2 + y^2 = 9$., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\text{intersect at}\,2\,\text{points}$.

Correct option: C

Final answer

$\text{intersect at}\,2\,\text{points}$

Worked Solutions **Circles**

Q201 MCQ

For the expression Classify $x^2 + y^2 = 9$ and $(x-3)^2 + (y-4)^2 = 4$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $B :$ touch externally.

Correct option: D

Final answer

$B :$ touch externally

Worked Solutions **Circles**

Q202 *MCQ*

For the expression Classify $(x-1)^2+y^2=5^2$ and $x^2+y^2+2x-6y=3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $E:$ intersect at 2 points.

Correct option: B

Final answer

$E:$ intersect at 2 points

Worked Solutions **Circles**

Q203 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x-1)^2 + y^2 = 1$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: A

Final answer

C : touch internally

Worked Solutions **Circles**

Q204 *MCQ*

For the expression Classify $x^2 + y^2 = 18$ and $(x+2)^2 + (y-2)^2 = 2$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: D

Final answer

C : touch internally

Worked Solutions **Circles**

Q205 MCQ

For the expression Classify $(x+3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $E\ :\ \text{intersect at}\,2\,\text{points}$.

Correct option: B

Final answer

$E\ :\ \text{intersect at}\,2\,\text{points}$

Worked Solutions **Circles**

Q206 MCQ

For the expression Classify $x^2 + (y-3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is D : one is inside the other.

Correct option: C

Final answer

D : one is inside the other

Worked Solutions **Circles**

Q207 Written

Find the circle with centre (3,1) touching $(x-1)^2+(y+3)^2=5$ internally.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $(x-3)^2+(y-1)^2=45$.

Final answer

$(x-3)^2+(y-1)^2=45$

Q208 Written

Find the circle with centre (-3,-9) touching $(x+6)^2+(y+5)^2=1$ externally.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $(x+3)^2+(y+9)^2=16$.

Final answer

$(x+3)^2+(y+9)^2=16$

Worked Solutions **Circles**

Q209 **Written**

Find the circle with centre (3,-9) touching $(x-3)^2+y^2=9$ internally.

Solution

1. Find the centre distance D and both radii.
2. Compare D with r_1+r_2 and $|r_1-r_2|$.
3. Relative position: $(x-3)^2+(y+9)^2=144$.

Final answer

$(x-3)^2+(y+9)^2=144$

Worked Solutions **Circles**

Q210 MCQ

Find the circle with centre (3,1) touching $(x-1)^2+(y+3)^2=5$ internally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-3)^2+(y-1)^2=45$.

Correct option: A

Final answer

$(x-3)^2+(y-1)^2=45$

Worked Solutions **Circles**

Q211 MCQ

Find the circle with centre (-3,-9) touching $(x+6)^2+(y+5)^2=1$ externally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x+3)^2+(y+9)^2=16$.

Correct option: D

Final answer

$(x+3)^2+(y+9)^2=16$

Worked Solutions **Circles**

Q212 MCQ

Find the circle with centre (3,-9) touching $(x-3)^2 + y^2 = 9$ internally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x-3)^2 + (y+9)^2 = 144$.

Correct option: A

Final answer

$(x-3)^2 + (y+9)^2 = 144$

Worked Solutions **Circles**

Q213 Challenge

Find the circle with centre (-1,-2) touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x + 1)^2 + (y + 2)^2 = 9$.

Final answer

$(x + 1)^2 + (y + 2)^2 = 9$

Worked Solutions **Circles**

Q214 Challenge

Find the circle with centre (-1,-2) touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 2)^2 = 9$.

Correct option: D

Final answer

$(x + 1)^2 + (y + 2)^2 = 9$

Worked Solutions **Circles**

Q215 **Written**

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $2x - 4y - 5 = 0$(equivalently $y = x/2 - 5/4$).

Final answer

$2x - 4y - 5 = 0$(equivalently $y = x/2 - 5/4$)

Q216 **Written**

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x + y - 1 = 0$.

Final answer

$x + y - 1 = 0$

Worked Solutions **Circles**

Q217 Written

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $12x - 6y - 7 = 0$.

Final answer

$12x - 6y - 7 = 0$

Q218 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x = 1/2$.

Final answer

$x = 1/2$

Worked Solutions **Circles**

Q219 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x - 4y - 5 = 0$(equivalently $y = x/2 - 5/4$).

Correct option: A

Final answer

$2x - 4y - 5 = 0$(equivalently $y = x/2 - 5/4$)

Worked Solutions **Circles**

Q220 MCQ

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: C

Final answer

$x + y - 1 = 0$

Worked Solutions **Circles**

Q221 MCQ

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $12x - 6y - 7 = 0$.

Correct option: B

Final answer

$12x - 6y - 7 = 0$

Worked Solutions **Circles**

Q222 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x-1)^2 + y^2 = 4$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x = 1/2$.

Correct option: D

Final answer

$x = 1/2$

Worked Solutions **Circles**

Q223 **Written**

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(-3, -1)$ and $(-1, 3)$.

Final answer

$(-3, -1)$ and $(-1, 3)$

Q224 **Written**

Find the intersections of $x^2 + y^2 = 4$ and $(x-4)^2 + (y-2)^2 = 16$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(0, 2)$ and $(8/5, -6/5)$.

Final answer

$(0, 2)$ and $(8/5, -6/5)$

Worked Solutions **Circles**

Q225 **Written**

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$.

Solution

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: (1, 2) and (2, 1).

Final answer

(1, 2) and (2, 1)

Worked Solutions **Circles**

Q226 *MCQ*

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-3, -1)$ and $(-1, 3)$.

Correct option: B

Final answer

$(-3, -1)$ and $(-1, 3)$

Worked Solutions **Circles**

Q227 MCQ

Find the intersections of $x^2 + y^2 = 4$ and $(x-4)^2 + (y-2)^2 = 16$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(0,\ 2)$ and $(8/5,\ -6/5)$.

Correct option: C

Final answer

$(0,\ 2)$ and $(8/5,\ -6/5)$

Worked Solutions **Circles**

Q228 *MCQ*

Find the intersections of $x^2 + y^2 = 5$ and $(x-2)^2 + (y-2)^2 = 1$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is (1, 2) and (2, 1).

Correct option: B

Final answer

(1, 2) and (2, 1)

Worked Solutions **Circles**

Q229 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(-4, 3)$ and $(3, -4)$.

Final answer

$(-4, 3)$ and $(3, -4)$

Worked Solutions **Circles**

Q230 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-4,\ 3)$ and $(3,\ -4)$.

Correct option: C

Final answer

$(-4,\ 3)$ and $(3,\ -4)$

Worked Solutions **Circles**

Q231 **Written**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 2x + 6y - 15 = 0$.

Final answer

$x^2 + y^2 + 2x + 6y - 15 = 0$

Q232 **Written**

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin.

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 4x + 2y = 0$.

Final answer

$x^2 + y^2 + 4x + 2y = 0$

Worked Solutions **Circles**

Q233 **Written**

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 10x + 20y - 39 = 0$.

Final answer

$x^2 + y^2 + 10x + 20y - 39 = 0$

Q234 **Written**

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through (-3,2).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 24x + 32y - 5 = 0$.

Final answer

$x^2 + y^2 + 24x + 32y - 5 = 0$

Worked Solutions **Circles**

Q235 **Written**

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 - 10x - 6y + 24 = 0$.

Final answer

$x^2 + y^2 - 10x - 6y + 24 = 0$

Worked Solutions **Circles**

Q236 **MCQ**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x + 6y - 15 = 0$.

Correct option: D

Final answer

$x^2 + y^2 + 2x + 6y - 15 = 0$

Worked Solutions **Circles**

Q237 MCQ

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 4x + 2y = 0$.

Correct option: A

Final answer

$x^2 + y^2 + 4x + 2y = 0$

Worked Solutions **Circles**

Q238 **MCQ**

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 10x + 20y - 39 = 0$.

Correct option: D

Final answer

$x^2 + y^2 + 10x + 20y - 39 = 0$

Worked Solutions **Circles**

Q239 MCQ

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through (-3,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 24x + 32y - 5 = 0$.

Correct option: C

Final answer

$x^2 + y^2 + 24x + 32y - 5 = 0$

Worked Solutions **Circles**

Q240 **MCQ**

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 10x - 6y + 24 = 0$.

Correct option: B

Final answer

$x^2 + y^2 - 10x - 6y + 24 = 0$

Worked Solutions **Circles**

Q241 **Challenge**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2).

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Final answer

$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Worked Solutions **Circles**

Q242 **Challenge**

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Correct option: A

Final answer

$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Worked Solutions **Circles**

Quiz MCQ 1 **Concept Quiz · MCQ**

For the circle $(x-2)^2+(y+3)^2=25$, determine its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is centre $(2,-3)$, radius 5.

Correct option: B

Final answer

centre $(2,-3)$, radius 5

Worked Solutions **Circles**

Quiz MCQ 2 **Concept Quiz · MCQ**

Determine the substitution discriminant when a line meets a circle at exactly one real point:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0.

Correct option: D

Final answer

0

Worked Solutions **Circles**

Quiz MCQ 3 Concept Quiz · MCQ

Determine the chord length for radius r and centre-to-chord distance d:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{r^2 - d^2}$.

Correct option: A

Final answer

$2\sqrt{r^2 - d^2}$

Worked Solutions **Circles**

Quiz MCQ 4 Concept Quiz · MCQ

Determine the relative position of two non-overlapping circles when $D = r1 + r2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is externally tangent.

Correct option: D

Final answer

externally tangent

Worked Solutions **Circles**

Quiz WRITTEN 5 Concept Quiz · Written

Find the circle with centre (1,-2) passing through (4,2).

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x-h)^2 + (y-k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x-1)^2 + (y+2)^2 = 25$.

Final answer

$(x-1)^2 + (y+2)^2 = 25$

Quiz WRITTEN 6 Concept Quiz · Written

Find the intersections of $x^2 + y^2 = 25$ and $y = 3$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(4,\ 3)\,,\ (-4,\ 3)$.

Final answer

$(4,\ 3)\,,\ (-4,\ 3)$

Worked Solutions **Circles**

Quiz WRITTEN 7 Concept Quiz · Written

Find the tangent to $x^2 + y^2 = 25$ at (3,4).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $3x + 4y = 25$.

Final answer

$3x + 4y = 25$

Quiz WRITTEN 8 Concept Quiz · Written

The circles $S_1 = x^2 + y^2 - 5 = 0$ and $S_2 = x^2 + y^2 - 2x - 2y - 4 = 0$ meet. Find the circle $S_1 + \lambda S_2 = 0$ passing through (1,1).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$.

Final answer

$\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$

Answer key

Use the question number shown on each page.

Q001 $(x-3)^2+(y+2)^2=25$

Q002 $(x+3)^2+(y-1)^2=4$

Q003 $x^2+y^2=16$

Q004 $(x-3)^2+(y-4)^2=36$

Q005 $(x+2)^2+(y-6)^2=12$

Q006 $x^2+y^2=25$

Q007 A

Q008 A

Q009 D

Q010 B

Q011 B

Q012 A

Q013 circle; centre $(3, -1)$, radius 2

Q014 circle; centre $(2, 3)$, radius 3

Q015 circle; centre $(2, -4)$, radius 4

Q016 circle; centre $(2, 0)$, radius 2

Q017 circle; centre $(-1, 2)$, radius 5

Q018 circle; centre $(-2, 3)$, radius $\sqrt{13}$

Q019 A

Q020 C

Q021 D

Q022 C

Q023 C

Q024 C

Q025 centre $(5, 3)$, radius$^2 = 13$; $(x-5)^2+(y-3)^2=13$

Q026 B

Q027 $(x-2)^2+(y-3)^2=13$

Q028 $(x+3)^2+(y-1)^2=58$

Q029 $(x-1)^2+(y+3)^2=25$

Q030 $(x+4)^2+(y-1)^2=58$

Q031 $(x-2)^2+(y+1)^2=5$

Q032 $x^2+y^2=25$

Q033 C

Q034 A

Q035 D

Q036 A

Q037 B

Q038 A

Q039 $(x-2)^2+(y+1)^2=17$

Q040 $(x-4)^2+(y-1)^2=10$

Q041 $(x-1)^2+(y-3)^2=25$

Q042 $(x+1)^2+(y+2)^2=25$

Q043 $(x+2)^2+(y-1)^2=2$

Q044 $x^2+(y-2)^2=25$

Q045 A

Q046 C

Q047 C

Q048 A

Q049 B

Q050 C

Q051 $x^2+y^2-5x-y+4=0$

Q052 $x^2+y^2-3x-7y+6=0$

Q053 $x^2+y^2-2x-4y=0$

Q054 $x^2+y^2-6y-16=0$

Q055 $x^2+y^2-2x+8y-8=0$

Q056 A

Q057 B

Q058 D

Q059 D

Q060 D

Q061 $x^2+y^2-6x-6y+8=0$; centre $(3, 3)$, radius $\sqrt{\ }$ [illegible]

Q062 A

Q063 $(x-4)^2+(y+2)^2=4$

Q064 $(x+2)^2+(y-3)^2=4$

Q065 $(x+1)^2+(y-4)^2=16$

Q066 $(x-3)^2+(y+4)^2=16$

Q067 $(x-5)^2+(y+4)^2=16$

Q068 $(x+1)^2+(y-6)^2=1$

Q069 $(x-2)^2+(y+5)^2=4$

Q070 $(x+3)^2+(y+4)^2=9$

Q071 B

Q072 B

Q073 B

Q074 D

Q075 D

Q076 D

Q077 B

Q078 C

Q079 $(x+1)^2+(y-1)^2=1$ or $(x+5)^2+(y-5)^2=25$

Q080 $(x-1)^2+(y-1)^2=1$ or $(x-5)^2+(y-5)^2=25$

Q081 $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Q082 $(x+1)^2+(y+1)^2=1$ or $(x+5)^2+(y+5)^2=25$

Q083 $(x-5)^2+(y+5)^2=25$ or $(x-13)^2+(y+13)^2=169$

Q084 C

Q085 D

Q086 C

Q087 C

Q088 B

Q089 $(x-1)^2+(y+1)^2=25$

Q090 $(x+5/2)^2+(y-5/2)^2=25/2$

Q091 $(x-2)^2+(y-3)^2=2$

Q092 $(x-4)^2+(y-2)^2=26$

Q093 D

Q094 C

Q095 C

Q096 D

Q097 $(x+1)^2+y^2=5$; centre $(-1, 0)$, radius $\sqrt{5}$

Q098 C

Q099 $k<-1$ or $k>3$

Q100 $k<-\sqrt{6}$ or $k>\sqrt{6}$

Q101 $k<5/4$

Q102 $k<-1/5$ or $k>3$

Q103 $-1<k<4$

Q104 C

Q105 B

Q106 C

Q107 B

Q108 D

Q109 $0 < k < 4$

Q110 B

Q111 $(1, -1)$ and $(-1, 3)$

Q112 $(0, 2)$ and $(2, 0)$

Q113 $(3, -1)$

Q114 $(-2, 1)$

Q115 $(-2, 1)$ and $(4, -1)$

Q116 C

Q117 D

Q118 C

Q119 D

Q120 B

Q121 0 points

Q122 2 points : $(1, -3)$ and $(3, 1)$

Q123 2 points : $\left(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2\right)$ and $\left(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2\right)$

Q124 0 points

Q125 1 point : $(-2, 0)$

Q126 D

Q127 A

Q128 C

Q129 C

Q130 D

Q131 $-5 < k < 5$

Q132 $r = \sqrt{5}$

Q133 $m = \pm\sqrt{3}$

Q134 $0 < r < \sqrt{10}$

Q135 $m = \pm 4$

Q136 $n < -10$ or $n > 10$

Q137 $-5 <= m <= 5$

Q138 $r = \sqrt{2}$

Q139 $r > \sqrt{17}$

Q140 D

Q141 B

Q142 B

Q143 B

Q144 C

Q145 A

Q146 C

Q147 B

Q148 D

Q149 $0 < r < 2$

Q150 D

Q151 $\sqrt{14}$

Q152 4

Q153 $2\sqrt{5}$

Q154 $3\sqrt{2}$

Q155 2

Q156 A

Q157 C

Q158 D

Q159 B

Q160 D

Q161 $k = \pm 2$

Q162 $k = \pm\sqrt{31}$

Q163 $k = \pm 2$

Q164 $k = 0$ or $k = -2$

Q165 C

Q166 D

Q167 A

Q168 B

Q169 $k = -1$ or $k = 7$

Q170 A

Q171 $4x + 3y = 25$

Q172 $5x + 3y = 34$

Q173 $3x - y = 10$

Q174 $-5x - 2y = 29$

Q175 $-2x + y = 5$

Q176 A

Q177 C

Q178 A

Q179 C

Q180 A

Q181 $2x + y = 5$ and $-x + 2y = 5$

Q182 $-3x + y = 10$ and $x + 3y = 10$

Q183 $-2x + y = 5$ and $x + 2y = 5$

Q184 $y = 1$ and $4x - 3y = 5$

Q185 $-x + 2y = 10$ and $2x + y = 10$

Q186 D

Q187 A

Q188 D

Q189 B

Q190 C

Q191 $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

Q192 B

Q193 intersect at 2 points

Q194 B : touch externally

Q195 E : intersect at 2 points

Q196 C : touch internally

Q197 C : touch internally

Q198 E : intersect at 2 points

Q199 D : one is inside the other

Q200 C

Q201 D

Q202 B

Q203 A

Q204 D

Q205 B

Q206 C

Q207 $(x - 3)^2 + (y - 1)^2 = 45$

Q208 $(x + 3)^2 + (y + 9)^2 = 16$

Q209 $(x - 3)^2 + (y + 9)^2 = 144$

Q210 A

Q211 D

Q212 A

Q213 $(x + 1)^2 + (y + 2)^2 = 9$

Q214 D

Q215 $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)

Q216 $x + y - 1 = 0$

Q217 $12x - 6y - 7 = 0$

Q218 $x = 1/2$

Q219 A

Q220 C

Q221 B

Q222 D

Q223 $(-3, -1)$ and $(-1, 3)$

Q224 $(0, 2)$ and $(8/5, -6/5)$

Q225 $(1, 2)$ and $(2, 1)$

Q226 B

Q227 C

Q228 B

Q229 $(-4, 3)$ and $(3, -4)$

Q230 C

Q231 $x^2 + y^2 + 2x + 6y - 15 = 0$

Q232 $x^2 + y^2 + 4x + 2y = 0$

Q233 $x^2 + y^2 + 10x + 20y - 39 = 0$

Q234 $x^2 + y^2 + 24x + 32y - 5 = 0$

Q235 $x^2 + y^2 - 10x - 6y + 24 = 0$

Q236 D

Q237 A

Q238 D

Q239 C

Q240 B

Q241 $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Q242 A

Quiz MCQ 1 B

Quiz MCQ 2 D

Quiz MCQ 3 A

Quiz MCQ 4 D

Quiz WRITTEN 5 $(x-1)^2 + (y+2)^2 = 25$

Quiz WRITTEN 6 $(4,\ 3)\,,\ (-4,\ 3)$

Quiz WRITTEN 7 $3x + 4y = 25$

Quiz WRITTEN 8 $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$